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(54) **DECOY-RESISTANT INTERLEUKIN 18 ARMORED CELLS AND RELATED METHODS**

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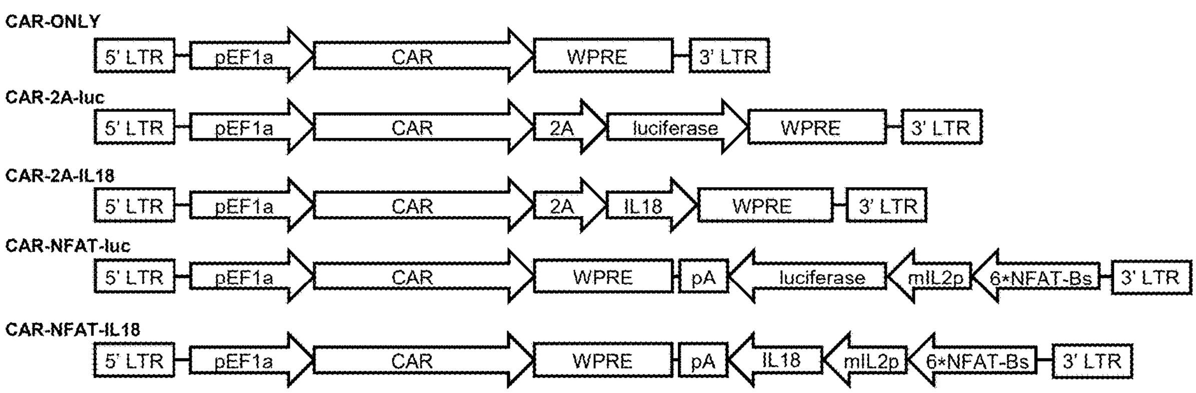
(52) **U.S. Cl.**

CPC ***C07K 14/54*** (2013.01); ***A61K 40/11*** (2025.01); ***A61K 40/31*** (2025.01); ***A61K 40/35*** (2025.01); ***A61K 40/4204*** (2025.01); ***A61K 40/4234*** (2025.01); ***A61P 35/00*** (2018.01); ***C07K 16/2863*** (2013.01); ***C12N 5/0636*** (2013.01); *C07K 2319/02* (2013.01); *C12N 2510/00* (2013.01)

(57) **ABSTRACT**

Provided are nucleic acids that encode decoy-resistant interleukin 18 (DR-18) polypeptides, as well as vectors and cells comprising such nucleic acids and cells that comprise such vectors. Cells encoding DR-18 polypeptides may be referred to as DR-18 armored cells. The nucleic acids may further encode a chimeric antigen receptor (CAR) or multiple CARs, including CARs that bind to antigens described herein. Methods are also provided, including methods of making the nucleic acids, vectors, and/or cells, as well as methods of use, such as employing the nucleic acids, vectors, and/or cells, in the treatment of a subject having cancer.

Specification includes a Sequence Listing.



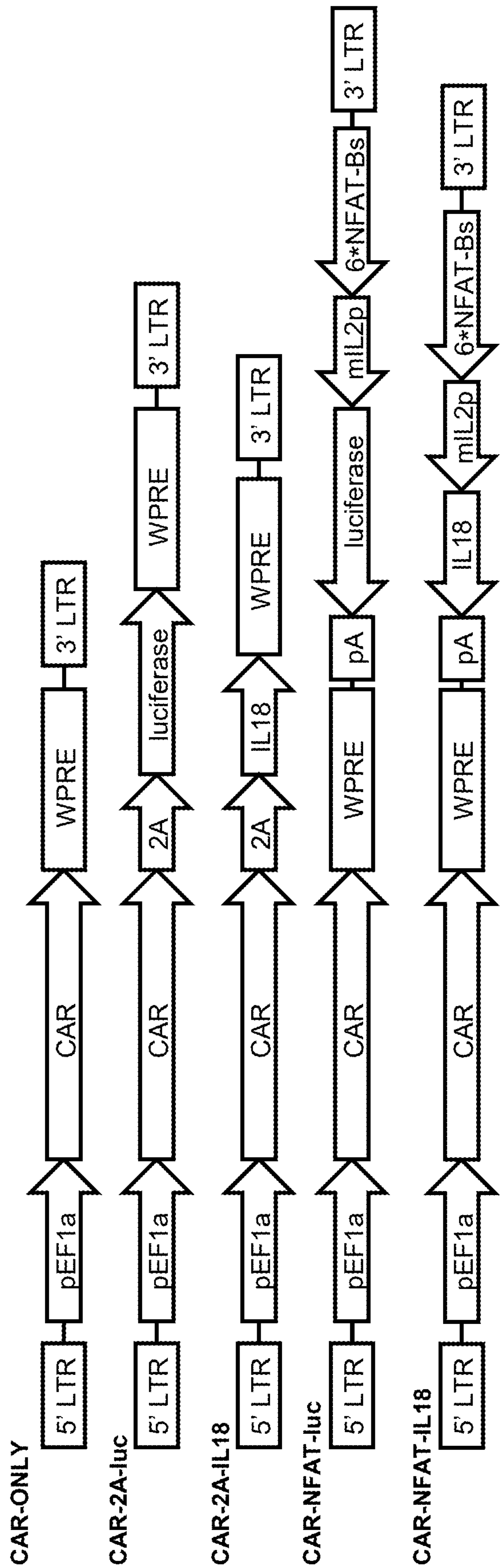

CAR-ONLY
5' LTR
pEF1a
CAR
WPRE
3' LTR
CAR-2A-luc
5' LTR
pEF1a
CAR
2A
luciferase
WPRE
3' LTR
CAR-2A-IL18
5' LTR
pEF1a
CAR
2A
IL18
WPRE
3' LTR
CAR-NFAT-luc
5' LTR
pEF1a
CAR
WPRE
pA
luciferase
mIL2p
6*NFAT-Bs
3' LTR
CAR-NFAT-IL18
5' LTR
pEF1a
CAR
WPRE
pA
IL18
mIL2p
6*NFAT-Bs
3' LTR

FIG. 1

DECOY-RESISTANT INTERLEUKIN 18 ARMORED CELLS AND RELATED METHODS

CROSS-REFERENCE

[0001] This application claims benefit of U.S. Provisional Patent Application No. 63/562,482 filed Mar. 7, 2024, which application is incorporated herein by reference in its entirety.

REFERENCE TO SEQUENCE LISTING SUBMITTED ELECTRONICALLY

[0002] The sequence listing of the present application is submitted electronically with a file name "ST-016-US1 Seqlist.xml", creation date of Mar. 5, 2025, and a size of 391 kb. This sequence listing submitted is part of the specification and is herein incorporated by reference in its entirety.

Introduction

[0003] According to the National Cancer Institute (NCI) Surveillance, Epidemiology, and End Results Program (SEER), approximately 606,520 people will have died of cancer in the United States (US) in 2020. Despite advances in cancer prevention, diagnosis, and treatment, an estimated 1,806,590 people in the US will have been diagnosed with a new cancer in 2020. Thus, cancer remains a continuing public health concern of major proportions.

[0004] Genetically modified T cells expressing chimeric antigen receptors (CARs) so far have mostly failed in the treatment of solid tumors owing to a number of limitations, including an immunosuppressive tumor microenvironment and insufficient CAR T cell activation and persistence. See Zimmerman et al. Cancers (Basel). 2020 February; 12(2): 375.

SUMMARY

[0005] Provided are nucleic acids that encode decoy-resistant interleukin 18 (DR-18) polypeptides, as well as vectors and cells comprising such nucleic acids and cells that comprise such vectors. Cells encoding DR-18 polypeptides may be referred to as DR-18 armored cells. The nucleic acids may further encode a chimeric antigen receptor (CAR) or multiple CARs, including CARs that bind to antigens described herein. Methods are also provided, including methods of making the nucleic acids, vectors, and/or cells, as well as methods of use, such as employing the nucleic acids, vectors, and/or cells, in the treatment of a subject having cancer.

[0006] Aspects of the present disclosure include a nucleic acid comprising a sequence encoding a decoy-resistant interleukin 18 (DR-18) polypeptide and a chimeric antigen receptor (CAR) sequence encoding a CAR, wherein the CAR is selected from: i) an anti-5T4 CAR comprising an antigen binding domain (ABD) that specifically binds 5T4; ii) an anti-AFP CAR comprising an ABD that specifically binds AFP; iii) an anti-ALPP CAR comprising an ABD that specifically binds ALPP; iv) an anti-ALPPL2 CAR comprising an ABD that specifically binds ALPPL2; v) an anti-APN (CD13) CAR comprising an ABD that specifically binds APN (CD13); vi) an anti-APRIL CAR comprising an ABD that specifically binds APRIL; vii) an anti-AXL CAR comprising an ABD that specifically binds AXL; viii) an anti-B7-H3 CAR comprising an ABD that specifically binds B7-H3; ix) an anti-B7-H4 CAR comprising an ABD that specifically binds B7-H4; x) an anti-BAFF-R CAR comprising an ABD that specifically binds BAFF-R; xi) an anti-BAFF-R CAR comprising an ABD that specifically binds BAFF-R; xii) an anti-BCMA CAR comprising an ABD that specifically binds BCMA; xiii) an anti-BSG CAR comprising an ABD that specifically binds BSG; xiv) an anti-CAIX CAR comprising an ABD that specifically binds CAIX; xv) an anti-CD123 CAR comprising an ABD that specifically binds CD123; xvi) an anti-CD133 CAR comprising an ABD that specifically binds CD133; xvii) an anti-CD138 CAR comprising an ABD that specifically binds CD138; xviii) an anti-CD19 CAR comprising an ABD that specifically binds CD19; xix) an anti-CD1A CAR comprising an ABD that specifically binds CD1A; xx) an anti-CD2 CAR comprising an ABD that specifically binds CD2; xxi) an anti-CD20 CAR comprising an ABD that specifically binds CD20; xxii) an anti-CD22 CAR comprising an ABD that specifically binds CD22; xxiii) an anti-CD24 CAR comprising an ABD that specifically binds CD24; xxiv) an anti-CD276 CAR comprising an ABD that specifically binds CD276; xxv) an anti-CD3 CAR comprising an ABD that specifically binds CD3; xxvi) an anti-CD30 CAR comprising an ABD that specifically binds CD30; xxvii) an anti-CD33 CAR comprising an ABD that specifically binds CD33; xxviii) an anti-CD38 CAR comprising an ABD that specifically binds CD38; xxix) an anti-CD4 CAR comprising an ABD that specifically binds CD4; xxx) an anti-CD43 CAR comprising an ABD that specifically binds CD43; xxxi) an anti-CD44 CAR comprising an ABD that specifically binds CD44; xxxii) an anti-CD44v6 CAR comprising an ABD that specifically binds CD44v6; xxxiii) an anti-CD45 CAR comprising an ABD that specifically binds CD45; xxxiv) an anti-CD5 CAR comprising an ABD that specifically binds CD5; xxxv) an anti-CD56 CAR comprising an ABD that specifically binds CD56; xxxvi) an anti-CD7 CAR comprising an ABD that specifically binds CD7; xxxvii) an anti-CD70 CAR comprising an ABD that specifically binds CD70; xxxviii) an anti-CD72 CAR comprising an ABD that specifically binds CD72; xxxix) an anti-CD83 CAR comprising an ABD that specifically binds CD83; xl) an anti-CD84 CAR comprising an ABD that specifically binds CD84; xli) an anti-CD99 CAR comprising an ABD that specifically binds CD99; xlii) an anti-CDCP1 CAR comprising an ABD that specifically binds CDCP1; xliii) an anti-CDH17 CAR comprising an ABD that specifically binds CDH17; xliv) an anti-CEA CAR comprising an ABD that specifically binds CEA; xlv) an anti-CEACAM5 CAR comprising an ABD that specifically binds CEACAM5; xlvi) an anti-CLDN18.2 CAR comprising an ABD that specifically binds CLDN18.2; xlvii) an anti-CLDN6 CAR comprising an ABD that specifically binds CLDN6; xlviii) an anti-CLEC4K (CD207) CAR comprising an ABD that specifically binds CLEC4K (CD207); xlix) an anti-CLL-1 CAR comprising an ABD that specifically binds CLL-1; l) an anti-CSPG4 CAR comprising an ABD that specifically binds CSPG4; li) an anti-DLL3 CAR comprising an ABD that specifically binds DLL3; lii) an anti-DR5 CAR comprising an ABD that specifically binds DR5; liii) an anti-EBV Protein CAR comprising an ABD that specifically binds EBV Protein; liv) an anti-EGFR CAR comprising an ABD that specifically binds EGFR; lv) an anti-EGFRvIII CAR comprising an ABD that specifically binds EGFRvIII; lvi) an anti-EMR1 CAR comprising an ABD that specifically binds EMR1; lvii) an anti-enkephalinase

(CD10) CAR comprising an ABD that specifically binds enkephalinase (CD10); lviii) an anti-EpCAM CAR comprising an ABD that specifically binds EpCAM; lix) an anti-EphA2 CAR comprising an ABD that specifically binds EphA2; lx) an anti-EphA3 CAR comprising an ABD that specifically binds EphA3; lxi) an anti-FAP CAR comprising an ABD that specifically binds FAP; lxii) an anti-FGFR4 CAR comprising an ABD that specifically binds FGFR4; lxiii) an anti-FLT3 CAR comprising an ABD that specifically binds FLT3; lxiv) an anti-FOLR1 CAR comprising an ABD that specifically binds FOLR1; lxv) an anti-FSHR CAR comprising an ABD that specifically binds FSHR; lxvi) an anti-GC-C CAR comprising an ABD that specifically binds GC-C; lxvii) an anti-GD2 CAR comprising an ABD that specifically binds GD2; lxviii) an anti-GFRA4 CAR comprising an ABD that specifically binds GFRA4; lxix) an anti-Globo H CAR comprising an ABD that specifically binds Globo H; lxx) an anti-GM2 (ganglioside M2) CAR comprising an ABD that specifically binds GM2 (ganglioside M2); lxxi) an anti-gp100 CAR comprising an ABD that specifically binds gp100; lxxii) an anti-GPC1 CAR comprising an ABD that specifically binds GPC1; lxxiii) an anti-GPC2 CAR comprising an ABD that specifically binds GPC2; lxxiv) an anti-GPC3 CAR comprising an ABD that specifically binds GPC3; lxxv) an anti-HAAH (ASPH) CAR comprising an ABD that specifically binds HAAH (ASPH); lxxvi) an anti-HER2 CAR comprising an ABD that specifically binds HER2; lxxvii) an anti-HGF CAR comprising an ABD that specifically binds HGF; lxxviii) an anti-HIV envelope protein gp120 CAR comprising an ABD that specifically binds HIV envelope protein gp120; lxxix) an anti-HIV-1 pol CAR comprising an ABD that specifically binds HIV-1 pol; lxxx) an anti-HLA-A2 CAR comprising an ABD that specifically binds HLA-A2; lxxxi) an anti-HLA-G CAR comprising an ABD that specifically binds HLA-G; lxxxii) an anti-HSP70 heat-shock protein CAR comprising an ABD that specifically binds HSP70 heat-shock protein; lxxxiii) an anti-ICAM-1 CAR comprising an ABD that specifically binds ICAM-1; lxxxiv) an anti-IL-10R CAR comprising an ABD that specifically binds IL-10R; lxxxv) an anti-IL13Rα2 CAR comprising an ABD that specifically binds IL13Rα2; lxxxvi) an anti-IL-13Rα2 CAR comprising an ABD that specifically binds IL-13Rα2; lxxxvii) an anti-integrin αvβ6 CAR comprising an ABD that specifically binds integrin αvβ6; lxxxviii) an anti-ITGB7 CAR comprising an ABD that specifically binds ITGB7; lxxxix) an anti-KKLC1 CAR comprising an ABD that specifically binds KKLC1; xc) an anti-KMA CAR comprising an ABD that specifically binds KMA; xci) an anti-L1CAM CAR comprising an ABD that specifically binds L1CAM; xcii) an anti-Lewis-Y antigen CAR comprising an ABD that specifically binds Lewis-Y antigen; xciii) an anti-LGR5 CAR comprising an ABD that specifically binds LGR5; xciv) an anti-LILRB4 CAR comprising an ABD that specifically binds LILRB4; xcv) an anti-LMP1 CAR comprising an ABD that specifically binds LMP1; xcvi) an anti-MAGEA3 CAR comprising an ABD that specifically binds MAGEA3; xcvii) an anti-M-CSF CAR comprising an ABD that specifically binds M-CSF; xcviii) an anti-MG7 CAR comprising an ABD that specifically binds MG7; xcix) an anti-MICA CAR comprising an ABD that specifically binds MICA; c) an anti-MMP2 CAR comprising an ABD that specifically binds MMP2; ci) an anti-MSLN CAR comprising an ABD that specifically binds MSLN; cii) an anti-MUC1/MUC16 CAR comprising an ABD that specifically binds MUC1/MUC16; ciii) an anti-NCR3LG1 CAR comprising an ABD that specifically binds NCR3LG1; civ) an anti-NECTIN2 CAR comprising an ABD that specifically binds NECTIN2; cv) an anti-nectin-4 CAR comprising an ABD that specifically binds nectin-4; cvi) an anti-NKG2D CAR comprising an ABD that specifically binds NKG2D; cvii) an anti-NKG2DL CAR comprising an ABD that specifically binds NKG2DL; cviii) an anti-NR2F6 CAR comprising an ABD that specifically binds NR2F6; cix) an anti-NY-ESO-1 CAR comprising an ABD that specifically binds NY-ESO-1; cx) an anti-OPCML CAR comprising an ABD that specifically binds OPCML; cxi) an anti-PD-1 CAR comprising an ABD that specifically binds PD-1; cxii) an anti-PDGFRα CAR comprising an ABD that specifically binds PDGFRα; cxiii) an anti-PDL1 CAR comprising an ABD that specifically binds PDL1; cxiv) an anti-PLA2R CAR comprising an ABD that specifically binds PLA2R; cxv) an anti-PR1 CAR comprising an ABD that specifically binds PR1; cxvi) an anti-PSCA CAR comprising an ABD that specifically binds PSCA; cxvii) an anti-PTK7 CAR comprising an ABD that specifically binds PTK7; cxviii) an anti-PVR CAR comprising an ABD that specifically binds PVR; cxix) an anti-ROBO1 CAR comprising an ABD that specifically binds ROBO1; cxx) an anti-ROR1 CAR comprising an ABD that specifically binds ROR1; cxxi) an anti-ROR2 CAR comprising an ABD that specifically binds ROR2; cxxii) an anti-SEMA4A CAR comprising an ABD that specifically binds SEMA4A; cxxiii) an anti-SLAMF7 CAR comprising an ABD that specifically binds SLAMF7; cxxiv) an anti-TAG-72 CAR comprising an ABD that specifically binds TAG-72; cxxv) an anti-TGF-β CAR comprising an ABD that specifically binds TGF-β; cxxvi) an anti-TM4SF1 CAR comprising an ABD that specifically binds TM4SF1; cxxvii) an anti-TRAIL CAR comprising an ABD that specifically binds TRAIL; cxxviii) an anti-TRBC1 CAR comprising an ABD that specifically binds TRBC1; cxxix) an anti-TRBC2 CAR comprising an ABD that specifically binds TRBC2; cxxx) an anti-Trop-2 CAR comprising an ABD that specifically binds Trop-2; cxxxi) an anti-TSHR CAR comprising an ABD that specifically binds TSHR; cxxxii) an anti-TSLPR CAR comprising an ABD that specifically binds TSLPR; cxxxiii) an anti-ULBP1 CAR comprising an ABD that specifically binds ULBP1; cxxxiv) an anti-αvβ3 integrin CAR comprising an ABD that specifically binds αvβ3 integrin; and cxxxv) an anti-K light chain of human immunoglobulin CAR comprising an ABD that specifically binds K light chain of human immunoglobulin.

[0007] Aspects of the present disclosure include a nucleic acid comprising a sequence encoding a decoy-resistant interleukin 18 (DR-18) polypeptide and a chimeric antigen receptor (CAR) sequence encoding a CAR, wherein the CAR is selected from: i) an anti-5T4 CAR comprising a means for specifically binding 5T4; ii) an anti-AFP CAR comprising a means for specifically binding AFP; iii) an anti-ALPP CAR comprising a means for specifically binding ALPP; iv) an anti-ALPPL2 CAR comprising a means for specifically binding ALPPL2; v) an anti-APN (CD13) CAR comprising a means for specifically binding APN (CD13); vi) an anti-APRIL CAR comprising a means for specifically binding APRIL; vii) an anti-AXL CAR comprising a means for specifically binding AXL; viii) an anti-B7-H3 CAR comprising a means for specifically binding B7-H3; ix) an anti-B7-H4 CAR comprising a means for specifically binding B7-H4; x) an anti-BAFF-R CAR comprising a means for specifically binding BAFF-R; xi) an anti-BAFF-R CAR comprising a means for specifically binding BAFF-R; xii) an anti-BCMA CAR comprising a means for specifically binding BCMA; xiii) an anti-BSG CAR comprising a means for specifically binding BSG; xiv) an anti-CAIX CAR comprising a means for specifically binding CAIX; xv) an anti-CD123 CAR comprising a means for specifically binding CD123; xvi) an anti-CD133 CAR comprising a means for specifically binding CD133; xvii) an anti-CD138 CAR comprising a means for specifically binding CD138; xviii) an anti-CD19 CAR comprising a means for specifically binding CD19; xix) an anti-CD1A CAR comprising a means for specifically binding CD1A; xx) an anti-CD2 CAR comprising a means for specifically binding CD2; xxi) an anti-CD20 CAR comprising a means for specifically binding CD20; xxii) an anti-CD22 CAR comprising a means for specifically binding CD22; xxiii) an anti-CD24 CAR comprising a means for specifically binding CD24; xxiv) an anti-CD276 CAR comprising a means for specifically binding CD276; xxv) an anti-CD3 CAR comprising a means for specifically binding CD3; xxvi) an anti-CD30 CAR comprising a means for specifically binding CD30; xxvii) an anti-CD33 CAR comprising a means for specifically binding CD33; xxviii) an anti-CD38 CAR comprising a means for specifically binding CD38; xxix) an anti-CD4 CAR comprising a means for specifically binding CD4; xxx) an anti-CD43 CAR comprising a means for specifically binding CD43; xxxi) an anti-CD44 CAR comprising a means for specifically binding CD44; xxxii) an anti-CD44v6 CAR comprising a means for specifically binding CD44v6; xxxiii) an anti-CD45 CAR comprising a means for specifically binding CD45; xxxiv) an anti-CD5 CAR comprising a means for specifically binding CD5; xxxv) an anti-CD56 CAR comprising a means for specifically binding CD56; xxxvi) an anti-CD7 CAR comprising a means for specifically binding CD7; xxxvii) an anti-CD70 CAR comprising a means for specifically binding CD70; xxxviii) an anti-CD72 CAR comprising a means for specifically binding CD72; xxxix) an anti-CD83 CAR comprising a means for specifically binding CD83; xl) an anti-CD84 CAR comprising a means for specifically binding CD84; xli) an anti-CD99 CAR comprising a means for specifically binding CD99; xlii) an anti-CDCP1 CAR comprising a means for specifically binding CDCP1; xliii) an anti-CDH17 CAR comprising a means for specifically binding CDH17; xliv) an anti-CEA CAR comprising a means for specifically binding CEA; xlv) an anti-CEACAM5 CAR comprising a means for specifically binding CEACAM5; xlvi) an anti-CLDN18.2 CAR comprising a means for specifically binding CLDN18.2; xlvii) an anti-CLDN6 CAR comprising a means for specifically binding CLDN6; xlviii) an anti-CLEC4K (CD207) CAR comprising a means for specifically binding CLEC4K (CD207); xlix) an anti-CLL-1 CAR comprising a means for specifically binding CLL-1; l) an anti-CSPG4 CAR comprising a means for specifically binding CSPG4; li) an anti-DLL3 CAR comprising a means for specifically binding DLL3; lii) an anti-DR5 CAR comprising a means for specifically binding DR5; liii) an anti-EBV Protein CAR comprising a means for specifically binding EBV Protein; liv) an anti-EGFR CAR comprising a means for specifically binding EGFR; lv) an anti-EGFRvIII CAR comprising a means for specifically binding EGFRvIII; lvi) an anti-EMR1 CAR comprising a means for specifically binding EMR1; lvii) an anti-enkephalinase (CD10) CAR comprising a means for specifically binding enkephalinase (CD10); lviii) an anti-EpCAM CAR comprising a means for specifically binding EpCAM; lix) an anti-EphA2 CAR comprising a means for specifically binding EphA2; lx) an anti-EphA3 CAR comprising a means for specifically binding EphA3; lxi) an anti-FAP CAR comprising a means for specifically binding FAP; lxii) an anti-FGFR4 CAR comprising a means for specifically binding FGFR4; lxiii) an anti-FLT3 CAR comprising a means for specifically binding FLT3; lxiv) an anti-FOLR1 CAR comprising a means for specifically binding FOLR1; lxv) an anti-FSHR CAR comprising a means for specifically binding FSHR; lxvi) an anti-GC-C CAR comprising a means for specifically binding GC-C; lxvii) an anti-GD2 CAR comprising a means for specifically binding GD2; lxviii) an anti-GFRA4 CAR comprising a means for specifically binding GFRA4; lxix) an anti-Globo H CAR comprising a means for specifically binding Globo H; lxx) an anti-GM2 (ganglioside M2) CAR comprising a means for specifically binding GM2 (ganglioside M2); lxxi) an anti-gp100 CAR comprising a means for specifically binding gp100; lxxii) an anti-GPC1 CAR comprising a means for specifically binding GPC1; lxxiii) an anti-GPC2 CAR comprising a means for specifically binding GPC2; lxxiv) an anti-GPC3 CAR comprising a means for specifically binding GPC3; lxxv) an anti-HAAH (ASPH) CAR comprising a means for specifically binding HAAH (ASPH); lxxvi) an anti-HER2 CAR comprising a means for specifically binding HER2; lxxvii) an anti-HGF CAR comprising a means for specifically binding HGF; lxxviii) an anti-HIV envelope protein gp120 CAR comprising a means for specifically binding HIV envelope protein gp120; lxxix) an anti-HIV-1 pol CAR comprising a means for specifically binding HIV-1 pol; lxxx) an anti-HLA-A2 CAR comprising a means for specifically binding HLA-A2; lxxxi) an anti-HLA-G CAR comprising a means for specifically binding HLA-G; lxxxii) an anti-HSP70 heat-shock protein CAR comprising a means for specifically binding HSP70 heat-shock protein; lxxxiii) an anti-ICAM-1 CAR comprising a means for specifically binding ICAM-1; lxxxiv) an anti-IL-10R CAR comprising a means for specifically binding IL-10R; lxxxv) an anti-IL13Rα2 CAR comprising a means for specifically binding IL13Rα2; lxxxvi) an anti-IL-13Rα2 CAR comprising a means for specifically binding IL-13Rα2; lxxxvii) an anti-integrin αvβ6 CAR comprising a means for specifically binding integrin αvβ6; lxxxviii) an anti-ITGB7 CAR comprising a means for specifically binding ITGB7; lxxxix) an anti-KKLC1 CAR comprising a means for specifically binding KKLC1; xc) an anti-KMA CAR comprising a means for specifically binding KMA; xci) an anti-L1CAM CAR comprising a means for specifically binding L1CAM; xcii) an anti-Lewis-Y antigen CAR comprising a means for specifically binding Lewis-Y antigen; xciii) an anti-LGR5 CAR comprising a means for specifically binding LGR5; xciv) an anti-LILRB4 CAR comprising a means for specifically binding LILRB4; xcv) an anti-LMP1 CAR comprising a means for specifically binding LMP1; xcvi) an anti-MAGEA3 CAR comprising a means for specifically binding MAGEA3; xcvii) an anti-M-CSF CAR comprising a means for specifically binding M-CSF; xcviii) an anti-MG7 CAR comprising a means for specifically binding MG7; xcix) an anti-MICA CAR comprising a means for specifically binding MICA; c) an anti-MMP2 CAR comprising a means for specifically binding MMP2; ci) an anti-MSLN CAR comprising a means for specifically binding MSLN; cii) an anti-MUC1/MUC16 CAR comprising a means for specifically binding MUC1/MUC16; ciii) an anti-NCR3LG1 CAR comprising a means for specifically binding NCR3LG1; civ) an anti-NECTIN2 CAR comprising a means for specifically binding NECTIN2; cv) an anti-nectin-4 CAR comprising a means for specifically binding nectin-4; cvi) an anti-NKG2D CAR comprising a means for specifically binding NKG2D; cvii) an anti-NKG2DL CAR comprising a means for specifically binding NKG2DL; cviii) an anti-NR2F6 CAR comprising a means for specifically binding NR2F6; cix) an anti-NY-ESO-1 CAR comprising a means for specifically binding NY-ESO-1; cx) an anti-OPCML CAR comprising a means for specifically binding OPCML; cxi) an anti-PD-1 CAR comprising a means for specifically binding PD-1; cxii) an anti-PDGFRα CAR comprising a means for specifically binding PDGFRα; cxiii) an anti-PDL1 CAR comprising a means for specifically binding PDL1; cxiv) an anti-PLA2R CAR comprising a means for specifically binding PLA2R; cxv) an anti-PR1 CAR comprising a means for specifically binding PR1; cxvi) an anti-PSCA CAR comprising a means for specifically binding PSCA; cxvii) an anti-PTK7 CAR comprising a means for specifically binding PTK7; cxviii) an anti-PVR CAR comprising a means for specifically binding PVR; cxix) an anti-ROBO1 CAR comprising a means for specifically binding ROBO1; cxx) an anti-ROR1 CAR comprising a means for specifically binding ROR1; cxxi) an anti-ROR2 CAR comprising a means for specifically binding ROR2; cxxii) an anti-SEMA4A CAR comprising a means for specifically binding SEMA4A; cxxiii) an anti-SLAMF7 CAR comprising a means for specifically binding SLAMF7; cxxiv) an anti-TAG-72 CAR comprising a means for specifically binding TAG-72; cxxv) an anti-TGF-β CAR comprising a means for specifically binding TGF-β; cxxvi) an anti-TM4SF1 CAR comprising a means for specifically binding TM4SF1; cxxvii) an anti-TRAIL CAR comprising a means for specifically binding TRAIL; cxxviii) an anti-TRBC1 CAR comprising a means for specifically binding TRBC1; cxxix) an anti-TRBC2 CAR comprising a means for specifically binding TRBC2; cxxx) an anti-Trop-2 CAR comprising a means for specifically binding Trop-2; cxxxi) an anti-TSHR CAR comprising a means for specifically binding TSHR; cxxxii) an anti-TSLPR CAR comprising a means for specifically binding TSLPR; cxxxiii) an anti-ULBP1 CAR comprising a means for specifically binding ULBP1; cxxxiv) an anti-αvβ3 integrin CAR comprising a means for specifically binding αvβ3 integrin; and cxxxv) an anti-K light chain of human immunoglobulin CAR comprising a means for specifically binding K light chain of human immunoglobulin

[0008] In some embodiments, the CAR is selected from: a) inaticabtagene autoleucel; b) actalycabtagene autoleucel; c) relmacabtagene autoleucel; d) lisocabtagene maraleucel; e) brexucabtagene autoleucel; f) axicabtagene ciloleucel; g) tisagenlecleucel; h) obecabatagene autoleucel; i) azercabtagene zapreleucel; j) rapcabtagene autoleucel; and k) idecabtagene vicleucel.

[0009] In some embodiments, the CAR is an anti-EGFRvIII CAR comprising an ABD that specifically binds EGFRvIII. In some embodiments, the CAR is an anti-EGFRvIII CAR comprising a means for specifically binding EGFRvIII. In some embodiments, the anti-EGFRvIII CAR comprises an antigen binding domain comprising one or more of light chain complementary determining region 1 (LC CDR1), light chain complementary determining region 2 (LC CDR2), and light chain complementary determining region 3 (LC CDR3) of SEQ ID NO: 110, and one or more of heavy chain complementary determining region 1 (HC CDR1), heavy chain complementary determining region 2 (HC CDR2), and heavy chain complementary determining region 3 (HC CDR3) of SEQ ID NO:111. In some embodiments, the antigen binding domain comprises LC CDR1 (SEQ ID NO:112), LC CDR2 (SEQ ID NO:113), and LC CDR3 (SEQ ID NO:114); and HC CDR1 (SEQ ID NO:115), HC CDR2 (SEQ ID NO:116), and HC CDR3 (SEQ ID NO:117). In some embodiments, the antigen binding domain comprises a light chain variable region comprising sequence SEQ ID NO:110 and a heavy chain variable region comprising sequence SEQ ID NO:111.

[0010] In some embodiments, the DR-18 polypeptide comprises the following amino acid sequence: XFGKXESXLSVIRNLNDQVLFIDQGNRPLFEDMTDSDXRDNAPRTIFIISXYXDXXXRXXAVTISV KXEKISTLSXXNKIISFKEMNPPDNIKDTKSDIIFFXRXVPGH XXKXQFESSSYEGYFLAXEKERD LFKLILKKEDELGDRSIMFTXQXED (SEQ ID NO: 66), wherein the X at pos. 1 is Y, R or H; the X at pos. 5 is L, H, I or Y; the X at pos. 8 is K, Q or R; the X at pos. 38 is C or S; the X at pos. 51 is M, T, K, D, N, E or R; the X at pos. 53 is K, R, G, S or T; the X at pos. 55 is S, K or R; the X at pos. 56 is Q, E, A, R, V, G, K, L or R; the X at pos. 57 is P, L, G, A or K; the X at pos. 59 is G, A or T; the X at pos. 60 is M, K, Q, R or L; the X at pos. 68 is C, S, G, A, V, D, E or N; the X at pos. 76 is C or S; the X at pos. 77 is E or D; the X at pos. 103 is Q, E, K, P, A or R; the X at pos. 105 is S, D, N, R, K or A; the X at pos. 110 is D, K, H, N, Q, E, S or G; the X at pos. 111 is N, H, Y, D, R, S or G; the X at pos. 113 is M, V, R, T or K; the X at pos. 127 is C or S; the X at pos. 153 is V, I, T or A; and the X at pos. 155 is N, K or H.

[0011] In some embodiments, the sequence encoding the DR-18 polypeptide is codon optimized for expression in humans. In some embodiments, the sequence encoding the DR-18 polypeptide comprises a DR-18 encoding sequence selected from SEQ ID NOs: 67-84.

[0012] In some embodiments, the nucleic acid is configured for in vivo expression of the DR-18 polypeptide and CAR.

[0013] In some embodiments, the nucleic acid further comprises a sequence encoding a cytokine selected from: interleukin 2, interleukin 21, interleukin 23, interleukin 27, interleukin 33, tumor necrosis factor (TNF), TNF ligand superfamily member 15, interferon (IFN) alpha, IFN beta, and IFN gamma.

[0014] In some embodiments, the nucleic acid comprises a plurality of CAR sequences encoding a plurality of different CARs that bind different antigens. In some embodiments, the plurality of CAR sequences encode a plurality of different CARs that bind to a combination of antigens selected from: CD19 and CD20, CD19 and CD22, BCMA and CD19, CD33 and CLL-1, CD19 and CD276, CD19 and CD20 and CD22, CD19 and IL15R, IL15R and MUC16, IL-15Rα and NKG2D, GPC3 and IL15R, CD16a and CD19 and IL-15Rα, BCMA and CD16a and NKp46, CD123 and CD33, CLDN18.2 and NKG2D, CD20 and CD22, BCMA and SLAMF7, CD19 and CD70, CD19 and IL18R1, GD2 and IL15R, CD70 and IL15R, CD19 and CD8, CD20 and CD79A, CD19 and CD22 and CD8, CD19 and EBV Protein, CD19 and CD7, BCMA and CD38, IL-12 and MUC16, CD33 and FLT3, CD19 and IL-2Rβ, BCMA and CD70, EGFR and IL-13Rα2, HER2 and IL15R, MICA and MICB, BCMA and TACI, GPC3 and IL-2Rβ, CD19 and DR5, CD123 and CD33 and CLL-1, CD3 and CD7, CEACAM5 and CEACAM6, CD20 and CD22 and CD38, CD19 and CLDN18.2, AGRE2 and CLL-1, CD123 and TIM3, CLDN18.2 and PDL1, CTLA4 and MSLN and PD-1, STEAP2 and TGFBR2, CEA and IL-15Rα and IL-21R, CD123 and CD33 and CD38 and CLL-1, CD133 and EGFR, BCMA and HPK1, CD8A and NY-ESO-1, CD38 and DR5, CD19 and EGFR, BCMA and CD19 and HER2 and Trop-2, CD19 and MUC1, PDL1 and VEGFR1, CD123 and CD33 and CD38 and CD56 and CLL-1 and MUC1, BCMA and CD138 and CD19 and CD38, BCMA and CD16a and IL-15Rα, NCR2 and NKG2D, CD38 and CLL-1, and CD276 and FGFR4.

[0015] Aspects of the present disclosure include an expression vector comprising such nucleic acids. In some embodiments, the expression vector comprises, in operable linkage to the sequence encoding the DR-18 polypeptide, a sequence encoding a signal peptide selected from: Human OSM signal peptide, VSV-G signal peptide, Mouse Ig Kappa signal peptide, Mouse Ig Heavy signal peptide, BM40 signal peptide, Secrecon signal peptide, Human IgKVIII signal peptide, CD33 signal peptide, tPA signal peptide, Human Chymotrypsinogen signal peptide, Human trypsinogen-2 signal peptide, Human IL-2 signal peptide, *Gaussia* luc signal peptide, Albumin (HSA) signal peptide, Influenza Haemagglutinin signal peptide, Human insulin signal peptide, Silkworm Fibroin LC signal peptide, Interleukin-18-binding protein signal peptide, Interleukin-12 subunit beta signal peptide, Interleukin-12 subunit alpha signal peptide, Interleukin-18 signal peptide, and CD8 signal peptide. In some embodiments, the expression vector comprises, in operable linkage to the sequence encoding the DR-18 polypeptide, a promoter sequence selected from: cytomegalovirus (CMV) promoter, elongation factor 1 alpha (EF1a) promoter, simian virus 40 (SV40) promoter, phosphoglycerate kinase (PGK) ubiquitin C (UBC) promoter, human beta actin promoter, CAG promoter, LC immunoglobulin promoter, HC immunoglobulin promoter, herpes simplex virus thymidine kinase promoter, and spleen focus-forming virus (SFFV) promoter. In some embodiments, the expression vector comprises, in operable linkage to the sequence encoding the DR-18 polypeptide, a human EF1a promoter. In some embodiments, the sequence encoding the DR-18 polypeptide and the sequence encoding the CAR are operably linked by a multicistronic element. In some embodiments, the multicistronic element is an internal ribosome entry site. In some embodiments, the multicistronic element is a sequence encoding 2A peptide selected from: T2A, P2A, E2A, and F2A.

[0016] Aspects of the present disclosure include a cell comprising such nucleic acids or such expression vectors. In some embodiments, the cell is selected from: a T cell, a Naïve T cell (TN), stem cell memory T cell (TSCM), a central memory T cell (TCM), a resident memory T cell (TRM), an effector T cell (TEFF), an effector memory cell (TEM), an alpha-beta (αβ) T cell, a gamma-delta (γδ) T cell, a Natural Killer (NK) cell, a CD4+ T cell, a CD4+Th1 cell, a CD4+Th2 cell, a CD4+Th9 cell, a CD4+Th17 cell, a CD4+Th22 cell, a CD4+T Regulatory Cell (Treg), a CD4+T follicular helper cell (Tfh), a resting Treg cell, an effector Treg cell, an effector memory Treg cell, a CD8+ T cell, a CD8+Tc1 cell, a CD8+Tc2 cell, a CD8+Tc9 cell, a CD8+ Th17 cell, a CD8+Treg cell, a macrophage, a B Cell, a dendritic cell, a stem cell, a hematopoietic stem cell (HSC), and an induced pluripotent stem cell (iPS cell).

[0017] Aspects of the present disclosure include a method of treating a subject for cancer, the method comprising administering to the subject an effective amount of such nucleic acids, expression vectors, or cells. In some embodiments, the cancer is selected from: Acute Lymphoblastic Leukemia (ALL), Acute Myeloid Leukemia (AML), Adrenocortical Carcinoma, an AIDS-Related Cancer, Anal Cancer, Appendix Cancer, Astrocytomas, Atypical Teratoid/Rhabdoid Tumor, Basal Cell Carcinoma, Bile Duct Cancer, Bladder Cancer, Bone Cancer, Brain Stem Glioma, Brain Tumor, Breast Cancer, Bronchial Tumors, Burkitt Lymphoma, Carcinoid Tumor, Carcinoma of Unknown Primary, Cardiac (Heart) Tumors, Central Nervous System Tumor, Cervical Cancer, Childhood Cancers, Chordoma, Chronic Lymphocytic Leukemia (CLL), Chronic Myelogenous Leukemia (CML), Chronic Myeloproliferative Neoplasms, Colon Cancer, Colorectal Cancer, Craniopharyngioma, Cutaneous T-Cell Lymphoma, Duct Cancer, Ductal Carcinoma In Situ (DCIS), Embryonal Tumors, Endometrial Cancer, Ependymoma, Esophageal Cancer, Esthesioneuroblastoma, Ewing Sarcoma, Extracranial Germ Cell Tumor, Extragonadal Germ Cell Tumor, Extrahepatic Bile Duct Cancer, Eye Cancer, Fibrous Histiocytoma of Bone, Gallbladder Cancer, Gastric (Stomach) Cancer, Gastrointestinal Carcinoid Tumor, Gastrointestinal Stromal Tumors (GIST), Germ Cell Tumor, Gestational Trophoblastic Disease, Glioma, Hairy Cell Leukemia, Head and Neck Cancer, Heart Cancer, Hepatocellular (Liver) Cancer, Histiocytosis, Hodgkin Lymphoma, Hypopharyngeal Cancer, Intraocular Melanoma, Islet Cell Tumors, Kaposi Sarcoma, Kidney Cancer, Langerhans Cell Histiocytosis, Laryngeal Cancer, Leukemia, Lip and Oral Cavity Cancer, Liver Cancer (Primary), Lobular Carcinoma In Situ (LCIS), Lung Cancer, Lymphoma, Macroglobulinemia, Male Breast Cancer, Malignant Fibrous Histiocytoma of Bone and Osteosarcoma, Melanoma, Merkel Cell Carcinoma, Mesothelioma, Metastatic Squamous Neck Cancer with Occult Primary, Midline Tract Carcinoma Involving NUT Gene, Mouth Cancer, Multiple Endocrine Neoplasia Syndromes, Multiple Myeloma/Plasma Cell Neoplasm, Mycosis Fungoides, Myelodysplastic Syndromes, Myelodysplastic/Myeloproliferative Neoplasms, Myelogenous Leukemia, Myeloid Leukemia, Myeloproliferative Neoplasms, Nasal Cavity and Paranasal Sinus Cancer, Nasopharyngeal Cancer, Neuroblastoma, Non-Hodgkin Lymphoma, Non-Small Cell Lung Cancer, Oral Cancer, Oral Cavity Cancer, Oropharyngeal Cancer, Osteosarcoma and Malignant Fibrous Histiocytoma of Bone, Ovarian Cancer, Pancreatic Cancer, Pancreatic Neuroendocrine Tumors, Papillomatosis, Paraganglioma, Paranasal Sinus and Nasal Cavity Cancer, Parathyroid Cancer, Penile Cancer, Pharyngeal Cancer, Pheochromocytoma, Pituitary Tumor, Pleuropulmonary Blastoma, Primary Central Nervous System (CNS) Lymphoma, Prostate Cancer, Rectal Cancer, Renal Cell (Kidney) Cancer, Renal Pelvis and Ureter, Transitional Cell Cancer, Retinoblastoma, Rhabdomyosarcoma, Salivary Gland Cancer, Sarcoma, Sezary Syndrome, Skin Cancer, Small Cell Lung Cancer, Small

Intestine Cancer, Soft Tissue Sarcoma, Squamous Cell Carcinoma, Squamous Neck Cancer, Stomach (Gastric) Cancer, T-Cell Lymphoma, Testicular Cancer, Throat Cancer, Thymoma and Thymic Carcinoma, Thyroid Cancer, Transitional Cell Cancer of the Renal Pelvis and Ureter, Ureter and Renal Pelvis Cancer, Urethral Cancer, Uterine Cancer, Uterine Sarcoma, Vaginal Cancer, Vulvar Cancer, Waldenstrom Macroglobulinemia, and Wilms Tumor.

[0018] In some embodiments, the subject does not undergo lymphodepletion during or prior to the administering. In some embodiments, the amount of the nucleic acid, the expression vector, or the cell administered to the subject is less than the amount of a corresponding nucleic acid, expression vector, or cell comprising the sequence encoding the CAR administered to the subject to treat the cancer without the sequence encoding the DR-18 polypeptide. In some embodiments, the administering comprises delivering multiple doses of the nucleic acid, the expression vector, or the cell to the subject and the dosing frequency is less than the dosing frequency of a corresponding nucleic acid, expression vector, or cell comprising the sequence encoding the CAR administered to the subject to treat the cancer without the sequence encoding the DR-18 polypeptide.

[0019] Aspects of the present disclosure include a method comprising contacting a human cell with the described nucleic acid or expression vector under conditions sufficient for delivery of the nucleic acid or expression vector into the human cell. In some embodiments, the contacting is performed under conditions sufficient for the expression by the human cell of the CAR from the CAR encoding sequence. In some embodiments, the contacting is performed under conditions sufficient for the expression by the human cell of the CAR from the CAR coding sequence, the DR-18 polypeptide from the DR-18 polypeptide coding sequence, or both.

[0020] Aspects of the present disclosure include a nucleic acid, or expression vector, comprising or consisting of the nucleic acid, comprising a sequence encoding a decoy-resistant interleukin 18 (DR-18) polypeptide, wherein the sequence encoding the DR-18 polypeptide is codon optimized for expression in human cells. In some embodiments, the nucleic acid is configured for in vivo expression of the DR-18 polypeptide. In some embodiments, the DR-18 polypeptide comprises the following amino acid sequence: XFGKXESXLSVIRNLNDQVLFIDQGNR-PLFEDMTDSDXRDNAPRTI-FIISXYXDXXXRXXAVTISV KXEKISTLSXXNKIIS-FKEMNPPDNIKDTKSDIIFFXRXVPGHXXKXQFESSS YEGYFLAXEKERD LFKLILKKEDELGDR-SIMFTXQXED (SEQ ID NO: 66), wherein the X at pos. 1 is Y, R or H; the X at pos. 5 is L, H, I or Y; the X at pos. 8 is K, Q or R; the X at pos. 38 is C or S; the X at pos. 51 is M, T, K, D, N, E or R; the X at pos. 53 is K, R, G, S or T; the X at pos. 55 is S, K or R; the X at pos. 56 is Q, E, A, R, V, G, K, L or R; the X at pos. 57 is P, L, G, A or K; the X at pos. 59 is G, A or T; the X at pos. 60 is M, K, Q, R or L; the X at pos. 68 is C, S, G, A, V, D, E or N; the X at pos. 76 is C or S; the X at pos. 77 is E or D; the X at pos. 103 is Q, E, K, P, A or R; the X at pos. 105 is S, D, N, R, K or A; the X at pos. 110 is D, K, H, N, Q, E, S or G; the X at pos. 111 is N, H, Y, D, R, S or G; the X at pos. 113 is M, V, R, T or K; the X at pos. 127 is C or S; the X at pos. 153 is V, I, T or A; and the X at pos. 155 is N, K or H.

[0021] In some embodiments, the sequence encoding the DR-18 polypeptide comprises a DR-18 encoding sequence selected from SEQ ID NOs: 67-84.

[0022] In some embodiments, the nucleic acid or expression vector further comprises a sequence encoding a cytokine selected from: interleukin 2, interleukin 21, interleukin 23, interleukin 27, interleukin 33, tumor necrosis factor (TNF), TNF ligand superfamily member 15, interferon (IFN) alpha, IFN beta, and IFN gamma.

[0023] In some embodiments, the nucleic acid or expression vector further comprises, in operable linkage to the sequence encoding the DR-18 polypeptide, a sequence encoding a signal peptide selected from: Human OSM signal peptide, VSV-G signal peptide, Mouse Ig Kappa signal peptide, Mouse Ig Heavy signal peptide, BM40 signal peptide, Secrecon signal peptide, Human IgKVIII signal peptide, CD33 signal peptide, tPA signal peptide, Human Chymotrypsinogen signal peptide, Human trypsinogen-2 signal peptide, Human IL-2 signal peptide, *Gaussia* luc signal peptide, Albumin (HSA) signal peptide, Influenza Haemagglutinin signal peptide, Human insulin signal peptide, Silkworm Fibroin LC signal peptide, Interleukin-18-binding protein signal peptide, Interleukin-12 subunit beta signal peptide, Interleukin-12 subunit alpha signal peptide, Interleukin-18 signal peptide, and CD8 signal peptide.

[0024] In some embodiments, the nucleic acid or expression vector further comprises, in operable linkage to the sequence encoding the DR-18 polypeptide, a promoter sequence selected from: cytomegalovirus (CMV) promoter, elongation factor 1 alpha (EF1a) promoter, simian virus 40 (SV40) promoter, phosphoglycerate kinase (PGK) ubiquitin C (UBC) promoter, human beta actin promoter, CAG promoter, LC immunoglobulin promoter, HC immunoglobulin promoter, herpes simplex virus thymidine kinase promoter, and spleen focus-forming virus (SFFV) promoter.

[0025] In some embodiments, the nucleic acid or expression vector further comprises a chimeric antigen receptor (CAR) sequence encoding a CAR. In some instances, the CAR is selected from: i) an anti-5T4 CAR comprising an antigen binding domain (ABD) that specifically binds, or a means for specifically binding, 5T4; ii) an anti-AFP CAR comprising an ABD that specifically binds, or a means for specifically binding, AFP; iii) an anti-ALPP CAR comprising an ABD that specifically binds, or a means for specifically binding, ALPP; iv) an anti-ALPPL2 CAR comprising an ABD that specifically binds, or a means for specifically binding, ALPPL2; v) an anti-APN (CD13) CAR comprising an ABD that specifically binds, or a means for specifically binding, APN (CD13); vi) an anti-APRIL CAR comprising an ABD that specifically binds, or a means for specifically binding, APRIL; vii) an anti-AXL CAR comprising an ABD that specifically binds, or a means for specifically binding, AXL; viii) an anti-B7-H3 CAR comprising an ABD that specifically binds, or a means for specifically binding, B7-H3; ix) an anti-B7-H4 CAR comprising an ABD that specifically binds, or a means for specifically binding, B7-H4; x) an anti-BAFF-R CAR comprising an ABD that specifically binds, or a means for specifically binding, BAFF-R; xi) an anti-BAFF-R CAR comprising an ABD that specifically binds, or a means for specifically binding, BAFF-R; xii) an anti-BCMA CAR comprising an ABD that specifically binds, or a means for specifically binding, BCMA; xiii) an anti-BSG CAR comprising an ABD that specifically binds, or a means for specifically binding, BSG;

xiv) an anti-CAIX CAR comprising an ABD that specifically binds, or a means for specifically binding, CAIX; xv) an anti-CD123 CAR comprising an ABD that specifically binds, or a means for specifically binding, CD123; xvi) an anti-CD133 CAR comprising an ABD that specifically binds, or a means for specifically binding, CD133; xvii) an anti-CD138 CAR comprising an ABD that specifically binds, or a means for specifically binding, CD138; xviii) an anti-CD19 CAR comprising an ABD that specifically binds, or a means for specifically binding, CD19; xix) an anti-CD1A CAR comprising an ABD that specifically binds, or a means for specifically binding, CD1A; xx) an anti-CD2 CAR comprising an ABD that specifically binds, or a means for specifically binding, CD2; xxi) an anti-CD20 CAR comprising an ABD that specifically binds, or a means for specifically binding, CD20; xxii) an anti-CD22 CAR comprising an ABD that specifically binds, or a means for specifically binding, CD22; xxiii) an anti-CD24 CAR comprising an ABD that specifically binds, or a means for specifically binding, CD24; xxiv) an anti-CD276 CAR comprising an ABD that specifically binds, or a means for specifically binding, CD276; xxv) an anti-CD3 CAR comprising an ABD that specifically binds, or a means for specifically binding, CD3; xxvi) an anti-CD30 CAR comprising an ABD that specifically binds, or a means for specifically binding, CD30; xxvii) an anti-CD33 CAR comprising an ABD that specifically binds, or a means for specifically binding, CD33; xxviii) an anti-CD38 CAR comprising an ABD that specifically binds, or a means for specifically binding, CD38; xxix) an anti-CD4 CAR comprising an ABD that specifically binds, or a means for specifically binding, CD4; xxx) an anti-CD43 CAR comprising an ABD that specifically binds, or a means for specifically binding, CD43; xxxi) an anti-CD44 CAR comprising an ABD that specifically binds, or a means for specifically binding, CD44; xxxii) an anti-CD44v6 CAR comprising an ABD that specifically binds, or a means for specifically binding, CD44v6; xxxiii) an anti-CD45 CAR comprising an ABD that specifically binds, or a means for specifically binding, CD45; xxxiv) an anti-CD5 CAR comprising an ABD that specifically binds, or a means for specifically binding, CD5; xxxv) an anti-CD56 CAR comprising an ABD that specifically binds, or a means for specifically binding, CD56; xxxvi) an anti-CD7 CAR comprising an ABD that specifically binds, or a means for specifically binding, CD7; xxxvii) an anti-CD70 CAR comprising an ABD that specifically binds, or a means for specifically binding, CD70; xxxviii) an anti-CD72 CAR comprising an ABD that specifically binds, or a means for specifically binding, CD72; xxxix) an anti-CD83 CAR comprising an ABD that specifically binds, or a means for specifically binding, CD83; xl) an anti-CD84 CAR comprising an ABD that specifically binds, or a means for specifically binding, CD84; xli) an anti-CD99 CAR comprising an ABD that specifically binds, or a means for specifically binding, CD99; xlii) an anti-CDCP1 CAR comprising an ABD that specifically binds, or a means for specifically binding, CDCP1; xliii) an anti-CDH17 CAR comprising an ABD that specifically binds, or a means for specifically binding, CDH17; xliv) an anti-CEA CAR comprising an ABD that specifically binds, or a means for specifically binding, CEA; xlv) an anti-CEACAM5 CAR comprising an ABD that specifically binds, or a means for specifically binding, CEACAM5; xlvi) an anti-CLDN18.2 CAR comprising an ABD that specifically binds, or a means for specifically binding, CLDN18.2; xlvii) an anti-CLDN6 CAR comprising an ABD that specifically binds, or a means for specifically binding, CLDN6; xlviii) an anti-CLEC4K (CD207) CAR comprising an ABD that specifically binds, or a means for specifically binding, CLEC4K (CD207); xlix) an anti-CLL-1 CAR comprising an ABD that specifically binds, or a means for specifically binding, CLL-1; l) an anti-CSPG4 CAR comprising an ABD that specifically binds, or a means for specifically binding, CSPG4; li) an anti-DLL3 CAR comprising an ABD that specifically binds, or a means for specifically binding, DLL3; lii) an anti-DR5 CAR comprising an ABD that specifically binds, or a means for specifically binding, DR5; liii) an anti-EBV Protein CAR comprising an ABD that specifically binds, or a means for specifically binding, EBV Protein; liv) an anti-EGFR CAR comprising an ABD that specifically binds, or a means for specifically binding, EGFR; lv) an anti-EGFRvIII CAR comprising an ABD that specifically binds, or a means for specifically binding, EGFRvIII; lvi) an anti-EMR1 CAR comprising an ABD that specifically binds, or a means for specifically binding, EMR1; lvii) an anti-enkephalinase (CD10) CAR comprising an ABD that specifically binds, or a means for specifically binding, enkephalinase (CD10); lviii) an anti-EpCAM CAR comprising an ABD that specifically binds, or a means for specifically binding, EpCAM; lix) an anti-EphA2 CAR comprising an ABD that specifically binds, or a means for specifically binding, EphA2; lx) an anti-EphA3 CAR comprising an ABD that specifically binds, or a means for specifically binding, EphA3; lxi) an anti-FAP CAR comprising an ABD that specifically binds, or a means for specifically binding, FAP; lxii) an anti-FGFR4 CAR comprising an ABD that specifically binds, or a means for specifically binding, FGFR4; lxiii) an anti-FLT3 CAR comprising an ABD that specifically binds, or a means for specifically binding, FLT3; lxiv) an anti-FOLR1 CAR comprising an ABD that specifically binds, or a means for specifically binding, FOLR1; lxv) an anti-FSHR CAR comprising an ABD that specifically binds, or a means for specifically binding, FSHR; lxvi) an anti-GC-C CAR comprising an ABD that specifically binds, or a means for specifically binding, GC-C; lxvii) an anti-GD2 CAR comprising an ABD that specifically binds, or a means for specifically binding, GD2; lxviii) an anti-GFRA4 CAR comprising an ABD that specifically binds, or a means for specifically binding, GFRA4; lxix) an anti-Globo H CAR comprising an ABD that specifically binds, or a means for specifically binding, Globo H; lxx) an anti-GM2 (ganglioside M2) CAR comprising an ABD that specifically binds, or a means for specifically binding, GM2 (ganglioside M2); lxxi) an anti-gp100 CAR comprising an ABD that specifically binds, or a means for specifically binding, gp100; lxxii) an anti-GPC1 CAR comprising an ABD that specifically binds, or a means for specifically binding, GPC1; lxxiii) an anti-GPC2 CAR comprising an ABD that specifically binds, or a means for specifically binding, GPC2; lxxiv) an anti-GPC3 CAR comprising an ABD that specifically binds, or a means for specifically binding, GPC3; lxxv) an anti-HAAH (ASPH) CAR comprising an ABD that specifically binds, or a means for specifically binding, HAAH (ASPH); lxxvi) an anti-HER2 CAR comprising an ABD that specifically binds, or a means for specifically binding, HER2; lxxvii) an anti-HGF CAR comprising an ABD that specifically binds, or a means for specifically binding, HGF; lxxviii) an anti-

HIV envelope protein gp120 CAR comprising an ABD that specifically binds, or a means for specifically binding, HIV envelope protein gp120; lxxix) an anti-HIV-1 pol CAR comprising an ABD that specifically binds, or a means for specifically binding, HIV-1 pol; lxxx) an anti-HLA-A2 CAR comprising an ABD that specifically binds, or a means for specifically binding, HLA-A2; lxxxi) an anti-HLA-G CAR comprising an ABD that specifically binds, or a means for specifically binding, HLA-G; lxxxii) an anti-HSP70 heat-shock protein CAR comprising an ABD that specifically binds, or a means for specifically binding, HSP70 heat-shock protein; lxxxiii) an anti-ICAM-1 CAR comprising an ABD that specifically binds, or a means for specifically binding, ICAM-1; lxxxiv) an anti-IL-10R CAR comprising an ABD that specifically binds, or a means for specifically binding, IL-10R; lxxxv) an anti-IL13Rα2 CAR comprising an ABD that specifically binds, or a means for specifically binding, IL13Rα2; lxxxvi) an anti-IL-13Rα2 CAR comprising an ABD that specifically binds, or a means for specifically binding, IL-13Rα2; lxxxvii) an anti-integrin αvβ6 CAR comprising an ABD that specifically binds, or a means for specifically binding, integrin αvβ6; lxxxviii) an anti-ITGB7 CAR comprising an ABD that specifically binds, or a means for specifically binding, ITGB7; lxxxix) an anti-KKLC1 CAR comprising an ABD that specifically binds, or a means for specifically binding, KKLC1; xc) an anti-KMA CAR comprising an ABD that specifically binds, or a means for specifically binding, KMA; xci) an anti-L1CAM CAR comprising an ABD that specifically binds, or a means for specifically binding, L1CAM; xcii) an anti-Lewis-Y antigen CAR comprising an ABD that specifically binds, or a means for specifically binding, Lewis-Y antigen; xciii) an anti-LGR5 CAR comprising an ABD that specifically binds, or a means for specifically binding, LGR5; xciv) an anti-LILRB4 CAR comprising an ABD that specifically binds, or a means for specifically binding, LILRB4; xcv) an anti-LMP1 CAR comprising an ABD that specifically binds, or a means for specifically binding, LMP1; xcvi) an anti-MAGEA3 CAR comprising an ABD that specifically binds, or a means for specifically binding, MAGEA3; xcvii) an anti-M-CSF CAR comprising an ABD that specifically binds, or a means for specifically binding, M-CSF; xcviii) an anti-MG7 CAR comprising an ABD that specifically binds, or a means for specifically binding, MG7; xcix) an anti-MICA CAR comprising an ABD that specifically binds, or a means for specifically binding, MICA; c) an anti-MMP2 CAR comprising an ABD that specifically binds, or a means for specifically binding, MMP2; ci) an anti-MSLN CAR comprising an ABD that specifically binds, or a means for specifically binding, MSLN; cii) an anti-MUC1/MUC16 CAR comprising an ABD that specifically binds, or a means for specifically binding, MUC1/MUC16; ciii) an anti-NCR3LG1 CAR comprising an ABD that specifically binds, or a means for specifically binding, NCR3LG1; civ) an anti-NECTIN2 CAR comprising an ABD that specifically binds, or a means for specifically binding, NECTIN2; cv) an anti-nectin-4 CAR comprising an ABD that specifically binds, or a means for specifically binding, nectin-4; cvi) an anti-NKG2D CAR comprising an ABD that specifically binds, or a means for specifically binding, NKG2D; cvii) an anti-NKG2DL CAR comprising an ABD that specifically binds, or a means for specifically binding, NKG2DL; cviii) an anti-NR2F6 CAR comprising an ABD that specifically binds, or a means for specifically binding, NR2F6; cix) an anti-NY-ESO-1 CAR comprising an ABD that specifically binds, or a means for specifically binding, NY-ESO-1; cx) an anti-OPCML CAR comprising an ABD that specifically binds, or a means for specifically binding, OPCML; cxi) an anti-PD-1 CAR comprising an ABD that specifically binds, or a means for specifically binding, PD-1; cxii) an anti-PDGFRα CAR comprising an ABD that specifically binds, or a means for specifically binding, PDGFRα; cxiii) an anti-PDL1 CAR comprising an ABD that specifically binds, or a means for specifically binding, PDL1; cxiv) an anti-PLA2R CAR comprising an ABD that specifically binds, or a means for specifically binding, PLA2R; cxv) an anti-PR1 CAR comprising an ABD that specifically binds, or a means for specifically binding, PR1; cxvi) an anti-PSCA CAR comprising an ABD that specifically binds, or a means for specifically binding, PSCA; cxvii) an anti-PTK7 CAR comprising an ABD that specifically binds, or a means for specifically binding, PTK7; cxviii) an anti-PVR CAR comprising an ABD that specifically binds, or a means for specifically binding, PVR; cxix) an anti-ROBO1 CAR comprising an ABD that specifically binds, or a means for specifically binding, ROBO1; cxx) an anti-ROR1 CAR comprising an ABD that specifically binds, or a means for specifically binding, ROR1; cxxi) an anti-ROR2 CAR comprising an ABD that specifically binds, or a means for specifically binding, ROR2; cxxii) an anti-SEMA4A CAR comprising an ABD that specifically binds, or a means for specifically binding, SEMA4A; cxxiii) an anti-SLAMF7 CAR comprising an ABD that specifically binds, or a means for specifically binding, SLAMF7; cxxiv) an anti-TAG-72 CAR comprising an ABD that specifically binds, or a means for specifically binding, TAG-72; cxxv) an anti-TGF-β CAR comprising an ABD that specifically binds, or a means for specifically binding, TGF-β; cxxvi) an anti-TM4SF1 CAR comprising an ABD that specifically binds, or a means for specifically binding, TM4SF1; cxxvii) an anti-TRAIL CAR comprising an ABD that specifically binds, or a means for specifically binding, TRAIL; cxxviii) an anti-TRBC1 CAR comprising an ABD that specifically binds, or a means for specifically binding, TRBC1; cxxix) an anti-TRBC2 CAR comprising an ABD that specifically binds, or a means for specifically binding, TRBC2; cxxx) an anti-Trop-2 CAR comprising an ABD that specifically binds, or a means for specifically binding, Trop-2; cxxxi) an anti-TSHR CAR comprising an ABD that specifically binds, or a means for specifically binding, TSHR; cxxxii) an anti-TSLPR CAR comprising an ABD that specifically binds, or a means for specifically binding, TSLPR; cxxxiii) an anti-ULBP1 CAR comprising an ABD that specifically binds, or a means for specifically binding, ULBP1; cxxxiv) an anti-αvβ3 integrin CAR comprising an ABD that specifically binds, or a means for specifically binding, αvβ3 integrin; and cxxxv) an anti-K light chain of human immunoglobulin CAR comprising an ABD that specifically binds, or a means for specifically binding, K light chain of human immunoglobulin.

[0026] In some embodiments, the CAR is selected from: a) inaticabtagene autoleucel; b) actalycabtagene autoleucel; c) relmacabtagene autoleucel; d) lisocabtagene maraleucel; e) brexucabtagene autoleucel; f) axicabtagene ciloleucel; g) tisagenlecleucel; h) obecabatagene autoleucel; i) azercabtagene zapreleucel; j) rapcabtagene autoleucel; and k) idecabtagene vicleucel.

[0027] In some embodiments, the sequence encoding the CAR and the sequence encoding the DR-18 polypeptide are operably linked by a multicistronic element. In some embodiments, the multicistronic element is an internal ribosome entry site. In some embodiments, the multicistronic element is a sequence encoding 2A peptide selected from: T2A, P2A, E2A, and F2A.

[0028] In some embodiments, the CAR is an anti-EGFRvIII CAR comprising a means for specifically binding EGFRvIII. In some embodiments, the anti-EGFRvIII CAR comprises an antigen binding domain comprising one or more of light chain complementary determining region 1 (LC CDR1), light chain complementary determining region 2 (LC CDR2), and light chain complementary determining region 3 (LC CDR3) of SEQ ID NO: 110, and one or more of heavy chain complementary determining region 1 (HC CDR1), heavy chain complementary determining region 2 (HC CDR2), and heavy chain complementary determining region 3 (HC CDR3) of SEQ ID NO:111. In some embodiments, the antigen binding domain comprises LC CDR1 (SEQ ID NO:112), LC CDR2 (SEQ ID NO:113), and LC CDR3 (SEQ ID NO:114); and HC CDR1 (SEQ ID NO:115), HC CDR2 (SEQ ID NO:116), and HC CDR3 (SEQ ID NO:117). In some embodiments, the antigen binding domain comprises a light chain variable region comprising sequence SEQ ID NO:110 and a heavy chain variable region comprising sequence SEQ ID NO:111.

[0029] Aspects of the present disclosure include a cell comprising the described nucleic acid or expression vector. In some embodiments, the cell is selected from: a T cell, a Naïve T cell (TN), stem cell memory T cell (TSCM), a central memory T cell (TCM), a resident memory T cell (TRM), an effector T cell (TEFF), an effector memory cell (TEM), an alpha-beta ($\alpha\beta$) T cell, a gamma-delta ($\gamma\alpha$) T cell, a Natural Killer (NK) cell, a CD4+ T cell, a CD4+Th1 cell, a CD4+Th2 cell, a CD4+Th9 cell, a CD4+Th17 cell, a CD4+Th22 cell, a CD4+T Regulatory Cell (Treg), a CD4+T follicular helper cell (Tfh), a resting Treg cell, an effector Treg cell, an effector memory Treg cell, a CD8+ T cell, a CD8+Tc1 cell, a CD8+Tc2 cell, a CD8+Tc9 cell, a CD8+ Th17 cell, a CD8+Treg cell, a macrophage, a B Cell, a dendritic cell, a stem cell, a hematopoietic stem cell (HSC), and an induced pluripotent stem cell (iPS cell).

[0030] Aspects of the present disclosure include a method of treating a subject for cancer, the method comprising administering to the subject an effective amount of the described nucleic acid, expression vector, or cells. In some embodiments, the subject does not undergo lymphodepletion during or prior to the administering. In some embodiments, the amount of the nucleic acid, the expression vector, or the cell administered to the subject is less than the amount of a corresponding nucleic acid, expression vector, or cell comprising the sequence encoding the CAR administered to the subject to treat the cancer without the sequence encoding the DR-18 polypeptide. In some embodiments, the administering comprises delivering multiple doses of the nucleic acid, the expression vector, or the cell to the subject and the dosing frequency is less than the dosing frequency of a corresponding nucleic acid, expression vector, or cell comprising the sequence encoding the CAR administered to the subject to treat the cancer without the sequence encoding the DR-18 polypeptide.

[0031] Aspects of the present disclosure include a method of making a population of therapeutic cells, the method comprising: a) contacting a human cell with a nucleic acid comprising a chimeric antigen receptor (CAR) sequence encoding a CAR and a sequence encoding a decoy-resistant interleukin 18 (DR-18) polypeptide; wherein the nucleic acid is configured to result in expression and secretion of the DR-18 by the human cell ex vivo; b) culturing the contacted human cell ex vivo in the presence of the secreted DR-18 polypeptide, thereby producing a population of therapeutic cells, wherein the population of therapeutic cells is enhanced as compared to a corresponding population of cells generated without the sequence encoding the DR-18 polypeptide.

[0032] In some embodiments, the population of therapeutic cells demonstrates enhanced ex vivo proliferation as compared to the corresponding population of cells. In some embodiments, the population of therapeutic cells demonstrates enhanced in vivo proliferation as compared to the corresponding population of cells. In some embodiments, the population of therapeutic cells demonstrates enhanced in vivo persistence as compared to the corresponding population of cells. In some embodiments, the population of therapeutic cells demonstrates enhanced tumor cell killing as compared to the corresponding population of cells.

INCORPORATION BY REFERENCE

[0033] All publications, patents, and patent applications mentioned in this specification are herein incorporated by reference to the same extent as if each individual publication, patent, or patent application was specifically and individually indicated to be incorporated by reference.

BRIEF DESCRIPTION OF THE DRAWINGS

[0034] FIG. **1** depicts the architecture of certain anti-epidermal growth factor variant Ill (EGFRvIII) chimeric antigen receptor (CAR) constructs as described herein. "IL-18" indicated in the drawing can refer to either wild-type human IL-18 or DR-18. Elements are not drawn to scale. Arrows indicate read-direction. The presence or absence of linking sequence between elements cannot necessarily be inferred from the existence of space or lack of space between individual elements.

DEFINITIONS

[0035] The terms "polynucleotide" and "nucleic acid," used interchangeably herein, refer to a polymeric form of nucleotides of any length, either ribonucleotides or deoxyribonucleotides. Thus, this term includes, but is not limited to, single-, double-, or multi-stranded DNA or RNA, genomic DNA, cDNA, DNA-RNA hybrids, or a polymer comprising purine and pyrimidine bases or other natural, chemically or biochemically modified, non-natural, or derivatized nucleotide bases.

[0036] As used herein, "codon" refers to a sequence of three nucleotides that together form a unit of genetic code in a DNA or RNA molecule. The term "codon-optimized" or "codon optimization" refers to genes or coding regions of nucleic acid molecules for transformation of various hosts, refers to the alteration of codons in the gene or coding regions of the nucleic acid molecules to reflect the typical codon usage of the host organism without altering the polypeptide encoded by the DNA. Such optimization includes replacing at least one, or more than one, or a significant number, of codons with one or more codons that are more frequently used in the genes of that organism. Codon usage tables are readily available, for example, at the "Codon Statistics Database" (see e.g., Subramanian et al.

(2022) Molecular Biology and Evolution 39(8)), "Codon Usage Database" at www(dot)kazusa(dot)or(dot)jp/codon/, High-performance Integrated Virtual Environment-Codon Usage Tables (HIVE-CUTs) (see e.g., Athey et al. (2017) BMC Bioinformatics 18(391)), and the like.

[0037] "Operably linked" refers to a juxtaposition wherein the components so described are in a relationship permitting them to function in their intended manner. For instance, a promoter is operably linked to a coding sequence if the promoter affects its transcription or expression. Operably linked nucleic acid sequences may but need not necessarily be adjacent. For example, in some instances a coding sequence operably linked to a promoter may be adjacent to the promoter. In some instances, a coding sequence operably linked to a promoter may be separated by one or more intervening sequences, including coding and non-coding sequences. Also, in some instances, more than two sequences may be operably linked including but not limited to e.g., where two or more coding sequences are operably linked to a single promoter.

[0038] A "vector" or "expression vector" is a replicon, such as plasmid, phage, virus, or cosmid, to which another DNA segment, i.e. an "insert", may be attached so as to bring about the replication of the attached segment in a cell.

[0039] Nucleic acids can be delivered into a cell in a variety of ways, including but not limited to e.g., as naked nucleic acid (e.g., naked DNA, naked plasmid, etc.), as a nucleic acid-protein complex (e.g., a DNA-protein complex, an RNA-protein complex such as a ribonucleoprotein (RNP) complex, and the like), as nucleic acid within a viral vector, as nucleic acid within a non-viral vector, and the like. In some instances, proteins may be incorporated into the delivery of a nucleic acid agent, such as but not limited to DNA-binding proteins, RNA-binding proteins, enzymes such as nucleases, and combinations thereof.

[0040] Useful viral vectors include non-replicating (i.e., replication deficient) viral vectors. Viral vectors may be integrating or non-integrating. Non-limiting examples of useful viral vectors include retroviral vectors, lentiviral vectors, adenoviral vectors, adeno-associated virus (AAV) vectors, and the like.

[0041] Nonviral vectors, which are delivery means that do not employ viral particles, may generally be considered to fall into three categories: naked nucleic acid, particle based (e.g., nanoparticles), or chemical based. Non-limiting examples of nonviral vectors include lipoplexes (e.g., cationic lipid-based lipoplexes), emulsions (such as e.g., lipid nano emulsions), lipid nanoparticles (LNPs), solid lipid nanoparticles, peptide based vectors, polymer based vectors (e.g., polymersomes, polyplexes, polyethylenimine (PEI)-based vectors, chitosan-based vectors, poly (DL-Lactide) (PLA) and poly (DL-Lactide-co-glycoside) (PLGA)-based vectors, dendrimers, vinyl based polymers (e.g., polymethacrylate-based vectors), and the like), inorganic nanoparticles, and the like.

[0042] "Heterologous nucleotide" and "heterologous polypeptide" as used herein, means a nucleotide or polypeptide sequence that is not found in the native (e.g., naturally-occurring) nucleic acid or protein, respectively. Heterologous nucleic acids or polypeptide may be derived from a different species as the organism or cell within which the nucleic acid or polypeptide is present or is expressed. Accordingly, a heterologous nucleic acids or polypeptide is generally of unlike evolutionary origin as compared to the cell or organism in which it resides.

[0043] The terms "polypeptide," "peptide," and "protein", used interchangeably herein, refer to a polymeric form of amino acids of any length, which can include genetically coded and non-genetically coded amino acids, chemically or biochemically modified or derivatized amino acids, and polypeptides having modified peptide backbones. The term includes fusion proteins, including, but not limited to, fusion proteins with a heterologous amino acid sequence, fusions with heterologous and homologous leader sequences, with or without N-terminal methionine residues; immunologically tagged proteins; and the like.

[0044] The terms "derivative" and "variant" refer without limitation to any compound such as nucleic acid or protein that has a structure or sequence derived from the compounds disclosed herein and whose structure or sequence is sufficiently similar to those disclosed herein such that it has the same or similar activities and utilities or, based upon such similarity, would be expected by one skilled in the art to exhibit the same or similar activities and utilities as the referenced compounds, thereby also interchangeably referred to "functionally equivalent" or as "functional equivalents." Modifications to obtain "derivatives" or "variants" may include, for example, addition, deletion and/or substitution of one or more of the nucleic acids or amino acid residues.

[0045] The terms "chimeric antigen receptor" and "CAR", used interchangeably herein, refer to artificial multi-module molecules capable of triggering or inhibiting the activation of an immune cell which generally but not exclusively comprise an extracellular domain (e.g., a ligand/antigen binding domain), a transmembrane domain and one or more intracellular signaling domains. The term CAR is not limited specifically to CAR molecules but also includes CAR variants. CAR variants include split CARs wherein the extracellular portion (e.g., the ligand binding portion) and the intracellular portion (e.g., the intracellular signaling portion) of a CAR are present on two separate molecules. CAR variants also include ON-switch CARs which are conditionally activatable CARs, e.g., comprising a split CAR wherein conditional hetero-dimerization of the two portions of the split CAR is pharmacologically controlled (e.g., as described in PCT publication no. WO 2014/127261 and US Patent Application No. 2015/0368342). CAR variants also include bispecific CARs, which include a secondary CAR binding domain that can either amplify or inhibit the activity of a primary CAR. CAR variants also include inhibitory chimeric antigen receptors (iCARs) which may, e.g., be used as a component of a bispecific CAR system, where binding of a secondary CAR binding domain results in inhibition of primary CAR activation. CAR molecules and derivatives thereof (i.e., CAR variants) are described, e.g., in PCT Application No. US2014/016527; Fedorov et al. *Sci Transl Med* (2013);5(215):215ra172; Glienke et al. *Front Pharmacol* (2015) 6:21; Kakarla & Gottschalk 52 *Cancer J* (2014) 20(2): 151-5; Riddell et al. *Cancer J* (2014) 20(2):141-4; Pegram et al. *Cancer J* (2014) 20(2):127-33; Cheadle et al. *Immunol Rev* (2014) 257(1):91-106; Barrett et al. *Annu Rev Med* (2014) 65:333-47; Sadelain et al. *Cancer Discov* (2013) 3(4):388-98; Cartellieri et al., *J Biomed Biotechnol* (2010) 956304.

[0046] The terms "domain" and "motif", used interchangeably herein, refer to both structured domains having one or more particular functions and unstructured segments of a polypeptide that, although unstructured, retain one or more particular functions. For example, a structured domain may encompass but is not limited to a continuous or discontinuous plurality of amino acids, or portions thereof, in a folded polypeptide that comprise a three-dimensional structure which contributes to a particular function of the polypeptide. In other instances, a domain may include an unstructured segment of a polypeptide comprising a plurality of two or more amino acids, or portions thereof, that maintains a particular function of the polypeptide unfolded or disordered. Also encompassed within this definition are domains that may be disordered or unstructured but become structured or ordered upon association with a target or binding partner.

[0047] As used herein, the term "antibody" refers to any form of immunoglobulin molecule that exhibits the desired biological or binding activity. Thus, it is used in the broadest sense and specifically covers, but is not limited to, monoclonal antibodies (including full length monoclonal antibodies), polyclonal antibodies, multispecific antibodies (e.g., bispecific antibodies), humanized, fully human antibodies, and chimeric antibodies, and may include post-translational modifications thereof (e.g., C-terminal Lysine clipping in the heavy chain, conversion of glutamine or glutamic acid to pyroglutamate) that may occur when an antibody is recombinantly expressed in host cells (e.g., CHO cells), or during purification/storage. "Parental antibodies" are antibodies obtained by exposure of an immune system to an antigen prior to modification of the antibodies for an intended use, such as humanization of an antibody for use as a human therapeutic. As used herein, the term "antibody" encompasses not only intact polyclonal or monoclonal antibodies, but also, unless otherwise specified, fusion proteins comprising an antigen binding fragment thereof that competes with the intact antibody for specific.

[0048] In general, the basic antibody structural unit comprises a tetramer. Each tetramer includes two identical pairs of polypeptide chains, each pair having one "light" (about 25 kDa) and one "heavy" chain (about 50-70 kDa). The amino-terminal portion of each chain includes a variable region of about 100 to 110 or more amino acids primarily responsible for antigen recognition. The variable regions of each light/heavy chain pair form the antibody binding site. Thus, in general, an intact antibody has two binding sites. The carboxy-terminal portion of the heavy chain may define a constant region primarily responsible for effector function. Typically, human light chains are classified as kappa and lambda light chains. Furthermore, human heavy chains are typically classified as mu, delta, gamma, alpha, or epsilon, and define the antibody's isotype as IgM, IgD, IgG, IgA, and IgE, respectively. Within light and heavy chains, the variable and constant regions are joined by a "J" region of about 12 or more amino acids, with the heavy chain also including a "D" region of about 10 more amino acids. See generally, Fundamental Immunology Ch. 7 (Paul, W., ed., 2nd ed. Raven Press, N.Y. (1989).

[0049] "Variable regions" or "V region" or "V chain" as used herein means the segment of IgG chains which is variable in sequence between different antibodies. A "variable region" of an antibody refers to the variable region of the antibody light chain or the variable region of the antibody heavy chain, either alone or in combination. The variable region of the heavy chain may be referred to as "VH." The variable region of the light chain may be referred to as "VL."

[0050] Typically, the variable regions of both the heavy and light chains comprise three hypervariable regions, also called complementarity determining regions (CDRs), which are located within relatively conserved framework regions (FR). The CDRs are usually aligned by the framework regions, enabling binding to a specific epitope. In general, from N-terminal to C-terminal, both light and heavy chains variable domains comprise FR1, CDR1, FR2, CDR2, FR3, CDR3, and FR4. As referred to herein the light chain CDRs are CDRL1, CDRL2 and CDRL3, respectively, and the heavy chain CDRs are CDRH1, CDRH2 and CDRH3, respectively. The assignment of amino acids to each domain is, generally, in accordance with the definitions of Sequences of Proteins of Immunological Interest, Kabat, et al.; National Institutes of Health, Bethesda, *Md.; 5th ed.; NIH Publ. No.* 91-3242 (1991); *Kabat* (1978) *Adv. Prot. Chem.* 32:1-75; Kabat, et al., (1977) *J. Biol. Chem.* 252:6609-6616; Chothia, et al., (1987) *J Mol. Biol.* 196:901-917 or Chothia, et al., (1989) *Nature* 342:878-883.

[0051] A "CDR" refers to one of three hypervariable regions (H1, H2, or H3) within the non-framework region of the antibody VH β-sheet framework, or one of three hypervariable regions (L1, L2, or L3) within the non-framework region of the antibody VL β-sheet framework. Accordingly, CDRs are variable region sequences interspersed within the framework region sequences. CDR regions are well known to those skilled in the art and have been defined by, for example, Kabat as the regions of most hypervariability within the antibody variable domains. CDR region sequences also have been defined structurally by Chothia as those residues that are not part of the conserved β-sheet framework, and thus are able to adapt to different conformations. Both terminologies are well recognized in the art. CDR region sequences have also been defined by AbM, Contact, and IMGT. The positions of CDRs within a canonical antibody variable region have been determined by comparison of numerous structures (Al-Lazikani et al., 1997, J. Mol. Biol. 273:927-48; Morea et al., 2000, *Methods* 20:267-79). Because the number of residues within a hypervariable region varies in different antibodies, additional residues relative to the canonical positions are conventionally numbered with a, b, c and so forth next to the residue number in the canonical variable region numbering scheme (Al-Lazikani et al., supra). Such nomenclature is similarly well known to those skilled in the art. Correspondence between the numbering system, including, for example, the Kabat numbering and the IMGT unique numbering system, is well known to one skilled in the art and shown below in the following table. In some embodiments, the CDRs are as defined by the Kabat numbering system. In other embodiments, the CDRs are as defined by the IMGT numbering system. In yet other embodiments, the CDRs are as defined by the AbM numbering system. In still other embodiments, the CDRs are as defined by the Chothia numbering system. In yet other embodiments, the CDRs are as defined by the Contact numbering system. Correspondence between the CDR Numbering Systems

	Kabat + Chothia	IMGT	Kabat	AbM	Chothia	Contact
VH CDR1	26-35	27-38	31-35	26-35	26-32	30-35
VH CDR2	50-65	56-65	50-65	50-58	52-56	47-58
VH CDR3	95-102	105-117	95-102	95-102	95-102	93-101
VL CDR1	24-34	27-38	24-34	24-34	24-34	30-36
VL CDR2	50-56	56-65	50-56	50-56	50-56	46-55
VL CDR3	89-97	105-117	89-97	89-97	89-97	89-96

[0052] "Chimeric antibody" refers to an antibody in which a portion of the heavy and/or light chain contains sequences derived from a particular species (e.g., human) or belonging to a particular antibody class or subclass, while the remainder of the chain(s) is derived from another species (e.g., mouse) or belonging to another antibody class or subclass, as well as fragments of such antibodies, so long as they exhibit the desired biological activity.

[0053] "Human antibody" refers to an antibody that comprises human immunoglobulin protein sequences or derivatives thereof. A human antibody may contain murine carbohydrate chains if produced in a mouse, in a mouse cell, or in a hybridoma derived from a mouse cell. Similarly, "mouse antibody" or "rat antibody" refer to an antibody that comprises only mouse or rat immunoglobulin sequences or derivatives thereof, respectively.

[0054] "Humanized antibody" refers to forms of antibodies that contain sequences from nonhuman (e.g., murine) antibodies as well as human antibodies. Such antibodies contain minimal sequence derived from non-human immunoglobulin. In general, the humanized antibody will comprise substantially all of at least one, and typically two, variable domains, in which all or substantially all of the hypervariable loops correspond to those of a non-human immunoglobulin and all or substantially all of the FR regions are those of a human immunoglobulin sequence. The humanized antibody optionally also will comprise at least a portion of an immunoglobulin constant region (Fc), typically that of a human immunoglobulin. The prefix "hum", "hu" or "h" may be added to antibody clone designations when necessary to distinguish humanized antibodies from parental rodent antibodies. The humanized forms of rodent antibodies will generally comprise the same CDR sequences of the parental rodent antibodies, although certain amino acid substitutions can be included to increase affinity, increase stability of the humanized antibody, or for other reasons.

[0055] "Monoclonal antibody" or "mAb" or "Mab", as used herein, refers to a population of substantially homogeneous antibodies, i.e., the antibody molecules comprising the population are identical in amino acid sequence except for possible naturally occurring mutations that may be present in minor amounts. In contrast, conventional (polyclonal) antibody preparations typically include a multitude of different antibodies having different amino acid sequences in their variable domains, particularly their CDRs, which are often specific for different epitopes. The modifier "monoclonal" indicates the character of the antibody as being obtained from a substantially homogeneous population of antibodies, and is not to be construed as requiring production of the antibody by any particular method. For example, the monoclonal antibodies to be used in accordance with the present disclosure may be made by the hybridoma method first described by Kohler et al. (1975) *Nature* 256: 495, or may be made by recombinant DNA methods (see, e.g., U.S. Pat. No. 4,816,567). The "monoclonal antibodies" may also be isolated from phage antibody libraries using the techniques described in Clackson et al. (1991) *Nature* 352: 624-628 and Marks et al. (1991) *J. Mol. Biol.* 222: 581-597, for example. See also Presta (2005) *J. Allergy Clin. Immunol.* 116:731.

[0056] As used herein, unless otherwise indicated, "antibody fragment" or "antigen binding fragment" refers to a fragment of an antibody that retains the ability to bind specifically to the antigen, e.g., fragments that retain one or more CDR regions and the ability to bind specifically to the antigen. An antibody that "specifically binds to" CD20 is an antibody that exhibits preferential binding to CD20 (as appropriate) as compared to other proteins, but this specificity does not require absolute binding specificity. An antibody is considered "specific" for its intended target if its binding is determinative of the presence of the target protein in a sample, e.g., without producing undesired results such as false positives. Antibodies, or binding fragments thereof, will bind to the target protein with an affinity that is at least two-fold greater, preferably at least ten times greater, more preferably at least 20-times greater, and most preferably at least 100-times greater than the affinity with non-target proteins.

[0057] Antigen binding portions include, for example, Fab, Fab', F(ab')2, Fd, Fv, fragments including CDRs, and single chain variable fragment antibodies (scFv), and polypeptides that contain at least a portion of an immunoglobulin that is sufficient to confer specific antigen binding to the antigen (e.g., CD20). An antibody includes an antibody of any class, such as IgG, IgA, or IgM (or sub-class thereof), and the antibody need not be of any particular class. Depending on the antibody amino acid sequence of the constant region of its heavy chains, immunoglobulins can be assigned to different classes. There are five major classes of immunoglobulins: IgA, IgD, IgE, IgG, and IgM, and several of these may be further divided into subclasses (isotypes), e.g., IgG1, IgG2, IgG3, IgG4, IgA1, and IgA2. The heavy-chain constant regions that correspond to the different classes of immunoglobulins are called alpha, delta, epsilon, gamma, and mu, respectively. The subunit structures and three-dimensional configurations of different classes of immunoglobulins are well known.

[0058] An "antigen" is a structure to which an antibody can selectively bind. A target antigen may be a polypeptide, carbohydrate, nucleic acid, lipid, hapten, or other naturally occurring or synthetic compound. In some embodiments, the target antigen is a polypeptide. In certain embodiments, an antigen is associated with a cell, for example, is present on or in a cell, for example, a cancer cell.

[0059] An "intact" antibody is one comprising an antigen-binding site as well as a CL and at least heavy chain constant regions, CH1, CH2 and CH3. The constant regions may include human constant regions or amino acid sequence variants thereof. In certain embodiments, an intact antibody has one or more effector functions.

[0060] As used herein, the term "affinity" refers to the equilibrium constant for the reversible binding of two agents (e.g., an antibody and an antigen) and is expressed as a dissociation constant (K_D). Affinity can be at least 1-fold greater, at least 2-fold greater, at least 3-fold greater, at least 4-fold greater, at least 5-fold greater, at least 6-fold greater, at least 7-fold greater, at least 8-fold greater, at least 9-fold greater, at least 10-fold greater, at least 20-fold greater, at least 30-fold greater, at least 40-fold greater, at least 50-fold greater, at least 60-fold greater, at least 70-fold greater, at least 80-fold greater, at least 90-fold greater, at least 100-fold greater, or at least 1,000-fold greater, or more, than the affinity of an antibody (or antigen-binding region of a CAR) for unrelated amino acid sequences. Affinity of an antibody (or antigen-binding region of a CAR) to a target protein can be, for example, from about 100 nanomolar (nM) to about 0.1 nM, from about 100 nM to about 1 picomolar (pM), or from about 100 nM to about 1 femtomolar (fM) or more. As used herein, the term "avidity" refers to the resistance of a complex of two or more agents to dissociation after dilution. The terms "immunoreactive" and "preferentially binds" are used interchangeably herein with respect to antibodies and/or antigen binding fragments.

[0061] The term "binding" refers to a direct association between two molecules, due to, for example, covalent, electrostatic, hydrophobic, and ionic and/or hydrogen-bond interactions, including interactions such as salt bridges and water bridges. In some cases, a specific binding member present in the extracellular domain of a chimeric polypeptide of the present disclosure binds specifically to its binding partner, such as an antigen. "Specific binding" refers to binding with an affinity of at least about 10^7 M or greater, e.g., 5×10^7 M, 10^8 M, 5×10^8 M, and greater. "Non-specific binding" refers to binding with an affinity of less than about 10^7 M, e.g., binding with an affinity of 10^6 M, 105 M, 104 M, etc.

[0062] As used herein, the terms "treatment," "treating," "treat" and the like, refer to obtaining a desired pharmacologic and/or physiologic effect. The effect can be prophylactic in terms of completely or partially preventing a disease or symptom thereof and/or can be therapeutic in terms of a partial or complete cure for a disease and/or adverse effect attributable to the disease. "Treatment," as used herein, covers any treatment of a disease in a mammal, particularly in a human, and includes: (a) preventing the disease from occurring in a subject which can be predisposed to the disease but has not yet been diagnosed as having it; (b) inhibiting the disease, i.e., arresting its development; and (c) relieving the disease, i.e., causing regression of the disease.

[0063] The term "subject in need thereof" as used herein refers to a subject diagnosed with or suspected of having a disease, such as, but not necessarily limited to, cancer.

[0064] A "therapeutically effective amount" or "efficacious amount" refers to the amount of an agent, or combined amounts of two agents, that, when administered to a mammal or other subject for treating a disease, is sufficient to effect such treatment for the disease. The "therapeutically effective amount" will vary depending on the agent(s), the disease and its severity and the age, weight, etc., of the subject to be treated.

[0065] The terms "individual," "subject," "host," and "patient," used interchangeably herein, refer to a mammal, including, but not limited to, murines (e.g., rats, mice), non-human primates, humans, canines, felines, ungulates (e.g., equines, bovines, ovines, porcines, caprines), lagomorphs, etc. In some cases, the individual is a human. In some cases, the individual is a non-human primate. In some cases, the individual is a rodent, e.g., a rat or a mouse. In some cases, the individual is a lagomorph, e.g., a rabbit.

[0066] As used herein, the term "immune cells" generally includes white blood cells (leukocytes) which are derived from hematopoietic stem cells (HSC) produced in the bone marrow. "Immune cells" includes, e.g., lymphocytes (T cells, B cells, natural killer (NK) cells) and myeloid-derived cells (neutrophil, eosinophil, basophil, monocyte, macrophage, dendritic cells).

[0067] The terms "stem cell", "stem cells", and the like, as used herein refer to cells that are capable of differentiating into two of more different cell types and proliferating. Non limiting examples of stem cells include but are not limited to embryonic stem cells, blastocyst derived stem cells, fetal stem cells, induced pluripotent stem cells, ectodermal derived stem cells, endodermal derived stem cells, mesodermal derived stem cells, neural crest cells, amniotic stem cells, cord blood stem cells, adult or somatic stem cells, neural stem cells, bone marrow stem cells, bone marrow stromal stem cells, hematopoietic stem cells, lymphoid progenitor cell, myeloid progenitor cell, mesenchymal stem cells, epithelial stem cells, adipose derived stem cells, skeletal muscle stem cells, muscle satellite cells, side population cells, intestinal stem cells, pancreatic stem cells, liver stem cells, hepatocyte stem cells, endothelial progenitor cells, hemangioblasts, gonadal stem cells, germline stem cells, and the like. Stem cells may be acquired from public or commercial sources or may be newly derived.

[0068] The term "synthetic" as used herein generally refers to an artificially derived polypeptide or polypeptide encoding nucleic acid that is not naturally occurring. Such synthetic polypeptides and/or nucleic acids may be assembled de novo from basic subunits including, e.g., single amino acids, single nucleotides, etc., or may be derived from pre-existing polypeptides or polynucleotides, whether naturally or artificially derived, e.g., as through recombinant methods.

[0069] The term "recombinant", as used herein describes a nucleic acid molecule, e.g., a polynucleotide of genomic, cDNA, viral, semisynthetic, and/or synthetic origin, which, by virtue of its origin or manipulation, is not associated with all or a portion of the polynucleotide sequences with which it is associated in nature. The term recombinant as used with respect to a protein or polypeptide means a polypeptide produced by expression from a recombinant polynucleotide. The term recombinant as used with respect to a host cell or a virus means a host cell or virus into which a recombinant polynucleotide has been introduced. Recombinant is also used herein to refer to, with reference to material (e.g., a cell, a nucleic acid, a protein, or a vector) that the material has been modified by the introduction of a heterologous material (e.g., a cell, a nucleic acid, a protein, or a vector).

[0070] The terms "control", "control reaction", "control assay", and the like, refer to a reaction, test, or other portion of an experimental or diagnostic procedure or experimental design for which an expected result is known with high certainty, e.g., in order to indicate whether the results obtained from associated experimental samples are reliable, indicate to what degree of confidence associated experimental results indicate a true result, and/or to allow for the calibration of experimental results. For example, in some instances, a control may be a "negative control" such that an essential component of the assay is excluded from the negative control reaction such that an experimenter may have high certainty that the negative control reaction will not produce a positive result. In some instances, a control may be "positive control" such that all components of a particular assay are characterized and known, when combined, to produce a particular result in the assay being performed such that an experimenter may have high certainty that the positive control reaction will not produce a positive negative result.

[0071] A "biological sample" encompasses a variety of sample types obtained from an individual or a population of individuals and can be used in a diagnostic, monitoring or screening assay. The definition encompasses blood and other liquid samples of biological origin, solid tissue samples such as a biopsy specimen or tissue cultures or cells derived therefrom and the progeny thereof. The definition also includes samples that have been manipulated in any way after their procurement, such as by mixing or pooling of individual samples, treatment with reagents, solubilization, or enrichment for certain components, such as cells, polynucleotides, polypeptides, etc. The term "biological sample" encompasses a clinical sample, and also includes cells in culture, cell supernatants, cell lysates, serum, plasma, biological fluid, and tissue samples. The term "biological sample" includes urine, saliva, cerebrospinal fluid, interstitial fluid, ocular fluid, synovial fluid, blood fractions such as plasma and serum, and the like. The term "biological sample" also includes solid tissue samples, tissue culture samples, and cellular samples.

[0072] The term "assessing" includes any form of measurement, and includes determining if an element is present or not. The terms "determining", "measuring", "evaluating", "assessing" and "assaying" are used interchangeably and include quantitative and qualitative determinations. Assessing may be relative or absolute. "Assessing the presence of includes determining the amount of something present, and/or determining whether it is present or absent. As used herein, the terms "determining," "measuring," and "assessing," and "assaying" are used interchangeably and include both quantitative and qualitative determinations.

DETAILED DESCRIPTION

[0073] Provided are nucleic acids that encode decoy-resistant interleukin 18 (DR-18) polypeptides, as well as vectors and cells comprising such nucleic acids and cells that comprise such vectors. Cells encoding DR-18 polypeptides may be referred to as DR-18 armored cells. The nucleic acids may further encode a chimeric antigen receptor (CAR) or multiple CARs, including CARs that bind to antigens described herein. Methods are also provided, including methods of making the nucleic acids, vectors, and/or cells, as well as methods of use, such as employing the nucleic acids, vectors, and/or cells, in the treatment of a subject having cancer.

[0074] Before the present invention is described in greater detail, it is to be understood that this invention is not limited to particular embodiments described, as such may, of course, vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting, since the scope of the present invention will be limited only by the appended claims.

[0075] Where a range of values is provided, it is understood that each intervening value, to the tenth of the unit of the lower limit unless the context clearly dictates otherwise, between the upper and lower limit of that range and any other stated or intervening value in that stated range, is encompassed within the invention. The upper and lower limits of these smaller ranges may independently be included in the smaller ranges and are also encompassed within the invention, subject to any specifically excluded limit in the stated range. Where the stated range includes one or both of the limits, ranges excluding either or both of those included limits are also included in the invention.

[0076] Certain ranges are presented herein with numerical values being preceded by the term "about." The term "about" is used herein to provide literal support for the exact number that it precedes, as well as a number that is near to or approximately the number that the term precedes. In determining whether a number is near to or approximately a specifically recited number, the near or approximating unrecited number may be a number which, in the context in which it is presented, provides the substantial equivalent of the specifically recited number.

[0077] Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. Although any methods and materials similar or equivalent to those described herein can also be used in the practice or testing of the present invention, representative illustrative methods and materials are now described.

[0078] All publications and patents cited in this specification are herein incorporated by reference as if each individual publication or patent were specifically and individually indicated to be incorporated by reference and are incorporated herein by reference to disclose and describe the methods and/or materials in connection with which the publications are cited. The citation of any publication is for its disclosure prior to the filing date and should not be construed as an admission that the present invention is not entitled to antedate such publication by virtue of prior invention. Further, the dates of publication provided may be different from the actual publication dates which may need to be independently confirmed.

[0079] It is noted that, as used herein and in the appended claims, the singular forms "a", "an", and "the" include plural referents unless the context clearly dictates otherwise. It is further noted that the claims may be drafted to exclude any optional element. As such, this statement is intended to serve as antecedent basis for use of such exclusive terminology as "solely," "only" and the like in connection with the recitation of claim elements, or use of a "negative" limitation.

[0080] As will be apparent to those of skill in the art upon reading this disclosure, each of the individual embodiments described and illustrated herein has discrete components and features which may be readily separated from or combined with the features of any of the other several embodiments without departing from the scope or spirit of the present invention. Any recited method can be carried out in the order of events recited or in any other order which is logically possible.

[0081] While the apparatus and method has or will be described for the sake of grammatical fluidity with functional explanations, it is to be expressly understood that the claims, unless expressly formulated under 35 U.S.C. § 112, are not to be construed as necessarily limited in any way by the construction of "means" or "steps" limitations, but are to be accorded the full scope of the meaning and equivalents of the definition provided by the claims under the judicial doctrine of equivalents, and in the case where the claims are expressly formulated under 35 U.S.C. § 112 are to be accorded full statutory equivalents under 35 U.S.C. § 112.

Systems & Compositions

[0082] The present disclosure provides compositions, including nucleic acid compositions for the expression of polypeptides from a cell into which the nucleic acids have been introduced. For example, a nucleic acid composition of the present disclosure may be contacted with a human cell under conditions sufficient for the nucleic acid composition to enter the cell and one or more encoded polypeptides to be expressed from the cell. The subject nucleic acid compositions may be contacted with cells ex vivo to introduce the nucleic acids into the cells, e.g., outside of a human body. The subject nucleic acid compositions may be contacted with the cells in vivo, e.g., through administration of a nucleic acid composition to a human subject, such that the nucleic acids are introduced in a cell of the subject within the human body.

[0083] Nucleic acid compositions of the present disclosure will include a sequence encoding a polypeptide, such as decoy-resistant interleukin 18 (DR-18) polypeptide, which polypeptides are described in more detail below. A cell or cells expressing a DR-18 polypeptide, or transduced or transfected with a DR-18 encoding nucleic acid, may be referred to herein as a DR-18 armored cell or cells. Nucleic acid compositions of the present disclosure may include a sequence encoding a chimeric antigen receptor (CAR) or multiple sequences encoding CARs, including e.g., where the nucleic acid includes multiple CAR-encoding sequences each encoding a different CAR. CARs, and their components, are described in more detail below. Sequences that encode polypeptides in such systems are not limited to DR-18 polypeptides and CAR polypeptides and may include various other polypeptide-encoding sequences such as, but not limited to e.g., various other cytokine-encoding sequences, and the like. A nucleic acid, construct, system, or cell involving the expression of a DR-18 polypeptide and a CAR, such as those configured with a DR-18 encoding nucleic acid and a CAR encoding nucleic acid, may be referred to herein as DR-18 armored, such as a DR-18 armored CAR or a DR-18 armored CAR nucleic acid, construct, vector, system, cell, or any combination thereof.

[0084] Systems of multiple nucleic acid sequences are also provided. At least one sequence of such a system will encode a polypeptide. Systems may include multiple polypeptide-encoding sequences each encoding a different polypeptide, such as e.g., one sequence encoding a DR-18 polypeptide and one sequence encoding a CAR. As described in more detail below, multiple polypeptide-encoding sequences in a nucleic acid may be configured in a multicistronic fashion, such as in a multicistronic expression vectors, to allow for co-expression of the multiple polypeptides from the nucleic acid. Accordingly, systems of the present disclosure include wherein multiple different polypeptides are encoded on and expressed from the same polynucleic acid molecule. Systems of the present disclosure include wherein multiple different polypeptides are encoded on and expressed from separate polynucleic acid molecules. Systems of the present disclosure also include combinations wherein one or more polypeptides are encoded on and expressed from a first polynucleic acid molecule and one or more different polypeptides are encoded on and expressed from a second polynucleic acid molecule.

[0085] Expressed polypeptides of the herein described systems may or may not directly or indirectly interact. For example, in the case of a DR-18 polypeptide and a CAR expressed from a nucleic acid the polypeptides may not directly bind to one another or may not share a common binding partner and thus will not directly or indirectly interact. In another example, in the case of a split polypeptide, such as a split CAR, the two or more polypeptides of the split polypeptide may be configured to directly bind to one another, and thus directly interact, or to directly bind to a common binding partner, and thus indirectly interact.

[0086] Polypeptides encoded in such systems may or may not be modified following translation. For example, an expressed polypeptide may be configured to include a heterologous cleavage site that is cut at some point following translation or an expressed polypeptide may include a portion, endogenous or heterologous, of the polypeptide that is cleaved following translation, such as e.g., an endogenous or heterologous signal peptide. Modification of an expressed polypeptide is not limited to cleavage and may include, e.g., essentially any post-translational modification.

[0087] Such systems may also include one or more sequences that do not code for a polypeptide, which may be referred to herein as non-coding sequences. Such non-coding sequences may provide for various functions within the system. For example, useful non-coding sequences include regulatory elements, such as DNA regulatory elements, post-transcriptional regulatory elements, and other functional non-coding elements, such as e.g., promoters, enhancers, introns, kozak sequences, slice acceptor sites, splice donor sites, internal ribosome entry sites, and the like.

Decoy Resistant IL-18 Polypeptides

[0088] As described herein, nucleic acid compositions and/or systems of the present disclosure will include a sequence encoding a DR-18 polypeptide. A DR-18 polypeptide-encoding sequence may be present on the same polynucleic acid molecule (such as e.g., the same vector) as a CAR-encoding sequence or the two encoding sequences may be present on separate polynucleic acid molecules (such as e.g., separate vectors). Useful DR-18 polypeptides that may be encoded by a sequence present in a nucleic acid will vary and are described in more detail below.

[0089] Decoy-resistant IL-18 polypeptides bind to, and signal through formation of, the IL-18Rα (IL-18 receptor α) and IL-18Rβ (IL-18 receptor β) receptor complex. Decoy-resistant IL-18 polypeptides do not bind to IL-18 binding protein (IL-18BP), or display substantially reduced binding to IL-18BP, such as substantially reduced binding to IL-18BP relative to wild-type (WT) human IL-18 (SEQ ID NO: 1) (i.e., as compared to the binding of IL-18BP to WT IL-18 (SEQ ID NO: 1)). In some embodiments, the DR IL-18 polypeptide does not bind to IL-18BP. In some embodiments, the DR IL-18 polypeptide has reduced binding to IL-18BP relative to WT IL-18. In some embodiments, the DR IL-18 polypeptide binds to IL-18Rα and does not bind to IL-18BP. In some embodiments, the DR IL-18 polypeptide binds to IL-18Rα and has reduced binding to IL-18BP relative to WT IL-18. In some instances, substantially reduced binding to IL-18BP may be represented by a DR IL-18 polypeptide having a Ko for IL-18BP of at least 10 nM or greater. In some instances, substantially reduced binding to IL-18BP may be represented by a DR IL-18 polypeptide having a Ko for IL-18BP that is at least twice, at least three times, at least four times, at least five times or greater, or at least ten times the Ko of WT human IL-18 for IL-18BP.

[0090] DR-18 polypeptides described herein are expressed from DR-18 encoding nucleic acids. Human WT IL-18 is expressed initially in a pre-processed form (SEQ ID NO: 2) that includes a 36 amino acid propeptide (SEQ ID NO: 3). DR-18 encoding nucleic acids may include or exclude sequence encoding the WT human IL-18 propeptide (SEQ ID NO: 3). As such, DR-18 polypeptides may be expressed with or without the WT human IL-18 propeptide. DR-18 polypeptides expressed with a WT human IL-18 propeptide are subsequently processed, similar to human WT IL-18, to produce the active mature form of the polypeptide. Unless clearly intended otherwise, references to WT IL-18 polypeptides and DR-18 polypeptides will generally refer to the active mature forms of the polypeptides (i.e., without a WT human IL-18 propeptide). For example, reference to a DR-18 polypeptide having X mutations relative to WT human IL-18 shall, unless specifically indicated otherwise, refer to X mutations occurring within amino acids 1 to 157 of SEQ ID NO: 1 or amino acids 37 to 193 of SEQ ID NO: 2.

[0091] In some embodiments, the DR-18 polypeptide has at least about two mutations, at least about three mutations, at least about four mutations, at least about five mutations, at least about six mutations, at least about seven mutations, at least about eight mutations, at least about nine mutations, or at least about ten mutations, relative to WT IL-18 as set forth in SEQ ID NO: 1. In some embodiments, the DR-18 polypeptide has about two mutations, about three mutations, about four mutations, about five mutations, about six mutations, about seven mutations, about eight mutations, about nine mutations, or about ten mutations, relative to WT IL-18 as set forth in SEQ ID NO: 1. In some instances, a DR-18 polypeptide may comprise an amino acid sequence having at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity with WT IL-18 as set forth in SEQ ID NO: 1.

[0092] Useful DR-18 polypeptides include but are not limited to e.g., a DR-18 polypeptide comprising any of the amino acid sequences of SEQ ID NOs.:4-65. In some instances, a DR-18 polypeptide may comprise an amino acid sequence having at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% with at least one of SEQ ID NOs: 4-65. In some instances, a DR-18 polypeptide may comprise an amino acid sequence having 100% or at least 95%, 96%, 97%, 98%, or 99% with SEQ ID NO:04. In some instances, a DR-18 polypeptide may comprise an amino acid sequence having 100% or at least 95%, 96%, 97%, 98%, or 99% with SEQ ID NO:19. In some instances, a DR-18 polypeptide may comprise an amino acid sequence having 100% or at least 95%, 96%, 97%, 98%, or 99% with SEQ ID NO:20. In some instances, a DR-18 polypeptide may comprise an amino acid sequence having 100% or at least 95%, 96%, 97%, 98%, or 99% with SEQ ID NO:21. In some instances, a DR-18 polypeptide may comprise an amino acid sequence having 100% or at least 95%, 96%, 97%, 98%, or 99% with SEQ ID NO:22. In some instances, a DR-18 polypeptide may comprise an amino acid sequence having 100% or at least 95%, 96%, 97%, 98%, or 99% with SEQ ID NO:23. In some instances, a DR-18 polypeptide may comprise an amino acid sequence having 100% or at least 95%, 96%, 97%, 98%, or 99% with SEQ ID NO:24.

[0093] In some instances, a useful DR-18 polypeptide comprises at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 13, 14, or 15 mutated residues relative to SEQ ID NO:1. In some instances, a useful DR-18 polypeptide comprises up to 15, 14, 13, 12, 11, or 10 mutated residues relative to SEQ ID NO:1. In some instances, a useful DR-18 polypeptide comprises from 1 to 20, from 2 to 20, from 3 to 20, from 4 to 20, from 5 to 20, from 6 to 20, from 7 to 20, from 8 to 20, from 9 to 20, from 10 to 20, from 11 to 20, from 12 to 20, from 13 to 20, from 14 to 20, from 15 to 20, from 1 to 18, from 2 to 18, from 3 to 18, from 4 to 18, from 5 to 18, from 6 to 18, from 7 to 18, from 8 to 18, from 9 to 18, from 10 to 18, from 11 to 18, from 12 to 18, from 13 to 18, from 14 to 18, from 15 to 18, from 1 to 16, from 2 to 16, from 3 to 16, from 4 to 16, from 5 to 16, from 6 to 16, from 7 to 16, from 8 to 16, from 9 to 16, from 10 to 16, from 11 to 16, from 12 to 16, from 13 to 16, from 14 to 16, from 1 to 14, from 2 to 14, from 3 to 14, from 4 to 14, from 5 to 14, from 6 to 14, from 7 to 14, from 8 to 14, from 9 to 14, from 10 to 14, from 11 to 14, from 12 to 14, from 1 to 12, from 2 to 12, from 3 to 12, from 4 to 12, from 5 to 12, from 6 to 12, from 7 to 12, from 8 to 12, from 9 to 12, from 10 to 12, from 1 to 10, from 2 to 10, from 3 to 10, from 4 to 10, from 5 to 10, from 6 to 10, from 7 to 10, or from 8 to 10 mutated residues relative to SEQ ID NO:1.

[0094] In some instances, a useful DR-18 polypeptide comprises an amino acid sequence of any one of SEQ ID NOs: 4-65 with 0, 1, 2, 3, 4, 5, or 6 additional mutated residues relative to SEQ ID NO:1.

[0095] In some instances, a useful DR-18 polypeptide comprises one or more mutations at positions Y1, L5, K8, C38, M51, K53, S55, Q56, P57, G59, M60, C68, E77, Q103, S105, D110, N111, M113, V153, and N155 relative to the WT IL-18 amino acid sequence set forth in SEQ ID NO:1, including e.g., at least 1, at least 2, at least 3, at least 4, at least 5, at least 6, at least 7, at least, 8, at least 9, at least 10, at least 11, or at least 12 of the listed mutated positions. Accordingly, useful DR-18 polypeptides can have one or any combination of mutations at the listed positions, with or without additional mutated residues at other positions. For example, useful polypeptides can have a mutation at position Y1, L5, K8, C38, M51, K53, S55, Q56, P57, G59, M60, C68, E77, Q103, S105, D110, N111, M113, V153, or N155 only; mutations at all positions Y1, L5, K8, C38, M51, K53, S55, Q56, P57, G59, M60, C68, E77, Q103, S105, D110, N111, M113, V153, and N155; or any combinations having more than one but less than all (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18 or 19) mutations at these positions.

[0096] In some instances, the DR-18 polypeptide comprises at least 12 mutations, at least 11 mutations, at least 10 mutations, at least 9 mutations, at least 8 mutations, at least 7 mutations, at least 6 mutations, at least 5 mutations, at least 4 mutations, at least 3 mutations, at least 2 mutations, or at least 1 mutation relative to the wild-type IL-18 amino acid sequence set forth in SEQ ID NO:1, optionally wherein the at least 10, 9, 8, 7, 6, 5, 4, 3, 2, or 1 mutations are at positions selected from Y1, L5, K8, C38, M51, K53, S55, Q56, P57, G59, M60, C68, E77, Q103, S105, D110, N111, M113, V153, and N155.

[0097] In some instances, the DR-18 polypeptide has less than 100%, 99%, 98%, 97%, 96%, 95%, 94%, 93%, 92%, 91%, 90%, 89%, 88%, 87%, 86%, or 85% sequence identity to WT human IL-18 (SEQ ID NO: 1) and comprises the following amino acid sequence: XFGKXESXLS-VIRNLNDQVLFIDQGNRPLFEDMTDSDXRDNAPRTI-FIISXYXDXXXRXXAVTISV KXEKISTLSXXNKIIS-FKEMNPPDNIKDTKSDIIFFXRXVPGHXXKXQFESSS YEGYFLAXEKERD LFKLILKKEDELGDR-SIMFTXQXED (SEQ ID NO: 66), wherein the X at pos. 1 is Y, R or H; the X at pos. 5 is L, H, I or Y; the X at pos. 8 is K, Q or R; the X at pos. 38 is C or S; the X at pos. 51 is M, T, K, D, N, E or R; the X at pos. 53 is K, R, G, S or T; the X at pos. 55 is S, K or R; the X at pos. 56 is Q, E, A, R, V, G, K, L or R; the X at pos. 57 is P, L, G, A or K; the X at pos. 59 is G, A or T; the X at pos. 60 is M, K, Q, R or L; the X at pos. 68 is C, S, G, A, V, D, E or N; the X at pos. 76 is C or S; the X at pos. 77 is E or D; the X at pos. 103 is Q, E, K, P, A or R; the X at pos. 105 is S, D, N, R, K or A; the X at pos. 110 is D, K, H, N, Q, E, S or G; the X at pos. 111 is N, H, Y, D, R, S or G; the X at pos. 113 is M, V, R, T or K; the X at pos. 127 is C or S; the X at pos. 153 is V, I, T or A; and the X at pos. 155 is N, K or H.

[0098] In some instances, the DR-18 polypeptide comprises one or more substitutions relative to WT human IL-18 (SEQ ID NO: 1), including 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more substitutions selected from: Y1H, Y1R, L5H, L51, L5Y, K8Q, K8R, M51T, M51K, M51D, M51N, M51E, M51R, K53R, K53G, K53S, K53T, S55K, S55R, Q56E, Q56A, Q56R, Q56V, Q56G, Q56K, Q56L, P57L, P57G, P57A, P57K, G59T, G59A, M60K, M60Q, M60R, M60L, E77D, Q103E, Q103K, Q103P, Q103A, Q103R, S105R, S105D, S105K, S105N, S105A, D110H, D110K, D110N, D110Q, D110E, D110S, D110G, N111H, N111Y, N111D, N111R, N111S, N111G, M113V, M113R, M113T, M113K, V1531, V153T, V153A, N155K, and N155H.

[0099] In some instances, the DR-18 polypeptide comprises one or more substitutions relative to WT human IL-18 (SEQ ID NO: 1) such that, e.g., the amino acid at pos. 1 is not Y and, e.g., may be R or H; the amino acid at pos. 5 is not L and, e.g., may be H, I or Y; the amino acid at pos. 8 is not K and, e.g., may be Q or R; the amino acid at pos. 38 is not C and, e.g., may be S; the amino acid at pos. 51 is not M and, e.g., may be T, K, D, N, E or R; the amino acid at pos. 53 is not K and, e.g., may be R, G, S or T; the amino acid at pos. 55 is not S and, e.g., may be K or R; the amino acid at pos. 56 is not Q and, e.g., may be E, A, R, V, G, K, L or R; the amino acid at pos. 57 is not P and, e.g., may be L, G, A or K; the amino acid at pos. 59 is not G and, e.g., may be A or T; the amino acid at pos. 60 is not M and, e.g., may be K, Q, R or L; the amino acid at pos. 68 is not C and, e.g., may be S, G, A, V, D, E or N; the amino acid at pos. 76 is not C and, e.g., may be S; the amino acid at pos. 77 is not E and, e.g., may be D; the amino acid at pos. 103 is not Q and, e.g., may be E, K, P, A or R; the amino acid at pos. 105 is not S and, e.g., may be D, N, R, K or A; the amino acid at pos. 110 is not D and, e.g., may be K, H, N, Q, E, S or G; the amino acid at pos. 111 is not N and, e.g., may be H, Y, D, R, S or G; the amino acid at pos. 113 is not M and, e.g., may be V, R, T or K; the amino acid at pos. 127 is not C and, e.g., may be S; the amino acid at pos. 153 is not V and, e.g., may be I, T or A; and/or the amino acid at pos. 155 is not N and, e.g., may be K or H.

[0100] In some instances, the DR-18 polypeptide comprises substitution mutations, relative to WT human IL-18 as set forth in SEQ ID NO:1, at positions: (i) M51, M60, S105, D110, and N111; (ii) M51, S55, G59, M60, S105, D110, N111, and V153; (iii) Y1, M51, M60, S105, D110, and N111; (iv) Y1, M51, K53, M60, S105, D110, and N111; (v) K8, M51, S55, G59, M60, S105, D110, and N155; (vi) K8, M51, S55, G59, M60, S105, D110, N111, and V153; (vii) L5, M51, K53, M60, S105, D110, and V153; (viii) L5, M51, S55, G59, M60, S105, D110, N111, and N155; (ix) L5, M51, S55, M60, Q103, S105, D110, N111, and V153; (x) L5, M51, S55, M60, S105, D110, N111, V153, and N155; (xi) L5, M51, S55, G59, M60, S105, D110, N111, V153, and N155; (xii) L5, K8, M51, S55, M60, S105, N111, V153, and N155; (xiii) L5, K8, M51, K53, M60, S105, D110, N111, and N155; (xiv) Y1, L5, M51, K53, M60, S105, D110, and N155; (xv) Y1, M51, K53, G59, M60, S105, D110, N111, V153, and N155; (xvi) Y1, K8, M51, K53, M60, Q103, S105, D110, N111, and N155; (xvii) Y1, K8, M51, M60, S105, D110, and N111; (xviii) Y1, L5, M51, K53, M60, Q103, S105, D110, and N111; (xix) Y1, K8, M51, K53, G59, M60, Q103, S105, D110, N111, V153, and N155; (xx) Y1, K8, M51, K53, G59, M60, S105, D110, N111, and N155; (xxi) Y1, K8, M51, G59, M60, Q103, S105, D110, N111, V153, and N155; (xxii) Y1, L5, M51, G59, M60, E77, S105, D110, and N111; (xxiii) M51, Q56, P57, M60, Q103, S105, D110, N111, and M113; (xxiv) M51, Q56, P57, M60, Q103, S105, D110, and M113; (xxv) M51, K53, Q56, P57, M60, D110, and N111; (xxvi) M51, K53, Q56, P57, M60, Q103, S105, D110, N111, and M113; (xxvii) M51, K53, Q56, M60, Q103, S105, D110, N111, and M113; (xxviii) M51, K53, Q56, P57, Q103, S105, D110, N111, and M113; (xxix) M51, K53, Q56, P57, M60, S105, D110, and N111; (xxx) M51, K53, Q56, P57, M60, Q103, D110, N111, and M113; (xxxi) M51, Q56, P57, M60, Q103, D110, N111, and M113; or (xxxii) M51, K53, Q56, S105, D110, and N111.

[0101] In some instances, the DR-18 polypeptide comprises substitution, relative to WT human IL-18 as set forth in SEQ ID NO:1, of: (i) M51T, M60K, S105D, D110K, and N111H; (ii) M51T, S55K, G59A, M60K, S105D, D110K, N111H, and V1531; (iii) Y1R, M51T, M60K, S105D, D110K, and N111H; (iv) Y1R, M51T, K53R, M60K, S105N, D110K, and N111Y; (v) K8Q, M51T, S55K, G59T, M60K, S105R, D110H, and N155K; (vi) K8R, M51K, S55K, G59A, M60Q, S105D, D110K, N111H, and V1531; (vii) K8R, M51D, S55K, G59A, M60X, S105D, D110K, N111H, and V1531; (viii) L5H, M51T, K53R, M60K, S105D, D110N, and V153T; (ix) L51, M51K, S55K, G59A, M60Q, S105K, D110Q, N111H, and N155K; (x) L51, M51T, S55R, M60K, Q103E, S105D, D110H, N111H, and V1531; (xi) L51, M51T, S55K, M60K, S105D, D110K, N111H, V153T, and N155H; (xii) L51, M51T, S55K, G59A, M60K, S105R, D110H, N111H, V1531, and N155K; (xiii) L51, K8R, M51T, S55K, M60K, S105D, N111Y, V1531, and N155K; (xiv) L5Y, K8R, M51T, K53R, M60K, S105D, D110E, N111H, and N155K; (xv) Y1H, L5Y, M51T, K53R, M60K, S105D, D110H, and N155K; (xvi) Y1R, M51T, K53R, G59A, M60K, S105D, D110Q, N111H, V153A, and N155K; (xvii) Y1R, K8R, M51D, K53R, M60R, Q103K, S105N, D110K, N111Y, and N155H; (xviii) Y1R, K8R, M51N, K53R, M60Q, Q103K, S105R, D110N, N111H, and N155K; (xix) Y1R, K8R, M51T, M60K, S105D, D110K, and N111H; (xx) Y1R, L5H, M51T, K53R, M60K, Q103E, S105N, D110K, and N111Y; (xxi) Y1R, K8R, M51T, K53R,

G59A, M60K, Q103E, S105D, D110Q, N111H, V1531, and N155X; (xxii) Y1R, K8R, M51T, K53R, G59T, M60K, S105N, D110H, N111D, and N155H; (xxiii) Y1R, K8R, M51T, G59A, M60K, Q103E, S105D, D110Q, N111H, V1531, and N155K; (xxiv) Y1R, L5Y, M51T, G59T, M60K, E77D, S105D, D110K, and N111H; (xxv) Y1R, K8R, M51T, K53R, G59T, M60K, S105K, D110N, N111H, and N155K; (xxvi) M51E, Q56E, P57L, M60R, Q103P, S105A, D110N, N111R, and M113V; (xxvii) M51K, Q56A, P57G, M60L, Q103E, S105D, D110S, and M113V; (xxviii) M51K, K53G, Q56A, P57A, M60L, D110K, and N111R; (xxix) M51K, K53G, Q56R, P57G, M60L, Q103E, S105D, D110N, N111S, and M113R; (xxx) M51K, K53G, Q56V, M60L, Q103A, S105A, D110S, N111R, and M113T; (xxxi) M51K, K53S, Q56G, P57A, M60L, Q103A, S105A, D110G, N111R, and M113T; (xxxii) M51K, K53S, Q56K, P57A, Q103A, S105D, D110S, N111S, and M113R; (xxxiii) M51K, K53S, Q56L, P57A, M60L, S105D, D110S, and N111R; (xxxiv) M51K, K53S, Q56R, P57A, M60L, S105N, D110G, and N111R; (xxxv) M51K, K53S, Q56R, P57A, M60L, Q103A, D110G, N111R, and M113T; (xxxvi) M51K, K53S, Q56R, P57A, M60L, Q103A, S105D, D110S, N111G, and M113R; (xxxvii) M51K, K53T, Q56R, M60L, Q103E, S105D, D110S, N111S, and M113K; (xxxviii) M51K, K53T, Q56R, P57A, Q103E, S105D, D110N, N111D, and M113R; (xxxix) M51R, Q56G, P57K, M60L, Q103R, D110S, N111R, and M113V; (xl) M51K, K53G, Q56G, P57A, M60L, Q103E, S105D, D110S, N111G, and M113V; or (xli) M51K, K53G, Q56R, S105A, D110N, and N111R.

[0102] In some embodiments, the DR IL-18 polypeptide comprises: (i) an amino acid sequence having 85% or more (e.g., 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97% or 98%) sequence identity with the any one of the amino sequences set forth in SEQ ID NO: 4-65; and (ii) one or more mutations at a wild-type cysteine present in WT IL-18 as set forth in SEQ ID NO: 1, such as e.g., C38, C68, C76, and/or C127. For example, in some instances, the DR IL-18 polypeptide comprises substitutions at amino acid positions Cysteine-38 and Cysteine-68 relative to WT human IL-18 SEQ ID NO: 1. In some instances, DR-18 polypeptide includes a stabilized DR-18 polypeptide. A stabilized DR-18 polypeptide can include mutations of two cysteine residues (e.g., C38 and C68) relative to human WT IL-18 (SEQ ID NO: 1). In some cases, the mutation is a C to S substitution and as such the stabilized IL-18 polypeptide can in some cases comprise the mutations C38S and C68S. In some cases a stabilized IL-18 polypeptide comprises the mutation pair C38S/C68S, C38S/C68G, C38S/C68A, C38S/C68V, C38S/C68D, C38S/C68E, or C38S/C68N (e.g., in some cases C38S/C68G, C38S/C68A, C38S/C68V, C38S/C68D, C38S/C68E, or C38S/C68N; in some cases C38S/C68S, C38S/C68G, C38S/C68A, C38S/C68D, or C38S/C68N; and in some cases C38S/C68G, C38S/C68A, C38S/C68D, or C38S/C68N).

[0103] DR-18 polypeptides can, additionally, be stabilized variants, through the presence of one or more stabilizing mutations. Useful DR-18 polypeptides, including in stabilized and non-stabilized forms, include those described in PCT Pub. Nos. WO2019/051015 and WO2022/094473, the disclosures of which are incorporated herein by reference in their entirety.

[0104] Useful IL-18 variants, as well as useful mutations and substitutions thereof, include but are not limited to those variants, mutations and substitutions described in WO2002101049 (such as e.g., variants having mutations at positions E42 and/or K89 and substitutions E42A and K89A (numbering according to SEQ ID NO:2)), US20080206189 (such as e.g., variants having mutations at one or more of C38, C68, C76, N78, E121, C127, L144, and D157 and substitutions, individually or in combination, of C38S, C68D, C68S, N78C, E121C, L144C, and D157C), WO2023118497 (such as e.g., variants having mutations at one or more of Y1, G3, S10, C38, M51, K53, S55, P57, M60, C68, Q103, M113, C127, and N155, and substitutions, individually or in combination, of Y1A, G3Y, S10R, S10K, C38S, M51A, M51Y, M51Q, K53L, K53H, K53A, S55L, S55Y, P57T, M60A, M60E, M60H, C68S, Q103S, M113A, M113E, C127S, and N155Y (inc. those at pos. K53 other than K53R, K53G, K53S, or K53T), WO2023114829 (such as e.g., variants having mutations at one or more of Y37, S43, L45, S46, V47, N50, F57, C74, T81, 185, S86, D90, S91, A97, V98, T99, S101, C104, 1107, T109, C112, N123, P124, Q139, R140, Q150, A162, C163, L174, 1185, V189, Q190, and E192, and substitutions, individually or in combination, of Y37C, S43C, L45C, S46C, V47C, N50C, F57C, C74S, T81C, 185C, S86C, D90C, S91C, A97C, V98C, T99C, S101C, C104S,1107C, T109C, C112S, N123C, P124C, Q139C, R140C, Q150C, A162C, C163S, L174C, 1185C, V189C, Q190C, E192C, (numbering according to SEQ ID NO:2)), WO2023056193 (such as e.g., variants having mutations at one or more of M51, K53, Q56, P57 and M60, and substitutions, individually or in combination, of M51A, M51S, M51G, K53G, K53A, K53S, Q56R, Q56K, Q56H, P57A, P57S, P57G, M60R, M60K, and M60H (and useful sequences encoding such variants, such as e.g., SEQ ID NOs: 7 and 8 of WO2023056193)), US20230146665 (such as e.g., variants having mutations at one or more of Y1, E6, S7, K8, S10, V11, N14, L15, D17, Q18, D23, R27, P28, L29, E31, M33, T34, D35, S36, D37, C38, R39, D40, N41, R44, 146, 149, S50, M51, K53, D54, S55, Q56, P57, M60, A61, V62, T63, S65, K67, C68, E69, 171, C76, E77, 180,181, N87, P88, D90, K93, T95, K96, S97, Q103, H109, D110, N111, M113, S119, A126, C127, D132, L136, L138, K139, E141, L144, D146, R147, 1149, M150, N155, E156, and D157, and substitutions, individually or in combination, of Y1F, Y1H, E6A, E6Q, S7C, S7P, K8E, K8Q, K8Y, S10C, V111, N14C, N14W, L15C, D17N, Q18L, D23N, D23S, R27Q, P28C, L29V, E31Q, M33C, T34P, D35N, D35E, S36D, S36N, D37N, C38S, C38Q, C38R, C38E, C38L, C381, C38V, C38K, C38D, R39S, R39T, D40N, N41Q, R44Q, 146V, 149C, S50C, S50Y, M511, M51K, M51Q, M51R, M51L, M51H, M51F, M51Y, K53A, K53D, K53E, K53G, K53H, K531, K53L, K53M, K53N, K53Q, K53R, K53S, K53T, K53V, K53Y, K53F, D54C, S55N, S55Q, S55D, S55E, S55T, Q561, Q56L, P57A, P57E, P57T, P57V, P57Q, P57D, P57Y, P57N, M601, M60L, M60K, M60Y, M60F, A61C, V62C, T63C, S65C, K67Q, C68S, C681, C68F, C68Y, C68D, C68N, C68E, C68Q, C68K, E69K, 171M, C76S, C76E, C76K, E77K, 180T, 181L, 181V, N87S, P88C, D90E, K93D, K93N, T95E, K96G, K96Q, S97N, Q103C, Q103E, Q1031, Q103L, H109W, H109Y, D110N, D110Q, D110R, N111D, N111Q, N111S, N111T, N111E, M1131, S119L, A126C, C127S, C127W, C127Y, C127F, C127D, C127E, C127K, D132Q, D132E, L136C, L138C, K139C, E141K, E141Q, L144N, D146F, D146L, D146Y, R147C, R147K, 1149V, M150F, M150T, N155C, E156Q, D157A, D157S, and D157N, as well as deletion mutants described therein), US20220056091 (such as e.g., variants having mutations at one or more of Y01, F02, E06, M33, C38, M51, K53, D54, S55, M60, T63, C68, E69, K70, C76, M86, M113, C127, and M150, and substitutions, individually or in combination, of Y01G, F02A, E06K, C38A, C38S, K53A, D54A, S55A, T63A, C68A, C68S, E69C, K70C, C76A, C76S, C127A, and C127S), US20230357342 (such as e.g., variants having mutations at one or more of Y1, F2, E6, K8, S10, V11, D17, E31, T34, D35, S36, D37, C38, D40, N41, 149, M51, K53, D54, S55, T63, C68, E69, K70, C76, Q103, S105, G108, H109, D110, C127, D132, and V153, and substitutions, individually or in combination, of Y1M, Y1G, F2A, E6K, E6R, K8E, K8L, K8R, S10T, V111, D17N, E31A, T34A, D35A, S36A, D37A, C38A, C38Q, C38S, D40A, N41A, 149E, 149M, 149R, M51G, K53A, D54A, S55A, S55H, S55R, S55T, T63A, C68A, C68S, E69C, K70C, C76A, C76S, Q103E, Q103K, Q103R, S1051, S105K, G108A, H109A, D110A, C127A, C127S, D132A, V153E, V153R, and V153Y), and WO2022172944 (such as e.g., variants having mutations at one or more of C38, C68, C76, C127, G3, A6, S7, T34, S50, M51, S72, K112, K119, and G145, and substitutions, individually or in combination, of C38S, C68S, C76S, C127S, G3Y, G3L, A6W, S7C, T34P, C38M, S50C, M51Y, S72Y, S72F, S72M, S72L, S72W, K112W, K119V, and G145N).

[0105] A nucleic acid sequence encoding a DR-18 polypeptide may be generated from the DR-18 polypeptide amino acid sequence, including e.g., any of the DR-18 polypeptide amino acid sequences described herein. Such DR-18 polypeptide-encoding sequences may be produced by conversion of each amino acid residue to a corresponding trinucleotide codon that codes for the residue. Such DR-18 polypeptide-encoding sequences may include a stop codon after the codon designating the terminal amino acid. In some instances, a stop codon may be added.

[0106] In some instances, DR-18 polypeptide encoding sequences of the present disclosure may be codon optimized, i.e., optimized for expression in a particular host cell or organism. In some instances, a DR-18 polypeptide encoding sequence of the present disclosure is optimized for expression in human cells, i.e., is human codon optimized. Useful methods for codon optimization, including human codon optimization, include but are not limited to e.g., those described in US20140244228, US20210366574, US20190325989, US20230245721, US20160259885, US20140377800, WO2020024917, WO2022221576, WO2024018050, and WO2009009743.

[0107] Useful human codon optimized DR-18 polypeptide-encoding sequences include human codon optimized versions encoding DR-18 polypeptides referred to herein as "hCS1" (SEQ ID NO:25), such as e.g., SEQ ID NOs:67 and 68; "hCS2" (SEQ ID NO:26), such as e.g., SEQ ID NOs:69 and 70; "hCS3" (SEQ ID NO:27), such as e.g., SEQ ID NOs:71 and 72; "hCS4" (SEQ ID NO:28), such as e.g., SEQ ID NOs:73 and 74; "6-12" (SEQ ID NO:57), such as e.g., SEQ ID NOs:75 and 76; "sq89v6" (SEQ ID NO:9), such as e.g., SEQ ID NOs:77 and 78; "C68v1" (SEQ ID NO:19), such as e.g., SEQ ID NOs:79 and 80; "C68v2" (SEQ ID NO:20), such as e.g., SEQ ID NOs:81 and 82; and "C68v4" (SEQ ID NO:22), such as e.g., SEQ ID NOs:83 and 84. The forgoing codon optimized sequences are merely examples and alternative codon optimized sequences, encoding these DR-18 polypeptides or any other DR-18 polypeptide, may be readily generated, e.g., by modifying the parameters of codon optimization algorithms that generate codon optimized nucleic acid sequences from input amino acid sequences. Common codon optimization parameters that may be modified include, but are not limited to, the allowance or prohibition of certain restriction enzyme recognition sites, the allowed or prohibited frequency of rare codons, the allowed presence or absence of out-of-frame cryptic translation start sites, the allowed or preferred overall GC concentration, the allowed or preferred GC distribution patterns, guanine and cytosine content at the third codon position (GC3), and the like.

[0108] In some embodiments, the sequence encoding the DR-18 polypeptide has at least 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%, or has 100%, sequence identity with any one of SEQ ID NOs: 67-84. In some embodiments, the sequence encoding the DR-18 polypeptide has less than 100% but at least 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity with any one of SEQ ID NOs: 67-84.

Chimeric Antigen Receptors (CARs)

[0109] As described herein, nucleic acid compositions and/or systems of the present disclosure may include a sequence encoding a CAR, or multiple sequences encoding multiple different CARs. A CAR-encoding sequence may be present on the same polynucleic acid molecule (such as e.g., the same vector) as a DR-18-encoding sequence or the two encoding sequences may be present on separate polynucleic acid molecules (such as e.g., separate vectors). Where multiple different CARs are employed, the multiple different CAR-encoding sequence may be present on the same polynucleic acid molecule (such as e.g., the same vector) or on separate polynucleic acid molecules (such as e.g., separate vectors).

[0110] A CAR is a synthetic receptor engineered to grant immune cells, such as T cells, the ability to target and destroy specific target cells, including cancer cells, as desired. A CAR is a fusion protein typically including an extracellular recognition domain (extracellular antigen-recognition domain), often but not exclusively derived from a monoclonal antibody, and an intracellular signaling domain that activates the immune cell upon binding of the extracellular recognition domain to its binding partner (e.g., antigen). The extracellular part of a CAR will specifically recognize, or bind to, one or more binding partners (e.g., antigens), often present on the surface of a target cell, such as a tumor cell. While useful binding partners of a CAR will vary widely, for simplicity, the binding partner bound by the extracellular recognition domain of a CAR may be generally referred to herein as an antigen. However, use of the term "antigen", except where expressly indicated otherwise, should not be interpreted as limiting, e.g., on the nature, type, identity, or function of the CAR or relevant extracellular recognition domain at least because an ordinarily skilled artisan will readily recognize that various types of extracellular recognition domains that have various different types of binding partners may be employed in a CAR.

[0111] The intracellular portion of the CAR, once engaged with the antigen, triggers signaling events that lead to cell activation, proliferation, and/or release of signaling molecules, such as endogenous cytokines. Activation enables immune cells modified to express the CAR, such as CAR T cells, e.g., to effectively locate and eliminate cancer cells.

CAR-based cell therapy, which often involves the genetic modification (ex vivo or in vivo) of a patient's own (i.e., autologous) cells to express one or more CARs, has shown significant success, resulting in numerous commercial and clinical stage CAR-based therapies. In some instances, the cells employed in a CAR-based cell therapy, i.e., administered to a subject in need thereof, may be derived from a heterologous (e.g., allogeneic) source, i.e., as related to the subject being treated.

[0112] CARs have been engineered into T cells; however, their use is not so limited. CARs may be engineered into a variety of cell types, including but not limited to, e.g., immune cells generally, T cells, Naïve T cells (T_N), stem cell memory T cells (T_{SCM}), central memory T cells (T_{CM}), resident memory T cells (T_{RM}), effector T cells (T_{EFF}), effector memory cells (T_{EM}), alpha-beta ($\alpha\beta$) T cells, gamma-delta ($\gamma\alpha$) T cells, Natural Killer (NK) cells, CD4+ T cells, CD4+ T cell subsets (e.g., Th1, Th2, Th9, Th17, Th22, T Regulatory Cells (Tregs), and Tfh), resting Treg cells, effector Treg cells, effector memory Treg cells, CD8+ T cells, CD8+T subsets (e.g., Tc1, Tc2, Tc9, Th17, and Treg), macrophages, B Cells, Dendritic Cells (DCs), stem cells, and the like.

[0113] Different populations and subpopulations of cells, including immune cells, stem cells, and the like, can be characterized, separated, and/or subdivided based on expression (e.g., presence/absence, level (inc. relative levels, e.g., high, low, etc.) of cellular (inc. extracellular (e.g., cell surface) and/or intracellular), epigenetic, and genetic markers, as well as expression, activation, or inactivity of transcription factors and metabolic pathways, and combinations thereof. Useful markers include e.g., CD3, CD4, CD8, CCR4, CCR5, CCR6, CCR7, CCR10, CD127, CD27, CD28, CD38, CD40, CD45, CD45RA, CD45RO, CD58, CD69, CD83, CD161, CTLA4, CXCR3, CXCR4, FA5, HLA-DR, IL2RA (CD25), IL2RB, ITGAE, ITGAL, KLRB1 (CD16), NCAM1 (CD56), PECAM1, PTGDR2, CD62L (SELL), IFNG, IL10, IL13, IL17A, IL2, IL21, IL22, IL25, IL26, IL4, IL5, IL9, IL-23R, ITGB1, TNF, AHR, EOMES, FOXO4, FOXP3, GATA3, IRF4, LEF1, PRDM1, RORC, STAT4, STAT5A, TBX21, TCF7, GZMA, alpha-beta ($\alpha\beta$) T cell receptor, gamma-delta ($\gamma\alpha$) T cell receptor, E-Cadherin, Dectin-1 (CLEC7A), Fas (TNFRSF6/CD95), Fas Ligand (TNFSF6), ICOS, NKG2D (CD314), NKG2E, Occludin, TLR2, TRAIL/TNFSF10, and the like. In some instances, cells may be characterized by their cytokine profiles, such as e.g., Th1: interferon gamma (IFN-γ), tumor necrosis factor (TNF); Th2: IL-1, IL-5, and IL-13; Th9: IL-9; Th17: IL-17, IL-21, IL-22, IL-25, and IL-26; Th22: IL-22; Treg: IL-10 and TGF-β; Tfh: IL-21; and the like. Examples of relevant markers of cytotoxic T cell populations include e.g., Tc1: TNF alpha, INF gamma, IL-2 CXCR3, and TBX21; Tc2: IL-4, IL-5, CCR4, and GATA3; Tc9: IL-9, IL-10, and IRF4; Tc17: CCR6, KLRB1, IL-17, IRF4, and RORC; and the like.

[0114] As noted above, a CAR is a fusion protein that generally includes one or more extracellular antigen-recognition domains, a transmembrane domain, and one or more intracellular signaling domains. In some instances, one or more domains of a CAR may be separated onto two or more polypeptide chains that may be joined, reversibly or irreversibly and intracellularly or intracellularly, to form a functional CAR (such as e.g., as described in WO2014127261; WO2016138034; Liu et al. *Ther Adv Med Oncol.* 2020, Cho et al. *Nature Communications.* 2021; and the like). Useful CAR domains, and methods of screening for useful combinations of CAR domains, include those described in WO2017040694.

[0115] Useful intracellular signaling domains can include one or more immunoreceptor tyrosine-based activation motif (ITAM) containing domains, such as e.g., those derived from the T-cell receptor (TCR) complex. Useful intracellular domains include chains of the CD3 protein complex or portions thereof, such as the CD3 zeta (CD3ζ), CD3 delta (CD3δ), CD3 gamma (CD3γ), and CD3 epsilon (CD3ε) chains and portions thereof, often referred to as CD3ζ, CD3δ, CD3γ, and CD3ε intracellular signaling domains. Examples of ITAM containing intracellular signaling domains that are of particular use include those of TCR zeta, FcR gamma, FcR beta, CD3 gamma, CD3 delta, CD3 epsilon, CD5, CD22, CD79a, CD79b, CD278 (also known as "ICOS"), FcεRI, DAP10, DAP12, and CD66d. Further examples of molecules containing a primary intracellular signaling domain that are of particular use include those of DAP10, DAP12, and CD32. In some embodiments, a CAR of the invention comprises an intracellular signaling domain, e.g., a primary signaling domain of CD3-zeta.The number of ITAMs present in an ITAM-containing intracellular signaling domain will vary and may include e.g., one ITAM or multiple ITAMs, such as e.g., 2, 3, 4, or more ITAMs, and may include the natural number of ITAMs occurring in the endogenous protein or a number that is more or fewer than the natural number of ITAMs occurring in the endogenous protein (e.g., by engineering the protein to introduce, duplicate, or remove ITAMs).

[0116] Useful intracellular signaling domains also include downstream molecules of TCR signaling including domains derived from proximal TCR signaling molecules, such as e.g., Lymphocyte-specific PTK (LCK), Tyrosine-protein kinase Fyn (FYN), 70 kDa zeta-chain associated protein (ZAP-70), Linker for activation of T-cells family member 1 (LAT), SH2 domain-containing leukocyte protein of 76 kDa (SLP-76), Phosphoinositide phospholipase C-gamma-1 (PLCy1), and the like. For example, in some instances, a ZAP-70-based intracellular signaling domain and/or a PLCy1-based intracellular signaling domain may be employed as the intracellular signaling domain of a CAR (see e.g., Tousley et al. *Nature* 2023 615(7952): 507-516).

[0117] In some embodiments, a primary signaling domain may be employed. In some embodiments, the primary signaling domain comprises a functional signaling domain derived from CD3 zeta, TCR zeta, FcR gamma, FcR beta, CD3 gamma, CD3 delta, CD3 epsilon, CD5, CD22, CD79a, CD79b, CD278 (ICOS), FcεRI, DAP10, DAP12, or CD66d.

[0118] CARs may include one or more co-stimulatory domains, e.g., in addition to an ITAM-containing intracellular signaling domain (e.g., from a CD3(chain) or other intracellular signaling domain (e.g., from ZAP-70). Useful co-stimulatory domains include but are not limited to those derived from CD28, 4-1BB (CD137), or OX40 (CD134). In some embodiments, the costimulatory domain is a functional signaling domain of a protein selected from: MHC class I molecule, TNF receptor proteins, Immunoglobulin-like proteins, cytokine receptors, integrins, signaling lymphocytic activation molecules (SLAM proteins), activating NK cell receptors, BTLA, a Toll ligand receptor, OX40, CD2, CD7, CD27, CD28, CD30, CD40, CDS, ICAM-1, LFA-1 (CD11a/CD18), 4-1BB (CD137), B7-H3, CDS, ICAM-1, ICOS (CD278), GITR, BAFFR, LIGHT, HVEM

(LIGHTR), KIRDS2, SLAMF7, NKp80 (KLRF1), NKp44, NKp30, NKp46, CD19, CD4, CD8alpha, CD8beta, IL2R beta, IL2R gamma, IL7R alpha, ITGA4, VLA1, CD49a, ITGA4, IA4, CD49D, ITGA6, VLA-6, CD49f, ITGAD, CD11d, ITGAE, CD103, ITGAL, CD11a, LFA-1, ITGAM, CD11b, ITGAX, CD11c, ITGB1, CD29, ITGB2, CD18, LFA-1, ITGB7, NKG2D, NKG2C, TNFR2, TRANCE/RANKL, DNAM1 (CD226), SLAMF4 (CD244, 2B4), CD84, CD96 (Tactile), CEACAMI, CRTAM, Ly9 (CD229), CD160 (BY55), PSGL1, CD100 (SEMA4D), CD69, SLAMF6 (NTB-A, Ly108), SLAM (SLAMFI, CD150, IPO-3), BLAME (SLAMF8), SELPLG (CD162), LTBR, LAT, GADS, SLP-76, PAG/Cbp, CD19a, and a ligand that specifically binds with CD83.

[0119] A CAR will further include a transmembrane (TM) region, linking the intracellular and extracellular portions of the CAR, and a hinge region. Common sources for the hinge region include segments from immunoglobulin G (IgG), such as the CH2-CH3 domains, including from IgG4 and IgG1, and segments from other proteins like CD8a or CD28. Common sources for the TM region include, but are not limited to, the TM regions of CD4, CD8, CD28, immunoglobulins (such as IgG1 and IgG4), and the like. In some embodiments, the transmembrane domain is transmembrane domain of a protein selected from: the alpha, beta or zeta chain of the T-cell receptor, CD28, CD3 epsilon, CD45, CD4, CD5, CD8, CD9, CD16, CD22, CD33, CD37, CD64, CD80, CD86, CD134, CD137 and CD154. The hinge domain and TM domain of a CAR may share a common source (such as e.g., CD28 hinge+CD28 TM, IgG4 hinge+IgG4 TM, etc.) or may be derived from different sources (such as e.g., IgG4 hinge+CD28 TM, IgG1 hinge+CD4 TM, IgG4 hinge+CD28 TM, etc.).

[0120] As previously noted, a CAR may be expressed as a single polypeptide chain (i.e., including extracellular, transmembrane, and intracellular domains in a single polypeptide chain). In some instances, a CAR may be expressed as multiple polypeptides (e.g., with at least one domain of the CAR on a polypeptide that is separate from a different polypeptide that contains one or more other domains of the CAR) that are joined reversibly or irreversibly to form the functional CAR. Useful architectures of a CAR split between two or more polypeptides vary widely.

[0121] The extracellular recognition domain of the CAR determines the specificity of the CAR and may be monospecific or multi-specific. Useful extracellular recognition domains, also referred to antigen-binding domains, antigen-recognition domains, or simply recognition domains, are commonly derived from antibodies and various forms can be used including but not limited to e.g., signal-chain variable fragment (scFv) based domains, VHH (nanobody) based domains, Fv fragment based domains, and the like. Useful antigen-recognition domains are not necessarily limited to those derived from antibodies and also include e.g., those derived from natural or engineered ligand-receptor interactions. For example, a receptor, or portion thereof, that binds a ligand may be employed as the recognition domain where the ligand is the target. Alternatively, a ligand, or portion thereof, that binds to a receptor may be employed as the recognition domain where the receptor is the target. Accordingly, essentially any binding partner, or binding portion thereof, that binds to the relevant target, whether natural or synthetic, may be employed including e.g., where the antigen recognition domain is (or is derived from) a ligand, a receptor, a peptide-MHC complex, an aptamer, or the like. A monospecific CAR will have an antigen-recognition domain that binds a single antigen. A multi-specific CAR may have a multi-specific antigen-recognition domain that binds multiple different antigens, multiple different antigen-recognition domains that each bind a single (but different) antigen, or any combination thereof.

[0122] Useful antigen-recognition domains include antigen-recognition domains that recognize the following antigens: CD19, BCMA, HER2, CD22, MSLN, CLDN18.2, CD20, CD7, GPC3, CD30, MUC1, CD123, CD276, EGFRvIII, NKG2D, GD2, CD70, CD33, CD5, EGFR, CD38, CEACAM5, ROR1, IL-13Rα2, CEA, FOLR1, PDL1, HLA-A2, PD-1, CLL-1, LGR5, BAFF-R, EpCAM, K light chain of human immunoglobulins, CDH17, DLL3, ALPP, ULBP1, ROR2, CD44v6, SLAMF7, DR5, PSCA, CD4, MMP2, EBV Protein, LILRB4, EphA2, GPC2, FLT3, TAG-72, HLA-G, GC-C, CD133, TRBC1, AFP, BAFF-R, ICAM-1, CD1A, B7-H3, L1CAM, CD44, CD83, CD84, Lewis-Y antigen, CD99, TSHR, TM4SF1, CLDN6, HIV-1 pol, NY-ESO-1, KMA, 5T4, ALPPL2, ROBO1, αvβ3 integrin, CD43, NKG2DL, GM2 (ganglioside M2), CD24, HAAH (ASPH), gp100, CD72, OPCML, FSHR, MAGEA3, NCR3LG1, CLEC4K (CD207), CAIX, BSG, CD45, AXL, ITGB7, FGFR4, PLA2R, MG7, CD138, FAP, Trop-2, APRIL, TRAIL, HSP70 heat-shock proteins, PR1, M-CSF, CSPG4, PVR, B7-H4, nectin-4, KKLC1, Globo H, CD56, LMP1, CD3, TSLPR, TGF-β, SEMA4A, integrin av36, NECTIN2, enkephalinase (CD10), HIV envelope protein gp120, EMR1, EphA3, APN (CD13), TRBC2, GPC1, CDCP1, HGF, IL-10R, GFRA4, CD2, NR2F6, MICA, PTK7, PDGFRα, IL13Rα2, and the like. Useful antigen recognition domain further include antigen-recognition domains that recognize the antigens described in US20220127373, US20230398216, WO2020097395, and WO2022087453; the disclosures of which are incorporated by reference in their entirety. Useful antigen recognition domains, as well as antibodies from which antigen recognition domains can be derived, that bind the aforementioned antigens, and methods of making additional antigen recognition domains to such antigens, are known to the relevant skilled artisan. Examples of useful antigen recognition domains and antibodies from which additional antigen recognition domains can be derived for a representative sampling of antigens are as follows:

[0123] CD123 antigen recognition domains are known to the relevant skilled artisan and include, e.g., the heavy and light chain domains of pivekimab, talacotuzumab, vibecotamab, and the like. Useful antigen recognition domains include those derived from the heavy chain, light chain, and/or CDRs of antibodies that bind to CD123, such as but not limited to e.g., pivekimab, talacotuzumab, vibecotamab, and the like.

[0124] CD19 antigen recognition domains are known to the relevant skilled artisan and include, e.g., the heavy and light chain domains of budoprutug, coltuximab, denintuzumab, inebilizumab, leronlimab, loncastuximab, mogamulizumab, obexelimab, plozalizumab, tafasitamab, and the like. Useful antigen recognition domains include those derived from the heavy chain, light chain, and/or CDRs of antibodies that bind to CD19, such as but not limited to e.g., budoprutug, coltuximab, denintuzumab, inebilizumab, leronlimab, loncastuximab, mogamulizumab, obexelimab, plozalizumab, tafasitamab, and the like.

[0125] CD20 antigen recognition domains are known to the relevant skilled artisan and include, e.g., the heavy and light chain domains of divozilimab, ibritumomab, obinutuzumab, ocaratuzumab, ocrelizumab, ofatumumab, ripertamab, rituximab, ublituximab, veltuzumab, zuberitamab, and the like. Useful antigen recognition domains include those derived from the heavy chain, light chain, and/or CDRs of antibodies that bind to CD20, such as but not limited to e.g., divozilimab, ibritumomab, obinutuzumab, ocaratuzumab, ocrelizumab, ofatumumab, ripertamab, rituximab, ublituximab, veltuzumab, zuberitamab, and the like.

[0126] CD22 antigen recognition domains are known to the relevant skilled artisan and include, e.g., the heavy and light chain domains of epratuzumab, inotuzumab, pinatuzumab, suciraslimab, and the like. Useful antigen recognition domains include those derived from the heavy chain, light chain, and/or CDRs of antibodies that bind to CD22, such as but not limited to e.g., epratuzumab, inotuzumab, pinatuzumab, suciraslimab, and the like.

[0127] CD30 antigen recognition domains are known to the relevant skilled artisan and include, e.g., the heavy and light chain domains of brentuximab, iratumumab, and the like. Useful antigen recognition domains include those derived from the heavy chain, light chain, and/or CDRs of antibodies that bind to CD30, such as but not limited to e.g., brentuximab, iratumumab, and the like.

[0128] CD33 antigen recognition domains are known to the relevant skilled artisan and include, e.g., the heavy and light chain domains of gemtuzumab, lintuzumab, vadastuximab, and the like. Useful antigen recognition domains include those derived from the heavy chain, light chain, and/or CDRs of antibodies that bind to CD33, such as but not limited to e.g., gemtuzumab, lintuzumab, vadastuximab, and the like.

[0129] CD38 antigen recognition domains are known to the relevant skilled artisan and include, e.g., the heavy and light chain domains of daratumumab, erzotabart, felzartamab, isatuximab, lumrotatug, mezagitamab, modakafusp, and the like. Useful antigen recognition domains include those derived from the heavy chain, light chain, and/or CDRs of antibodies that bind to CD38, such as but not limited to e.g., daratumumab, erzotabart, felzartamab, isatuximab, lumrotatug, mezagitamab, modakafusp, and the like.

[0130] CD70 antigen recognition domains are known to the relevant skilled artisan and include, e.g., the heavy and light chain domains of cusatuzumab, vorsetuzumab, and the like. Useful antigen recognition domains include those derived from the heavy chain, light chain, and/or CDRs of antibodies that bind to CD70, such as but not limited to e.g., cusatuzumab, vorsetuzumab, and the like.

[0131] CLDN18.2 antigen recognition domains are known to the relevant skilled artisan and include e.g., the heavy and light chain domains of osemitamab, zolbetuximab, and the like. Useful antigen recognition domains include those derived from the heavy chain, light chain, and/or CDRs of antibodies that bind to CLDN18.2, such as but not limited to e.g., osemitamab, zolbetuximab, and the like.

[0132] CLL-1 antigen recognition domains are known to the relevant skilled artisan and include, e.g., the heavy and light chain domains of tepoditamab, and the like. Useful antigen recognition domains include those derived from the heavy chain, light chain, and/or CDRs of antibodies that bind to CLL-1, such as but not limited to e.g., tepoditamab, and the like.

[0133] DLL3 antigen recognition domains are known to the relevant skilled artisan and include, e.g., the heavy and light chain domains of rovalpituzumab, tarlatamab, and the like. Useful antigen recognition domains include those derived from the heavy chain, light chain, and/or CDRs of antibodies that bind to DLL3, such as but not limited to e.g., rovalpituzumab, tarlatamab, and the like.

[0134] EGFR antigen recognition domains are known to the relevant skilled artisan and include, e.g., the heavy and light chain domains of becotatug, cetuximab, demupitamab, futuximab, imgatuzumab, matuzumab, modotuximab, necitumumab, nimotuzumab, panitumumab, pimurutamab, tomuzotuximab, zalutumumab, and the like. Useful antigen recognition domains include those derived from the heavy chain, light chain, and/or CDRs of antibodies that bind to EGFR, such as but not limited to e.g., becotatug, cetuximab, demupitamab, futuximab, imgatuzumab, matuzumab, modotuximab, necitumumab, nimotuzumab, panitumumab, pimurutamab, tomuzotuximab, zalutumumab, and the like.

[0135] EGFRvII antigen recognition domains are known to the relevant skilled artisan and include e.g., the heavy and light chain domains of umizortamig, and the like. Useful antigen recognition domains include those derived from the heavy chain, light chain, and/or CDRs of antibodies that bind to EGFRvIII, such as but not limited to e.g., umizortamig, and the like.

[0136] FLT3 antigen recognition domains are known to the relevant skilled artisan and include, e.g., the heavy and light chain domains of emirodatamab, and the like. Useful antigen recognition domains include those derived from the heavy chain, light chain, and/or CDRs of antibodies that bind to FLT3, such as but not limited to e.g., emirodatamab, and the like.

[0137] GD2 antigen recognition domains are known to the relevant skilled artisan and include, e.g., the heavy and light chain domains of dinutuximab, lorukafusp, naxitamab, and the like. Useful antigen recognition domains include those derived from the heavy chain, light chain, and/or CDRs of antibodies that bind to GD2, such as but not limited to e.g., dinutuximab, lorukafusp, naxitamab, and the like.

[0138] GPC3 antigen recognition domains are known to the relevant skilled artisan and include, e.g., the heavy and light chain domains of codrituzumab, and the like. Useful antigen recognition domains include those derived from the heavy chain, light chain, and/or CDRs of antibodies that bind to GPC3, such as but not limited to e.g., codrituzumab, and the like.

[0139] HER2 antigen recognition domains are known to the relevant skilled artisan and include, e.g., the heavy and light chain domains of anvatabart, coprelotamab, disitamab, gancotamab, margetuximab, marstacimab, pertuzumab, timigutuzumab, trastuzumab, and the like. Useful antigen recognition domains include those derived from the heavy chain, light chain, and/or CDRs of antibodies that bind to HER2, such as but not limited to e.g., anvatabart, coprelotamab, disitamab, gancotamab, margetuximab, marstacimab, pertuzumab, timigutuzumab, and the like.

[0140] MSLN antigen recognition domains are known to the relevant skilled artisan and include, e.g., the heavy and light chain domains of amatuximab, anetumab, inezetamab, and the like. Useful antigen recognition domains include those derived from the heavy chain, light chain, and/or CDRs of antibodies that bind to MSLN, such as but not limited to e.g., amatuximab, anetumab, inezetamab, and the like.

[0141] MUC1 and MUC16 antigen recognition domains are known to the relevant skilled artisan and include, e.g., the heavy and light chain domains of cantuzumab, clivatuzumab, gatipotuzumab, abagovomab, sofituzumab, ubamatamab, and the like. Useful antigen recognition domains include those derived from the heavy chain, light chain, and/or CDRs of antibodies that bind to MUC1 and MUC16, such as but not limited to e.g., cantuzumab, clivatuzumab, gatipotuzumab, abagovomab, sofituzumab, ubamatamab, and the like.

[0142] NKG2D antigen recognition domains are known to the relevant skilled artisan and include, e.g., the heavy and light chain domains of tesnatilimab, and the like. Useful antigen recognition domains include those derived from the heavy chain, light chain, and/or CDRs of antibodies that bind to NKG2D, such as but not limited to e.g., tesnatilimab, and the like.

[0143] PD-1 antigen recognition domains are known to the relevant skilled artisan and include, e.g., the heavy and light chain domains of acrixolimab, balstilimab, budigalimab, camrelizumab, cemiplimab, cetrelimab, dostarlimab, enlonstobart, ezabenlimab, finotonlimab, geptanolimab, iparomlimab, lipustobart, nivolumab, nofazinlimab, penpulimab, peresolimab, pidilizumab, pimivalimab, pradusinstobart, prolgolimab, pucotenlimab, retifanlimab, rosnilimab, rulonilimab, sasanlimab, serplulimab, sintilimab, spartalizumab, tislelizumab, toripalimab, zeluvalimab, zimberelimab, and the like. Useful antigen recognition domains include those derived from the heavy chain, light chain, and/or CDRs of antibodies that bind to PD-1, such as but not limited to e.g., acrixolimab, balstilimab, budigalimab, camrelizumab, cemiplimab, cetrelimab, dostarlimab, enlonstobart, ezabenlimab, finotonlimab, geptanolimab, iparomlimab, lipustobart, nivolumab, nofazinlimab, penpulimab, peresolimab, pidilizumab, pimivalimab, pradusinstobart, prolgolimab, pucotenlimab, retifanlimab, rosnilimab, rulonilimab, sasanlimab, serplulimab, sintilimab, spartalizumab, tislelizumab, toripalimab, zeluvalimab, zimberelimab, and the like.

[0144] ROR1 antigen recognition domains are known to the relevant skilled artisan and include, e.g., the heavy and light chain domains of nebratamig, zilovertamab, and the like. Useful antigen recognition domains include those derived from the heavy chain, light chain, and/or CDRs of antibodies that bind to ROR1, such as but not limited to e.g., nebratamig, zilovertamab, and the like.

[0145] SLAMF7 antigen recognition domains are known to the relevant skilled artisan and include, e.g., the heavy and light chain domains of azintuxizumab, elotuzumab, and the like. Useful antigen recognition domains include those derived from the heavy chain, light chain, and/or CDRs of antibodies that bind to SLAMF7, such as but not limited to e.g., azintuxizumab, elotuzumab, and the like.

[0146] Useful CLDN6 antigen recognition domains include, e.g., the heavy and light chain domains of those antibodies described in U.S. Pat. Nos. 9,487,584, 9,274,119, 10,745,477, 10,233,253, and 10,053,511. Useful antigen recognition domains include those derived from the heavy chain, light chain, and/or CDRs of antibodies that bind to CLDN6, including e.g., those described in U.S. Pat. Nos. 9,487,584, 9,274,119, 10,745,477, 10,233,253, and 10,053,511.

[0147] Useful multi-specific CARs include antigen-recognition domains, or combinations of antigen-recognition domains, targeting the following combinations of antigens: CD19+CD20, CD19+CD22, BCMA+CD19, CD33+CLL-1, CD19+CD276, CD19+CD20+CD22, CD19+IL15R, IL15R+MUC16, IL-15Rα+NKG2D, GPC3+IL15R, CD16a+CD19+IL-15Rα, BCMA+CD16a+NKp46, CD123+CD33, CLDN18.2+NKG2D, CD20+CD22, BCMA+SLAMF7, CD19+CD70, CD19+IL18R1, GD2+IL15R, CD70+IL15R, CD19+CD8, CD20+CD79A, CD19+CD22+CD8, CD19+EBV Protein, CD19+CD7, BCMA+CD38, IL-12+MUC16, CD33+FLT3, CD19+IL-2Rβ, BCMA+CD70, EGFR+IL-13Rα2, HER2+IL15R, MICA+MICB, BCMA+TACI, GPC3+IL-2Rβ, CD19+DR5, CD123+CD33+CLL-1, CD3+CD7, CEACAM5+CEACAM6, CD20+CD22+CD38, CD19+CLDN18.2, AGRE2+CLL-1, CD123+TIM3, CLDN18.2+PDL1, CTLA4+MSLN+PD-1, STEAP2+TGFBR2, CEA+IL-15Rα+IL-21R, CD123+CD33+CD38+CLL-1, CD133+EGFR, BCMA+HPK1, CD8A+NY-ESO-1, CD38+DR5, CD19+EGFR, BCMA+CD19+HER2+Trop-2, CD19+MUC1, PDL1+VEGFR1, CD123+CD33+CD38+CD56+CLL-1+MUC1, BCMA+CD138+CD19+CD38, BCMA+CD16a+IL-15Rα, NCR2+NKG2D, CD38+CLL-1, CD276+FGFR4, and the like.

[0148] In some instances, useful CARs (or useful component parts thereof) will include a monospecific CAR targeting CD19, including but not limited to e.g., the CARs of CAR therapies Inaticabtagene Autoleucel, Actalycabtagene autoleucel, Relmacabtagene autoleucel, Lisocabtagene maraleucel, Brexucabtagene Autoleucel, Axicabtagene Ciloleucel, Tisagenlecleucel, Obecabatagene autoleucel, Azercabtagene Zapreleucel, Rapcabtagene autoleucel, and the like.

[0149] In some instances, useful CARs (or useful component parts thereof) will include a CAR targeting CD19, including but not limited to e.g., the CARs of CAR therapies ALLO-501A (Cellectis SA|Servier Pharmaceuticals LLC|Allogene Therapeutics, Inc.), CTL-119 (University of Pennsylvania|Novartis Pharma AG), IM19CAR-T (Beijing Imunopharm Technology Co., Ltd.), JCAR-014 (Celgene Corp.|Juno Therapeutics, Inc.), JCAR-015 (Celgene Corp. |Juno Therapeutics, Inc.), MB-CART19.1 (Miltenyi Biotec, Inc.), pCAR-19B (Chongqing Precision Biotechnology Co., Ltd.), TI-1007 (Timmune|Shenzhen Beike Biotechnology Co., Ltd.), UCART-19 (Cellectis SA|Pfizer Inc.|Allogene Therapeutics, Inc.), AT-101 (AbClon, Inc.), CLIC-1901 (Ottawa Hospital Research Institute), CT-032 (CARsgen Therapeutics Co.,Ltd), CTX-112 (CRISPR Therapeutics AG), GC007g (Gracell Biotechnologies, Inc.), GLPG-5101 (Galapagos NV), KYV-101 (Kyverna Therapeutics, Inc.), MB CART19.1 (Miltenyi GmbH), MC150 (Chongqing Precision Biotechnology Co., Ltd.), PL-001 (PELL Bio-Med Technology Co., Ltd.), SCRI-huCAR19v1 (Seattle Children's Hospital), SCRI-huCAR19v2 (Seattle Children's Hospital), 19(T2)28z1xx TRAC T cell (Takeda Pharmaceutical Co., Ltd.), ATA-3219 (Atara Biotherapeutics, Inc.), ATLCAR.CD19 (Bellicum Pharmaceuticals, Inc.), BRL-301 (BRL Medicine Inc.), CC-97540 (Juno Therapeutics, Inc.|Bristol Myers Squibb Co.), CLBR-001/SWI-019 (California Institute for Biomedical Research|AbbVie, Inc.), CNTY-101 (Century Therapeutics, Inc.), CT-RD06 (Nanjing Bioheng Biotech Co., Ltd.), ET019003-T Cells (Eureka Therapeutics, Inc.), ET190L1-ARTEMIS T cells (Eureka Therapeutics, Inc.), FT-522 (Fate Therapeutics, Inc.), FT-819 (Fate Therapeutics, Inc.), PBCAR-19B (Precision BioSciences, Inc.), PCAR-019 (PersonGenAnke Cellular Therapeutics Co., Ltd.), SC-291 (Sana Biotechnology, Inc.), TAK 940 (Takeda Pharmaceutical Co., Ltd.), TBI-2001 (Takara Bio Inc.), and the like.

[0150] In some instances, useful CARs (or useful component parts thereof) will include a multi-specific CAR targeting CD19 and at least one other antigen, including but not limited to e.g., the CARs of CAR therapies MB-CART2019.1 (Miltenyi Biotec, Inc.), KITE-363 (Kite Pharma, Inc.), BYC-104 (BRL Medicine Inc.), Anbalcabtagene autoleucel (Curocell, Inc.), Bi-4SCAR CD19/22 T cells (Shenzhen Geno-Immune Medical Institute), Prizloncabtagene autoleucel (AbelZeta Pharma, Inc.), KITE-753 (Kite Pharma, Inc.), CB-010 (Caribou Biosciences, Inc.), CTA-101 (Nanjing Bioheng Biotech Co., Ltd.), Denocabtagene Ciloleucel (Immunotech Applied Science Limited), IMJ-995 (Novartis Pharmaceuticals Corp.), LB1909 (Nanjing Legend Biotechnology Co., Ltd.), LCAR-AIO (Nanjing Legend Biotechnology Co., Ltd.), NKX-019 (Nkarta, Inc.), SYN-CAR-001 (Synthekine, Inc.), GC-022 (Gracell Biotechnologies (Shanghai) Co., Ltd.), GC-502 (Gracell Biotechnologies (Shanghai) Co., Ltd.), EPC-001 (Elpis Biopharmaceuticals Corp.), API-192 (Appia Bio, Inc.|Kite Pharma, Inc.), ATA-3431 (Atara Biotherapeutics, Inc.), AVC-203 (AvenCell Therapeutics, Inc.), CNTY-102 (Century Therapeutics, Inc.), CRG-023 (Cargo Therapeutics, Inc.), CTA302 (Nanjing Bioheng Biotech Co., Ltd.), GF-CARTO1 (Genomefrontier Therapeutics, Inc.), MB-CART 2219.1 (Miltenyi Biomedicine GmbH), MTB-002 (Medtherapy Biotechnology, Inc.), ORGCAR19.22 (Orgenesis-|Beijing Win Win Technology Co., Ltd.), RG-1970 (Hangzhou Ronggu Biotechnology Co., Ltd.), SC276 (Sana Biotechnology, Inc.), SG-299 (Sana Biotechnology, Inc.), SG233 (Sana Biotechnology, Inc.), MTB-004 (Medtherapy Biotechnology, Inc.), HY-003 (Juventas Cell Therapy Ltd.), ONKT-101 (ONK Therapeutics Ltd.), YT-19/22 (Huaxia Yingtai Beijing Biotechnology Co Ltd.), CABA-201 (Cabaletta Bio, Inc.), GC-012F (Gracell Biotechnologies, Inc.), GLPG-5201 (Galapagos NV), KQ-2003 (Shanghai Keqi Pharmaceutical Technology Co., Ltd.), FT-596 (Fate Therapeutics, Inc.), P-BCMACD19-ALLO1 (Poseida Therapeutics, Inc.), MVR-T7011 (ImmVira, Co., Ltd), MVR-T7012 (ImmVira, Co., Ltd), MVR-T7013 (ImmVira, Co., Ltd), LCAR-AIO (Legend Biotech USA, Inc.), and the like.

[0151] In some instances, useful CARs (or useful component parts thereof) will include a multi-specific CAR targeting at least CD19 and CD20, including but not limited to e.g., the CARs of CAR therapies MB-CART2019.1 (Miltenyi Biotec, Inc.), KITE-363 (Kite Pharma, Inc.), HY-004 (Juventas) (Juventas Cell Therapy Ltd.), IMPT-314 (Immpact Bio USA, Inc.), IMPT-514 (Immpact Bio USA, Inc.), Prizloncabtagene autoleucel (AbelZeta Pharma, Inc.), KITE-753 (Kite Pharma, Inc.), CAR20.19.22 T-cells (Miltenyi Biomedicine GmbH), P-CD19CD20-ALLO1 (Poseida Therapeutics, Inc.), HX-s001 (Huaxia Yingtai Beijing Biotechnology Co Ltd.), HXYT-001 (Huaxia Yingtai Beijing Biotechnology Co Ltd.), LCAR-AIO (Nanjing Legend Biotechnology Co., Ltd.), Lcaraio (Legend Biotech USA, Inc.), P-CD19CD20-AL-LO1 (Poseida Therapeutics, Inc.), YT-19/20 (Huaxia Yingtai Beijing Biotechnology Co Ltd.), ICG 132 or ICG 134 (Icar Bio Therapeutics Ltd), API-192 (Appia Bio, Inc.|Kite Pharma, Inc.), ATA-3431 (Atara Biotherapeutics, Inc.), AVC-203 (AvenCell Therapeutics, Inc.), CRG-023 (Cargo Therapeutics, Inc.), and GF-CARTO1 (Genomefrontier Therapeutics, Inc.).

[0152] In some instances, useful CARs (or useful component parts thereof) will include a multi-specific CAR targeting at least CD19 and BCMA, including but not limited to e.g., the CARs of CAR therapies GC-012F (Gracell Biotechnologies (Shanghai) Co., Ltd.), KQ-2003 (Shanghai Keqi Pharmaceutical Technology Co., Ltd.), and P-BCMACD19-ALLO1 (Poseida Therapeutics, Inc.).

[0153] In some instances, useful CARs (or useful component parts thereof) will include a multi-specific CAR targeting at least CD19 and CD22, including but not limited to e.g., the CARs of CAR therapies CD19/CD22 dual CART Chongqing Precision Biotech (Chongqing Precision Biotechnology Co., Ltd.), Bi-4SCAR CD19/22 T cells (Shenzhen Geno-Immune Medical Institute), CT-120 (Nanjing IASO Biotherapeutics Co., Ltd.), CART2219.1 (KK Women's & Children's Hospital Pte Ltd.), LCAR-L10D (Xi'an Jiaotong University), CD19/CD22 CAR T cells (Orca Biosystems, Inc.), CTA-101 (Nanjing Bioheng Biotech Co., Ltd.), IMJ-995 (Novartis Pharmaceuticals Corp.), LB1909 (Nanjing Legend Biotechnology Co., Ltd.), GC-022 (Gracell Biotechnologies (Shanghai) Co., Ltd.), EPC-001 (Elpis Biopharmaceuticals Corp.), B-019 (Shanghai Pharmaceuticals Holding Co., Ltd.), CTA302 (Nanjing Bioheng Biotech Co., Ltd.), MB-CART 2219.1 (Miltenyi Biomedicine GmbH), MTB-002 (Medtherapy Biotechnology, Inc.), ORGCAR19.22 (Beijing Win Win Technology Co., Ltd.), SC276 (Sana Biotechnology, Inc.), MTB-004 (Medtherapy Biotechnology, Inc.), and YT-19/22 (Huaxia Yingtai Beijing Biotechnology Co Ltd.).

[0154] In some instances, useful CARs (or useful component parts thereof) will include a monospecific CAR targeting BCMA, including but not limited to e.g., the CARs of CAR therapies Inaticabtagene Autoleucel, Actalycabtagene autoleucel, Equecabtagene Autoleucel, Ciltacabtagene autoleucel, Relmacabtagene autoleucel, Idecabtagene Vicleucel, Lisocabtagene maraleucel, Brexucabtagene Autoleucel, Axicabtagene Ciloleucel, Tisagenlecleucel, Obecabatagene autoleucel, Zevorcabtagene autoleucel, and the like.

[0155] In some instances, useful CARs (or useful component parts thereof) will include a monospecific CAR targeting BCMA, including but not limited to e.g., the CARs of CAR therapies CART-ddBCMA (Arcellx, Inc.), ET-140 or FCARH-143 or ET140-CAR or JCARH-125 or MCARH-171 (Juno Therapeutics, Inc.), Descartes 011 (Cartesian Therapeutics, Inc. (United States)), Descartes-08 (Cartesian Therapeutics, Inc. (United States)), KITE-585 (Kite Pharma, Inc.), PBCAR-269A (Precision BioSciences, Inc.), PHE-885 (Novartis Pharmaceuticals Corp.), PRG-1801 (Shenzhen Puruijin Biological Pharmaceutical Co., Ltd.), ALLO-605 (Allogene Therapeutics, Inc.), Anti-BCMA CAR-NK cell therapy (Asclepius) (Go Better Studio Accessibility Technology (SuZhou) Co., Ltd.), ARI0002H (Instituto De SAlud Carlos Iii De Madrid), C-CARO88 (AbelZeta Pharma, Inc.), GLPG-5301 (Galapagos NV), NXC-201 (Nexcella, Inc.), P-BCMA-101 (Poseida Therapeutics, Inc.), ACLX-001 (Arcellx, Inc.), ALLO-715 (Cellectis SA), BB21217 (bluebird bio, Inc.), BCMA-CS1 cCAR-T cell therapy (iCell Gene Therapeutics, Inc.), CB-011 (Caribou Biosciences, Inc.), CBG002 (Zhejiang Kangbaiyu Biotechnology Co., Ltd.), CC-98633 (Celgene Corp.), CT 0590 or KJ-C2111 (CARsgen Therapeutics Co., Ltd), CTX-120 (CRISPR Therapeutics AG), CYAD-211 (Celyad Oncology SA), IBI-346 (Innovent Biologics, Inc.), IM21 CAR-T cells (Beijing Imunopharm Technology Co., Ltd.), JWCAR-129 (Juno Therapeutics, Inc.), LCAR-B4822M (Nanjing Legend Biotechnology Co., Ltd.), LCAR-BCDR (Nanjing Legend Biotechnology Co., Ltd.), LUCAR-B68 Cells (Nanjing Legend Biotechnology Co., Ltd.), MCARH125 (Memorial Sloan Kettering Cancer Center), MCM998 (Novartis AG), MDC-CAR-BCMA001 (–), P-BCMA-ALLO1 (Poseida Therapeutics, Inc.), RG-6538 (Janssen Biotech, Inc.|Transposagen Biopharmaceuticals, Inc.), SENL-103 (Hebei Senlangbio Biotechnology Co., Ltd.), SA-102 (Wuhan Si'an Medical Technology Co., Ltd.), SKB-394 (Sichuan Kelun-Biotech Biopharmaceutical Co., Ltd.), BCMA CAR iNKT (Agenus, Inc.), BCMA GoCAR-NK (Bellicum Pharmaceuticals, Inc.), BG BCMA (Guangzhou Bio-gene Technology Co., Ltd.), CB-020 (Caribou Biosciences, Inc.), GB-5010 (Shanghai Genechem Co., Ltd.), HY-029 (Juventas Cell Therapy Ltd.), MiNK-413 (MiNK Therapeutics, Inc.), Ori-CAR-002 (Yuanqi Biotechnology Shanghai Co Ltd.), RDO9 (Nanjing Bioheng Biotech Co., Ltd.), SA-BCMA-CS1-102 (SA Science, Inc.), SC-255 (Sana Biotechnology, Inc.), SENL302 (Hebei Senlangbio Biotechnology Co., Ltd.), XYF-B08 (Xi'An Yufan Biotechnology), CTX-121 (CRISPR Therapeutics AG), CART-BCMA (University of Pennsylvania), HY-015 (Juventas Cell Therapy Ltd.), and HY-027 (Juventas Cell Therapy Ltd.).

[0156] In some instances, useful CARs (or useful component parts thereof) will include a CAR targeting at least CD20, including but not limited to e.g., the CARs of CAR therapies 4SCAR-CD20 (Shenzhen Geno-Immune Medical Institute), bbT-369 (bluebird bio, Inc.), MB-106 CD20 CAR (Fred Hutchinson Cancer Research Center), ELC-301 (Elicera Therapeutics AB), UCART20×22 (Cellectis SA), ADI-001 (Regeneron Pharmaceuticals, Inc.|Adicet Therapeutics, Inc.), LB1905 (Nanjing Legend Biotechnology Co., Ltd.), LY007 (Shanghai Longyao Biotech Co., Ltd.), MB-CART20.1 (Miltenyi Biotec, Inc.), ASP2802 (Astellas Pharma, Inc.), LUCAR-20SD (Nanjing Legend Biotechnology Co., Ltd.), UB-VV300 (Umoja Biopharma, Inc.), UB-VV310 (Umoja Biopharma, Inc.), and PBCAR-20A (Precision BioSciences, Inc.).

[0157] In some instances, useful CARs (or useful component parts thereof) will include a CAR targeting at least CD22, including but not limited to e.g., the CARs of CAR therapies CRG-022 (Cargo Therapeutics, Inc.), 4SCAR-CD22 (Shenzhen Geno-Immune Medical Institute), RD102 (Nanjing IASO Biotherapeutics Co., Ltd.), SL22P CART cells (Hebei Senlangbio Biotechnology Co., Ltd.), JCAR-018 (National Cancer Institute), ThisCART 22 (Fundamenta Therapeutics, Ltd.), UCART-22 OR BALLI-01 (Cellectis SA), Anti-CD22-CAR m972 (National Cancer Institute), ICTCAR017 (Innovative Cellular Therapeutics Co., Ltd.), LQ-031 (Shanghai Novamab Biopharmaceuticals Co., Ltd.), OC-2 (Onechain Immunotherapeutics), and SC262 (Sana Biotechnology, Inc.).

[0158] In some instances, useful CARs (or useful component parts thereof) will include a CAR targeting at least CD123, including but not limited to e.g., the CARs of CAR therapies MB-102 (Mustang Bio, Inc.), ACLX-002 (Arcellx, Inc.), Allo-RevCAR01-T-CD123 (AvenCell Europe GmbH), CD123CAR-41 BB-CD3zeta-EGFRt (Hrain Biotechnology Co. Ltd.), AVC-201 (AvenCell Therapeutics, Inc.), IM-23 (Beijing Imunopharm Technology Co., Ltd.), UCART-123 (Cellectis SA), UniCAR-T-CD-123 (GEMoaB Monoclonals GmbH), Unicar-T-CD123 OR AVC-101 (AvenCell Therapeutics, Inc.), JD-023 (Beijing Jingda Biotechnology Co., Ltd.), AUTO9 (University College London|Autolus Therapeutics Plc), SENL401 (Hebei Senlangbio Biotechnology Co., Ltd.), EPC-005 (Elpis Biopharmaceuticals Corp.), ARD-102 (Arce Therapeutics, Inc.), HY-007 (Juventas Cell Therapy Ltd.), and 4SCAR-CD123 (Shenzhen Geno-Immune Medical Institute).

[0159] In some instances, useful CARs (or useful component parts thereof) will include a CAR targeting at least CD30, including but not limited to e.g., the CARs of CAR therapies ATLCAR.CD30 cells (UNC Lineberger Comprehensive Cancer Center), CD30 CAR-T cell therapy (Tessa Therapeutics Pte Ltd, TT11 CD30 CAR T (The University of North Carolina at Chapel Hill|Baylor College of Medicine), HSP-CAR30 (Josep Carreras Leukaemia Research Institute), ATLCAR.CD30.CCR4 cells (UNC Lineberger Comprehensive Cancer Center) HU30-CD28Z (National Cancer Institute), CD30.CARIH-27721 (Baylor College of Medicine), EBV-specific-CAR.CD30 (Baylor College of Medicine), TT-11X (Tessa Therapeutics Ltd), and RD 111 (Nanjing IASO Biotherapeutics Co., Ltd.).

[0160] In some instances, useful CARs (or useful component parts thereof) will include a CAR targeting at least CD33, including but not limited to e.g., the CARs of CAR therapies VCAR-33 (National Cancer Institute), VCAR33ALLO (Vor Biopharma, Inc.), QN-023A (Qihan Biotech), BECAR33 (Great Ormond Street Hospital for Children NHS Foundation Trust), CART33 (University of Pennsylvania), CC-96191 (Dragonfly Therapeutics, Inc.), CLL1-CD33 cCART cell therapy (iCell Gene Therapeutics, Inc.), ICG-136 (iCell Gene Therapeutics, Inc.), LB1910 (Nanjing Legend Biotechnology Co., Ltd.), LCAR-AMDR Cells (Nanjing Legend Biotechnology Co., Ltd.), PRGN-3006 (Precigen, Inc.), SC-DARIC33 (bluebird bio, Inc.), SENTI-202 (Senti Sub|, Inc.), GCK-02 (Shanghai Jingshan Biotechnology Co., Ltd.), HY-002 (Juventas Cell Therapy Ltd.), AntiCD33 CARNK92 cells (PersonGen BioTherapeutics (Suzhou) Co. Ltd), INXN-3004 (Precigen, Inc.|Alaunos Therapeutics, Inc.), AUTO9 (University College London|Autolus Therapeutics Plc), and HY-007 (Juventas Cell Therapy Ltd.).

[0161] In some instances, useful CARs (or useful component parts thereof) will include a CAR targeting at least CD38, including but not limited to e.g., the CARs of CAR therapies CART-38 CellIs (University of Pennsylvania), CD38 CAR-T (Sorrento Therapeutics, Inc.), STI-5171 (Sorrento Therapeutics, Inc.), CYT-538 (Cytovia Therapeutics, Inc.), FT-555 (Fate Therapeutics, Inc.), LQ-032 (Shanghai Novamab Biopharmaceuticals Co., Ltd.), ONKT-102 (Universitstsklinikum Hamburg-Eppendorf|ONK Therapeutics Ltd.), GDA-601 (Gamida Cell Ltd.), and 4SCAR-CD38 (Shenzhen Geno-Immune Medical Institute).

[0162] In some instances, useful CARs (or useful component parts thereof) will include a CAR targeting at least CD70, including but not limited to e.g., the CARs of CAR therapies 4SCAR-CD70 (Shenzhen Geno-Immune Medical Institute), CAR.70/IL15-transduced CB-NK cells (The University of Texas MD Anderson Cancer Center), CTX-131 (CRISPR Therapeutics AG), ALLO-316 (Allogene Therapeutics, Inc.), HR010 (Hrain Biotechnology Co. Ltd.), ADI-270 (Adicet Bio, Inc.), CAT-248 (Catamaran Bio, Inc.), CD70 CAR-NK cell therapy (Nkarta, Inc.|CRISPR Therapeutics AG), P-CD70-ALLO1 (Poseida Therapeutics, Inc.), CTX-130 (CRISPR Therapeutics AG), C-4-29 (Chongqing Precision Biotechnology Co., Ltd.), and RG-1970 (Hangzhou Ronggu Biotechnology Co., Ltd.).

[0163] In some instances, useful CARs (or useful component parts thereof) will include a CAR targeting at least CLDN18.2, including but not limited to e.g., the CARs of CAR therapies CT-041 (CARsgen Therapeutics Co., Ltd), IMC-002 (Suzhou Immunofoco Biotechnology Co. Ltd.), AZD-6422 (AstraZeneca PLC), LY011 (Shanghai Longyao Biotech Co., Ltd.), CLDN18.2 UCAR-T (T-Maximum Pharmaceutical (Suzhou) Co., Ltd.), Dual-targeting CLDN18.2 and PD-L1 CAR-T cells (Sichuan University), IBI-345 (Innovent Biologics, Inc.), IMC-008 (Suzhou Immunofoco Biotechnology Co. Ltd.), KD-022 (Nanjing KAEDI Biotech, Inc.), KD-496 (Nanjing KAEDI Biotech, Inc.), KJ-C1807 (CARsgen Therapeutics Co., Ltd), LB-1904 (Nanjing Legend Biotechnology Co., Ltd.), LB1908 (Nanjing Legend Biotechnology Co., Ltd.), HEC-016 (Sunshine Lake Pharma Co. Ltd.), CTB001 (Nanjing Bioheng Biotech Co., Ltd.), iPD-1-Claudin18.2-CAR-T OR XKDCT086 (The Affiliated Hospital of Qingdao University), CTB001 (Nanjing Bioheng Biotech Co., Ltd.), ALLO-182 (Allogene Therapeutics, Inc.), BNT-212 (BioNTech SE), CLDN18.2-CAR-iNKT (Arovella Therapeutics Ltd.), CT-007 (Guangdong Xiankangda Biotechnology Co., Ltd.), CT-076 (Guangdong Xiankangda Biotechnology Co., Ltd.), CT-086 (Guangdong Xiankangda Biotechnology Co., Ltd.), CT-253 (Guangdong Xiankangda Biotechnology Co., Ltd.), CTD-101 (Nanjing Bioheng Biotech Co., Ltd.), DCTY-1502 (Beijing Dingcheng Peptide Source Biotechnology Co., Ltd.), GB70041 (Shanghai Genechem Co., Ltd.), GC-506 (Gracell Biotechnologies (Shanghai) Co., Ltd.), IM92 (Beijing Imunopharm Technology Co., Ltd.), KD-593 (Nanjing KAEDI Biotech, Inc.), KD-U182 (Nanjing KAEDI Biotech, Inc.), P-2003 (Shandong Boan Biotechnology Co., Ltd.), PM-3023 (Pumis Biotechnology (Zhuhai) Co., Ltd.), PR-401 (Shandong Boan Biotechnology Co., Ltd.), SENL202 (Hebei Senlangbio Biotechnology Co., Ltd.), and YT-claudin18.2 (Huaxia Yingtai Beijing Biotechnology Co Ltd.).

[0164] In some instances, useful CARs (or useful component parts thereof) will include a CAR targeting at least CLL-1, including but not limited to e.g., the CARs of CAR therapies CB-012 (Caribou Biosciences), CD371-YSNVZ-IL18 CAR T cells (Memorial Sloan Kettering Cancer Center), ICG-136 (iCell Gene Therapeutics, Inc.), LB1910 (Nanjing Legend Biotechnology Co., Ltd.), ARD103 (Arce Therapeutics, Inc.), HY-032 (Juventas Cell Therapy Ltd.), GCK-02 (Shanghai Jingshan Biotechnology Co., Ltd.), and AUTO9 (University College London|Autolus Therapeutics Plc).

[0165] In some instances, useful CARs (or useful component parts thereof) will include a CAR targeting at least DLL3, including but not limited to e.g., the CARs of CAR therapies LB2102 (Nanjing Legend Biotechnology Co., Ltd.), ALLO-213 (Allogene Therapeutics, Inc.), and AMG-119 (Amgen, Inc.).

[0166] In some instances, useful CARs (or useful component parts thereof) will include a CAR targeting at least EGFR or EGFRvIII, including but not limited to e.g., the CARs of CAR therapies Anti-EGFR CAR-T cell therapy (AbelZeta Pharma, Inc.), Anti-EGFRvIII-CAR (Kite Pharma, Inc.), Anti-EGFRvII-directed CAR-T cell therapy (Dehe Biotech Energae), LEU-001 (Leucid Bio Ltd.), Anti-EGFR/IL-13R alpha 2 CAR-T cell therapy (University of Pennsylvania| Tmunity Therapeutics, Inc.), EGFR IL12 CART (Shenzhen Puruijin Biological Pharmaceutical Co., Ltd.), EGFR806 CAR-T Cell (Seattle Children's Hospital), KJ-C21120 (CARsgen Therapeutics Co., Ltd), LXF-821 (Novartis AG), DCTY-0801 (Beijing Dingcheng Peptide Source Biotechnology Co., Ltd.), CYT-501 (Cytoimmune Therapeutics, Inc.), EGFRt-IL21 (Asher Biotherapeutics, Inc.), EGFRvII synNotch CAR-T (University of California), GCT-02 (Myrio Therapeutics Pty Ltd.), HY-026 (Juventas Cell Therapy Ltd.), MT-029 (T-Maximum Pharmaceutical (Suzhou) Co., Ltd.), GCK-03 (Shanghai Jingshan Biotechnology Co., Ltd.), LQ-103 (Shanghai Novamab Biopharmaceuticals Co., Ltd.), Anti-EGFRvIII CAR-T cell therapy (Beijing Marion Biotechnology Co., Ltd.|Beijing Sanbo Brain Hospital Co., Ltd.), and EGFRvIII-directed CAR-T-cell tharapy (University of Pennsylvania). Useful EGFRvII targeting CARs further include those described in WO2014130657 and the like.

[0167] In some instances, useful CARs (or useful component parts thereof) will include a CAR targeting at least FLT3, including but not limited to e.g., the CARs of CAR therapies Anti-FLT3 CAR-T cell (PersonGen BioTherapeutics (Suzhou)), TAA5 CAR T (PersonGen BioTherapeutics (Suzhou) Co. Ltd), AMG-553 (Amgen, Inc.), CYTO NK-201 (Cytoimmune Therapeutics, Inc.), Anti-FLT3 CAR-T (City of Hope National Medical Center|Cytoimmune Therapeutics, Inc.), HEMO-CAR-T cell therapy (Hemogenyx), and SENTI-202 (Senti Sub|, Inc.).

[0168] In some instances, useful CARs (or useful component parts thereof) will include a CAR targeting at least GD2, including but not limited to e.g., the CARs of CAR therapies 4SCAR-GD2 (Shenzhen Geno-Immune Medical Institute), GD2-CART01 (Bambino Gesu Hospital), 1RG-CART (Cancer Research UK), Anti-GD2 CART (Zhujiang Hospital), AUTO6NG (University College London), AUTO 6 (University College London), C7R-GD2.CART cell therapy (Baylor College of Medicine) iC9-GD2-CAR-IL-15 T-cells (UNC Lineberger Comprehensive Cancer Center), VZV-specific-GD2-CAR (Baylor College of Medicine), iC9-GD2.CAR.IL-15 T-cells (UNC Lineberger Comprehensive Cancer Center), AUTO6-NG (University College London|Autolus Ltd.), IKT-703 (AnoVac Inc.), GINAKIT-Cells OR KUR-501 OR CAR.GD2-IL-15 NKTs (Baylor College of Medicine), andGD2.CAR-triVSTs (Baylor College of Medicine).

[0169] In some instances, useful CARs (or useful component parts thereof) will include a CAR targeting at least GPC3, including but not limited to e.g., the CARs of CAR therapies GB-5011 (Shanghai Genechem Co., Ltd.), AZD-5851 (AstraZeneca PLC), BOXR-1030 (Cogent Biosciences, Inc.), gpc3-car-ori2 (Yuanqi Biotechnology Shanghai Co Ltd.), C-CAR031 (Zhejiang University), CT-011 (CARsgen Therapeutics Co., Ltd), CT-0181 (CARsgen Therapeutics Co., Ltd), GLYCAR T cell therapy (Baylor College of Medicine), IM 83 (Beijing Imunopharm Technology Co., Ltd.), LCAR-H93T Cells (Nanjing Legend Biotechnology Co., Ltd.), GKL-006 (Beijing Gene Key Life Technology Co., Ltd), ADI-002 (Regenacy Pharmaceuticals LLC|Adicet Therapeutics, Inc.), CYT-150 (Cytovia Therapeutics, Inc.), CYT-503 (National Cancer Institute), KUR-503 (Baylor College of Medicine), SYNCAR002 (Synthekine, Inc.), SA-GPC3-103 (SA Science, Inc.), SENTI-301 (Senti Sub|, Inc.), TT-14 (Tessa Therapeutics Ltd), Anti-GPC3 autologous CART cell therapy (CRISPR Therapeutics AG), CT-017 (CARsgen Therapeutics Co., Ltd), LQ-102 (Shanghai Novamab Biopharmaceuticals Co., Ltd.), andHX-s002 (Huaxia Yingtai Beijing Biotechnology Co Ltd.).

[0170] In some instances, useful CARs (or useful component parts thereof) will include a CAR targeting at least HER2, including but not limited to e.g., the CARs of CAR therapies ALETA-002 (Aleta Biotherapeutics, Inc.), AU-101 (Aurora Biopharma, Inc.), AU-105 (Aurora Biopharma, Inc.), AB-201 (Artiva Biotherapeutics, Inc.), BPX-603 (Bellicum Pharmaceuticals, Inc.), CCT3-HER2-0406 (EXUMA Biotech Corp.), BP-2301 (Shinshu University), CIDeCAR (Bellicum Pharmaceuticals, Inc.), CT-0508 (CTx Operations, Inc.), HER2(EQ)BBzeta/CD19 T cells (Mustang Bio, Inc.), HER2.taNK (NantKwest, Inc.), MB-103 (Mustang Bio, Inc.), TT-16 (Tessa Therapeutics Ltd), CT-0525 (Carisma Therapeutics, Inc.), AT-501 (AbClon, Inc.), BM-1901 (Kunshi Biotechnology (Shenzhen) Co., Ltd.), CAT-179 (Catamaran Bio, Inc.), FT825/ONO-8250 (Fate Therapeutics, Inc.|Ono Pharmaceutical Co., Ltd.), MT-028 (T-Maximum Pharmaceutical (Suzhou) Co., Ltd.), OmniCAR HER2 (Prescient Therapeutics Ltd.), UWHer2T (UWELL Biopharma, Inc.), RB-H21 (Refuge Biotechnologies, Inc.), GDA-501 (Gamida Cell Ltd.), GCK-03 (Shanghai Jingshan Biotechnology Co., Ltd.), and MVR-T7 Shenzhen Yinuowei Medical Technology Co., Ltd.).

[0171] In some instances, useful CARs (or useful component parts thereof) will include a CAR targeting at least MSLN, including but not limited to e.g., the CARs of CAR therapies A2B-694 (A2 Biotherapeutics, Inc.), GC-008T (Gracell Biotechnologies (Shanghai) Co., Ltd.), iCasp9M28z (Memorial Sloan Kettering Cancer Center), KD-021 (Nanjing KAEDI Biotech, Inc.), LB1902 (Nanjing Legend Biotechnology Co., Ltd.), MCY-M11 (MaxCyte, Inc.), UCLM-802 (UTC Therapeutics, Inc.), KT032 (Nanjing Kati Medical Technology Co., Ltd.), LD-013 (Nanjing Blue Shield Biotechnology Co., Ltd.), TMMSTNO2 (Tmunity Therapeutics, Inc.), ATA-3271 (Atara Biotherapeutics, Inc.), CT 1119 (CTx Operations, Inc.), IM81 (Beijing Imunopharm Technology Co., Ltd.), KBISM Kiromic OR ALEXISISO1 (Kiromic Biopharma, Inc.), KJ-C2113 (CARsgen Therapeutics Co., Ltd), m-28z-T2 (GIBH), MSLN CART(LVV) (UTC Therapeutics, Inc.), PM-3006 (Pumis Biotechnology (Zhuhai) Co., Ltd.), RD11 (Nanjing Bioheng Biotech Co., Ltd.), ATA-2271 (Memorial Sloan Kettering Cancer Center), GC-503 (Gracell Biotechnologies (Shanghai) Co., Ltd.), and UCARTMESO (Cellectis SA).

[0172] In some instances, useful CARs (or useful component parts thereof) will include a CAR targeting at least MUC1 or MUC16, including but not limited to e.g., the CARs of CAR therapies TmTNMUC 01 CART (University of Pennsylvania), huMNC2-CAR22 (Minerva Biotechnologies Corp.), huMNC2-CAR44 T cells (Minerva Biotechnologies Corp.), P-MUC1C-ALLO1 (Poseida Therapeutics, Inc.), TmTnMUC1-01 (Tmunity Therapeutics, Inc.), bbT-4015 (2seventy Bio, Inc.), ONKT-103 (ONK Therapeutics Ltd.), TB-201 (Therabest, INC.), LQ-101 (Shanghai Novamab Biopharmaceuticals Co., Ltd.), UCARTMUC1 (Cellectis SA), JCAR-020 (Eureka Therapeutics, Inc.|Memorial Sloan Kettering Cancer Center), and PRGN-3005 (Precigen, Inc.).

[0173] In some instances, useful CARs (or useful component parts thereof) will include a CAR targeting at least NKG2D, including but not limited to e.g., the CARs of CAR therapies CYAD-01 (Dartmouth College), NKG2D-ACE2 CAR-NK cell therapy (Sidemu Biotechnology Technology), KD-025 (Nanjing KAEDI Biotech, Inc.), CYAD-203 (Celyad Oncology SA), LEU-011 (Leucid Bio Ltd.), KD-U25 (Nanjing KAEDI Biotech, Inc.), LEU-005 (Leucid Bio Ltd.), LEU-006 (Leucid Bio Ltd.), CYAD-101 (Celyad Oncology SA), KD-496 (Nanjing KAEDI Biotech, Inc.), NKX-101 (Nkarta, Inc.), andCYTO NK 301 (Cytoimmune Therapeutics, Inc.).

[0174] In some instances, useful CARs (or useful component parts thereof) will include a CAR targeting at least ROR1, including but not limited to e.g., the CARs of CAR therapies ROR1 CAR-T (Oncternal Therapeutics), JCAR-024 (Fred Hutchinson Cancer Research Center), LYL-797 (Lyell Immunopharma, Inc.), and LYL119 (Lyell Immunopharma, Inc.), ROR1R-CAR (The University of Texas MD Anderson Cancer Center).

[0175] In some instances, useful CARs (or useful component parts thereof) will include a CAR targeting at least SLAMF7, including but not limited to e.g., the CARs of CAR therapies IC9-Luc90-CD828Z (National Cancer Institute), UCART-CS1 (Cellectis SA), and CD319 chimeric antigen T cell therapy (Wuhan Bio Raid Biotechnology).

[0176] In some instances, useful CARs (or useful component parts thereof) will include a CAR targeting at least CLDN6, including but not limited to e.g., the CAR of CAR therapy BNT-211 (BioNTech SE). In some instances, useful anti-CLDN6 CARs or useful components thereof (including the antigen binding domains and/or CDRs thereof) will include those described in US20220306711, US20220184119, and WO2023105005.

[0177] In some instances, useful CARs (or useful component parts thereof, including e.g., the antigen binding domains and/or CDRs thereof) will include, but are not limited to, those described in the following U.S. and PCT patent publications: US20240050568, US20240041921, US20240041929, WO2024023124, US20240018256, U.S. Ser. No. 11/872,249B2, US20240009242, US20240000839, US20240002505, WO2024003786, US20230416390, WO2023248126, WO2023250489, WO2023246578, US20230406953, US20230405122, WO2023241141, US20230399412, US20230390336, WO2023192908, WO2023215746, U.S. Ser. No. 11/834,509B2, WO2023164256, US20230381228, U.S. Ser. No. 11/827,889B2, US20230365699, WO2023220560, US20230357427, US20230355673, US20230357355, WO2023208157, US20230340146, WO2023154890, WO2023205148, U.S. Ser. No. 11/795,238B2, WO2023199069, WO2023201314, WO2023201221, WO2023169555A9, US20230312675, US20230312708, WO2023147293, US20230295319, U.S. Ser. No. 11/753,457B2, WO2023161846, WO2023137069, WO2023158986, US20230256017, US20230250175, WO2023114777, WO2023143497, US20230242666, WO2023137291, US20230226182, WO2023134718, US20230220090, US20230220103, WO2023131285, WO2023131657, WO2023131063, US20230212255, WO2023125975, US20230203160, WO2023122337, US20230192803, US20230190804, US20230192842, WO2023114980, WO2023104099, US20230183371, US20230181634, US20230181640, U.S. Ser. No. 11/673,964B2, US20230174654, US20230174933, US20230167184, WO2023093811, WO2023088482, US20230159644, WO2023086900, WO2023086829, US20230151094, US20230149462, US20230147657, WO2023079135, U.S. Ser. No. 11/643,468B2, US20230136252, WO2023076811, WO2023047098, WO2023044350, WO2023028494, US20230114854, WO2023056429, US20230101046, US20230099646, WO2023047100, US20230087953, US20230084763, US20230074145, US20230072955, WO2023030539, US20230060292, US20230055426, US20230058044, WO2023021477, WO2023021494, U.S. Ser. No. 11/578,126B2, US20230039030, US20230029341, US20230000964, WO2023279095, WO2022240360A9, WO2022216811, WO2022242710, US20220364055, US20220362298, US20220356247, WO2022235662, WO2022236049, US20220348655, US20220348689, WO2022228579, US20220340671, WO2022224241, US20220324964, WO2022216813, US20220315665, WO2022212879, US20220305056, WO2022164886, US20220281994, US20220273710, US20220249563, WO2022166365, US20220242948, US20220233590, WO2022157673, US20220227874, WO2022151960, WO2022147075, US20220204930, US20220204609, US20220193232, US20220193133, US20220184126, US20220184127, US20220184125, US20220184129, US20220177573, WO2022120010, US20220168389, WO2022104424, US20220162301, US20220152106, US20220125840, WO2022078286, WO2022076898, US20220096651, US20220089678, WO2022048523, US20220064254, US20220064316, US20220048978, US20220047633, US20220033509, US20210393689, US20210395329, US20210393690, US20210363272, U.S. Ser. No. 11/180,553B2, US20210347851, U.S. Ser. No. 11/161,908B2, US20210322477, US20210324100, U.S. Ser. No. 11/149,076B2, US20210317209, US20210315985, WO2021207709, US20210309716, US20210284752, WO2021178695, US20210269537, U.S. Ser. No. 11/090,334B2, US20210238309, US20210221880, US20210220404, WO2021127428, US20210177896, US20210171909, U.S. Ser. No. 11/028,177B2, US20210163612, WO2021099944, US20210139595, US20210113618, U.S. Ser. No. 10/981,970B2, WO2021067290, US20210046155, US20210038646, US20210017277, US20210000870, US20200407461, U.S. Ser. No. 10/851,149B2, U.S. Ser. No. 10/815,301 B2, WO2020214937, US20200306304, US20200297760, WO2020191293, U.S. Ser. No. 10/752,684B2, U.S. Ser. No. 10/752,670B2, US20200247867, US20200237821, US20200231686, WO2020123691, US20200147134, US20200079864, US20200055948, US20200023010, U.S. Ser. No. 10/526,406B2, US20190367621, US20190345256, US20190336504, US20190328784, US20190307797, WO2019178078, WO2019152660, WO2019152742, US20190231819, WO2019122875, U.S. Ser. No. 10/316,101B2, WO2019108932, WO2019085102, US20190125840, WO2019067015, US20190085081, US20190048061, US20190030073, WO2018169922, US20180230225, WO2018126369, WO2018068766, WO2017216562, US20170334991, WO2017149515, US20170226183, WO2017091546, US20170066827, US20170066838, WO2016094304, US20160280798, US20160096902, US20160051651, WO2015164739, WO2015063069, and WO2014186469; the disclosures of which are incorporated herein by reference in their entirety.

Additional Components

[0178] Nucleic acid constructs and vectors may include various components in addition to the DR-18 polypeptide and/or CAR encoding sequences such as, but not limited to, e.g., a signal sequence (i.e., signal peptides or signal peptide sequences) to facilitate secretion, polyadenylation signals, transcription terminator sequences (e.g., from Bovine Growth Hormone (BGH) gene), an element allowing episomal replication and replication in prokaryotes (e.g. SV40 origin and ColE1 or others known in the art) and/or elements to allow selection (e.g., ampicillin resistance gene, zeocin marker, and/or a detectable reporter). Any of the individual additional components described herein may be combined with any other individual additional components, including e.g., where a construct or vector may include 2, 3, 4, 5, 6, 7, 8, 9, 10, or more individual additional components, in various combinations, as desired.

[0179] In some instances, coding sequences, e.g., a DR-18 polypeptide encoding sequence, may be preceded, including fused in-frame to, a sequence encoding a signal peptide. Useful signal peptide sequences include but are not limited to e.g., the signal peptide sequences of Human OSM, VSV-G, Mouse Ig Kappa, Mouse Ig Heavy, BM40, Secrecon, Human IgKVIII, CD33, tPA, Human Chymotrypsinogen, Human trypsinogen-2, Human IL-2, Gaussia luc, Albumin (HSA), Influenza Haemagglutinin, Human insulin, Silkworm Fibroin LC, IL-18BP signal peptide, Interleukin-12 subunit beta, Interleukin-12 subunit alpha, Interleukin-18 propeptide, Human CD8, and the like. In some instances, the coding sequence, e.g., DR-18 polypeptide coding sequence, will include an endogenous signal peptide sequence, e.g., the signal peptide sequence of the wild-type IL-18 propeptide. In some instances, the coding sequence, e.g., DR-18 polypeptide coding sequence, will include a heterologous signal peptide sequence, e.g., an IL-2, IL-12, or IL-18BP signal peptide sequence, or the like. In some instances, a coding sequence may not include a signal peptide sequence.

[0180] Coding sequences of the present disclosure, e.g., DR-18 coding sequences, CAR coding sequences, and the like, will generally be operably linked to one or more regulatory elements. Useful regulatory elements to which coding sequences of the present disclosure may be operably linked include, but are not limited to e.g., enhancers, introns, polyadenylation signals, kozak consensus sequences, splice acceptor/donor sequences, response elements, such as woodchuck hepatitis virus (WPRE) elements, NFAT response element(s), and the like, promoters, and the like.

[0181] Nonlimiting examples of useful promoters include e.g., endogenous promoters, heterologous promoters, cytomegalovirus (CMV) promoter, elongation factor 1 alpha (EF1a) promoter, simian virus 40 (SV40) promoter, phosphoglycerate kinase (PGK) (e.g., human or mouse) promoter, ubiquitin C (UBC) promoter, human beta actin promoter, CAG promoter, TRE promoter, UAS promoter, LC immunoglobulin promoter, HC immunoglobulin promoter, herpes simplex virus thymidine kinase promoter, spleen focus-forming virus (SFFV) promoter, synthetic inducible promoters, inducible promoters, chemically/biochemicallyregulated promoter, physically-regulated promoters, alcohol-regulated promoters, tetracycline-regulated promoters, steroid-regulated promoters (e.g., promoters based on the rat glucocorticoid receptor, human estrogen receptor, moth ecdysone receptors, and promoters from the steroid/retinoid/thyroid receptor superfamily), metal-regulated promoters (e.g., promoters derived from metallothionein (proteins that bind and sequester metal ions) genes from yeast, mouse and human), pathogenesis-regulated promoters (e.g., induced by salicylic acid, ethylene or benzothiadiazole (BTH)), temperature/heat-inducible promoters (e.g., heat shock promoters), light-regulated promoters (e.g., light responsive promoters from plant cells)), immune cell promoters, CD8 cell-specific promoter, CD4 cell-specific promoter, neutrophil-specific promoter, NK-specific promoter, B29 gene promoter, CD14 gene promoter, CD43 gene promoter, CD45 gene promoter, CD68 gene promoter, IFN-b gene promoter, WASP gene promoter, T-cell receptor b-chain gene promoter, V9 g (TRGV9) gene promoter, V2 d (TRDV2) gene promoter, and the like.

[0182] Various types of promoters may be employed in the nucleic acid, compositions, and systems of the present disclosure, including constitutive promoters, inducible promoters, repressible promoters, tissue specific promoters, cell-type specific promoters, and the like.

[0183] An example of a promoter that is capable of driving expression of a DR-18 polypeptide and/or a CAR transgene in a mammalian T cell is the EF1a promoter. The native EF1a promoter drives expression of the alpha subunit of the elongation factor-1 complex, which is responsible for the enzymatic delivery of aminoacyl tRNAs to the ribosome. The EF1a promoter has been extensively used in mammalian expression plasmids and has been shown to be effective in driving CAR expression from transgenes cloned into a lentiviral vector. See, e.g., Milone et al., *Mol. Ther.* 17(8): 1453-1464 (2009). In some embodiments, an EF1a promoter used in the described nucleic acids and vectors comprises the sequence provided as SEQ ID NO: 87.

[0184] Another example of a promoter is the immediate early cytomegalovirus (CMV) promoter sequence. This promoter sequence is a strong constitutive promoter sequence capable of driving high levels of expression of any polynucleotide sequence operatively linked thereto. However, other constitutive promoter sequences may also be used, including, but not limited to the simian virus 40 (SV40) early promoter, mouse mammary tumor virus (MMTV), human immunodeficiency virus (HIV) long terminal repeat (LTR) promoter, MoMuLV promoter, an avian leukemia virus promoter, an Epstein-Barr virus immediate early promoter, a Rous sarcoma virus promoter, as well as human gene promoters such as, but not limited to, the actin promoter, the myosin promoter, the elongation factor-1a promoter, the hemoglobin promoter, and the creatine kinase promoter.

[0185] Polypeptide encoding nucleic acid sequences of the present disclosure may be expressed from a constitutive promoter. Polypeptide encoding nucleic acid sequences of the present disclosure may be expressed from an inducible promoter. The use of an inducible promoter provides a molecular switch capable of turning on expression of the polynucleotide sequence which it is operatively linked in certain contexts, e.g., such as when expression is desired, or turning off the expression in certain contexts, e.g., such as when expression is not desired. Examples of inducible promoters include, but are not limited to a metallothionine promoter, a glucocorticoid promoter, a progesterone promoter, and a tetracycline promoter. In some instances, promoters responsive to immune cell activation may be employed, such as e.g., a promoter induced upon T cell activation, such as but not limited to e.g., a minimal interleukin 2 promoter (IL-2p) operably linked to one or more NFAT response elements, such as e.g., 2, 3, 4, 5, 6, or more NFAT response elements.

[0186] As described herein, nucleic acids of the present disclosure may be configured to include a one coding sequence per nucleic acid molecule or multiple coding sequences per nucleic acid molecule. For example, where at least one nucleic acid molecule includes a single coding sequence, multiple different nucleic acid molecules, including e.g., at least one molecule encoding a DR-18 polypeptide and at least one molecule encoding a CAR, may be employed to deliver all desired coding sequences into a cell. Where a nucleic acid molecule includes multiple coding sequences (e.g., a DR-18 polypeptide coding sequence and a CAR coding sequence) one nucleic acid molecule may be employed to deliver all desired coding sequences into a cell. In some instances, where a nucleic acid molecule includes multiple coding sequences one or more elements for multicistronic expression may be employed (also referred to as multicistronic elements).

[0187] Multicistronic expression includes but is not necessarily limited to bicistronic expression and tricistronic expression, which may, in some instances, be achieved through the use of multicistronic vectors, such as bicistronic and tricistronic vectors. Non-limiting examples of useful multicistronic elements include an internal ribosome entry site (IRES), a 2A peptide (e.g., T2A peptide, P2A peptide, E2A peptide, or F2A peptide) coding sequence, and the like. Multicistronic elements may be operably linked to one or more coding sequences.

[0188] Nucleic acids of the present disclosure may encode for one or more additional cytokines, where the term "additional cytokine" as used herein generally refers to a cytokine other than IL-18 (including wt and engineered forms of IL-18, such as DR-18). Inclusion of one or more additional cytokines in the nucleic acids, constructs, and/or vectors of the present disclosure provides for multi-armoring of a CAR construct, i.e., armoring the CAR construct with a DR-18 polypeptide and at least one more polypeptide with cytokine activity. Useful additional cytokines include but are not limited to e.g., IL-2, IL-7, IL-10, IL-12, IL-15, IL-21, IL-23, IL-27, IL-33, TNF, TL1A, IFNa, IFNb, and IFNg. Useful additional cytokines include wild-type cytokines and engineered versions thereof.

[0189] In order to assess the expression of a polypeptide or portions thereof, the expression vector to be introduced into a cell can also contain either a selectable marker gene or a reporter gene or both to facilitate identification and selection of expressing cells from the population of cells sought to be transduced, transfected or infected through viral or non-viral vectors. In other aspects, the selectable marker may be carried on a separate piece of DNA and used in a co-transfection procedure. Both selectable markers and reporter genes may be flanked with appropriate regulatory sequences to enable expression in the host cells. Useful reporters and selectable markers include, for example, antibiotic-resistance genes, such as neo and the like, fluorescent proteins such as green fluorescent protein or the like, enzymatic reporters such as luciferase, beta-galactosidase, chloramphenicol acetyl transferase, secreted alkaline phosphatase and the like.

Methods

[0190] Methods of the present disclosure include methods of administering nucleic acids encoding a DR-18 polypeptide and at least one CAR to a subject in need thereof. Such subjects in need thereof, include subjects that may benefit from treatment with an expressed DR-18 and/or a CAR, including subjects having cancer, including where the cancer is a cancer to which an antigen binding domain of the CAR is directed. Useful methods include administering a DR-18 armored CAR where the CAR is any CAR described herein or a CAR directed to any antigen described herein. Useful methods include administering a DR-18 armored CAR where the DR-18 is any DR-18 described herein and any DR-18 encoding nucleic acid, including those described herein, may be employed. Nucleic acids of the present disclosure may be administered in such methods by a variety of different means, including e.g., where the nucleic acids are administered in vivo to the subject (e.g., through the use of a vector, such as a viral or non-viral vector), where the nucleic acids are contacted ex vivo with cells and the cells are subsequently administered to the subject. Cells employed for the delivery of DR-18 armored CAR nucleic acids may be derived from the subject receiving treatment (e.g., autologous) or from a donor that is not the subject receiving treatment (i.e., allogeneic). Nucleic acids of the present disclosure are administered to subjects in need thereof in a manner sufficient for the encoded elements of the nucleic acids to be expressed.

[0191] Allogeneic cells may include one or more genetic modifications rendering the cells less susceptible to clearance from a hosts immune response, such as but not limited to e.g., modification to disrupt T cell receptor alpha constant (TRAC) gene, modification to disrupt or prevent expression of 32 microglobulin, modification to disrupt endogenous human leukocyte antigen (HLA), modification to express HLA-E, and the like. Allogeneic cells, and methods of modifying cells for allogeneic cell therapy, include but are not limited to e.g., those described in WO2023078287, U.S. Ser. No. 11/389,481 B2, WO2022007784, US20230158070, WO2023093763, WO2019118475, US20230190809, WO2022192895, US20210268028, US20230068949, WO2021202581, US20230011889, WO2023279112, WO2022115864, US20210040449, WO2022018262, US20220041984, WO2023147776, US20230181641, WO2023010018, US20220023344, US20210355230, WO2023023515, US20220251505, US20180360883, WO2023070059, WO2021113759, WO2019089650, WO2020191378, WO2023196980, U.S. Ser. No. 10/538,574B2, WO2021228177, US20220118014, WO2021173630, WO2022187663, WO2019243835, US20220409665, US20220313736, and WO2022132720.

[0192] Methods of introducing genes for expression into a cell are known in the art. In the context of an expression vector, the vector can be readily introduced into a host cell, e.g., mammalian, by any method in the art. For example, the expression vector can be transferred into a host cell by physical, chemical, or biological means. Biological methods for introducing a polynucleotide of interest into a host cell include the use of DNA and RNA vectors.

[0193] The nucleic acid sequences coding for the desired molecules can be obtained using recombinant methods known in the art, such as, for example by screening libraries from cells expressing the gene, by deriving the gene from a vector known to include the same, or by isolating directly from cells and tissues containing the same, using standard techniques. Alternatively, the nucleic acid of interest can be produced synthetically, rather than cloned. Nucleic acid coding sequences can be derived from desired amino acid sequences, and such nucleic acid coding sequences can further be codon optimized for expression in a desired host, such as a human.

[0194] Nucleic acids of the present disclosure may be introduced into or inserted into vectors. The nucleic acid can be cloned into a number of types of vectors. For example, the nucleic acid can be cloned into a vector including, but not limited to a plasmid, a phagemid, a phage derivative, an animal virus, and a cosmid. Vectors of interest include expression vectors, replication vectors, transfer vectors, shuttle vectors, donor vectors, and the like. Viruses, which are useful as vectors include, but are not limited to, retroviruses, adenoviruses, adeno-associated viruses, herpes viruses, and lentiviruses.

[0195] Vectors derived from retroviruses such as the lentivirus are suitable tools to achieve long-term gene transfer since they allow long-term, stable integration of a transgene and its propagation in daughter cells. They also have the added advantage of low immunogenicity. A retroviral vector may also be, e.g., a gammaretroviral vector. A gammaretroviral vector may include, e.g., a promoter, a packaging signal (tp), a primer binding site (PBS), one or more (e.g., two) long terminal repeats (LTR), and a transgene of interest, e.g., a gene encoding a DR-18 polypeptide and/or a CAR. A gammaretroviral vector may lack viral structural genes such as gag, pol, and env. Exemplary gammaretroviral vectors include Murine Leukemia Virus (MLV), Spleen-Focus Forming Virus (SFFV), and Myeloproliferative Sarcoma Virus (MPSV), and vectors derived therefrom.

[0196] In another embodiment, the vector comprising a nucleic acid of the present disclose is an adenoviral vector. In another embodiment, the expression of nucleic acids encoding DR-18 polypeptides and/or CARs can be accomplished using transposons such as sleeping beauty, CRISPR nucleases (e.g., CAS9), and zinc finger nucleases. See e.g., June et al. 2009 *Nature Reviews Immunology* 9.10: 704-716.

[0197] Non-viral delivery system may be utilized, where an exemplary delivery vehicle is a nanoparticle, e.g., a liposome, lipid nanoparticle (LNP) or other suitable submicron sized delivery system. Lipid formulations are useful for the introduction of the nucleic acids, including DNA or RNA, into a host cell (in vitro, ex vivo or in vivo).

[0198] The instant disclosure also includes RNA constructs, encoding elements described herein, that can be directly transfected into a cell. A method for generating mRNA for use in transfection involves in vitro transcription (IVT) of a template with specially designed primers, followed by poly A addition, to produce a construct containing 3' and 5' untranslated sequence ("UTR"), a 5 cap and/or Internal Ribosome Entry Site (IRES), 2A sequence (P2A, T2A, etc.), etc., the sequence(s) encoding polypeptide(s) to be expressed, and a poly A tail. RNA so produced can efficiently transfect different kinds of cells. In one embodiment, the template includes sequences for the CAR and/or the DR-18. In an embodiment, an RNA armored CAR vector is transduced into a T cell by electroporation. In an embodiment, an armored CAR vector is associated with a lipid, such as encapsulated in the aqueous interior of a liposome, LNP, or the like, interspersed within the lipid bilayer of a liposome, LNP, or the like, attached to a lipid, liposome, or LNP, via a linking molecule, entrapped in a liposome, LNP, or the like, complexed with an LNP or the like, dispersed in a solution containing a lipid, mixed with a lipid, combined with a lipid, contained as a suspension in a lipid, contained or complexed with a micelle, or otherwise associated with a lipid.

[0199] The expression of natural or synthetic nucleic acids encoding armored CARs may be achieved by operably linking a nucleic acid encoding the CAR polypeptide or portions thereof and/or the DR-18 polypeptide to a promoter and incorporating the construct into an expression vector. The vectors can be suitable for replication and integration in eukaryotes. Typical cloning vectors contain transcription and translation terminators, initiation sequences, and promoters useful for regulation of the expression of the desired nucleic acid sequence.

[0200] Methods of making and using CAR encoding cells, which are applicable to making and using DR-18 polypeptide armored cells of the present disclosure, including methods of making vectors, transfection, transduction, cell culture and expansion, lymphodepletion, patient preparation, dose preparation and deliver, dosing, co-administration of other agents, patient monitoring, patient evaluation, and like, are described in e.g., the following U.S. and PCT patents and applications: U.S. Ser. No. 11/872,249B2, US20230416390, US20230381228, WO2023154890, US20230312708, WO2023147293, US20230256017, WO2023114777, US20230220090, US20230220103, U.S. Ser. No. 11/673,964B2, US20230174933, WO2023086900, US20230151094, WO2023044350, US20230084763, WO2023021477, US20230000964, US20220364055, US20220315665, US20220305056, US20220227874, US20220184129, US20220168389, US20220064254, US20220064316, US20220047633, US20210393689, US20210347851, US20210324100, U.S. Ser. No. 11/149,076B2, US20210284752, US20210220404, US20210177896, US20210171909, U.S. Ser. No. 11/028,177B2, US20210163612, WO2021099944, US20210139595, WO2021067290, US20200407461, US20200306304, U.S. Ser. No. 10/752,684B2, U.S. Ser. No. 10/752,670B2, US20200231686, US20200055948, U.S. Ser. No. 10/526,406B2, US20190336504, WO2019152660, U.S. Ser. No. 10/316,101B2, WO2017149515, US20170226183, and US20160051651; the disclosures of which are incorporated herein by reference in their entirety.

[0201] Cells into which nucleic acids of the present disclosure are introduced may first be obtained, e.g., prior to expansion and/or genetic modification, from a mammalian subject. Useful mammalian subjects include living mammalian subjects from which an immune response can be elicited, such as e.g., humans, primates, non-human primates, dogs, cats, mice, rats, and transgenic species thereof. Cells can be obtained from a number of sources, including peripheral blood mononuclear cells, bone marrow, lymph node tissue, cord blood, thymus tissue, tissue from a site of infection, ascites, pleural effusion, spleen tissue, and tumors. In some embodiments, cells are obtained from a subject, prepared, including genetically modified, expanded, etc., and used to treat the subject from which they were obtained. In some embodiments, cells are obtained from a first subject, prepared, including genetically modified, expanded, etc., and used to treat a second subject other than the subject from which they were obtained. In some embodiments, any number of available cell lines, including available immune cell lines, such as available T cell lines, may be employed. Various useful methods of may be employed in cell collection, alone or in combination, such as but not limited to e.g., blood collection kits, density gradients (with or without centrifugation), apheresis, cell washing, isolation of desired cell types, positive selection, depletion of unwanted cell types, negative selection, and the like.

[0202] In some instances, cells of the present disclosure may be expanded, e.g., before, after, or during activation; or before, after, or during genetic modification. Any useful and appropriate method for expansion of the relevant cell type may be employed for expansion. Cells may be expanded by contact with one or more agents that stimulates proliferation of the cells, including e.g., where one or more of the agents is bound to a surface. For example, with respect to T cells, the cells may be expanded by contact with one or more agents such as antibodies, such as e.g., an anti-CD3 antibody or binding fragment thereof, an anti-CD2 antibody or binding fragment thereof, an anti-CD28 antibody or binding fragment thereof, and the like, including where one or more such antibodies are in solution, bound to a surface, such as a bead, a plate, a well, or the like, or a combination thereof.

[0203] In some embodiments, T cells may be activated and expanded generally using methods as described, for example, in U.S. Pat. Nos. 6,352,694; 6,534,055; 6,905,680; 6,692,964; 5,858,358; 6,887,466; 6,905,681; 7,144,575; 7,067,318; 7,172,869; 7,232,566; 7,175,843; 5,883,223; 6,905,874; 6,797,514; 6,867,041; and U.S. Patent Application Publication Nos. 20060121005 and 20230416390.

[0204] Agents for the activation and/or expansion of cells of the present disclosure may be cultured together for any useful and appropriate amount of time, including e.g., hours, such as e.g., about 1-2 hours, about 2-4 hours, about 4-6 hours, about 6-12 hours, about 12-24 hours, about 1-24 hours, about 1-12 hours, about 1-6 hours, etc., days, such as e.g., about 1 day, about 2 days, about 1-2 days, about 2-3 days, about 2-4 days, about 3-5 days, about 4-6 days, about 5-7 days, about 7 days, about 6-10 days, about 8 days, about 7-14 days, about 10 days, about 14 days, about 10-20 days, about 15 days, about 18 days, about 20 days, about 14-21 days, about 21 days, etc., weeks, such as e.g., about 1 week, about 2 weeks, about 3 weeks, about 4 weeks, etc. In some instances, multiple cycles of activation and/or stimulation may be employed.

[0205] Compositions and methods of the present disclosure may be used to treat a subject who has been characterized as having cells or tissues expressing one or more antigens to which the antigen-binding portion of the relevant CAR binds, or is suspected of having cells or tissues expressing one or more antigens to which the antigen-binding portion of the relevant CAR binds. For example, subjects benefiting from treatment according to the invention include subjects with a cancer, or subjects suspected of having a cancer, for example, as evidenced by the presence of one or more symptoms, clinical manifestations, or diagnostic indicators of the caner, including where the cancer is known or suspected to express one or more antigens to which the antigen-binding portion of the relevant CAR binds.

[0206] The present disclosure provides methods for inhibiting the proliferation or reducing a cell population expressing an antigen to which the antigen binding region of the CAR binds (i.e., a target cell), the methods comprising contacting a population of cells comprising a target cell with an armored CAR-expressing cell (or nucleic acids encoding the armored CAR components) described herein, e.g., a T cell, that binds to an antigen expressed by the target cell. In an embodiment, the present disclosure provides methods for inhibiting the proliferation or reducing the population of cancer cells expressing a target cell antigen, the methods comprising contacting the target cell-expressing cancer cell population with armored CAR-expressing cells (or nucleic acids encoding the armored CAR components) described herein, e.g., a T cell, that binds to the target antigen-expressing cell.

[0207] Cancers and tumors, the treatment of which may include the use of a DR-18 armored CAR of the instant disclosure, will vary and may include but are not limited to e.g. Acute Lymphoblastic Leukemia (ALL), Acute Myeloid Leukemia (AML), Adrenocortical Carcinoma, AIDS-Related Cancers (e.g., Kaposi Sarcoma, Lymphoma, etc.), Anal Cancer, Appendix Cancer, Astrocytomas, Atypical Teratoid/Rhabdoid Tumor, Basal Cell Carcinoma, Bile Duct Cancer (Extrahepatic), Bladder Cancer, Bone Cancer (e.g., Ewing Sarcoma, Osteosarcoma and Malignant Fibrous Histiocytoma, etc.), Brain Stem Glioma, Brain Tumors (e.g., Astrocytomas, Central Nervous System Embryonal Tumors, Central Nervous System Germ Cell Tumors, Craniopharyngioma, Ependymoma, etc.), Breast Cancer (e.g., female breast cancer, male breast cancer, childhood breast cancer, etc.), Bronchial Tumors, Burkitt Lymphoma, Carcinoid Tumor (e.g., Childhood, Gastrointestinal, etc.), Carcinoma of Unknown Primary, Cardiac (Heart) Tumors, Central Nervous System (e.g., Atypical Teratoid/Rhabdoid Tumor, Embryonal Tumors, Germ Cell Tumor, Lymphoma, etc.), Cervical Cancer, Childhood Cancers, Chordoma, Chronic Lymphocytic Leukemia (CLL), Chronic Myelogenous Leukemia (CML), Chronic Myeloproliferative Neoplasms, Colon Cancer, Colorectal Cancer, Craniopharyngioma, Cutaneous T-Cell Lymphoma, Duct (e.g., Bile Duct, Extrahepatic, etc.), Ductal Carcinoma In Situ (DCIS), Embryonal Tumors, Endometrial Cancer, Ependymoma, Esophageal Cancer, Esthesioneuroblastoma, Ewing Sarcoma, Extracranial Germ Cell Tumor, Extragonadal Germ Cell Tumor, Extrahepatic Bile Duct Cancer, Eye Cancer (e.g., Intraocular Melanoma, Retinoblastoma, etc.), Fibrous Histiocytoma of Bone (e.g., Malignant, Osteosarcoma, etc.), Gallbladder Cancer, Gastric (Stomach) Cancer, Gastrointestinal Carcinoid Tumor, Gastrointestinal Stromal Tumors (GIST), Germ Cell Tumor (e.g., Extracranial, Extragonadal, Ovarian, Testicular, etc.), Gestational Trophoblastic Disease, Glioma, Hairy Cell Leukemia, Head and Neck Cancer, Heart Cancer, Hepatocellular (Liver) Cancer, Histiocytosis (e.g., Langerhans Cell, etc.), Hodgkin Lymphoma, Hypopharyngeal Cancer, Intraocular Melanoma, Islet Cell Tumors (e.g., Pancreatic Neuroendocrine Tumors, etc.), Kaposi Sarcoma, Kidney Cancer (e.g., Renal Cell, Wilms Tumor, Childhood Kidney Tumors, etc.), Langerhans Cell Histiocytosis, Laryngeal Cancer, Leukemia (e.g., Acute Lymphoblastic (ALL), Acute Myeloid (AML), Chronic Lymphocytic (CLL), Chronic Myelogenous (CML), Hairy Cell, etc.), Lip and Oral Cavity Cancer, Liver Cancer (Primary), Lobular Carcinoma In Situ (LCIS), Lung Cancer (e.g., Non-Small Cell, Small Cell, etc.), Lymphoma (e.g., AIDS-Related, Burkitt, Cutaneous T-Cell, Hodgkin, Non-Hodgkin, Primary Central Nervous System (CNS), etc.), Macroglobulinemia (e.g., Waldenstrom, etc.), Male Breast Cancer, Malignant Fibrous Histiocytoma of Bone and Osteosarcoma, Melanoma, Merkel Cell Carcinoma, Mesothelioma, Metastatic Squamous Neck Cancer with Occult Primary, Midline Tract Carcinoma Involving NUT Gene, Mouth Cancer, Multiple Endocrine Neoplasia Syndromes, Multiple Myeloma/Plasma Cell Neoplasm, Mycosis Fungoides, Myelodysplastic Syndromes, Myelodysplastic/Myeloproliferative Neoplasms, Myelogenous Leukemia (e.g., Chronic (CML), etc.), Myeloid Leukemia (e.g., Acute (AML), etc.), Myeloproliferative Neoplasms (e.g., Chronic, etc.), Nasal Cavity and Paranasal Sinus Cancer, Nasopharyngeal Cancer, Neuroblastoma, Non-Hodgkin Lymphoma, Non-Small Cell Lung Cancer, Oral Cancer, Oral Cavity Cancer (e.g., Lip, etc.), Oropharyngeal Cancer, Osteosarcoma and Malignant Fibrous Histiocytoma of Bone, Ovarian Cancer (e.g., Epithelial, Germ Cell Tumor, Low Malignant Potential Tumor, etc.), Pancreatic Cancer, Pancreatic Neuroendocrine Tumors (Islet Cell Tumors), Papillomatosis, Paraganglioma, Paranasal Sinus and Nasal Cavity Cancer, Parathyroid Cancer, Penile Cancer, Pharyngeal Cancer, Pheochromocytoma, Pituitary Tumor, Pleuropulmonary Blastoma, Primary Central Nervous System (CNS) Lymphoma, Prostate Cancer, Rectal Cancer, Renal Cell (Kidney) Cancer, Renal Pelvis and Ureter, Transitional Cell Cancer, Retinoblastoma, Rhabdomyosarcoma, Salivary Gland Cancer, Sarcoma (e.g., Ewing, Kaposi, Osteosarcoma, Rhabdomyosarcoma, Soft Tissue, Uterine, etc.), Sezary Syndrome, Skin Cancer (e.g., Childhood, Melanoma, Merkel Cell Carcinoma, Nonmelanoma, etc.), Small Cell Lung Cancer, Small Intestine Cancer, Soft Tissue Sarcoma, Squamous Cell Carcinoma, Squamous Neck Cancer (e.g., with Occult Primary, Metastatic, etc.), Stomach (Gastric) Cancer, T-Cell Lymphoma, Testicular Cancer, Throat Cancer, Thymoma and Thymic Carcinoma, Thyroid Cancer, Transitional Cell Cancer of the Renal Pelvis and Ureter, Ureter and Renal Pelvis Cancer, Urethral Cancer, Uterine Cancer (e.g., Endometrial, etc.), Uterine Sarcoma, Vaginal Cancer, Vulvar Cancer, Waldenstrom Macroglobulinemia, Wilms Tumor, and the like.

[0208] In some instances, methods of treatment of the present disclosure will include treating a blood cancer. In some instances, methods of treatment of the present disclosure will include treating a solid tumor. In some instances, methods of treatment of the present disclosure will include treating a subject for a naïve cancer. In some instances, methods of treatment of the present disclosure will include treating a subject for a cancer that has been previously treated, including previously treated with one or more of chemotherapy, radiation, immunotherapy, or any combination thereof. In some instances, methods of treatment of the present disclosure will include treating a primary tumor. In some instances, methods of treatment of the present disclosure will include treating a metastatic tumor. In some instances, methods of treatment of the present disclosure will include treating a recurrent cancer or tumor.

[0209] Methods of the present disclosure find use in treating treatment naïve cancers, treatment refractory cancers, primary tumors, metastatic tumors, and the like. In some instances, methods of the present disclosure find use in treating and/or managing a cancer that includes cells expressing a target antigen. In some instances, methods of the present disclosure find use in preventing relapse of a cancer that includes cells expressing a target antigen. Methods of the present disclosure include administering an effective amount of armored CAR cells (or nucleic acids encoding the armored CAR components), described herein, to a patient in need thereof, including where the subject in need thereof is a subject, including a human, having cancer, suspected of having cancer, at risk of developing cancer, at risk of relapsing, or the like. In some instances, administration of the armored CAR cells (or nucleic acids encoding the armored CAR components) may be the only therapy administered to the subject during the course of treatment (i.e., monotherapy). In some instances, the armored CAR cells (or nucleic acids encoding the armored CAR components) may be administered in combination with one or more other agents active against the subject's cancer during the course of therapy as part of a combination therapy.

[0210] The armored CAR-modified cells (or nucleic acids encoding the armored CAR components) of the present invention may be administered either alone, or as a pharmaceutical composition in combination with diluents and/or with other components such as other cytokines or cell populations or other drug treatments, e.g., described herein. Additional cancer-treating agents, e.g., for use in combination therapies with nucleic acids and/or cells of the present disclosure, are known to the relevant ordinarily skilled artisan. For example, in some instances, armored CAR immune cells (or a vector for armored CAR in vivo therapy) may be administered in combination with one or more cytokines, such as e.g., IL-2, IL-7, IL-10, IL-12, IL-15, IL-21, IL-23, IL-27, IL-33, TNF, TL1A, IFNa, IFNb, and IFNg. In some instances, the additional cytokine administered is not IL-7, IL15, or IL-10.

[0211] Methods of the present disclosure may include administering to a subject a nucleic acid that includes a sequence encoding a DR-18 polypeptide and one or more sequences encoding at least one additional cytokine. The term "additional cytokine" as used herein refers to cytokines providing a cytokine function in addition to the IL-18 signaling function provided by the encoded DR-18 polypeptide (i.e., a cytokine other than IL-18). Such additional cytokines include e.g., IL-2, IL-7, IL-10, IL-12, IL-15, IL-21, IL-23, IL-27, IL-33, TNF, TL1A, IFNa, IFNb, and IFNg. In some instances, the additional cytokine is not IL-7, IL15, or IL-10. Accordingly, in some instances, a cell or a treatment of the present disclosure may be referred to a multi- (e.g., dual-, triple-, etc.) armored where such multi-armoring includes armoring with DR-18 polypeptide expression and expression of at least one additional cytokine, such as, e.g., IL-2, IL-7, IL-10, IL-12, IL-15, IL-21, IL-23, IL-27, IL-33, TNF, TL1A, IFNa, IFNb, and IFNg. In some instances, the additional cytokine of the multi-armored DR-18 polypeptide is not (or is a cytokine other than) one or more of IL-2, IL-7, IL-10, IL-12, IL-15, IL-21, IL-23, IL-27, IL-33, TNF, TL1A, IFNa, IFNb, and IFNg, for example, the cytokine is not one or more of IL-7, IL15, or IL-10 or is a cytokine other than one or more of IL-7, IL15, or IL-10.

[0212] In one embodiment, a cell or a CAR therapy may be multi-armored with expression of a DR-18 polypeptide and expression of an IL-12, including e.g., an IL-12 with a wild-type IL12p35 with (SEQ ID NO: 88) or without (SEQ ID NO: 89) endogenous signal peptide, an IL-12 with a wild-type IL12p40 with (SEQ ID NO: 90) or without (SEQ ID NO: 91) endogenous signal peptide, an IL-12 partial agonist comprising a variant IL12p35, an IL-12 partial agonist comprising a variant IL12p40, a detuned IL-12, combinations thereof or the like.

[0213] As used herein, a "partial agonist" refers to a variant of an agonist that has submaximal signaling through one or more of its receptors (e.g., as compared to the wild-type agonist) and, without being bound by theory, is capable of selectively biasing the activation of cells and/or signaling pathways that possess differential activation thresholds. For example, an IL-12 partial agonist will include a variant IL-12p40 subunit, variant IL-12p35 subunit, or variants of both subunits, that have submaximal signaling (as compared to wild-type IL-12) through one or more of its receptors or receptor subunits. Without being bound by theory, such submaximal signaling may have the effect of selectively activating desired cell populations while avoiding the activation (or minimally activating) undesired cell populations. As used herein, a "detuned" version of a cytokine refers to a variant that has reduced affinity for one or more of its receptors.

[0214] In some instances, a subject may be administered one or more nucleic acids including sequences that encode for at least a DR-18 polypeptide, an IL-12 partial agonist or detuned IL-12, and a CAR. Any of the herein described DR-18 polypeptides and CARs, or components thereof, may be employed in an at least dual-armored DR-18 and IL-12 partial agonist or detuned IL-12 CAR therapy. Useful IL-12 partial agonists include but are not limited to those described in PCT Pat. Pub. Nos. WO2023023503 and WO2021212083. Useful IL-12 partial agonists (including variant IL12p40 subunits, variant IL12p35 subunits, or both variant IL12p40 and variant IL12p35 subunits) that may be encoded by one or more sequences included the nucleic acids of the presented disclosure, include e.g., SEQ ID NOs: 92-108. Such IL-12 coding sequences may encode for a wt IL-12p35 subunit, a IL-12p35 variant subunit, a wt IL-12p40 subunit, a IL-12p40 variant subunit, a IL-12p35 subunit (variant or wt) with its endogenous signal peptide, a IL-12p35 subunit (variant or wt) without its endogenous signal peptide (e.g., with a heterologous signal peptide), a IL-12p40 subunit (variant or wt) with its endogenous signal peptide, a IL-12p40 subunit (variant or wt) without its endogenous signal peptide (e.g., with a heterologous signal peptide), or any combination thereof. In some instances, where an endogenous signal peptide is not present the coding sequence may include sequence encoding a heterologous signal peptide.

[0215] Methods of the present disclosure that include administering a DR-18 armored CAR immune cell or a vector for in vivo DR-18 armored CAR immune cell therapy will provide for certain advantage as compared to corresponding CAR therapies not employing DR-18 armoring. For example, in some instances, methods of the present disclosure may be performed without lymphodepletion of the subject prior to or during treatment with the DR-18 armored CAR therapy. For example, in the case of autologous CAR T or NK cell therapy, cells may be collected from the subject, modified ex vivo to express the DR-18 polypeptide and CAR, activated and/or expanded as desired, and administered to the subject without the subject having undergone lymphodepletion. In the case of allogeneic CAR T or NK cell therapy, allogeneic DR-18 armored CAR T or NK cells may be administered to the subject without the subject having undergone lymphodepletion. In some instances, the DR-18 armored CAR immune cell therapy without lymphodepletion achieves at least equivalent engraftment, efficacy, and/or persistence of the DR-18 armored CAR immune cells as compared to an equivalent CAR immune cell therapy without DR-18 armoring and with lymphodepletion.

[0216] Lymphodepletion is a conditioning step used in autologous and allogeneic CAR immune cell therapies that depletes and modulates endogenous lymphocytes in the recipient to prepare the host to receive the CAR immune cell therapy. Various methods using various agents, combinations of agents, radiation, and combinations thereof are employed in lymphodepletion regimens. Non-limiting examples of agents useful in lymphodepletion include fludarabine, cyclophosphamide, bendamustine, alemtuzumab, oxaliplatin, clofarabine, and combinations thereof. Lymphodepletion and useful lymphodepletion regimens are described e.g., in Lickefett et al. (2023) *Front. Immunol.* 14:1303935.

[0217] In some instances, treatment of a subject with a DR-18 armored CAR immune cell, such as a T cell or NK cell, or a vector for in vivo DR-18 armored CAR immune cell therapy will achieve an equivalent or better effect as compared to the corresponding CAR immune cell therapy performed without DR-18 armoring, including e.g., where the DR-18 armored CAR therapy is administered at a lower dose than the CAR therapy without DR-18 armoring, where the DR-18 armored CAR therapy is administered less frequently as compared to the CAR therapy without DR-18 armoring, or where the DR-18 armored CAR therapy is administered at a lower dose and less frequently as compared to the CAR therapy without DR-18 armoring. In some instances, DR-18 armored CAR immune cells (whether generated ex vivo or in vivo) will persist and/or remain active longer in a subject than corresponding CAR immune cells without DR-18 armoring, and thus require less frequent and/or lower dosing. In some instances, DR-18 armored CAR immune cells (whether generated ex vivo or in vivo) will be less susceptible to exhaustion as compared to corresponding CAR immune cells without DR-18 armoring, and thus require less frequent and/or lower dosing.

[0218] Methods of the present disclosure further include methods of making DR-18 armored CAR immune cells. DR-18 armored CAR immune cells, containing at least a sequence encoding a DR-18 polypeptide and a CAR encoding sequence, may be made using any of the components and/or processes described herein. In some instances, methods of the present disclosure may include introducing a DR-18 polypeptide encoding sequence into a cell ex vivo and culturing the cell following introduction of the DR-18 polypeptide. The introduced DR-18 polypeptide encoding sequence may be configured to express the DR-18 polypeptide during culture of the modified cell. In some instances, the presence of DR-18 polypeptide during the cell manufacturing process provides for an enhanced manufacturing process as compared to a corresponding process in the absence of expressed DR-18 polypeptide.

[0219] In some instances, culturing a population of cells wherein at least some portion of the cells (e.g., at least 10%, 20%, 30%, 40%, 50%, or more) express the DR-18 polypeptide will provide certain advantages as compared to culturing a corresponding population of cells that do not express the DR-18 polypeptide during culturing. For example, the presence of expressed DR-18 polypeptide during the culturing process may provide for enhanced proliferation or expansion of the cells as compared to corresponding cells cultured in the absence of expressed DR-18 polypeptide. In some instances, the presence of expressed DR-18 polypeptide during the culturing process may provide for enhanced activation of the cells as compared to corresponding cells cultured in the absence of expressed DR-18 polypeptide. In some instances, cells or cell populations produced in the presence of expressed DR-18 polypeptide during the culturing process will have an enhanced phenotype as compared to corresponding cells or cell populations produced in the absence of expressed DR-18 polypeptide. Such enhanced phenotypes may include e.g., enhanced in vivo activity, enhanced persistence in culture, enhanced persistence in vivo, enhanced proliferation in culture, enhanced proliferation in vivo, enhanced tumor cell killing, combinations thereof, and the like.

SEQUENCES

SEQ ID NO:	Ref. name	Sequence
1	wt hIL-18	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISMYKDSQPRGMAVTI SVKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRSVPGHDNKMQFESSSYEGYFLACEK ERDLFKLILKKEDELGDRSIMFTVQNED
2	wt hIL-18 (unprocessed)	MAAEPVEDNCINFVAMKFIDNTLYFIAEDDENLESDYFGKLESKLSVIRNLNDQVLFIDQGNRP LFEDMTDSDCRDNAPRTIFIISMYKDSQPRGMAVTISVKCEKISTLSCENKIISFKEMNPPDNIK DTKSDIIFFQRSVPGHDNKMQFESSSYEGYFLACEKERDLFKLILKKEDELGDRSIMFTVQNED
3	wt hIL-18 propeptide	MAAEPVEDNCINFVAMKFIDNTLYFIAEDDENLESD
4	sq89v1	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDSRDNAPRTIFIISKYSDSLARGLAVTIS VKSEKISTLSSENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHSRKMQFESSSYEGYFLASEKE RDLFKLILKKEDELGDRSIMFTVQNED
5	sq89v2	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISKYSDSLARGLAVTIS VKSEKISTLSSENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHSRKMQFESSSYEGYFLASEKE RDLFKLILKKEDELGDRSIMFTVQNED

-continued

		SEQUENCES
6	sq89v3	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDSRDNAPRTIFIISKYSDSLARGLAVTIS VKCEKISTLSSENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHSRKMQFESSSYEGYFLASEKE RDLFKLILKKEDELGDRSIMFTVQNED
7	sq89v4	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDSRDNAPRTIFIISKYSDSLARGLAVTIS VKSEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHSRKMQFESSSYEGYFLASEKE RDLFKLILKKEDELGDRSIMFTVQNED
8	sq89v5	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDSRDNAPRTIFIISKYSDSLARGLAVTIS VKSEKISTLSSENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHSRKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED
9	sq89v6	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDSRDNAPRTIFIISKYSDSLARGLAVTIS VKSEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHSRKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED
10	sq89v7	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDSRDNAPRTIFIISKYSDSLARGLAVTIS VKCEKISTLSSENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHSRKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED
11	sq89v8	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDSRDNAPRTIFIISKYSDSLARGLAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHSRKMQFESSSYEGYFLASEKE RDLFKLILKKEDELGDRSIMFTVQNED
12	sq89v9	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISKYSDSLARGLAVTIS VKSEKISTLSSENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHSRKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED
13	q89v10	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISKYSDSLARGLAVTIS VKSEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHSRKMQFESSSYEGYFLASEKE RDLFKLILKKEDELGDRSIMFTVQNED
14	q89v11	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISKYSDSLARGLAVTIS VKCEKISTLSSENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHSRKMQFESSSYEGYFLASEKE RDLFKLILKKEDELGDRSIMFTVQNED
15	q89v12	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDSRDNAPRTIFIISKYSDSLARGLAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHSRKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED
16	q89v13	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISKYSDSLARGLAVTIS VKSEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHSRKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED
17	q89v14	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISKYSDSLARGLAVTIS VKCEKISTLSSENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHSRKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED
18	q89v15	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISKYSDSLARGLAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHSRKMQFESSSYEGYFLASEKE RDLFKLILKKEDELGDRSIMFTVQNED
19	C68v1	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDSRDNAPRTIFIISKYSDSLARGLAVTIS VKGEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHSRKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED
20	C68v2	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDSRDNAPRTIFIISKYSDSLARGLAVTIS VKAEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHSRKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED
21	C68v3	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDSRDNAPRTIFIISKYSDSLARGLAVTIS VKVEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHSRKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED
22	C68v4	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDSRDNAPRTIFIISKYSDSLARGLAVTIS VKDEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHSRKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED
23	C68v5	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDSRDNAPRTIFIISKYSDSLARGLAVTIS VKEEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHSRKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED
24	C68v6	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDSRDNAPRTIFIISKYSDSLARGLAVTIS VKNEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHSRKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED

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SEQUENCES		
25	hCS1	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISTYKDSQPRGKAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHKHKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED
26	hCS2	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISTYKDKQPRAKAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHKHKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTIQNED
27	hCS3	RFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISTYKDSQPRGKAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHKHKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED
28	hCS4	RFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISTYRDSQPRGKAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRNVPGHKYKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED
29	hC4	YFGKLESQLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISTYKDKQPRTKAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRRVPGHHNKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQKED
30	hA8	YFGKLESRLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISKYKDKQPRAQAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHKHKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTIQNED
31	hD6	YFGKLESRLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISDYKDKQPRAXAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHKHKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTIQNED
32	hH12	YFGKHESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISTYRDSQPRGKAVTI SVKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHNNKMQFESSSYEGYFLACEK ERDLFKLILKKEDELGDRSIMFTTQNED
33	hB11	YFGKIESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISKYKDKQPRAQAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRKVPGHQHKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQKED
34	hC3	YFGKIESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISTYKDRQPRGKAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFERDVPGHHHKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTIQNED
35	hC2	YFGKIESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISTYKDKQPRGKAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHKHKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTTQHED
36	hG10	YFGKIESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISTYKDKQPRAKAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRRVPGHHHKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTIQKED
37	hG1	YFGKIESRLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISTYKDKQPRGKAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHDYKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTIQKED
38	hF1	YFGKYESRLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISTYRDSQPRGKAVTI SVKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHEHKMQFESSSYEGYFLACEK ERDLFKLILKKEDELGDRSIMFTVQKED
39	hD2	HFGKYESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISTYRDSQPRGKAVTI SVKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHHNKMQFESSSYEGYFLACEK ERDLFKLILKKEDELGDRSIMFTVQKED
40	hA1	RFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISTYRDSQPRAKAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHQHKMQFESSSYEGYFLACEK ERDLFKLILKKEDELGDRSIMFTAQKED
41	hB3	RFGKLESRLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISDYRDSQPRGRAVTI SVKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFKRNVPGHKYKMQFESSSYEGYFLACEK ERDLFKLILKKEDELGDRSIMFTVQHED
42	hB4	RFGKLESRLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISNYRDSQPRGQAVTI SVKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFKRRVPGHNHKMQFESSSYEGYFLACEK ERDLFKLILKKEDELGDRSIMFTVQKED

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SEQUENCES		
43	hH3	RFGKLESRLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISTYKDSQPRGKAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHKHKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED
44	hH5	RFGKHESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISTYRDSQPRGKAVTI SVKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFERNVPGHKYKMQFESSSYEGYFLACEK ERDLFKLILKKEDELGDRSIMFTVQNED
45	hH4	RFGKLESRLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISTYRDSQPRAKAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFERDVPGHQHKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTIQXED
46	hE1	RFGKLESRLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISTYRDSQPRTKAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRNVPGHHDKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQHED
47	hG2	RFGKLESRLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISTYKDSQPRAKAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFERDVPGHQHKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTIQKED
48	hE12	RFGKYESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISTYKDSQPRTKAVTIS VKCEKISTLSCDNKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHKHKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED
49	hC5	RFGKLESRLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISTYRDSQPRTKAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRKVPGHNHKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQKED
50	5-18	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISEYKDSELRGRAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFPRAVPGHNRKVQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED
51	5-29	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISKYKDSAGRGLAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFERDVPGHSNKVQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED
52	5-8	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISKYGDSAARGLAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRSVPGHKRKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED
53	5-6	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISKYGDSRGRGLAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFERDVPGHNSKRQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED
54	5-27	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISKYGDSVPRGLAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFARAVPGHSRKTQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED
55	5-20	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISKYSDSGARGLAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFARAVPGHGRKTQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED
56	5-2	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISKYSDSKARGMAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFARDVPGHSSKRQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED
57	6-12	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISKYSDSLARGLAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRDVPGHSRKMQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED
58	5-42	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISKYSDSRARGLAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRNVPGHGRKMQFESSSYEGYFLACEK ERDLFKLILKKEDELGDRSIMFTVQNED
59	5-13	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISKYSDSRARGLAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFARSVPGHGRKTQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED
60	5-12	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISKYSDSRARGLAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFARDVPGHSGKRQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED
61	5-1	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISKYTDSRPRGLAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFERDVPGHSSKKQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED

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SEQUENCES		
62	5-33	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISKYTDSRARGMAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFERDVPGHNDKRQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED
63	5-21	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISRYKDSGKRGLAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFRRSVPGHSRKVQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED
64	6-31	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISKYGDSGARGLAVTIS VKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFERDVPGHSGKVQFESSSYEGYFLACEKE RDLFKLILKKEDELGDRSIMFTVQNED
65	6-20	YFGKLESKLSVIRNLNDQVLFIDQGNRPLFEDMTDSDCRDNAPRTIFIISKYGDSRPRGMAVTI SVKCEKISTLSCENKIISFKEMNPPDNIKDTKSDIIFFQRAVPGHNRKMQFESSSYEGYFLACEK ERDLFKLILKKEDELGDRSIMFTVQNED
66		XFGKXESXLSVIRNLNDQVLFIDQGNRPLFEDMTDSDXRDNAPRTIFIISXYXDXXXRXXAVTIS VKXEKISTLSXXNKIISFKEMNPPDNIKDTKSDIIFFXRXVPGHXXKXQFESSSYEGYFLAXEKE RDLFKLILKKEDELGDRSIMFTXQXED
67	hCS1 coding1	TATTTCGGCAAACTGGAAAGCAAGCTCAGCGTGATCCGGAACCTGAACGACCAAGTGCT GTTCATCGACCAGGGCAACCGGCCTCTGTTTGAGGACATGACAGACAGCGACTGCAGAG ATAATGCCCCTCGGACCATCTTCATCATCTCTACCTACAAGGATTCTCAGCCTAGAGGCAA GGCCGTGACCATCTCTGTCAAGTGCGAGAAGATCAGCACACTGAGCTGCGAGAACAAGA TCATTAGCTTCAAAGAAATGAACCCCCCCGACAATATCAAGGACACCAAGTCCGACATCA TCTTTTTCCAGAGAGATGTGCCAGGCCACAAGCACAAAATGCAGTTCGAGAGCAGCTCCT ACGAGGGCTACTTCCTGGCTTGTGAAAAGGAAAGAGACCTGTTCAAGCTGATCCTGAAAA AGGAAGATGAGCTGGGAGATAGAAGCATCATGTTCACCGTGCAGAACGAGGAC
68	hCS1 coding2	TACTTCGGCAAACTGGAGTCCAAGCTGAGCGTGATCAGGAATCTGAACGATCAGGTGCTC TTCATCGATCAGGGCAATAGACCTCTGTTTGAGGACATGACTGACTCCGACTGTCGGGAC AACGCCCCCCGAACAATCTTTATTATCAGCACCTACAAGGACAGCCAGCCCAGGGGGAA GGCCGTGACTATCTCCGTGAAGTGCGAAAAGATCAGCACCCTGTCATGTGAGAACAAGAT TATCTCCTTTAAGGAGATGAACCCCCCCGATAACATTAAGGACACCAAGAGCGACATCAT CTTCTTCCAGCGCGACGTGCCAGGACACAAGCACAAAATGCAGTTTGAGAGCAGCAGCT ACGAAGGGTATTTCCTGGCCTGTGAGAAGGAAAGAGATCTGTTCAAGCTGATCCTGAAGA AGGAGGACGAGCTGGGCGACCGGAGCATCATGTTTACCGTGCAGAACGAGGACTGA
69	hCS2 coding1	TACTTCGGCAAGCTGGAAAGCAAGCTGAGCGTGATCCGGAACCTGAACGACCAAGTGCT GTTCATCGACCAGGGAAATAGACCTCTGTTCGAGGACATGACCGACAGCGACTGCAGAG ATAATGCCCCTAGAACAATCTTCATCATCAGCACATACAAAGATAAGCAGCCTCGGGCTAA GGCCGTGACCATCAGCGTCAAGTGCGAGAAAATCTCTACCCTGTCTTGTGAAAACAAGAT CATCTCCTTCAAGGAAATGAACCCCCCCGACAACATCAAGGACACCAAGTCCGACATCAT TTTCTTCCAGCGGGACGTGCCAGGCCACAAGCACAAAATGCAGTTCGAGAGCAGCTCTTA TGAGGGCTACTTTCTGGCCTGCGAGAAGGAAAGAGATCTGTTTAAGCTCATCCTGAAGAA AGAAGATGAGCTGGGCGACAGAAGCATCATGTTCACCATCCAGAACGAGGAT
70	hCS2 coding2	TACTTTGGCAAGCTGGAGTCCAAACTGTCCGTGATCCGCAACCTGAATGATCAGGTGCTG TTCATCGACCAGGGCAACCGCCCCCTGTTCGAGGACATGACCGATAGCGACTGCAGGGA TAACGCCCCTCGGACTATCTTTATCATCAGCACATACAAGGACAAACAGCCCAGAGCCAA GGCCGTGACCATCAGCGTCAAGTGTGAGAAGATTAGCACTCTGTCCTGTGAGAACAAGAT CATTAGCTTTAAGGAAATGAACCCACCTGATAATATCAAGGACACCAAGAGCGACATTATC TTCTTTCAGAGAGATGTGCCCGGCCACAAGCACAAGATGCAGTTCGAGAGCTCTAGCTAC GAGGGCTACTTCCTGGCTTGCGAAAAGGAGAGAGACCTGTTTAAACTCATTCTGAAGAAG GAGGACGAACTGGGCGATAGATCCATCATGTTCACCATCCAGAACGAGGATTGA
71	hCS3 coding1	CGGTTCGGCAAACTGGAATCTAAGCTGTCTGTGATCCGGAACCTGAACGACCAAGTGCT GTTCATCGACCAGGGCAATCGCCCCCTGTTTGAGGATATGACCGACAGCGACTGCAGAG ATAACGCCCCTAGAACAATCTTCATCATCAGCACCTACAAGGATAGCCAGCCTCGGGGCA AGGCCGTGACAATCAGCGTCAAGTGCGAAAAAATCTCCACCCTGAGCTGTGAAAACAAGA TCATCTCTTTTAAGGAAATGAACCCACCTGATAATATCAAGGACACCAAGTCCGACATCAT TTTCTTCCAGAGAGATGTGCCCGGACACAAGCACAAAATGCAGTTCGAGAGCAGCAGCTA CGAGGGCTACTTCCTGGCTTGCGAGAAGGAAAGAGACCTGTTCAAGCTCATCCTGAAGA AAGAGGACGAGCTGGGCGACAGAAGCATCATGTTCACCGTGCAGAACGAGGAC
72	hCS3 coding2	CGCTTCGGCAAACTGGAGTCCAAGCTGAGCGTGATCAGGAATCTGAACGATCAGGTGCT CTTCATCGATCAGGGCAATAGACCTCTGTTTGAGGACATGACTGACTCCGACTGTCGGGA CAACGCCCCCCGAACAATCTTTATTATCAGCACCTACAAGGACAGCCAGCCCAGGGGGA AGGCCGTGACTATCTCCGTGAAGTGCGAAAAGATCAGCACCCTGTCATGTGAGAACAAGA TTATCTCCTTTAAGGAGATGAACCCCCCCGATAACATTAAGGACACCAAGAGCGACATCA TCTTCTTCCAGCGCGACGTGCCAGGACACAAGCACAAAATGCAGTTTGAGAGCAGCAGC TACGAAGGGTATTTCCTGGCCTGTGAGAAGGAAAGAGATCTGTTCAAGCTGATCCTGAAG AAGGAGGACGAGCTGGGCGACCGGAGCATCATGTTTACCGTGCAGAACGAGGACTGA

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		SEQUENCES
73	hCS4 coding1	AGGTTCGGCAAACTGGAAAGCAAGCTCTCCGTGATCCGGAACCTGAACGACCAGGTGCT GTTTATCGACCAGGGAAATAGACCTCTGTTCGAGGACATGACAGATAGCGACTGCAGAGA TAATGCCCCTCGGACCATCTTCATCATCTCTACCTACCGGGACAGCCAGCCTAGAGGCAA GGCCGTGACCATCTCTGTCAAGTGCGAGAAGATCAGCACCCTGAGCTGCGAAAACAAGA TCATCAGCTTCAAGGAAATGAACCCCCCAGACAACATCAAGGACACAAAGTCCGACATCA TTTTCTTCCAGAGAAACGTGCCCGGCCACAAGTACAAGATGCAATTTGAGTCTAGCAGCT ACGAGGGCTATTTCCTGGCTTGTGAAAAGGAAAGAGATCTGTTCAAACTGATCCTGAAGA AAGAGGACGAGCTGGGCGATAGAAGCATCATGTTCACCGTGCAGAACGAGGAT
74	hCS4 coding2	CGCTTCGGCAAACTGGAGTCCAAGCTGAGCGTGATCAGGAATCTGAACGATCAGGTGCT CTTCATCGATCAGGGCAATAGACCTCTGTTTGAGGACATGACTGACTCCGACTGTCGGGA CAACGCCCCCCGAACAATCTTTATTATCAGCACCTACCGGGACAGCCAGCCCAGGGGGA AGGCCGTGACTATCTCCGTGAAGTGCGAAAAGATCAGCACCCTGTCATGTGAGAACAAGA TTATCTCCTTTAAGGAGATGAACCCCCCCGATAACATTAAGGACACCAAGAGCGACATCA TCTTCTTCCAGCGCAACGTGCCAGGACACAAGTACAAAATGCAGTTTGAGAGCAGCAGCT ACGAAGGGTATTTCCTGGCCTGTGAGAAGGAAAGAGATCTGTTCAAGCTGATCCTGAAGA AGGAGGACGAGCTGGGCGACCGGAGCATCATGTTTACCGTGCAGAACGAGGACTGA
75	6-12 coding1	TATTTCGGCAAGCTGGAAAGCAAGCTCTCTGTGATCCGGAACCTGAACGACCAAGTGCTG TTCATCGACCAGGGCAACCGGCCTCTGTTCGAGGATATGACCGACAGCGACTGCAGAGA TAATGCCCCTCGCACAATCTTCATCATTTCTAAATACAGCGATAGCCTGGCCAGAGGCCT GGCTGTGACCATCAGCGTCAAGTGCGAGAAAATCTCCACCCTGAGCTGCGAGAACAAGA TCATCTCTTTCAAGGAAATGAACCCCCCTGACAATATCAAGGACACAAAGTCCGACATCAT CTTTTTCCAGAGAGATGTGCCAGGACACAGCAGAAAGATGCAGTTCGAGAGCAGCTCCTA CGAGGGCTACTTCCTGGCCTGTGAAAAGGAACGGGACCTGTTTAAGCTGATCCTGAAGA AAGAAGATGAGCTGGGCGACAGAAGCATCATGTTCACCGTGCAGAACGAGGAC
76	6-12 coding2	TATTTCGGCAAACTGGAGAGCAAGCTGAGCGTGATTCGGAACCTGAATGACCAGGTGCT GTTCATTGACCAGGGCAATCGCCCTCTGTTCGAAGACATGACCGACTCCGATTGTAGAGA CAACGCCCCTCGGACCATCTTCATTATCAGCAAGTACAGCGACAGCCTGGCCCGCGGCC TGGCTGTGACAATCAGCGTGAAGTGCGAGAAGATCAGCACACTGTCCTGCGAGAACAAA ATCATCAGCTTTAAAGAGATGAATCCCCCTGACAATATCAAAGATACTAAATCTGACATCA TCTTCTTCCAGCGGGACGTGCCTGGCCACTCTAGAAAAATGCAGTTCGAGTCCTCCAGTT ACGAGGGGTACTTCCTGGCCTGCGAGAAGGAGCGGGACCTGTTCAAACTGATCCTGAAA AAAGAAGACGAGCTGGGCGATCGGAGCATCATGTTCACTGTGCAGAACGAGGATTGA
77	sq89v6 coding1	TACTTCGGCAAGCTGGAAAGCAAGCTCAGCGTCATCCGGAACCTGAACGACCAGGTGCT GTTCATCGACCAAGGCAACCGGCCTCTGTTTGAGGACATGACCGACTCTGATTCTAGAGA TAATGCCCCTAGAACCATCTTCATCATCTCCAAGTACAGCGATAGCCTGGCCAGAGGCCT GGCTGTGACAATCTCTGTGAAGAGCGAGAAAATCAGCACCCTGTCTTGTGAAAACAAGAT CATCAGCTTCAAGGAAATGAACCCTCCAGACAATATCAAGGACACAAAGTCCGACATCAT TTTCTTCCAGAGAGATGTGCCCGGCCACAGCAGGAAGATGCAGTTCGAGTCCAGCAGCT ACGAAGGATATTTTCTGGCCTGCGAGAAGGAACGGGACCTGTTCAAACTGATCCTGAAGA AAGAGGACGAGCTGGGCGACAGAAGCATCATGTTCACCGTGCAGAACGAGGAT
78	sq89v6 coding2	TATTTTGGGAAGCTGGAGAGCAAACTGAGCGTGATCCGGAATCTGAATGACCAGGTGCT GTTCATTGACCAGGGCAATCGACCACTGTTCGAAGATATGACCGATAGCGACAGCAGGG ACAATGCCCCCAGGACCATCTTTATCATCAGCAAGTACAGCGATAGCCTGGCCAGGGGA CTGGCCGTGACAATCAGCGTGAAGTCCGAGAAGATCAGCACCCTGTCCTGCGAGAACAA GATCATTTCTTTCAAGGAGATGAACCCCCCTGACAACATCAAGGACACAAAGAGCGACAT CATCTTCTTTCAGAGAGACGTGCCCGGACACAGCCGCAAGATGCAGTTTGAATCTTCCAG CTACGAAGGGTACTTCCTGGCCTGTGAAAAAGAGCGCGATCTGTTCAAACTGATCCTCAA GAAGGAGGACGAGCTGGGCGATCGCTCTATCATGTTCACAGTGCAGAACGAGGACTGA
79	C68v1 coding1	TATTTCGGCAAGCTGGAAAGCAAGCTCTCTGTGATCCGGAACCTGAACGACCAAGTGCTG TTTATCGACCAGGGAAATAGACCTCTGTTCGAGGACATGACCGACAGCGACAGCCGGGA TAACGCCCCTCGCACCATCTTCATCATCTCCAAGTACAGCGATAGCCTGGCTAGAGGCCT GGCCGTGACCATCAGCGTCAAGGGCGAGAAGATCTCTACACTGAGCTGTGAAAACAAGA TCATCAGCTTCAAGGAAATGAACCCCCCCGACAATATCAAGGACACAAAGTCCGACATCA TTTTCTTCCAGAGAGATGTGCCTGGCCACAGCAGAAAGATGCAGTTCGAGTCCAGCAGCT ACGAGGGCTACTTTCTGGCCTGCGAGAAAGAACGGGACCTGTTCAAACTGATCCTGAAAA AGGAAGACGAGCTGGGCGATAGATCTATCATGTTCACCGTGCAGAACGAGGAT
80	C68v1 coding2	TATTTTGGGAAGCTGGAGAGCAAACTGAGCGTGATCCGGAATCTGAATGACCAGGTGCT GTTCATTGACCAGGGCAATCGACCACTGTTCGAAGATATGACCGATAGCGACAGCAGGG ACAATGCCCCCAGGACCATCTTTATCATCAGCAAGTACAGCGATAGCCTGGCCAGGGGA CTGGCCGTGACAATCAGCGTGAAGGGCGAGAAGATCAGCACCCTGTCCTGCGAGAACAA GATCATTTCTTTCAAGGAGATGAACCCCCCTGACAACATCAAGGACACAAAGAGCGACAT CATCTTCTTTCAGAGAGACGTGCCCGGACACAGCCGCAAGATGCAGTTTGAATCTTCCAG CTACGAAGGGTACTTCCTGGCCTGTGAAAAAGAGCGCGATCTGTTCAAACTGATCCTCAA GAAGGAGGACGAGCTGGGCGATCGCTCTATCATGTTCACAGTGCAGAACGAGGACTGA

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SEQUENCES		
81	C68v2 coding1	TATTTCGGCAAGCTGGAAAGCAAGCTGAGCGTCATCCGGAACCTCAACGACCAAGTGCT GTTTATCGACCAGGGCAACCGGCCTCTGTTCGAGGACATGACAGACAGCGACAGCAGAG ATAATGCCCCTAGAACCATCTTCATCATTTCCAAGTACAGCGATTCTCTGGCCAGAGGCCT GGCCGTGACCATCTCTGTGAAGGCCGAGAAAATCTCCACCCTGAGCTGCGAAAACAAGA TCATCAGCTTCAAGGAGATGAACCCTCCAGACAATATCAAGGACACCAAGAGCGACATCA TCTTTTTCCAGCGGGACGTGCCCGGCCACTCTCGCAAGATGCAGTTCGAGTCCAGCAGC TACGAGGGCTACTTCCTGGCTTGTGAAAAGGAAAGAGATCTGTTCAAACTGATCCTGAAA AAGGAAGATGAGCTGGGAGATAGAAGCATCATGTTCACAGTGCAGAACGAGGAC
82	C68v2 coding2	TATTTTGGGAAGCTGGAGAGCAAACTGAGCGTGATCCGGAATCTGAATGACCAGGTGCT GTTCATTGACCAGGGCAATCGACCACTGTTCGAAGATATGACCGATAGCGACAGCAGGG ACAATGCCCCCAGGACCATCTTTATCATCAGCAAGTACAGCGATAGCCTGGCCAGGGGA CTGGCCGTGACAATCAGCGTGAAGGCCGAGAAGATCAGCACCCTGTCCTGCGAGAACAA GATCATTTCTTTCAAGGAGATGAACCCCCCTGACAACATCAAGGACACAAAGAGCGACAT CATCTTCTTTCAGAGAGACGTGCCCGGACACAGCCGCAAGATGCAGTTTGAATCTTCCAG CTACGAAGGGTACTTCCTGGCCTGTGAAAAAGAGCGCGATCTGTTCAAACTGATCCTCAA GAAGGAGGACGAGCTGGGCGATCGCTCTATCATGTTCACAGTGCAGAACGAGGACTGA
83	C68v4 coding1	TATTTCGGCAAGCTGGAATCTAAACTCTCCGTGATCCGGAACCTGAACGACCAAGTGCTG TTCATCGACCAGGGAAATAGACCTCTGTTCGAGGATATGACAGACAGCGATAGCCGCGAT AATGCCCCTAGAACCATCTTTATCATTAGCAAGTACAGCGACAGCCTGGCCAGAGGCCTG GCCGTGACAATCAGCGTCAAGGACGAGAAAATCTCCACCCTGAGCTGCGAGAACAAGAT CATCAGCTTCAAGGAAATGAACCCTCCAGACAACATCAAGGACACCAAGTCCGATATCAT CTTCTTCCAGCGGGACGTGCCCGGCCACTCTAGAAAGATGCAGTTCGAGAGCTCTAGCT ACGAGGGCTACTTCCTGGCTTGTGAAAAAGAACGGGACCTGTTTAAGCTGATCCTGAAGA AGGAAGATGAGCTGGGCGACAGAAGCATCATGTTCACCGTGCAGAACGAGGAC
84	C68v4 coding2	TATTTTGGGAAGCTGGAGAGCAAACTGAGCGTGATCCGGAATCTGAATGACCAGGTGCT GTTCATTGACCAGGGCAATCGACCACTGTTCGAAGATATGACCGATAGCGACAGCAGGG ACAATGCCCCCAGGACCATCTTTATCATCAGCAAGTACAGCGATAGCCTGGCCAGGGGA CTGGCCGTGACAATCAGCGTGAAGGACGAGAAGATCAGCACCCTGTCCTGCGAGAACAA GATCATTTCTTTCAAGGAGATGAACCCCCCTGACAACATCAAGGACACAAAGAGCGACAT CATCTTCTTTCAGAGAGACGTGCCCGGACACAGCCGCAAGATGCAGTTTGAATCTTCCAG CTACGAAGGGTACTTCCTGGCCTGTGAAAAAGAGCGCGATCTGTTCAAACTGATCCTCAA GAAGGAGGACGAGCTGGGCGATCGCTCTATCATGTTCACAGTGCAGAACGAGGACTGA
85	wt hIL-18 coding1	TATTTCGGCAAGCTGGAAAGCAAGCTCTCCGTGATCAGAAACCTGAACGACCAAGTGCTG TTCATCGACCAGGGCAACAGACCTCTGTTTGAGGATATGACCGACTCTGATTGCAGAGAT AATGCCCCTAGAACCATCTTTATCATCAGCATGTACAAGGACAGCCAGCCTCGGGGAATG GCCGTGACCATCAGCGTCAAGTGCGAGAAGATCTCTACCCTGAGCTGCGAAAACAAAATC ATCTCCTTCAAGGAAATGAACCCCCCAGACAATATCAAGGATACAAAGAGCGACATCATTT TCTTCCAGCGGAGCGTGCCCGGCCACGACAACAAGATGCAGTTCGAGAGCTCTAGCTAC GAGGGCTACTTCCTGGCTTGTGAAAAGGAACGGGACCTGTTCAAACTGATCCTGAAGAAA GAGGACGAGCTGGGCGATAGAAGCATCATGTTCACAGTGCAGAACGAGGAC
86	wt hIL-18 coding2	TACTTCGGCAAACTGGAGTCCAAGCTGAGCGTGATCAGGAATCTGAACGATCAGGTGCTC TTCATCGATCAGGGCAATAGACCTCTGTTTGAGGACATGACTGACTCCGACTGTCGGGAC AACGCCCCCCGAACAATCTTTATTATCAGCATGTACAAGGACAGCCAGCCCAGGGGGAT GGCCGTGACTATCTCCGTGAAGTGCGAAAAGATCAGCACCCTGTCATGTGAGAACAAGAT TATCTCCTTTAAGGAGATGAACCCCCCCGATAACATTAAGGACACCAAGAGCGACATCAT CTTCTTCCAGCGCTCCGTGCCAGGACACGATAACAAAATGCAGTTTGAGAGCAGCAGCTA CGAAGGGTATTTCCTGGCCTGTGAGAAGGAAAGAGATCTGTTCAAGCTGATCCTGAAGAA GGAGGACGAGCTGGGCGACCGGAGCATCATGTTTACCGTGCAGAACGAGGACTGA
87	EF1a promoter	CGTGAGGCTCCGGTGCCCGTCAGTGGGCAGAGCGCACATCGCCCACAGTCCCCGAGAA GTTGGGGGGAGGGGTCGGCAATTGAACCGGTGCCTAGAGAAGGTGGCGCGGGGTAAAC TGGGAAAGTGATGTCGTGTACTGGCTCCGCCTTTTTCCCGAGGGTGGGGGAGAACCGTA TATAAGTGCAGTAGTCGCCGTGAACGTTCTTTTTCGCAACGGGTTTGCCGCCAGAACACA GGTAAGTGCCGTGTGTGGTTCCCGCGGGCCTGGCCTCTTTACGGGTTATGGCCCTTGCG TGCCTTGAATTACTTCCACCTGGCTGCAGTACGTGATTCTTGATCCCGAGCTTCGGGTTG GAAGTGGGTGGGAGAGTTCGAGGCCTTGCGCTTAAGGAGCCCCTTCGCCTCGTGCTTGA GTTGAGGCCTGGCCTGGGCGCTGGGGCCGCCGCGTGCGAATCTGGTGGCACCTTCGCG CCTGTCTCGCTGCTTTCGATAAGTCTCTAGCCATTTAAAATTTTTGATGACCTGCTGCGAC GCTTTTTTTCTGGCAAGATAGTCTTGTAAATGCGGGCCAAGATCTGCACACTGGTATTTCG GTTTTTGGGGCCGCGGGCGGCGACGGGGCCCGTGCGTCCCAGCGCACATGTTCGGCGA GGCGGGGCCTGCGAGCGCGGCCACCGAGAATCGGACGGGGGTAGTCTCAAGCTGGCC GGCCTGCTCTGGTGCCTGGCCTCGCGCCGCCGTGTATCGCCCCGCCCTGGGCGGCAAG GCTGGCCCGGTCGGCACCAGTTGCGTGAGCGGAAAGATGGCCGCTTCCCGGCCCTGCT GCAGGGAGCTCAAAATGGAGGACGCGGCGCTCGGGAGAGCGGGCGGGTGAGTCACCC ACACAAAGGAAAAGGGCCTTTCCGTCCTCAGCCGTCGCTTCATGTGACTCCACGGAGTAC CGGGCGCCGTCCAGGCACCTCGATTAGTTCTCGAGCTTTTGGAGTACGTCGTCTTTAGGT TGGGGGGAGGGGTTTTATGCGATGGAGTTTCCCCACACTGAGTGGGTGGAGACTGAAGT TAGGCCAGCTTGGCACTTGATGTAATTCTCCTTGGAATTTGCCCTTTTTGAGTTTGGATCT TGGTTCATTCTCAAGCCTCAGACAGTGGTTCAAAGTTTTTTTCTTCCATTTCAGGTGTCGT GA

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SEQUENCES		
88	wtIL12p35	MCPARSLLLVATLVLLDHLSLARNLPVATPDPGMFPCLHHSQNLLRAVSNMLQKARQTLEFYP CTSEEIDHEDITKDKTSTVEACLPLELTKNESCLNSRETSFITNGSCLASRKTSFMMALCLSSIY EDLKMYQVEFKTMNAKLLMDPKRQIFLDQNMLAVIDELMQALNFNSETVPQKSSLEEPDFYKT KIKLCILLHAFRIRAVTIDRVMSYLNAS
89	wtIL12p35_ noSP	RNLPVATPDPGMFPCLHHSQNLLRAVSNMLQKARQTLEFYPCTSEEIDHEDITKDKTSTVEAC LPLELTKNESCLNSRETSFITNGSCLASRKTSFMMALCLSSIYEDLKMYQVEFKTMNAKLLMD PKRQIFLDQNMLAVIDELMQALNFNSETVPQKSSLEEPDFYKTKIKLCILLHAFRIRAVTIDRVM SYLNAS
90	wtIL12p40	MCHQQLVISWFSLVFLASPLVAIWELKKDVYVVELDWYPDAPGEMVVLTCDTPEEDGITWTL DQSSEVLGSGKTLTIQVKEFGDAGQYTCHKGGEVLSHSLLLLHKKEDGIWSTDILKDQKEPKN KTFLRCEAKNYSGRFTCWWLTTISTDLTFSVKSSRGSSDPQGVTCGAATLSAERVRGDNKEY EYSVECQEDSACPAAEESLPIEVMVDAVHKLKYENYTSSFFIRDIIKPDPPKNLQLKPLKNSRQ VEVSWEYPDTWSTPHSYFSLTFCVQVQGKSKREKKDRVFTDKTSATVICRKNASISVRAQDR YYSSSWSEWASVPCS
91	wtIL12p40_ noSP	IWELKKDVYVVELDWYPDAPGEMVVLTCDTPEEDGITWTLDQSSEVLGSGKTLTIQVKEFGD AGQYTCHKGGEVLSHSLLLLHKKEDGIWSTDILKDQKEPKNKTFLRCEAKNYSGRFTCWWLT TISTDLTFSVKSSRGSSDPQGVTCGAATLSAERVRGDNKEYEYSVECQEDSACPAAEESLPIE VMVDAVHKLKYENYTSSFFIRDIIKPDPPKNLQLKPLKNSRQVEVSWEYPDTWSTPHSYFSLT FCVQVQGKSKREKKDRVFTDKTSATVICRKNASISVRAQDRYYSSSWSEWASVPCS
92	H216A	MCHQQLVISWFSLVFLASPLVAIWELKKDVYVVELDWYPDAPGEMVVLTCDTPEEDGITWTL DQSSEVLGSGKTLTIQVKEFGDAGQYTCHKGGEVLSHSLLLLHKKEDGIWSTDILKDQKEPKN KTFLRCEAKNYSGRFTCWWLTTISTDLTFSVKSSRGSSDPQGVTCGAATLSAERVRGDNKEY EYSVECQEDSACPAAEESLPIEVMVDAVAKLKYENYTSSFFIRDIIKPDPPKNLQLKPLKNSRQ VEVSWEYPDTWSTPHSYFSLTFCVQVQGKSKREKKDRVFTDKTSATVICRKNASISVRAQDR YYSSSWSEWASVPCS
93	K217A	MCHQQLVISWFSLVFLASPLVAIWELKKDVYVVELDWYPDAPGEMVVLTCDTPEEDGITWTL DQSSEVLGSGKTLTIQVKEFGDAGQYTCHKGGEVLSHSLLLLHKKEDGIWSTDILKDQKEPKN KTFLRCEAKNYSGRFTCWWLTTISTDLTFSVKSSRGSSDPQGVTCGAATLSAERVRGDNKEY EYSVECQEDSACPAAEESLPIEVMVDAVHALKYENYTSSFFIRDIIKPDPPKNLQLKPLKNSRQ VEVSWEYPDTWSTPHSYFSLTFCVQVQGKSKREKKDRVFTDKTSATVICRKNASISVRAQDR YYSSSWSEWASVPCS
94	K219A	MCHQQLVISWFSLVFLASPLVAIWELKKDVYVVELDWYPDAPGEMVVLTCDTPEEDGITWTL DQSSEVLGSGKTLTIQVKEFGDAGQYTCHKGGEVLSHSLLLLHKKEDGIWSTDILKDQKEPKN KTFLRCEAKNYSGRFTCWWLTTISTDLTFSVKSSRGSSDPQGVTCGAATLSAERVRGDNKEY EYSVECQEDSACPAAEESLPIEVMVDAVHKLAYENYTSSFFIRDIIKPDPPKNLQLKPLKNSRQ VEVSWEYPDTWSTPHSYFSLTFCVQVQGKSKREKKDRVFTDKTSATVICRKNASISVRAQDR YYSSSWSEWASVPCS
95	HK217	MCHQQLVISWFSLVFLASPLVAIWELKKDVYVVELDWYPDAPGEMVVLTCDTPEEDGITWTL DQSSEVLGSGKTLTIQVKEFGDAGQYTCHKGGEVLSHSLLLLHKKEDGIWSTDILKDQKEPKN KTFLRCEAKNYSGRFTCWWLTTISTDLTFSVKSSRGSSDPQGVTCGAATLSAERVRGDNKEY EYSVECQEDSACPAAEESLPIEVMVDAVAALKYENYTSSFFIRDIIKPDPPKNLQLKPLKNSRQ VEVSWEYPDTWSTPHSYFSLTFCVQVQGKSKREKKDRVFTDKTSATVICRKNASISVRAQDR YYSSSWSEWASVPCS
96	HK219	MCHQQLVISWFSLVFLASPLVAIWELKKDVYVVELDWYPDAPGEMVVLTCDTPEEDGITWTL DQSSEVLGSGKTLTIQVKEFGDAGQYTCHKGGEVLSHSLLLLHKKEDGIWSTDILKDQKEPKN KTFLRCEAKNYSGRFTCWWLTTISTDLTFSVKSSRGSSDPQGVTCGAATLSAERVRGDNKEY EYSVECQEDSACPAAEESLPIEVMVDAVAKLAYENYTSSFFIRDIIKPDPPKNLQLKPLKNSRQ VEVSWEYPDTWSTPHSYFSLTFCVQVQGKSKREKKDRVFTDKTSATVICRKNASISVRAQDR YYSSSWSEWASVPCS
97	KK	MCHQQLVISWFSLVFLASPLVAIWELKKDVYVVELDWYPDAPGEMVVLTCDTPEEDGITWTL DQSSEVLGSGKTLTIQVKEFGDAGQYTCHKGGEVLSHSLLLLHKKEDGIWSTDILKDQKEPKN KTFLRCEAKNYSGRFTCWWLTTISTDLTFSVKSSRGSSDPQGVTCGAATLSAERVRGDNKEY EYSVECQEDSACPAAEESLPIEVMVDAVHALAYENYTSSFFIRDIIKPDPPKNLQLKPLKNSRQ VEVSWEYPDTWSTPHSYFSLTFCVQVQGKSKREKKDRVFTDKTSATVICRKNASISVRAQDR YYSSSWSEWASVPCS
98	HKK	MCHQQLVISWFSLVFLASPLVAIWELKKDVYVVELDWYPDAPGEMVVLTCDTPEEDGITWTL DQSSEVLGSGKTLTIQVKEFGDAGQYTCHKGGEVLSHSLLLLHKKEDGIWSTDILKDQKEPKN KTFLRCEAKNYSGRFTCWWLTTISTDLTFSVKSSRGSSDPQGVTCGAATLSAERVRGDNKEY EYSVECQEDSACPAAEESLPIEVMVDAVAALAYENYTSSFFIRDIIKPDPPKNLQLKPLKNSRQ VEVSWEYPDTWSTPHSYFSLTFCVQVQGKSKREKKDRVFTDKTSATVICRKNASISVRAQDR YYSSSWSEWASVPCS

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SEQUENCES		
99	WO202121 2083_SEQ: 3	MCHQQLVISWFSLVFLASPLVAIWELKKDVYVVELDWYPDAPGEMVVLTCDTPEEDGITWTL DQSSEVLGSGKTLTIQVKAFGDAGQYTCHKGGEVLSHSLLLLHKKEDGIWSTDILKDQKEPKN KTFLRCEAKNYSGRFTCWWLTTISTDLTFSVKSSRGSSDPQGVTCGAATLSAERVRGDNKEY EYSVECQEDSACPAAEESLPIEVMVDAVHKLKYENYTSSFFIRDIIKPDPPKNLQLKPLKNSRQ VEVSWEYPDTWSTPHSYFSLTFCVQVQGKSKREKKDRVFTDKTSATVICRKNASISVRAQDR YYSSSWSEWASVPCS
100	WO202121 2083_SEQ: 4	MCHQQLVISWFSLVFLASPLVAIWELKKDVYVVELDWYPDAPGEMVVLTCDTPEEDGITWTL DQSSEVLGSGKTLTIQVKEAGDAGQYTCHKGGEVLSHSLLLLHKKEDGIWSTDILKDQKEPKN KTFLRCEAKNYSGRFTCWWLTTISTDLTFSVKSSRGSSDPQGVTCGAATLSAERVRGDNKEY EYSVECQEDSACPAAEESLPIEVMVDAVHKLKYENYTSSFFIRDIIKPDPPKNLQLKPLKNSRQ VEVSWEYPDTWSTPHSYFSLTFCVQVQGKSKREKKDRVFTDKTSATVICRKNASISVRAQDR YYSSSWSEWASVPCS
101	WO202121 2083_SEQ: 5	MCHQQLVISWFSLVFLASPLVAIWELKKDVYVVELDWYAAAPGEMVVLTCDTPEEDGITWTLD QSSEVLGSGKTLTIQVKAAGDAGQYTCHKGGEVLSHSLLLLHKKEDGIWSTDILKDQKEPKNK TFLRCEAKNYSGRFTCWWLTTISTDLTFSVKSSRGSSDPQGVTCGAATLSAERVRGDNKEYE YSVECQEDSACPAAEESLPIEVMVDAVHKLKYENYTSSFFIRDIIKPDPPKNLQLKPLKNSRQV EVSWEYPDTWSTPHSYFSLTFCVQVQGKSKREKKDRVFTDKTSATVICRKNASISVRAQDRY YSSSWSEWASVPCS
102	WO202121 2083_SEQ: 6	MCHQQLVISWFSLVFLASPLVAIWELKKDVYVVELDWYPDAPGEMVVLTCDTPEEDGITWTL DQSSEVLGSGKTLTIQVKEFGDAGQYTCHKGGEVLSHSLLLLHAKEDGIWSTDILKDQKEPKN KTFLRCEAKNYSGRFTCWWLTTISTDLTFSVKSSRGSSDPQGVTCGAATLSAERVRGDNKEY EYSVECQEDSACPAAEESLPIEVMVDAVHKLKYENYTSSFFIRDIIKPDPPKNLQLKPLKNSRQ VEVSWEYPDTWSTPHSYFSLTFCVQVQGKSKREKKDRVFTDKTSATVICRKNASISVRAQDR YYSSSWSEWASVPCS
103	WO202121 2083_SEQ: 7	MCHQQLVISWFSLVFLASPLVAIWELKKDVYVVELDWYPDAPGEMVVLTCDTPEEDGITWTL DQSSEVLGSGKTLTIQVKEFGDAGQYTCHKGGEVLSHSLLLLHKKEDGIWSTDILKDQKEPKN KTFLRCEAKNYSGRFTCWWLTTISTDLTFSVKSSRGSSDPQGVTCGAATLSAERVRGDNKEY EYSVECQEDSACPAAEESLPIEVMVDAVHALKYENYTSSFFIRDIIKPDPPKNLQLKPLKNSRQ VEVSWEYPDTWSTPHSYFSLTFCVQVQGKSKREKKDRVFTDKTSATVICRKNASISVRAQDR YYSSSWSEWASVPCS
104	WO202121 2083_SEQ: 8	MCHQQLVISWFSLVFLASPLVAIWELKKDVYVVELDWYPDAPGEMVVLTCDTPEEDGITWTL DQSSEVLGSGKTLTIQVKEFGDAGQYTCHKGGEVLSHSLLLLHKKEDGIWSTDILKDQKEPKN KTFLRCEAKNYSGRFTCWWLTTISTDLTFSVKSSRGSSDPQGVTCGAATLSAERVRGDNKEY EYSVECQEDSACPAAEESLPIEVMVDAVHKLAYENYTSSFFIRDIIKPDPPKNLQLKPLKNSRQ VEVSWEYPDTWSTPHSYFSLTFCVQVQGKSKREKKDRVFTDKTSATVICRKNASISVRAQDR YYSSSWSEWASVPCS
105	WO202121 2083_SEQ: 13	MCHQQLVISWFSLVFLASPLVAIWELKKDVYVVELDWYPDAPGEMVVLTCDTPEEDGITWTL DQSSEVLGSGKTLTIQVKAAGDAGQYTCHKGGEVLSHSLLLLHKKEDGIWSTDILKDQKEPKN KTFLRCEAKNYSGRFTCWWLTTISTDLTFSVKSSRGSSDPQGVTCGAATLSAERVRGDNKEY EYSVECQEDSACPAAEESLPIEVMVDAVHKLKYENYTSSFFIRDIIKPDPPKNLQLKPLKNSRQ VEVSWEYPDTWSTPHSYFSLTFCVQVQGKSKREKKDRVFTDKTSATVICRKNASISVRAQDR YYSSSWSEWASVPCS
106	WO202121 2083_SEQ: 14	MCHQQLVISWFSLVFLASPLVAIWELKKDVYVVELDWYPDAPGEMVVLTCDTPEEDGITWTL DQSSEVLGSGKTLTIQVKAAGDAGQYTCHKGGEVLSHSLLLLHAKEDGIWSTDILKDQKEPKN KTFLRCEAKNYSGRFTCWWLTTISTDLTFSVKSSRGSSDPQGVTCGAATLSAERVRGDNKEY EYSVECQEDSACPAAEESLPIEVMVDAVHKLKYENYTSSFFIRDIIKPDPPKNLQLKPLKNSRQ VEVSWEYPDTWSTPHSYFSLTFCVQVQGKSKREKKDRVFTDKTSATVICRKNASISVRAQDR YYSSSWSEWASVPCS
107	WO202121 2083_SEQ: 15	MCHQQLVISWFSLVFLASPLVAIWELKKDVYVVELDAYPDAPGEMVVLTCDTPEEDGITWTLD QSSEVLGSGKTLTIQVKAAGDAGQYTCHKGGEVLSHSLLLLHKKEDGIWSTDILKDQKEPKNK TFLRCEAKNYSGRFTCWWLTTISTDLTFSVKSSRGSSDPQGVTCGAATLSAERVRGDNKEYE YSVECQEDSACPAAEESLPIEVMVDAVHKLKYENYTSSFFIRDIIKPDPPKNLQLKPLKNSRQV EVSWEYPDTWSTPHSYFSLTFCVQVQGKSKREKKDRVFTDKTSATVICRKNASISVRAQDRY YSSSWSEWASVPCS
108	WO202121 2083_SEQ: 16	MCHQQLVISWFSLVFLASPLVAIWELKKDVYVVELDWYPDAPGEMVVLTCDTPEEDGITWTL DQSSEVLGSGKTLTIQVKAAGDAGQYTCHKGGEVLSHSLLLLHAKADGIWSTAILKDQKEPKN KTFLRCEAKNYSGRFTCWWLTTISTDLTFSVKSSRGSSDPQGVTCGAATLSAERVRGDNKEY EYSVECQEDSACPAAEESLPIEVMVDAVHKLKYENYTSSFFIRDIIKPDPPKNLQLKPLKNSRQ VEVSWEYPDTWSTPHSYFSLTFCVQVQGKSKREKKDRVFTDKTSATVICRKNASISVRAQDR YYSSSWSEWASVPCS
109	EGFRvIII CAR	MALPVTALLLPLALLLHAARPDVVMTQSPDSLAVSLGERATINCKSSQSLLDSDGKTYLNWLQ QKPGQPPKRLISLVSKLDSGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCWQGTHFPGTFG GGTKVEIKGGGGGGGGSGGGGSEIQLVQSGAEVKKPGESLRISCKGSGFNIEDYYIHWVR QMPGKGLEWMGRIDPENDETKYGPIFQGHVTISADTSINTVYLQWSSLKASDTAMYYCAFRG

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SEQUENCES		
		GVYWGQGTTVTVSSTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIW APLAGTCGVLLLSLVITRSKRSRLLHSDYMNMTPRRPGPTRKHYQPYAPPRDFAAYRSRVKF SRSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQK DKMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDTYDALHMQALPPR
110	EGFRvIII CAR vL	DVVMTQSPDSLAVSLGERATINCKSSQSLLDSDGKTYLNWLQQKPGQPPKRLISLVSKLDSG VPDRFSGSGSGTDFTLTISSLQAEDVAVYYCWQGTHFPGTFGGGTKVEIK
111	EGFRvIII CAR vH	EIQLVQSGAEVKKPGESLRISCKGSGFNIEDYYIHWVRQMPGKGLEWMGRIDPENDETKYGPI FQGHVTISADTSINTVYLQWSSLKASDTAMYYCAFRGGVYWGQGTTVTVSS
112	EGFRvIII CAR vL CDR1	KSSQSLLDSDGKTYLN
113	EGFRvIII CAR vL CDR2	LVSKLDS
114	EGFRvIII CAR vL CDR3	WQGTHFPGT
115	EGFRVIII CAR vH CDR1	DYYIH
116	EGFRvIII CAR vH CDR2	RIDPENDETKYGPIFQG
117	EGFRvIII CAR vH CDR3	RGGVY
118	Human OSM signal peptide	MGVLLTQRTLLSLVLALLFPSMASM
119	VSV-G signal peptide	MKCLLYLAFLFIGVNC
120	Mouse Ig Kappa signal peptide	METDTLLLWVLLLWVPGSTGD
121	Mouse Ig Heavy signal peptide	MGWSCIILFLVATATGVHS
122	BM40 signal peptide	MRAWIFFLLCLAGRALA
123	Secrecon signal peptide	MWWRLWWLLLLLLLLWPMVWA
124	Human IgKVIII signal peptide	MDMRVPAQLLGLLLLWLRGARC
125	CD33 signal peptide	MPLLLLLPLLWAGALA
126	tPA signal peptide	MDAMKRGLCCVLLLCGAVFVSPS
127	Human Chymotrypsinogen signal peptide	MAFLWLLSCWALLGTTFG
128	Human trypsinogen-2 signal peptide	MNLLLILTFVAAAVA
129	Human IL-2 signal peptide	MYRMQLLSCIALSLALVTNS
130	Gaussia luc signal peptide	MGVKVLFALICIAVAEA
131	Albumin (HSA) signal peptide	MKWVTFISLLFSSAYS
132	Influenza Haemagglutinin signal peptide	MKTIIALSYIFCLVLG
133	Human insulin signal peptide	MALWMRLLPLLALLALWGPDPAAA
134	Silkworm Fibroin LC signal peptide	MKPIFLVLLVVTSAYA
135	Interleukin-18-binding protein signal peptide	MRHNWTPDLSPLWVLLLCAHVVTLLVRA

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SEQUENCES		
136	Interleukin-12 subunit beta signal peptide	MCHQQLVISWFSLVFLASPLVA
137	Interleukin-12 subunit alpha signal peptide	MCPARSLLLVATLVLLDHLSLA
138	Interleukin-18 signal peptide	MAAEPVEDNCINFVAMKFIDNTLYFIAEDDENLESD
139	CD8 signal peptide	MALPVTALLLPLALLLHAARP
140	WO2023023503A1 #3 signal peptide	MEFGLSWVFLVALFRGVQS
141	hIL2 variant signal peptide	MGYRMQLLSCIALSLALVTNSG
142	EMCV IRES	acgttactggccgaagccgcttggaataaggccggtgtgcgtttgtctatatgttattttccacc atattgccgtcttttggcaatgtgagggcccggaaacctggccctgtcttcttgacgagcattcc taggggtctttcccctctcgccaaaggaatgcaaggtctgttgaatgtcgtgaaggaagcagttc ctctggaagcttcttgaagacaaacaacgtctgtagcgaccctttgcaggcagcggaacccccca cctggcgacaggtgcctctgcggccaaaagccacgtgtataagatacacctgcaaaggcggcaca accccagtgccacgttgtgagttggatagttgtggaaagagtcaaatggctctcctcaagcgtat tcaacaaggggctgaaggatgcccagaaggtaccccattgtatgggatctgatctggggcctcgg tgcacatgctttacatgtgtttagtcgaggttaaaaaaacgtctaggccccccgaaccacgggga cgtggttttcctttgaaaaacacgatgataata
143	T2A	(GSG) EGRGSLLTCGDVEENPGP
144	P2A	(GSG) ATNFSLLKQAGDVEENPGP
145	E2A	(GSG) QCTNYALLKLAGDVESNPGP
146	F2A	(GSG) VKQTLNFDLLKLAGDVESNPGP

Antibody	Heavy Chain sequence	SEQ ID NO:	Light chain sequence	SEQ ID NO:
pivekimab	QVQLVQSGAEVKKPGASVKVSCKASGYIFTS SIMHWVRQAPGQGLEWIGYIKPYNDGTKYNE KFKGRATLTSDRSTSTAYMELSSLRSEDTAV YYCAREGGNDYYDTMDYWGQGTLVTVSS	147	EIQMTQSPSSLSASVGDRVTITCRASQ DINSYLSWFQQKPGKAPKTLIYRVNRL VDGVPSRFSGSGSGNDYTLTISSLQP EDFATYYCLQYDAFPYTFGQGTKVEIK	148
talacotuzumab	EVQLVQSGAEVKKPGESLKISCKGSGYSFTD YYMKWARQMPGKGLEWMGDIIPSNGATFYN QKFKGQVTISADKSISTTYLQWSSLKASDTAM YYCARSHLLRASWFAYWGQGTMVTVSS	149	DIVMTQSPDSLAVSLGERATINCESSQ SLLNSGNQKNYLTWYQQKPGQPPKPL IYWASTRESGVPDRFSGSGSGTDFTL TISSLQAEDVAVYYCQNDYSYPYTFG QGTKLEIK	150
vibecotamab	QVQLQQSGAEVKKPGASVKVSCKASGYTFT DYYMKWVKQSHGKSLEWMGDIIPSNGATFY NQKFKGKATLTVDRSTSTAYMELSSLRSEDT AVYYCARSHLLRASWFAYWGQGTLVTVSS	151	DFVMTQSPDSLAVSLGERATINCKSS QSLLNTGNQKNYLTWYQQKPGQPPK LLIYWASTRESGVPDRFTGSGSGTDF TLTISSLQAEDVAVYYCQNDYSYPYTF GGGTKLEIK	152
budoprutug	QVQLEQPGAEVVKPGASVKVSCKTSGYTFTS NWMHWVKQTPGKGLEWIGEIDPSDSYTNYN QKFDGKAKLTVDKSSSTAYMEVSDLTAEDSA TYYCARGSNPYYYAMDYWGQGTSVTVSS	153	QIVLTQSPATLSASPGEKATMTCSASS GVNYMHWYQQKPGTSPKRWIYDTDK TASGVPARFSGSGSGTSYSLTISSMEA EDAATYYCHQRGSYTFGGGTKLEIK	154
coltuximab	QVQLVQPGAEVVKPGASVKLSCKTSGYTFTS NWMHWVKQAPGQGLEWIGEIDPSDSYTNYN QNFQGKAKLTVDKSTSTAYMEVSSLRSDDTA VYYCARGSNPYYYAMDYWGQGTSVTVSS	155	EIVLTQSPAIMSASPGERVTMTCSASS GVNYMHWYQQKPGTSPRRWIYDTSK LASGVPARFSGSGSGTDYSLTISSMEP EDAATYYCHQRGSYTFGGGTKLEIK	156
denintuzumab	QVQLQESGPGLVKPSQTLSLTCTVSGGSIST SGMGVGWIRQHPGKGLEWIGHIWWDDDKRY NPALKSRVTISVDTSKNQFSLKLSSVTAADTA VYYCARMELWSYYFDYWGQGTLVTVSS	157	EIVLTQSPATLSLSPGERATLSCSASS SVSYMHWYQQKPGQAPRLLIYDTSKL ASGIPARFSGSGSGTDFTLTISSLEPE DVAVYYCFQGSVYPFTFGQGTKLEIK	158
inebilizumab	EVQLVESGGGLVQPGGSLRLSCAASGFTFSS SWMNWVRQAPGKGLEWVGRIYPGDGDTNY NVKFKGRFTISRDDSKNSLYLQMNSLKTEDTA VYYCARSGFITTVRDFDYWGQGTLVTVSS	159	EIVLTQSPDFQSVTPKEKVTITCRASES VDTFGISFMNWFQQKPDQSPKLLIHEA SNQGSGVPSRFSGSGSGTDFTLTINS LEAEDAATYYCQQSKEVPFTFGGGTK VEIK	160

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SEQUENCES				
leronlimab	EVQLVESGGGLVKPGGSLRLSCAASGYTFSN YWIGWVRQAPGKGLEWIGDIYPGGNYIRNNE KFKDKTTLSADTSKNTAYLQMNSLKTEDTAVY YCGSSFGSNYVFAWFTYWGQGTLVTVSS	161	DIVMTQSPLSLPVTPGEPASISCRSSQ RLLSSYGHTYLHWYLQKPGQSPQLLIY EVSNRFSGVPDRFSGSGSGTDFTLKIS RVEAEDVGVYYCSQSTHVPLTFGQGT KVEIK	162
loncastuximab	QVQLVQPGAEVVKPGASVKLSCKTSGYTFTS NWMHWVKQAPGQGLEWIGEIDPSDSYTNYN QNFQGKAKLTVDKSTSTAYMEVSSLRSDDTA VYYCARGSNPYYYAMDYWGQGTSVTVSS	163	EIVLTQSPAIMSASPGERVTMTCSASS GVNYMHWYQQKPGTSPRRWIYDTSK LASGVPARFSGSGSGTSYSLTISSMEP EDAATYYCHQRGSYTFGGGTKLEIK	164
mogamulizumab	EVQLVESGGDLVQPGRSLRLSCAASGFIFSN YGMSWVRQAPGKGLEWVATISSASTYSYYP DSVKGRFTISRDNAKNSLYLQMNSLRVEDTA LYYCGRHSDGNFAFGYWGQGTLVTVSS	165	DVLMTQSPLSLPVTPGEPASISCRSSR NIVHINGDTYLEWYLQKPGQSPQLLIY KVSNRFSGVPDRFSGSGSGTDFTLKIS RVEAEDVGVYYCFQGSLLPWTFGQG TKVEIK	166
obexelimab	EVQLVESGGGLVKPGGSLKLSCAASGYTFTS YVMHWVRQAPGKGLEWIGYINPYNDGTKYN EKFQGRVTISSDKSISTAYMELSSLRSEDTAM YYCARGTYYYGTRVFDYWGQGTLVTVSS	167	DIVMTQSPATLSLSPGERATLSCRSSK SLQNVNGNTYLYWFQQKPGQSPQLLI YRMSNLNSGVPDRFSGSGSGTEFTLT ISSLEPEDFAVYYCMQHLEYPITFGAG TKLEIK	168
plozalizumab	EVQLVESGGGLVKPGGSLRLSCAASGFTFSA YAMNWVRQAPGKGLEWVGRIRTKNNNYATY YADSVKDRFTISRDDSKNTLYLQMNSLKTEDT AVYYCTTFYGNGVWGQGTLVTVSS	169	DVVMTQSPLSLPVTLGQPASISCKSSQ SLLDSDGKTFLNWFQQRPGQSPRRLI YLVSKLDSGVPDRFSGSGSGTDFTLKI SRVEAEDVGVYYCWQGTHFPYTFGQ GTRLEIK	170
tafasitamab	EVQLVESGGGLVKPGGSLKLSCAASGYTFTS YVMHWVRQAPGKGLEWIGYINPYNDGTKYN EKFQGRVTISSDKSISTAYMELSSLRSEDTAM YYCARGTYYYGTRVFDYWGQGTLVTVSS	171	DIVMTQSPATLSLSPGERATLSCRSSK SLQNVNGNTYLYWFQQKPGQSPQLLI YRMSNLNSGVPDRFSGSGSGTEFTLT ISSLEPEDFAVYYCMQHLEYPITFGAG TKLEIK	172
divozilimab	EVQLVQPGAEVVKPGASVKVSCKASGYTFTS YNMHWVRQAPGRGLEWMGAIYPGNGDTSY NQKFKGRVTMTRDKSTSTVYMELSSLRSEDT AVYYCARSTYYGGDWYFNVWGQGTLVTVSS	173	QIVLSQSPAILSASPGERVTLTCRASSS VSYIHWFQQKPGKAPKPLIYATSNLAS GVPSRFSGSGSGTDFSLTISRVEPEDF AVYYCQQWTSNPPTFGGGTKVEIK	174
ibritumomab	QAYLQQSGAELVRPGASVKMSCKASGYTFT SYNMHWVKQTPRQGLEWIGAIYPGNGDTSY NQKFKGKATLTVDKSSSTAYMQLSSLTSEDS AVYFCARVVYYSNSYWYFDVWGTGTTVTVS A	175	QIVLSQSPAILSASPGEKVTMTCRASS SVSYMHWYQQKPGSSPKPWIYAPSN LASGVPARFSGSGSGTSYSLTISRVEA EDAATYYCQQWSFNPPTFGAGTKLEL K	176
ocaratuzumab	EVQLVQSGAEVKKPGESLKISCKGSGRTFTS YNMHWVRQMPGKGLEWMGAIYPLTGDTSYN QKSKLQVTISADKSISTAYLQWSSLKASDTAM YYCARSTYVGGDWQFDVWGKGTTVTVSS	177	EIVLTQSPGTLSLSPGERATLSCRASS SVPYIHWYQQKPGQAPRLLIYATSALA SGIPDRFSGSGSGTDFTLTISRLEPED FAVYYCQQWLSNPPTFGQGTKLEIK	178
ocrelizumab	EVQLVESGGGLVQPGGSLRLSCAASGYTFTS YNMHWVRQAPGKGLEWVGAIYPGNGDTSYN QKFKGRFTISVDKSKNTLYLQMNSLRAEDTAV YYCARVVYYSNSYWYFDVWGQGTLVTVSS	179	DIQMTQSPSSLSASVGDRVTITCRASS SVSYMHWYQQKPGKAPKPLIYAPSNL ASGVPSRFSGSGSGTDFTLTISSLQPE DFATYYCQQWSFNPPTFGQGTKVEIK	180
ofatumumab	EVQLVESGGGLVQPGRSLRLSCAASGFTFND YAMHWVRQAPGKGLEWVSTISWNSGSIGYA DSVKGRFTISRDNAKKSLYLQMNSLRAEDTAL YYCAKDIQYGNYYYGMDVWGQGTTVTVSS	181	EIVLTQSPATLSLSPGERATLSCRASQ SVSSYLAWYQQKPGQAPRLLIYDASN RATGIPARFSGSGSGTDFTLTISSLEPE DFAVYYCQQRSNWPITFGQGTRLEIK	182
ripertamab	QVQLQQPGAELVKPGASVKMSCKASGYTFT SYNMHWVKQTPGRGLEWIGAIYPGNGDTSY NQKFKGKATLTADKSSSTAYMQLSSLTSEDS AVYYCARSTYYGGDWYFNVWGAGTTVTVSA	183	QIVLSQSPAILSASPGEKVTMTCRASS SVSYIHWFQQKPGSSPKPWIYATSNL ASGVPVRFSGSGSGTSYSLTISRVEAE DAATYYCQQWTSNPPTFGGGTKLEIK	184
rituximab	QVQLQQPGAELVKPGASVKMSCKASGYTFT SYNMHWVKQTPGRGLEWIGAIYPGNGDTSY NQKFKGKATLTADKSSSTAYMQLSSLTSEDS AVYYCARSTYYGGDWYFNVWGAGTTVTVSA	185	QIVLSQSPAILSASPGEKVTMTCRASS SVSYIHWFQQKPGSSPKPWIYATSNL ASGVPVRFSGSGSGTSYSLTISRVEAE DAATYYCQQWTSNPPTFGGGTKLEIK	186

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SEQUENCES				
ublituximab	QAYLQQSGAELVRPGASVKMSCKASGYTFTSYNMHWVKQTPRQGLEWIGGIYPGNGDTSYNQKFKGKATLTVGKSSSTAYMQLSSLTSEDSAVYFCARYDYNYAMDYWGQGTSVTVSS	187	QIVLSQSPAILSASPGEKVTMTCRASSSVSYMHWFQQKPGSSPKPWIYATSNLASGVPARFSGSGSGTSYSFTISRVEAEDAATYYCQQWTFNPPTFGGGTRLEIK	188
veltuzumab	QVQLQQSGAEVKKPGSSVKVSCKASGYTFTSYNMHWVKQAPGQGLEWIGAIYPGMGDTSYNQKFKGKATLTADESTNTAYMELSSLRSEDTAFYYCARSTYYGGDWYFDVWGQGTTVTVSS	189	DIQLTQSPSSLSASVGDRVTMTCRASSSVSYIHWFQQKPGKAPKPWIYATSNLASGVPVRFSGSGSGTDYTFTISSLQPEDIATYYCQQWTSNPPTFGGGTKLEIK	190
zuberitamab	EVQLQQSGAELVRPGASVKMSCKASGYTFTSYNMHWVKQTPRQGLEWIGAIYPGNGDTSYNQKFKGKATLTVDKSSSTAYMQLSSLTSEDSAVYFCARVVYYSNSYWYFDVWGTGTTVTVSS	191	DIELSQSPAILSASPGEKVTMTCRASSSVSYMHWYQQKPGSSPKPWIYAPSNLASGVPARFSGSGSGTSYSLTISRVEAEDAATYYCQQWSFNPPTFGAGTKLEIK	192
epratuzumab	QVQLVQSGAEVKKPGSSVKVSCKASGYTFTSYWLHWVRQAPGQGLEWIGYINPRNDYTEYNQNFKDKATITADESTNTAYMELSSLRSEDTAFYFCARRDITTFYWGQGTTVTVSS	193	DIQLTQSPSSLSASVGDRVTMSCKSSQSVLYSANHKNYLAWYQQKPGKAPKLLIYWASTRESGVPSRFSGSGSGTDFTFTISSLQPEDIATYYCHQYLSSWTFGGGTKLEIK	194
inotuzumab	EVQLVQSGAEVKKPGASVKVSCKASGYRFTNYWIHWVRQAPGQGLEWIGGINPGNNYATYRRKFQGRVTMTADTSTSTVYMELSSLRSEDTAVYYCTREGYGNYGAWFAYWGQGTLVTVSS	195	DVQVTQSPSSLSASVGDRVTITCRSSQSLANSYGNTFLSWYLHKPGKAPQLLIYGISNRFSGVPDRFSGSGSGTDFTLTISSLQPEDFATYYCLQGTHQPYTFGQGTKVEIK	196
pinatuzumab	EVQLVESGGGLVQPGGSLRLSCAASGYEFSRSWMNWVRQAPGKGLEWVGRIYPGDGDTNYSGKFKGRFTISADTSKNTAYLQMNSLRAEDTAVYYCARDGSSWDWYFDVWGQGTLVTVSS	197	DIQMTQSPSSLSASVGDRVTITCRSSQSIVHSVGNTFLEWYQQKPGKAPKLLIYKVSNRFSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCFQGSQFPYTFGQGTKVEIK	198
suciraslimab	QVQLQESGGGLVKPGGSLKLSCAASGFAFSIYDMSWVRQTPEKRLEWVAYISSGGGTTYYPDTVKGRFTISRDNAKNTLYLQMSSLKSEDTAMYYCARHSGYGSSYGVLFAYWGQGTLVTVSS	199	DIQLTQTTSSLSASLGDRVTISCRASQDISNYLNWYQQKPDGTVKLLIYYTSILHSGVPSRFSGSGSGTDYSLTISNLEQEDFATYFCQQGNTLPWTFGGGTKLEIK	200
brentuximab	QIQLQQSGPEVVKPGASVKISCKASGYTFTDYYITWVKQKPGQGLEWIGWIYPGSGNTKYNEKFKGKATLTVDTSSSTAFMQLSSLTSEDTAVYFCANYGNYWFAYWGQGTQVTVSA	201	DIVLTQSPASLAVSLGQRATISCKASQSVDFDGDSYMNWYQQKPGQPPKVLIYAASNLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNEDPWTFGGGTKLEIK	202
iratumumab	QVQLQQWGAGLLKPSETLSLTCAVYGGSFSAYYWSWIRQPPGKGLEWIGDINHGGGTNYNPSLKSRVTISVDTSKNQFSLKLNSVTAADTAVYYCASLTAYWGQGSLVTVSS	203	DIQMTQSPTSLSASVGDRVTITCRASQGISSWLTWYQQKPEKAPKSLIYAASSLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQYDSYPITFGQGTRLEIK	204
gemtuzumab	EVQLVQSGAEVKKPGSSVKVSCKASGYTITDSNIHWVRQAPGQSLEWIGYIYPYNGGTDYNQKFKNRATLTVDNPTNTAYMELSSLRSEDTAFYYCVNGNPWLAYWGQGTLVTVSS	205	DIQLTQSPSTLSASVGDRVTITCRASESLDNYGIRFLTWFQQKPGKAPKLLMYAASNQGSGVPSRFSGSGSGTEFTLTISSLQPDDFATYYCQQTKEVPWSFGQGTKVEVK	206
lintuzumab	QVQLVQSGAEVKKPGSSVKVSCKASGYTFTDYNMHWVRQAPGQGLEWIGYIYPYNGGTGYNQKFKSKATITADESTNTAYMELSSLRSEDTAVYYCARGRPAMDYWGQGTLVTVSS	207	DIQMTQSPSSLSASVGDRVTITCRASESVDNYGISFMNWFQQKPGKAPKLLIYAASNQGSGVPSRFSGSGSGTDFTLTISSLQPDDFATYYCQQSKEVPWTFGQGTKVEIK	208
vadastuximab	QVQLVQSGAEVKKPGASVKVSCKASGYTFTNYDINWVRQAPGQGLEWIGWIYPGDGSTKYNEKFKAKATLTADTSTSTAYMELRSLRSDDTAVYYCASGYEDAMDYWGQGTTVTVSS	209	DIQMTQSPSSLSASVGDRVTINCKASQDINSYLSWFQQKPGKAPKTLIYRANRLVDGVPSRFSGSGSGQDYTLTISSLQPEDFATYYCLQYDEFPLTFGGGTKVEIK	210
daratumumab	EVQLLESGGGLVQPGGSLRLSCAVSGFTFNSFAMSWVRQAPGKGLEWVSAISGSGGGTYYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYFCAKDKILWFGEPVFDYWGQGTLVTVSS	211	EIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDASNRATGIPARFSGSGSGTDFTLTISSLEPEDFAVYYCQQRSNWPPTFGQGTKVEIK	212

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SEQUENCES				
erzotabart	QVQLVQSGAEVKKPGSSVKVSCKAFGGTFS SYAISWVRQAPGQGLEWMGRIIRFLGIANYA QKFQGRVTLIADKSTNTAYMELSSLRSEDTAV YYCAGEPGERDPDAVDIWGQGTMVTVSS	213	DIQMTQSPSSLSASVGDRVTITCRASQ GIRSWLAWYQQKPEKAPKSLIYAASSL QSGVPSRFSGSGSGTDFTLTISSLQPE DFATYYCQQYNSYPLTFGGGTKVEIK	214
felzartamab	QVQLVESGGGLVQPGGSLRLSCAASGFTFSS YYMNWVRQAPGKGLEWVSGISGDPSNTYYA DSVKGRFTISRDNSKNTLYLQMNSLRAEDTA VYYCARDLPLVYTGFAYWGQGTLVTVSS	215	DIELTQPPSVSVAPGQTARISCSGDNL RHYYVYWYQQKPGQAPVLVIYGDSKR PSGIPERFSGSNSGNTATLTISGTQAE DEADYYCQTYTGGASLVFGGGTKLTV L	216
isatuximab	QVQLVQSGAEVAKPGTSVKLSCKASGYTFTD YWMQWVKQRPGQGLEWIGTIYPGDGDTGYA QKFQGKATLTADKSSKTVYMHLSSLASEDSA VYYCARGDYYGSNSLDYWGQGTSVTVSS	217	DIVMTQSHLSMSTSLGDPVSITCKASQ DVSTVVAWYQQKPGQSPRRLIYSASY RYIGVPDRFTGSGAGTDFTFTISSVQA EDLAVYYCQQHYSPPYTFGGGTKLEIK	218
lumrotatug	QVQLQESGPGLVKPSETLSLTCTVSGFSLTS YGIHWLRQPPGKGLEWIGVIWRGGSTDYNPS LKSRVTISKDTSKSQVSLKLSSVTAADTAVYY CAKGKVTTGFYFDFWGQGTLVTVSS	219	EIVMTQSPPTLSLSPGERVTLSCRASE DIYNRLVWYQQKPGQAPRLLISGVTSL ETSIPARFSGSGSGTDYTLTISSLQPE DFAVYYCQQYWSTPYTFGQGTKVEIK	220
mezagitamab	EVQLLESGGGLVQPGGSLRLSCAASGFTFDD YGMSWVRQAPGKGLEWVSDISWNGGKTHY VDSVKGQFTISRDNSKNTLYLQMNSLRAEDT AVYYCARGSLFHDSSGFYFGHWGQGTLVTV SS	221	QSVLTQPPSASGTPGQRVTISCSGSS SNIGDNYVSWYQQLPGTAPKLLIYRDS QRPSGVPDRFSGSKSGTSASLAISGL RSEDEADYYCQSYDSSLSGSVFGGGT KLTVL	222
modakafusp	EVQLVQSGAEVKKPGATVKISCKVSGYTFTD SVMNWVQQAPGKGLEWMGWIDPEYGRTDV AEKFQGRVTITADTSTDTAYMELSSLRSEDTA VYYCARTKYNSGYGFPYWGQGTTVTVSS	223	DIQMTQSPSSLSASVGDRVTITCKASQ NVDSDVDWYQQKPGKAPKLLIYKASN DYTGVPSRFSGSGSGTDFTFTISSLQP EDIATYYCMQSNTHPRTFGGGTKVEIK	224
cusatuzumab	EVQLVESGGGLVQPGGSLRLSCAASGFTFSV YYMNWVRQAPGKGLEWVSDINNEGGTTYYA DSVKGRFTISRDNSKNSLYLQMNSLRAEDTA VYYCARDAGYSNHVPIFDSWGQGTLVTVSS	225	QAVVTQEPSLTVSPGGTVTLTCGLKS GSVTSDNFPTWYQQTPGQAPRLLIYN TNTRHSGVPDRFSGSILGNKAALTITG AQADDEAEYFCALFISNPSVEFGGGT QLTVL	226
vorsetuzumab	QVQLVQSGAEVKKPGASVKVSCKASGYTFTN YGMNWVRQAPGQGLKWMGWINTYTGEPTY ADAFKGRVTMTRDTSISTAYMELSRLRSDDT AVYYCARDYGDYGMDYWGQGTTVTVSS	227	DIVMTQSPDSLAVSLGERATINCRASK SVSTSGYSFMHWYQQKPGQPPKLLIY LASNLESGVPDRFSGSGSGTDFTLTIS SLQAEDVAVYYCQHSREVPWTFGQG TKVEIK	228
osemitamab	QVQLVQSGAEVKKPGASVKVSCKASGYTFT GYNMNWVRQAPGQGLEWMGNIDPYYGGTS YNQKFKGRVTMTIDKSTSTVYMELSSLRSED TAVYYCARMYHGNAFDYWGQGTTVTVSS	229	DIVMTQSPDSLAVSLGERATINCKSSQ SLLNSGNLKNYLTWYQQKPGQPPKLLI YWASTRKSGVPDRFSGSGSGTDFTLT ISSLQAEDVAVYYCQNDYSYPLTFGG GTKVEIK	230
zolbetuximab	QVQLQQPGAELVRPGASVKLSCKASGYTFTS YWINWVKQRPGQGLEWIGNIYPSDSYTNYNQ KFKDKATLTVDKSSSTAYMQLSSPTSEDSAV YYCTRSWRGNSFDYWGQGTTLTVSS	231	DIVMTQSPSSLTVTAGEKVTMSCKSS QSLLNSGNQKNYLTWYQQKPGQPPK LLIYWASTRESGVPDRFTGSGSGTDF TLTISSVQAEDLAVYYCQNDYSYPFTF GSGTKLEIK	232
tepoditamab	QVQLVQSGAEVKKPGASVKVSCKASGYTFTS YYMHWVRQAPGQGLEWMGIINPSGGSTSYA QKFQGRVTMTRDTSTSTVYMELSSLRSEDTA VYYCAKGTTGDWFDYWGQGTLVTVSS	233	DIQMTQSPSSLSASVGDRVTITCRASQ SISSYLNWYQQKPGKAPKLLIYAASSL QSGVPSRFSGSGSGTDFTLTISSLQPE DFATYYCQQSYSTPPTFGQGTKVEIK	234
roval- pituzumab	QVQLVQSGAEVKKPGASVKVSCKASGYTFTN YGMNWVRQAPGQGLEWMGWINTYTGEPTY ADDFKGRVTMTTDTSTSTAYMELRSLRSDDT AVYYCARIGDSSPSDYWGQGTLVTVSS	235	EIVMTQSPATLSVSPGERATLSCKASQ SVSNDVVWYQQKPGQAPRLLIYYASN RYTGIPARFSGSGSGTEFTLTISSLQS EDFAVYYCQQDYTSPWTFGQGTKLEI K	236
tarlatamab	QVQLQESGPGLVKPSETLSLTCTVSGGSISSY YWSWIRQPPGKCLEWIGYVYYSGTTNYNPSL KSRVTISVDTSKNQFSLKLSSVTAADTAVYYC ASIAVTGFYFDYWGQGTLVTVSS	237	EIVLTQSPGTLSLSPGERVTLSCRASQ RVNNNYLAWYQQRPGQAPRLLIYGAS SRATGIPDRFSGSGSGTDFTLTISRLE PEDFAVYYCQQYDRSPLTFGCGTKLEI K	238

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SEQUENCES				
becotatug	QVQLQESGPGLVKPSETLSLTCTVSGFSLSN YDVHWVRQAPGKGLEWLGVIWSGGNTDYNT PFTSRLTISVDTSKNQFSLKLSSVTAADTAVY YCARALDYYDYEFAYWGQGTLVTVSS	239	EIVLTQSPDFQSVTPKEKVTITCRASQ SIGTNIHWYQQKPDQSPKLLIKYASESI SGIPSRFSGSGSGTDFTLTINSLEAED AATYYCQQNNEWPTSFGQGTKLEIK	240
cetuximab	QVQLKQSGPGLVQPSQSLSITCTVSGFSLTN YGVHWVRQSPGKGLEWLGVIWSGGNTDYNT PFTSRLSINKDNSKSQVFFKMNSLQSNDTAIY YCARALTYYDYEFAYWGQGTLVTVSA	241	DILLTQSPVILSVSPGERVSFSCRASQ SIGTNIHWYQQRTNGSPRLLIKYASESI SGIPSRFSGSGSGTDFTLSINSVESEDI ADYYCQQNNNWPTTFGAGTKLELK	242
demupitamab	QVQLQESGPGLVKPSETLSLTCTVSGGSVSS GDYYWTWIRQSPGKGLEWIGHIYYSGNTNYN PSLKSRLTISIDTSKTQFSLKLSSVTAADTAIY YCVRDRVTGAFDIWGQGTMVTVSS	243	DIQMTQSPSTLSASVGDRVTITCQASQ DISNYLNWYQQKPGKAPKLLIYDASNL ETGVPSRFSGSGSGTDFTFTISSLQPE DIATYFCQHFDHLPLAFGGGTKVEIK	244
futuximab	EVQLQQPGSELVRPGASVKLSCKASGYTFTS YWMHWVKQRPGQGLEWIGNIYPGSRSTNYD EKFKSKATLTVDTSSSTAYMQLSSLTSEDSAV YYCTRNGDYYVSSGDAMDYWGQGTSVTVSS	245	DIQMTQTTSSLSASLGDRVTISCRTSQ DIGNYLNWYQQKPDGTVKLLIYYTSRL HSGVPSRFSGSGSGTDFSLTINNVEQ EDVATYFCQHYNTVPPTFGGGTKLEIK	246
imgatuzumab	QVQLVQSGAEVKKPGSSVKVSCKASGFTFTD YKIHWVRQAPGQGLEWMGYFNPNSGYSTYA QKFQGRVTITADKSTSTAYMELSSLRSEDTAV YYCARLSPGGYYVMDAWGQGTTVTVSS	247	DIQMTQSPSSLSASVGDRVTITCRASQ GINNYLNWYQQKPGKAPKRLIYNTNNL QTGVPSRFSGSGSGTEFTLTISSLQPE DFATYYCLQHNSFPTFGQGTKLEIK	248
matuzumab	QVQLVQSGAEVKKPGASVKVSCKASGYTFTS HWMHWVRQAPGQGLEWIGEFNPSNGRTNY NEKFKSKATMTVDTSTNTAYMELSSLRSEDT AVYYCASRDYDYDGRYFDYWGQGTLVTVSS	249	DIQMTQSPSSLSASVGDRVTITCSASS SVTYMYWYQQKPGKAPKLLIYDTSNL ASGVPSRFSGSGSGTDYTFTISSLQPE DIATYYCQQWSSHIFTFGQGTKVEIK	250
modotuximab	QVQLQQPGAELVEPGGSVKLSCKASGYTFTS HWMHWVKQRPGQGLEWIGEINPSSGRNNY NEKFKSKATLTVDKSSSTAYMQFSSLTSEDS AVYYCVRYYGYDEAMDYWGQGTSVTVSS	251	DIVMTQAAFSNPVTLGTSASISCRSSK SLLHSNGITYLYWYLQKPGQSPQLLIY QMSNLASGVPDRESSSGSGTDFTLRI SRVEAEDVGVYYCAQNLELPYTFGGG TKLEIK	252
necitumumab	QVQLQESGPGLVKPSQTLSLTCTVSGGSISS GDYYWSWIRQPPGKGLEWIGYIYYSGSTDYN PSLKSRVTMSVDTSKNQFSLKVNSVTAADTA VYYCARVSIFGVGTFDYWGQGTLVTVSS	253	EIVMTQSPATLSLSPGERATLSCRASQ SVSSYLAWYQQKPGQAPRLLIYDASN RATGIPARFSGSGSGTDFTLTISSLEPE DFAVYYCHQYGSTPLTFGGGTKAEIK	254
nimotuzumab	QVQLQQSGAEVKKPGSSVKVSCKASGYTFT NYYIYWVRQAPGQGLEWIGGINPTSGGSNFN EKFKTRVTITADESSTTAYMELSSLRSEDTAF YFCTRQGLWFDSDGRGFDFWGQGTTVTVSS	255	DIQMTQSPSSLSASVGDRVTITCRSSQ NIVHSNGNTYLDWYQQTPGKAPKLLIY KVSNRFSGVPSRFSGSGSGTDFTFTIS SLQPEDIATYYCFQYSHVPWTFGQGT KLQIT	256
panitumumab	QVQLQESGPGLVKPSETLSLTCTVSGGSVSS GDYYWTWIRQSPGKGLEWIGHIYYSGNTNYN PSLKSRLTISIDTSKTQFSLKLSSVTAADTAIYY CVRDRVTGAFDIWGQGTMVTVSS	257	DIQMTQSPSSLSASVGDRVTITCQASQ DISNYLNWYQQKPGKAPKLLIYDASNL ETGVPSRFSGSGSGTDFTFTISSLQPE DIATYFCQHFDHLPLAFGGGTKVEIK	258
pimurutamab	EVQLVESGGGLVQPGGSLRLSCAASGFSLTN YGVHWVRQAPGKGLEWLGVIWSGGNTDYG NEFTSRFTISRDNAKNSLYLQMNSLRAEDTAV YYCARALDYYDYEFAYWGQGTMVTVSS	259	EIVLTQSPATLSLSPGERATLSCRASQ SIGTNIHWYQQKPGQAPRLLIKYASESI SGIPARFSGSGSGTDFTLTISSLEPEDF AVYYCQQNNNWPTSFGGGTKVEIK	260
tomuzotuximab	QVQLKQSGPGLVQPSQSLSITCTVSGFSLTN YGVHWVRQSPGKGLEWLGVIWSGGNTDYNT PFTSRLSINKDNSKSQVFFKMNSLQSNDTAIY YCARALTYYDYEFAYWGQGTLVTVST	261	DILLTQSPVILSVSPGERVSFSCRASQ SIGTNIHWYQQRTNGSPRLLIKYASESI SGIPSRFSGSGSGTDFTLSINSVESEDI ADYYCQQNNNWPTTFGAGTKLELK	262
zalutumumab	QVQLVESGGGVVQPGRSLRLSCAASGFTFST YGMHWVRQAPGKGLEWVAVIWDDGSYKYY GDSVKGRFTISRDNSKNTLYLQMNSLRAEDT AVYYCARDGITMVRGVMKDYFDYWGQGTLV TVSS	263	AIQLTQSPSSLSASVGDRVTITCRASQ DISSALVWYQQKPGKAPKLLIYDASSL ESGVPSRFSGSESGTDFTLTISSLQPE DFATYYCQQFNSYPLTFGGGTKVEIK	264
umizortamig	EVQLVESGGGLVQPGGSLRLSCAASGFTIST NAMSWVRQAPGKGLEWIGVITGRDITYYASW AKGRFTISRDNSKNTLYLQMNSLRAEDTAVY YCARDGGSSAITSNNIWGQGTLVTVSS	265	DVVMTQSPSTLSASVGDRVTINCQAS ESISSWLAWYQQKPGKAPKLLIYEASK LASGVPSRFSGSGSGTEFTLTISSLQP DDFATYYCQGYFYFISRTYVNSFGGG TKVEIK	266

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SEQUENCES				
emirodatamab	QVTLKESGPTLVKPTETLTLTCTLSGFSLNNA RMGVSWIRQPPGKCLEWLAHIFSNDEKSYST SLKNRLTISKDSSKTQVVLTMTNVDPVDTATY YCARIVGYGSGWYGFFDYWGQGTLVTVSS	267	DIQMTQSPSSLSASVGDRVTITCRASQ GIRNDLGWYQQKPGKAPKRLIYAASTL QSGVPSRFSGSGSGTEFTLTISSLQPE DFATYYCLQHNSYPLTFGCGTKVEIK	268
dinutuximab	EVQLLQSGPELEKPGASVMISCKASGSSFTG YNMNWVRQNIGKSLEWIGAIDPYYGGTSYNQ KFKGRATLTVDKSSSTAYMHLKSLTSEDSAV YYCVSGMEYWGQGTSVTVSS	269	EIVMTQSPATLSVSPGERATLSCRSSQ SLVHRNGNTYLHWYLQKPGQSPKLLI HKVSNRFSGVPDRFSGSGSGTDFTLK ISRVEAEDLGVYFCSQSTHVPPLTFGA GTKLELK	270
lorukafusp	EVQLVQSGAEVEKPGASVKISCKASGSSFTG YNMNWVRQNIGKSLEWIGAIDPYYGGTSYNQ KFKGRATLTVDKSTSTAYMHLKSLRSEDTAV YYCVSGMEYWGQGTSVTVSS	271	DVVMTQTPLSLPVTPGEPASISCRSSQ SLVHRNGNTYLHWYLQKPGQSPKLLI HKVSNRFSGVPDRFSGSGSGTDFTLK ISRVEAEDLGVYFCSQSTHVPPLTFGA GTKLELK	272
naxitamab	QVQLVESGPGVVQPGRSLRISCAVSGFSVTN YGVHWVRQPPGKGLEWLGVIWAGGITNYNS AFMSRLTISKDNSKNTVYLQMNSLRAEDTAM YYCASRGGHYGYALDYWGQGTLVTVSS	273	EIVMTQTPATLSVSAGERVTITCKASQ SVSNDVTWYQQKPGQAPRLLIYSASN RYSGVPARFSGSGYGTEFTFTISSVQS EDFAVYFCQQDYSSFGQGTKLEIK	274
codrituzumab	QVQLVQSGAEVKKPGASVKVSCKASGYTFTD YEMHWVRQAPGQGLEWMGALDPKTGDTAY SQKFKGRVTLTADKSTSTAYMELSSLTSEDTA VYYCTRFYSYTYWGQGTLVTVSS	275	DVVMTQSPLSLPVTPGEPASISCRSSQ SLVHSNRNTYLHWYLQKPGQSPQLLI YKVSNRFSGVPDRFSGSGSGTDFTLKI SRVEAEDVGVYYCSQNTHVPPTFGQ GTKLEIK	276
anvatabart	EVQLVESGGGLVQPGGSLRLSCAASGFNIKD TYIHWVRQAPGKGLEWVARIYPTNGYTRYAD SVKGRFTISADTSKNTAYLQMNSLRAEDTAVY YCSRWGGDGFYAMDYWGQGTLVTVSS	277	DIQMTQSPSSLSASVGDRVTITCRASQ DVNTAVAWYQQKPGKAPKLLIYSASFL YSGVPSRFSGSRSGTDFTLTISSLQPE DFATYYCQQHYTTPPTFGQGTKVEIK	278
coprelotamab	EVQLVESGGGLVQPGGSLRLSCAASGFNIKD TYIHWVRQAPGKGLEWVARIYPTNGYTRYAD SVKGRFTISADTSKNTAYLQMNSLRAEDTAVY YCSRWGGDGFYAMDYWGQGTLVTVSS	279	DIQMTQSPSSLSASVGDRVTITCRASQ DVNTAVAWYQQKPGKAPKLLIYSASFL YSGVPSRFSGSRSGTDFTLTISSLQPE DFATYYCQQHYTTPPTFGQGTKVEIK	280
disitamab	EVQLVQSGAEVKKPGATVKISCKVSGYTFTD YYIHWVQQAPGKGLEWMGRVNPDHGDSYY NQKFKDKATITADKSTDTAYMELSSLRSEDTA VYFCARNYLFDHWGQGTLVTVSS	281	DIQMTQSPSSVSASVGDRVTITCKASQ DVGTAVAWYQQKPGKAPKLLIYWASI RHTGVPSRFSGSGSGTDFTLTISSLQP EDFATYYCHQFATYTFGGGTKVEIK	282
gancotamab	QVQLVESGGGLVQPGGSLRLSCAASGFTFR SYAMSWVRQAPGKGLEWVSAISGRGDNTYY ADSVKGRFTISRDNSKNTLYLQMNSLRAEDT AVYYCAKMTSNAFAFDYWGQGTLVTVSS	283	QSVLTQPPSVSGAPGQRVTISCTGSS SNIGAGYGVHWYQQLPGTAPKLLIYG NTNRPSGVPDRFSGFKSGTSASLAITG LQAEDEADYYCQSYDSSLSGWVFGG GTKLTVL	284
margetuximab	QVQLQQSGPELVKPGASLKLSCTASGFNIKD TYIHWVKQRPEQGLEWIGRIYPTNGYTRYDP KFQDKATITADTSSNTAYLQVSRLTSEDTAVY YCSRWGGDGFYAMDYWGQGASVTVSS	285	DIVMTQSHKFMSTSVGDRVSITCKASQ DVNTAVAWYQQKPGHSPKLLIYSASF RYTGVPDRFTGSRSGTDFTFTISSVQA EDLAVYYCQQHYTTPPTFGGGTKVEIK	286
marstacimab	EVQLLESGGGLVQPGGSLRLSCAASGFTFSS YAMSWVRQAPGKGLEWVSAISGSGGSTYYA DSVKGRFTISRDNSKNTLYLQMNSLRAEDTA VYYCAILGATSLSAFDIWGQGTMVTVSS	287	QSVLTQPPSVSGAPGQRVTISCTGSS SNIGAGYDVHWYQQLPGTAPKLLIYG NSNRPSGVPDRFSGSKSGTSASLAIT GLQAEDEADYYCQSYDSSLSGSGVFG GGTKLTVL	288
pertuzumab	EVQLVESGGGLVQPGGSLRLSCAASGFTFTD YTMDWVRQAPGKGLEWVADVNPNSGGSIYN QRFKGRFTLSVDRSKNTLYLQMNSLRAEDTA VYYCARNLGPSFYFDYWGQGTLVTVSS	289	DIQMTQSPSSLSASVGDRVTITCKASQ DVSIGVAWYQQKPGKAPKLLIYSASYR YTGVPSRFSGSGSGTDFTLTISSLQPE DFATYYCQQYYIYPYTFGQGTKVEIK	290
timigutuzumab	EVQLVESGGGLVQPGGSLRLSCAASGFNIKD TYIHWVRQAPGKGLEWVARIYPTNGYTRYAD SVKGRFTISADTSKNTAYLQMNSLRAEDTAVY YCSRWGGDGFYAMDYWGQGTLVTVSS	291	DIQMTQSPSSLSASVGDRVTITCRASQ DVNTAVAWYQQKPGKAPKLLIYSASFL YSGVPSRFSGSRSGTDFTLTISSLQPE DFATYYCQQHYTTPPTFGQGTKVEIK	292
trastuzumab	EVQLVESGGGLVQPGGSLRLSCAASGFNIKD TYIHWVRQAPGKGLEWVARIYPTNGYTRYAD SVKGRFTISADTSKNTAYLQMNSLRAEDTAVY YCSRWGGDGFYAMDYWGQGTLVTVSS	293	DIQMTQSPSSLSASVGDRVTITCRASQ DVNTAVAWYQQKPGKAPKLLIYSASFL YSGVPSRFSGSRSGTDFTLTISSLQPE DFATYYCQQHYTTPPTFGQGTKVEIK	294

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amatuximab	QVQLQQSGPELEKPGASVKISCKASGYSFTG YTMNWVKQSHGKSLEWIGLITPYNGASSYNQ KFRGKATLTVDKSSSTAYMDLLSLTSEDSAVY FCARGGYDGRGFDYWGSGTPVTVSS	295	DIELTQSPAIMSASPGEKVTMTCSASS SVSYMHWYQQKSGTSPKRWIYDTSKL ASGVPGRFSGSGSGNSYSLTISSVEA EDDATYYCQQWSKHPLTFGSGTKVEI K	296
anetumab	QVELVQSGAEVKKPGESLKISCKGSGYSFTS YWIGWVRQAPGKGLEWMGIIDPGDSRTRYS PSFQGQVTISADKSISTAYLQWSSLKASDTAM YYCARGQLYGGTYMDGWGQGTLVTVSS	297	DIALTQPASVSGSPGQSITISCTGTSSD IGGYNSVSWYQQHPGKAPKLMIYGVN NRPSGVSNRFSGSKSGNTASLTISGL QAEDEADYYCSSYDIESATPVFGGGT KLTVL	298
inezetamab	QVQLVESGGGLVKPGGSLRLSCAASGFTFSD YYMSWIRQAPGKCLEWISYISSSESIIYYVDAV KGRFTISRDNAKNSLYLQMNSLRAEDTAVYY CARDVGSHFDYWGQGTLVTVSS	299	DIQMTQSPSSVSASVGDRVTITCRASQ DISRWLAWYQQKPGKAPKLLISAASRL QSGVPSRFSGSGSGTDFTLTISSLQPE DFAIYYCQQAKSFPRTFGCGTKVEIK	300
cantuzumab	QVQLVQSGAEVKKPGETVKISCKASDYTFTY YGMNWVKQAPGQGLKWMGWIDTTTGEPTY AQKFQGRIAFSLETSASTAYLQIKSLKSEDTAT YFCARRGPYNWYFDVWGQGTTVTVSS	301	DIVMTQSPLSVPVTPGEPVSISCRSSK SLLHSNGNTYLYWFLQRPGQSPQLLIY RMSNLVSGVPDRFSGSGSGTAFTLRI SRVEAEDVGVYYCLQHLEYPFTFGPG TKLELK	302
clivatuzumab	QVQLQQSGAEVKKPGASVKVSCEASGYTFP SYVLHWVKQAPGQGLEWIGYINPYNDGTQY NEKFKGKATLTRDTSINTAYMELSRLRSDDTA VYYCARGFGGSYGFAYWGQGTLVTVSS	303	DIQLTQSPSSLSASVGDRVTMTCSASS SVSSSYLYWYQQKPGKAPKLWIYSTS NLASGVPARFSGSGSGTDFTLTISSLQ PEDSASYFCHQWNRYPYTFGGGTRL EIK	304
gatipotuzumab	EVQLVESGGGLVQPGGSMRLSCVASGFPFS NYWMNWVRQAPGKGLEWVGEIRLKSNNYTT HYAESVKGRFTISRDDSKNSLYLQMNSLKTE DTAVYYCTRHYYFDYWGQGTLVTVSS	305	DIVMTQSPLSNPVTPGEPASISCRSSK SLLHSNGITYFFWYLQKPGQSPQLLIY QMSNLASGVPDRFSGSGSGTDFTLRI SRVEAEDVGVYYCAQNLELPPTFGQG TKVEIK	306
abagovomab	QVKLQESGAELARPGASVKLSCKASGYTFTN YWMQWVKQRPGQGLDWIGAIYPGDGNTRYT HKFKGKATLTADKSSSTAYMQLSSLASEDSG VYYCARGEGNYAWFAYWGQGTTVTVSS	307	DIELTQSPASLSASVGETVTITCQASEN IYSYLAWHQQKQGKSPQLLVYNAKTL AGGVSSRFSGSGSGTHFSLKIKSLQP EDFGIYYCQHHYGILPTFGGGTKLEIK	308
sofituzumab	EVQLVESGGGLVQPGGSLRLSCAASGYSITN DYAWNWVRQAPGKGLEWVGYISYSGYTTYN PSLKSRFTISRDTSKNTLYLQMNSLRAEDTAV YYCARWTSGLDYWGQGTLVTVSS	309	DIQMTQSPSSLSASVGDRVTITCKASD LIHNWLAWYQQKPGKAPKLLIYGATSL ETGVPSRFSGSGSGTDFTLTISSLQPE DFATYYCQQYWTTPFTFGQGTKVEIK	310
ubamatamab	QVQLVESGGGLVKPGGSLRLSCAASGFTFSN YYMSWVRQAPGKGLEWISYISGRGSTIFYAD SVKGRITISRDNAKNSLFLQMNSLRAEDTAVY FCVKDRGGYSPYWGQGTLVTVSS	311	DIQMTQSPSSLSASVGDRVTITCRASQ SISTYLNWYQQKPGKAPKLLIYTASSL QSGVPSRFSGSGSGTDFTLTISSLQPE DFATYYCQQSYSTPPITFGQGTRLEIK	312
tesnatilimab	QVHLQESGPGLVKPSETLSLTCTVSDDSISSY YWSWIRQPPGKGLEWIGHISYSGSANYNPSL KSRVTISVDTSKNQFSLKLSSVTAADTAVYYC ANWDDAFNIWGQGTMVTVSS	313	EIVLTQSPGTLSLSPGERATLSCRASQ SVSSSYLAWYQQKPGQAPRLLIYGAS SRATGIPDRFSGSGSGTDFTLTISRLE PEDFAVYYCQQYGSSPWTFGQGTKV EIK	314
acrixolimab	QVQLVESGGGVVQPGRSLRLSCAASGFTFLR YAMHWVRQAPGKGLEWVAVISYDGRYKYYA DSVKGRFTISRDNSKNTLYLQMNSLRAEDTA VYYCTTTTFDSWGQGTLVTVSS	315	DIVMTQTPLSLPVTPGEAASISCRSSQ SLLDSEDGNTYLDWYLQKPGQSPQLLI YTLSHRASGVPDRFSGSGSGTDFTLEI SRVEAEDVGVYYCMQRRDFPFTFGQ GTKVDIK	316
balstilimab	QVQLVESGGGVVQPGRSLRLSCAASGFTFS SYGMHWVRQAPGKGLEWVAVIWYDGSNKY YADSVKGRFTISRDNSKNTLYLQMNSLRAED TAVYYCASNGDHWGQGTLVTVSS	317	EIVMTQSPATLSVSPGERATLSCRASQ SVSSNLAWYQQKPGQAPRLLIYGAST RATGIPARFSGSGSGTEFTLTISSLQS EDFAVYYCQQYNNWPRTFGQGTKVEI K	318
budigalimab	EIQLVQSGAEVKKPGSSVKVSCKASGYTFTH YGMNWVRQAPGQGLEWVGWVNTYTGEPTY ADDFKGRLTFTLDTSTSTAYMELSSLRSEDTA VYYCTREGEGLGFGDWGQGTTVTVSS	319	DVVMTQSPLSLPVTPGEPASISCRSSQ SIVHSHGDTYLEWYLQKPGQSPQLLIY KVSNRFSGVPDRFSGSGSGTDFTLKIS RVEAEDVGVYYCFQGSHIPVTFGQGT KLEIK	320

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camrelizumab	EVQLVESGGGLVQPGGSLRLSCAASGFTFSS YMMSWVRQAPGKGLEWVATISGGGANTYYP DSVKGRFTISRDNAKNSLYLQMNSLRAEDTA VYYCARQLYYFDYWGQGTTVTVSS	321	DIQMTQSPSSLSASVGDRVTITCLASQ TIGTWLTWYQQKPGKAPKLLIYTATSL ADGVPSRFSGSGSGTDFTLTISSLQPE DFATYYCQQVYSIPWTFGGGTKVEIK	322
cemiplimab	EVQLLESGGVLVQPGGSLRLSCAASGFTFSN FGMTWVRQAPGKGLEWVSGISGGGRDTYFA DSVKGRFTISRDNSKNTLYLQMNSLKGEDTA VYYCVKWGNIYFDYWGQGTLVTVSS	323	DIQMTQSPSSLSASVGDSITITCRASLS INTFLNWYQQKPGKAPNLLIYAASSLH GGVPSRFSGSGSGTDFTLTIRTLQPED FATYYCQQSSNTPFTFGPGTVVDFR	324
cetrelimab	QVQLVQSGAEVKKPGSSVKVSCKASGGTFS SYAISWVRQAPGQGLEWMGGIIPIFDTANYA QKFQGRVTITADESTSTAYMELSSLRSEDTAV YYCARPGLAAAYDTGSLDYWGQGTLVTVSS	325	EIVLTQSPATLSLSPGERATLSCRASQ SVRSYLAWYQQKPGQAPRLLIYDASN RATGIPARFSGSGSGTDFTLTISSLEPE DFAVYYCQQRNYWPLTFGQGTKVEIK	326
dostarlimab	EVQLLESGGGLVQPGGSLRLSCAASGFTFSS YDMSWVRQAPGKGLEWVSTISGGGSYTYYQ DSVKGRFTISRDNSKNTLYLQMNSLRAEDTA VYYCASPYYAMDYWGQGTTVTVSS	327	DIQLTQSPSFLSAYVGDRVTITCKASQ DVGTAVAWYQQKPGKAPKLLIYWAST LHTGVPSRFSGSGSGTEFTLTISSLQP EDFATYYCQHYSSYPWTFGQGTKLEI K	328
enlonstobart	QVQLVESGGGVVQPGRSLRLTCKASGLTFSS SGMHWVRQAPGKGLEWVAVIWYDGSKRYY ADSVKGRFTISRDNSKNTLFLQMNSLRAEDT AVYYCATNNDYWGQGTLVTVSS	329	EIVLTQSPATLSLSPGERATLSCRASQ SVSSYLAWYQQKPGQAPRLLIYTASN RATGIPARFSGSGSGTDFTLTISSLEPE DFAVYYCQQYSNWPRTFGQGTKVEIK	330
ezabenlimab	EVMLVESGGGLVQPGGSLRLSCTASGFTFSK SAMSWVRQAPGKGLEWVAYISGGGGDTYYS SSVKGRFTISRDNAKNSLYLQMNSLRAEDTA VYYCARHSNVNYYAMDYWGQGTLVTVSS	331	EIVLTQSPATLSLSPGERATMSCRASE NIDVSGISFMNWYQQKPGQAPKLLIYV ASNQGSGIPARFSGSGSGTDFTLTISR LEPEDFAVYYCQQSKEVPWTFGQGTK LEIK	332
finotonlimab	EVQLVESGGGLVKPGGSLRLSCAASGFTFSS YGMSWVRQAPGKRLEWVATISGGGRDTYYS DSVKGRFTISRDNAKNNLYLQMNSLRAEDTA VYYCSRQYGTVWFFNWGQGTLVTVSS	333	EIVLTQSPATLSLSPGERATLSCRASE SVDSYGNSFMHWYQQKPGQPPRLLIY AASNQGSGVPARFSGSGSGTDFTLTI SSLEPEDFAMYFCQQSKEVPWTFGQ GTKVEIK	334
geptanolimab	QIQLVQSGSELKKPGASVKVSCKASGYTFTN FGMNWVRQAPGQGLKWMGWISGYTREPTY AADFKGRFVISLDTSVSTAYLQISSLKAEDTAV YYCARDVFDYWGQGTLVTVSS	335	DIVLTQSPASLAVSPGQRATITCRASE SVDNYGYSFMNWFQQKPGQPPKLLIY RASNLESGVPARFSGSGSRTDFTLTIN PVEADDTANYYCQQSNADPTFGQGT KLEIK	336
iparomlimab	QVQLVQSGAEVKKPGASVKVSCKASGYTFTN YWIHWVRQAPGQGLEWMGEIDPYDSYTNYN QKFKGRVTMTVDKSTSTVYMELSSLRSEDTA VYYCARPGFTYGGMDFWGQGTLVTVSS	337	DIQMTQSPSSLSASVGDRVTITCKSSQ SLFNSGNQKNYLAWYQQKPGKVPKLL IYGASTRDSGVPYRFSGSGSGTDFTLT ISSLQPEDVATYYCQNDHYYPYTFGG GTKVEIK	338
lipustobart	QVQLVQSGAEVKKPGASVKVSCKASGYTFTS YYMYWVRQAPGQGLEWMGGVNPSNGGTNF NEKFKSRVTITADKSTSTAYMELSSLRSEDTA VYYCARRDYRYDMGFDYWGQGTTVTVSS	339	EIVLTQSPATLSLSPGERATISCRASKG VSTSGYSYLHWYQQKPGQAPRLLIYL ASYLESGVPARFSGSGSGTDFTLTISS LEPEDFATYYCQHSRELPLTFGTGTKV EIK	340
nivolumab	QVQLVESGGGVVQPGRSLRLDCKASGITFSN SGMHWVRQAPGKGLEWVAVIWYDGSKRYY ADSVKGRFTISRDNSKNTLFLQMNSLRAEDT AVYYCATNDDYWGQGTLVTVSS	341	EIVLTQSPATLSLSPGERATLSCRASQ SVSSYLAWYQQKPGQAPRLLIYDASN RATGIPARFSGSGSGTDFTLTISSLEPE DFAVYYCQQSSNWPRTFGQGTKVEIK	342
nofazinlimab	QVQLVQSGAEVKKPGSSVKVSCKASGFTFTT YYISWVRQAPGQGLEYLGYINMGSGGTNYNE KFKGRVTITADKSTSTAYMELSSLRSEDTAVY YCAIIGYFDYWGQGTMVTVSS	343	DVVMTQSPLSLPVTLGQPASISCRSSQ SLLDSDGGTYLYWFQQRPGQSPRRLI YLVSTLGSGVPDRFSGSGSGTDFTLKI SRVEAEDVGVYYCMQLTHWPYTFGQ GTKLEIK	344
penpulimab	EVQLVESGGGLVQPGGSLRLSCAASGFAFSS YDMSWVRQAPGKGLDWVATISGGGRYTYYP DSVKGRFTISRDNSKNNLYLQMNSLRAEDTA LYYCANRYGEAWFAYWGQGTLVTVSS	345	DIQMTQSPSSMSASVGDRVTFTCRAS QDINTYLSWFQQKPGKSPKTLIYRANR LVSGVPSRFSGSGSGQDYTLTISSLQP EDMATYYCLQYDEFPLTFGAGTKLELK	346

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peresolimab	QVQLVQSGAEVKKPGASVKVSCKVSGYSLSK YDMSWVRQAPGKGLEWMGIIYTSGYTDYAQ KFQGRVTMTEDTSTDTAYMELSSLRSEDTAV YYCATGNPYYTNGFNSWGQGTLVTVSS	347	DIQMTQSPSSLSASVGDRVTITCQASQ SPNNLLAWYQQKPGKAPKLLIYGASDL PSGVPSRFSGSGSGTDFTLTISSLQPE DFATYYCQNNYYVGPVSYAFGGGTKV EIK	348
pidilizumab	QVQLVQSGSELKKPGASVKISCKASGYTFTN YGMNWVRQAPGQGLQWMGWINTDSGESTY AEEFKGRFVFSLDTSVNTAYLQITSLTAEDTG MYFCVRVGYDALDYWGQGTLVTVSS	349	EIVLTQSPSSLSASVGDRVTITCSARSS VSYMHWFQQKPGKAPKLWIYRTSNLA SGVPSRFSGSGSGTSYCLTINSLQPE DFATYYCQQRSSFPLTFGGGTKLEIK	350
pimivalimab	QVQLVQSGAEVKKPGASVKVSCKASGYTFPS YYMHWVRQAPGQGLEWMGIINPEGGSTAYA QKFQGRVTMTRDTSTSTVYMELSSLRSEDTA VYYCARGGTYYDYTYWGQGTLVTVSS	351	DIQMTQSPSTLSASVGDRVTITCRASQ SISSWLAWYQQKPGKAPKLLIYEASSL ESGVPSRFSGSGSGTEFTLTISSLQPD DFATYYCQQYNSFPPTFGGGTKVEIK	352
pradusin-stobart	QVTLKESGPALVKPTQTLTLTCTFSGFSLSTS GTCVSWIRQPPGKALEWLATICWEDSKGYNP SLKSRLTISKDTSKNQAVLTMTNMDPVDTATY YCARREDSGYFWFPYWGQGTLVTVSS	353	NIQMTQSPSSLSASVGDRVTITCKAGQ NVNNYLAWYQQKPGKAPKVLIFNANS LQTGVPSRFSGSGSGTDFTLTISSLQP EDFATYYCQQYNSWTTFGGGTKVEIK	354
prolgolimab	QVQLVQSGGGLVQPGGSLRLSCAASGFTFS SYWMYWVRQVPGKGLEWVSAIDTGGGRTY YADSVKGRFAISRVNAKNTMYLQMNSLRAED TAVYYCARDEGGGTGWGVLKDWPYGLDAW GQGTLVTVSS	355	QPVLTQPLSVSVALGQTARITCGGNNI GSKNVHWYQQKPGQAPVLVIYRDSN RPSGIPERFSGSNSGNTATLTISRAQA GDEADYYCQVWDSSTAVFGTGTKLTV L	356
pucotenlimab	EVQLVQSGGGLVQPGGSLKLSCAASGFTFSS YGMSWVRQAPGKGLDWVATISGGGRDTYYP DSVKGRFTISRDNSKNNLYLQMNSLRAEDTA LYYCARQKGEAWFAYWGQGTLVTVSS	357	DIVLTQSPASLAVSPGQRATITCRASE SVDNYGISFMNWFQQKPGQPPKLLIY AASNKGTGVPARFSGSGSGTDFTLNI NPMEENDTAMYFCQQSKEVPWTFGG GTKLEIK	358
retifanlimab	QVQLVQSGAEVKKPGASVKVSCKASGYSFTS YWMNWVRQAPGQGLEWIGVIHPSDSETWLD QKFKDRVTITVDKSTSTAYMELSSLRSEDTAV YYCAREHYGTSPFAYWGQGTLVTVSS	359	EIVLTQSPATLSLSPGERATLSCRASE SVDNYGMSFMNWFQQKPGQPPKLLI HAASNQGSGVPSRFSGSGSGTDFTLT ISSLEPEDFAVYFCQQSKEVPYTFGGG TKVEIK	360
rosnilimab	QVQLVQSGSELKKPGASVKVSCKASNYTFTD YSMHWVRQAPGQGLEWMGWINIETYYPTYA DQFKGRFAFSLDTSVSTAYLQISSLKAEDTAV YYCARDYYGRFYYAMDYWGQGTTVTVSS	361	EIVLTQSPATLSLSPGERATLSCTASSS VSSSYFHWYQQKPGQAPRLLIYSTSN LASGIPARFSGSGSGTDFTLTISRLEPE DFAVYYCHQYHRSPLTFGGGTKVEIK	362
rulonilimab	EVQLVESGGGLVKPGGSLRLSCAASGFTFSS YGMSWVRQTPEKRLEWVATISGGGRDTYYP DSVKGRFTISRDNAKNNLYLQMSSLRSEDTA LYYCARQKDTSWFVHWGQGTLVTVSS	363	EIVLTQSPATLAVSPGERATISCRASES VDDYGISFMNWFQQKPGQPPKLLIYV ASNQGSGVPARFSGSGSGTDFTLNIH PMEEDDTAMYFCQQSKEVPWTFGGG TKLEIK	364
sasanlimab	QVQLVQSGAEVKKPGASVKVSCKASGYTFTS YWINWVRQAPGQGLEWMGNIYPGSSLTNYN EKFKNRVTMTRDTSTSTVYMELSSLRSEDTA VYYCARLSTGTFAYWGQGTLVTVSS	365	DIVMTQSPDSLAVSLGERATINCKSSQ SLWDSGNQKNFLTWYQQKPGQPPKL LIYWTSYRESGVPDRFSGSGSGTDFT LTISSLQAEDVAVYYCQNDYFYPHTFG GGTKVEIK	366
serplulimab	QVQLVESGGGLVKPGGSLRLSCAASGFTFSN YGMSWIRQAPGKGLEWVSTISGGGSNIYYAD SVKGRFTISRDNAKNSLYLQMNSLRAEDTAV YYCVSYYYGIDFWGQGTSVTVSS	367	DIQMTQSPSSLSASVGDRVTITCKASQ DVTTAVAWYQQKPGKAPKLLIYWAST RHTGVPSRFSGSGSGTDFTLTISSLQP EDFATYYCQQHYTIPWTFGGGTKLEIK	368
sintilimab	QVQLVQSGAEVKKPGSSVKVSCKASGGTFS SYAISWVRQAPGQGLEWMGLIIPMFDTAGYA QKFQGRVAITVDESTSTAYMELSSLRSEDTAV YYCARAEHSSTGTFDYWGQGTLVTVSS	369	DIQMTQSPSSVSASVGDRVTITCRASQ GISSWLAWYQQKPGKAPKLLISAASSL QSGVPSRFSGSGSGTDFTLTISSLQPE DFATYYCQQANHLPFTFGGGTKVEIK	370
spartalizumab	EVQLVQSGAEVKKPGESLRISCKGSGYTFTT YWMHWVRQATGQGLEWMGNIYPGTGGSNF DEKFKNRVTITADKSTSTAYMELSSLRSEDTA VYYCTRWTTGTGAYWGQGTTVTVSS	371	EIVLTQSPATLSLSPGERATLSCKSSQ SLLDSGNQKNFLTWYQQKPGQAPRLL IYWASTRESGVPSRFSGSGSGTDFTF TISSLEAEDAATYYCQNDYSYPYTFGQ GTKVEIK	372

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SEQUENCES				
tislelizumab	QVQLQESGPGLVKPSETLSLTCTVSGFSLTS YGVHWIRQPPGKGLEWIGVIYADGSTNYNPS LKSRVTISKDTSKNQVSLKLSSVTAADTAVYY CARAYGNYWYIDVWGQGTTVTVSS	373	DIVMTQSPDSLAVSLGERATINCKSSE SVSNDVAWYQQKPGQPPKLLINYAFH RFTGVPDRFSGSGYGTDFTLTISSLQA EDVAVYYCHQAYSSPYTFGQGTKLEIK	374
toripalimab	QGQLVQSGAEVKKPGASVKVSCKASGYTFT DYEMHWVRQAPIHGLEWIGVIESETGGTAYN QKFKGRVTITADKSTSTAYMELSSLRSEDTAV YYCAREGITTVATTYYWYFDVWGQGTTVTVS S	375	DVVMTQSPLSLPVTLGQPASISCRSSQ SIVHSNGNTYLEWYLQKPGQSPQLLIY KVSNRFSGVPDRFSGSGSGTDFTLKIS RVEAEDVGVYYCFQGSHVPLTFGQGT KLEIK	376
zeluvalimab	EVQLLESGGGLVQPGGSLRLSCAASGFTFSS YDMSWVRQAPGKGLEWVSLISGGGSQTYYA ESVKGRFTISRDNSKNTLYLQMNSLRAEDTA VYFCASPSGHYFYAMDVWGQGTTVTVSS	377	DIQMTQSPSSVSASVGDRVTITCRASQ GISNWLAWYQQKPGKAPKLLIFAASSL QSGVPSRFSGSGSGTDFTLTISSLQPE DFATYYCQQAESFPHTFGGGTKVEIK	378
zimberelimab	QLQLQESGPGLVKPSETLTLTCTVSADSISST TYYWVWIRQPPGKGLEWIGSISYSGSTYYNP SLKSRVTVSVDTSKNQFSLKLNSVAATDTALY YCARHLGYNGRYLPFDYWGQGTLVTVSS	379	QSALTQPASVSGSPGQSITISCTGTSS DVGFYNYVSWYQQHPGKAPELMIYDV SNRPSGVSDRFSGSKSGNTASLTISGL QAEDEADYYCSSYTSISTWVFGGGTK LTVL	380
nebratamig	EVQLLESGGGLVQPGGSLRLSCAASGFTISR YHMTWVRQAPGKGLEWIGHIYVNNDDTDYA SSAKGRFTISRDNSKNTLYLQMNSLRAEDTAT YFCARLDVGGGGAYIGDIWGQGTLVTVSS	381	DIQMTQSPSSLSASVGDRVTITCQSSQ SVYNNNDLAWYQQKPGKVPKLLIYYA STLASGVPSRFSGSGSGTDFTLTISSL QPEDVATYYCAGGYDTDGLDTFAFGG GTKVEIK	382
zilovertamab	QVQLQESGPGLVKPSQTLSLTCTVSGYAFTA YNIHWVRQAPGQGLEWMGSFDPYDGGSSY NQKFKDRLTISKDTSKNQVVLTMTNMDPVDT ATYYCARGWYYFDYWGHGTLVTVSS	383	DIVMTQTPLSLPVTPGEPASISCRASK SISKYLAWYQQKPGQAPRLLIYSGSTL QSGIPPRFSGSGYGTDFTLTINNIESED AAYYFCQQHDESPYTFGEGTKVEIK	384
azintuxizumab	EVQLVESGGGLVQPGGSLRLSCAASGFTFSD YYMAWVRQAPGKGLEWVASINYDGSSTYYV DSVKGRFTISRDNAKNSLYLQMNSLRAEDTA VYYCARDRGYYFDYWGQGTTVTVSS	385	DVVMTQTPLSLSVTPGQPASISCRSS QSLVHSNGNTYLHWYLQKPGQSPQLL IYKVSNRFSGVPDRFSGSGSGTDFTLK ISRVEAEDVGVYFCSQSTHVPPFTFG GGTKVEIK	386
elotuzumab	EVQLVESGGGLVQPGGSLRLSCAASGFDFS RYWMSWVRQAPGKGLEWIGEINPDSSTINYA PSLKDKFIISRDNAKNSLYLQMNSLRAEDTAV YYCARPDGNYWYFDVWGQGTLVTVSS	387	DIQMTQSPSSLSASVGDRVTITCKASQ DVGIAVAWYQQKPGKVPKLLIYWAST RHTGVPDRFSGSGSGTDFTLTISSLQP EDVATYYCQQYSSYPYTFGQGTKVEI K	388

Nonlimiting Embodiments

[0220] Notwithstanding the appended claims, the present disclosure is also defined by the following embodiments:

[0221] 1. A nucleic acid comprising a sequence encoding a decoy-resistant interleukin 18 (DR-18) polypeptide and a chimeric antigen receptor (CAR) sequence encoding a CAR, wherein the CAR is selected from the group consisting of:

[0222] i) an anti-5T4 CAR comprising an antigen binding domain (ABD) that specifically binds 5T4; ii) an anti-AFP CAR comprising an ABD that specifically binds AFP;

[0223] iii) an anti-ALPP CAR comprising an ABD that specifically binds ALPP;

[0224] iv) an anti-ALPPL2 CAR comprising an ABD that specifically binds ALPPL2;

[0225] v) an anti-APN (0013) CAR comprising an ABD that specifically binds APN (0013);

[0226] vi) an anti-APRIL CAR comprising an ABD that specifically binds APRIL;

[0227] vii) an anti-AXL CAR comprising an ABD that specifically binds AXL;

[0228] viii) an anti-B7-1H3 CAR comprising an ABD that specifically binds 1B7-H3;

[0229] ix) an anti-B7-H4 CAR comprising an ABD that specifically binds B7-H4;

[0230] x) an anti-BAFF-R CAR comprising an ABD that specifically binds BAFF-R;

[0231] xi) an anti-BCMA CAR comprising an ABD that specifically binds BCMA;

[0232] xii) an anti-BSG CAR comprising an ABD that specifically binds BSG;

[0233] xiii) an anti-CAIX CAR comprising an ABD that specifically binds CAIX;

[0234] xiv) an anti-CD123 CAR comprising an ABD that specifically binds CD123;

[0235] xv) an anti-CD133 CAR comprising an ABD that specifically binds CD133;

[0236] xvi) an anti-CD138 CAR comprising an ABD that specifically binds CD138;

[0237] xvii) an anti-CD19 CAR comprising an ABD that specifically binds CD19;

[0238] xviii) an anti-CD1A CAR comprising an ABD that specifically binds CD1A;

[0239] xix) an anti-CD2 CAR comprising an ABD that specifically binds CD2;

[0240] xx) an anti-CD20 CAR comprising an ABD that specifically binds CD20;

[0241] xxi) an anti-CD22 CAR comprising an ABD that specifically binds CD22;

[0242] xxii) an anti-CD24 CAR comprising an ABD that specifically binds CD24;

[0243] xxiii) an anti-CD276 CAR comprising an ABD that specifically binds CD276;

[0244] xxiv) an anti-CD3 CAR comprising an ABD that specifically binds CD3;

[0245] xxv) an anti-CD30 CAR comprising an ABD that specifically binds CD30;

[0246] xxvi) an anti-CD33 CAR comprising an ABD that specifically binds CD33;

[0247] xxvii) an anti-CD38 CAR comprising an ABD that specifically binds CD38;

[0248] xxviii) an anti-CD4 CAR comprising an ABD that specifically binds CD4;

[0249] xxix) an anti-CD43 CAR comprising an ABD that specifically binds CD43;

[0250] xxx) an anti-CD44 CAR comprising an ABD that specifically binds CD44;

[0251] xxxi) an anti-CD44v6 CAR comprising an ABD that specifically binds CD44v6; xxxii) an anti-CD45 CAR comprising an ABD that specifically binds CD45;

[0252] xxxiii) an anti-CD5 CAR comprising an ABD that specifically binds CD5;

[0253] xxxiv) an anti-CD56 CAR comprising an ABD that specifically binds CD56;

[0254] xxxv) an anti-CD7 CAR comprising an ABD that specifically binds CD7;

[0255] xxxvi) an anti-CD70 CAR comprising an ABD that specifically binds CD70;

[0256] xxxvii) an anti-CD72 CAR comprising an ABD that specifically binds CD72;

[0257] xxxviii) an anti-CD83 CAR comprising an ABD that specifically binds CD83;

[0258] xxxix) an anti-CD84 CAR comprising an ABD that specifically binds CD84;

[0259] xl) an anti-CD99 CAR comprising an ABD that specifically binds CD99;

[0260] xli) an anti-CDCP1 CAR comprising an ABD that specifically binds CDCP1;

[0261] xlii) an anti-CDH17 CAR comprising an ABD that specifically binds CDH17;

[0262] xliii) an anti-CEA CAR comprising an ABD that specifically binds CEA;

[0263] xliv) an anti-CEACAM5 CAR comprising an ABD that specifically binds CEACAM5;

[0264] xlv) an anti-CLDN18.2 CAR comprising an ABD that specifically binds CLDN18.2;

[0265] xlvi) an anti-CLDN6 CAR comprising an ABD that specifically binds CLDN6;

[0266] xlvii) an anti-CLEC4K (CD207) CAR comprising an ABD that specifically binds CLEC4K (CD207);

[0267] xlviii) an anti-CLL-1 CAR comprising an ABD that specifically binds CLL-1;

[0268] xlix) an anti-CSPG4 CAR comprising an ABD that specifically binds CSPG4;

[0269] l) an anti-DLL3 CAR comprising an ABD that specifically binds DLL3;

[0270] li) an anti-DR5 CAR comprising an ABD that specifically binds DR5;

[0271] lii) an anti-EBV Protein CAR comprising an ABD that specifically binds EBV Protein;

[0272] liii) an anti-EGFR CAR comprising an ABD that specifically binds EGFR;

[0273] liv) an anti-EGFRvIII CAR comprising an ABD that specifically binds EGFRvII;

[0274] lv) an anti-EMR1 CAR comprising an ABD that specifically binds EMR1;

[0275] lvi) an anti-enkephalinase (CD10) CAR comprising an ABD that specifically bindsenkephalinase (CD10);

[0276] lvii) an anti-EpCAM CAR comprising an ABD that specifically binds EpCAM;

[0277] lviii) an anti-EphA2 CAR comprising an ABD that specifically binds EphA2;

[0278] lix) an anti-EphA3 CAR comprising an ABD that specifically binds EphA3;

[0279] lx) an anti-FAP CAR comprising an ABD that specifically binds FAP;

[0280] lxi) an anti-FGFR4 CAR comprising an ABD that specifically binds FGFR4;

[0281] lxii) an anti-FLT3 CAR comprising an ABD that specifically binds FLT3;

[0282] lxiii) an anti-FOLR1 CAR comprising an ABD that specifically binds FOLR1;

[0283] lxiv) an anti-FSHR CAR comprising an ABD that specifically binds FSHR;

[0284] lxv) an anti-GC-C CAR comprising an ABD that specifically binds GC-C;

[0285] lxvi) an anti-GD2 CAR comprising an ABD that specifically binds GD2;

[0286] lxvii) an anti-GFRA4 CAR comprising an ABD that specifically binds GFRA4;

[0287] lxviii) an anti-Globo H CAR comprising an ABD that specifically binds Globo H;

[0288] lxix) an anti-GM2 (ganglioside M2) CAR comprising an ABD that specifically binds GM2 (ganglioside M2);

[0289] lxx) an anti-gp100 CAR comprising an ABD that specifically binds gp100;

[0290] lxxi) an anti-GPC1 CAR comprising an ABD that specifically binds GPC1;

[0291] lxxii) an anti-GPC2 CAR comprising an ABD that specifically binds GPC2;

[0292] lxxiii) an anti-GPC3 CAR comprising an ABD that specifically binds GPC3;

[0293] lxxiv) an anti-HAAH (ASPH) CAR comprising an ABD that specifically binds HAAH (ASPH);

[0294] lxxv) an anti-HER2 CAR comprising an ABD that specifically binds HER2;

[0295] lxxvi) an anti-HGF CAR comprising an ABD that specifically binds HGF;

[0296] lxxvii) an anti-HIV envelope protein gp120 CAR comprising an ABD that specifically binds HIV envelope protein gp120;

[0297] lxxviii) an anti-HIV-1 pol CAR comprising an ABD that specifically binds HIV-1 pol;

[0298] lxxix) an anti-HLA-A2 CAR comprising an ABD that specifically binds HLA-A2;

[0299] lxxx) an anti-HLA-G CAR comprising an ABD that specifically binds HLA-G;

[0300] lxxxi) an anti-HSP70 heat-shock protein CAR comprising an ABD that specifically binds HSP70 heat-shock protein;

[0301] lxxxii) an anti-ICAM-1 CAR comprising an ABD that specifically binds ICAM-1;

[0302] lxxxiii) an anti-IL-10R CAR comprising an ABD that specifically binds IL-10R;

[0303] lxxxiv) an anti-IL-13Rα2 CAR comprising an ABD that specifically binds IL-13Rα2;

[0304] lxxxv) an anti-integrin αvβ6 CAR comprising an ABD that specifically binds integrin αvβ6;

[0305] lxxxvi) an anti-ITGB7 CAR comprising an ABD that specifically binds ITGB7;

[0306] lxxxvii) an anti-KKLC1 CAR comprising an ABD that specifically binds KKLC1;

[0307] lxxxviii) an anti-KMA CAR comprising an ABD that specifically binds KMA;

[0308] lxxxix) an anti-L1CAM CAR comprising an ABD that specifically binds L1CAM;

[0309] xc) an anti-Lewis-Y antigen CAR comprising an ABD that specifically binds Lewis-Y antigen;

[0310] xci) an anti-LGR5 CAR comprising an ABD that specifically binds LGR5;

[0311] xcii) an anti-LILRB4 CAR comprising an ABD that specifically binds LILRB4;

[0312] xciii) an anti-LMP1 CAR comprising an ABD that specifically binds LMP1;

[0313] xciv) an anti-MAGEA3 CAR comprising an ABD that specifically binds MAGEA3;

[0314] xcv) an anti-M-CSF CAR comprising an ABD that specifically binds M-CSF;

[0315] xcvi) an anti-MG7 CAR comprising an ABD that specifically binds MG7;

[0316] xcvii) an anti-MICA CAR comprising an ABD that specifically binds MICA;

[0317] xcviii) an anti-MMP2 CAR comprising an ABD that specifically binds MMP2;

[0318] xcix) an anti-MSLN CAR comprising an ABD that specifically binds MSLN;

[0319] c) an anti-MUC1/MUC16 CAR comprising an ABD that specifically binds MUC1/MUC16;

[0320] ci) an anti-NCR3LG1 CAR comprising an ABD that specifically binds NCR3LG1;

[0321] cii) an anti-NECTIN2 CAR comprising an ABD that specifically binds NECTIN2;

[0322] ciii) an anti-nectin-4 CAR comprising an ABD that specifically binds nectin-4;

[0323] civ) an anti-NKG2D CAR comprising an ABD that specifically binds NKG2D;

[0324] cv) an anti-NKG2DL CAR comprising an ABD that specifically binds NKG2DL;

[0325] cvi) an anti-NR2F6 CAR comprising an ABD that specifically binds NR2F6;

[0326] cvii) an anti-NY-ESO-1 CAR comprising an ABD that specifically binds NY-ESO-1;

[0327] cviii) an anti-OPCML CAR comprising an ABD that specifically binds OPCML;

[0328] cix) an anti-PD-1 CAR comprising an ABD that specifically binds PD-1;

[0329] cx) an anti-PDGFRα CAR comprising an ABD that specifically binds PDGFRα;

[0330] cxi) an anti-PDL1 CAR comprising an ABD that specifically binds PDL1;

[0331] cxii) an anti-PLA2R CAR comprising an ABD that specifically binds PLA2R;

[0332] cxiii) an anti-PR1 CAR comprising an ABD that specifically binds PR1;

[0333] cxiv) an anti-PSCA CAR comprising an ABD that specifically binds PSCA;

[0334] cxv) an anti-PTK7 CAR comprising an ABD that specifically binds PTK7;

[0335] cxvi) an anti-PVR CAR comprising an ABD that specifically binds PVR;

[0336] cxvii) an anti-ROBO1 CAR comprising an ABD that specifically binds ROBO1;

[0337] cxviii) an anti-ROR1 CAR comprising an ABD that specifically binds ROR1;

[0338] cxix) an anti-ROR2 CAR comprising an ABD that specifically binds ROR2;

[0339] cxx) an anti-SEMA4A CAR comprising an ABD that specifically binds SEMA4A;

[0340] cxxi) an anti-SLAMF7 CAR comprising an ABD that specifically binds SLAMF7;

[0341] cxxii) an anti-TAG-72 CAR comprising an ABD that specifically binds TAG-72;

[0342] cxxiii) an anti-TGF-β CAR comprising an ABD that specifically binds TGF-β;

[0343] cxxiv) an anti-TM4SF1 CAR comprising an ABD that specifically binds TM4SF1;

[0344] cxxv) an anti-TRAIL CAR comprising an ABD that specifically binds TRAIL;

[0345] cxxvi) an anti-TRBC1 CAR comprising an ABD that specifically binds TRBC1;

[0346] cxxvii) an anti-TRBC2 CAR comprising an ABD that specifically binds TRBC2;

[0347] cxxviii) an anti-Trop-2 CAR comprising an ABD that specifically binds Trop-2;

[0348] cxxix) an anti-TSHR CAR comprising an ABD that specifically binds TSHR;

[0349] cxxx) an anti-TSLPR CAR comprising an ABD that specifically binds TSLPR;

[0350] cxxxi) an anti-ULBP1 CAR comprising an ABD that specifically binds ULBP1;

[0351] cxxxii) an anti-αvβ3 integrin CAR comprising an ABD that specifically binds αvβ3 integrin; and cxxxiii) an anti-K light chain of human immunoglobulin CAR comprising an ABD that specifically binds K light chain of human immunoglobulin.

[0352] 2. The nucleic acid of embodiment 1, wherein the CAR is selected from the group consisting of: a) inaticabtagene autoleucel; b) actalycabtagene autoleucel; c) relmacabtagene autoleucel; d) lisocabtagene maraleucel; e) brexucabtagene autoleucel; f) axicabtagene ciloleucel; g) tisagenlecleucel; h) obecabatagene autoleucel; i) azercabtagene zapreleucel; j) rapcabtagene autoleucel; and k) idecabtagene vicleucel.

[0353] 3. The nucleic acid of embodiment 1, wherein the CAR is an anti-EGFRvIII CAR comprising a means for specifically binding EGFRvIII.

[0354] 4. The nucleic acid of embodiment 3, wherein the anti-EGFRvIII CAR comprises an antigen binding domain comprising one or more of light chain complementary determining region 1 (LC CDR1), light chain complementary determining region 2 (LC CDR2), and light chain complementary determining region 3 (LC CDR3) of SEQ ID NO:110, and one or more of heavy chain complementary determining region 1 (HC

CDR1), heavy chain complementary determining region 2 (HC CDR2), and heavy chain complementary determining region 3 (HC CDR3) of SEQ ID NO:111.

[0355] 5. The nucleic acid of embodiment 4, wherein the antigen binding domain comprises LC CDR1 (SEQ ID NO:112), LC CDR2 (SEQ ID NO:113), and LC CDR3 (SEQ ID NO:114); and HC CDR1 (SEQ ID NO:115), HC CDR2 (SEQ ID NO:116), and HC CDR3 (SEQ ID NO:117).

[0356] 6. The nucleic acid of embodiment 5, wherein the antigen binding domain comprises a light chain variable region comprising sequence SEQ ID NO:110 and a heavy chain variable region comprising sequence SEQ ID NO:111.

[0357] 7. The nucleic acid of any one of the preceding embodiments, wherein the DR-18 polypeptide comprises the following amino acid sequence: XFGKXESXLSVIRNLNDQVLFIDQGNRPLFEDMTDSDXRDNAPRTIFIISXYXDXXXRXXAVTISV KXEKISTLSXXNKIISFKEMNPPDNIKDTKSDIIFFXRXVPGHXXKXQFESSSYEGYFLAXEKERD LFKLILKKEDELGDRSIMFTXQXED (SEQ ID NO: 66), wherein the X at pos. 1 is Y, R or H; the X at pos. 5 is L, H, I or Y; the X at pos. 8 is K, Q or R; the X at pos. 38 is C or S; the X at pos. 51 is M, T, K, D, N, E or R; the X at pos. 53 is K, R, G, S or T; the X at pos. 55 is S, K or R; the X at pos. 56 is Q, E, A, R, V, G, K, L or R; the X at pos. 57 is P, L, G, A or K; the X at pos. 59 is G, A or T; the X at pos. 60 is M, K, Q, R or L; the X at pos. 68 is C, S, G, A, V, D, E or N; the X at pos. 76 is C or S; the X at pos. 77 is E or D; the X at pos. 103 is Q, E, K, P, A or R; the X at pos. 105 is S, D, N, R, K or A; the X at pos. 110 is D, K, H, N, Q, E, S or G; the X at pos. 111 is N, H, Y, D, R, S or G; the X at pos. 113 is M, V, R, T or K; the X at pos. 127 is C or S; the X at pos. 153 is V, I, T or A; and the X at pos. 155 is N, K or H.

[0358] 8. The nucleic acid of any one of the preceding embodiments, wherein the sequence encoding the DR-18 polypeptide is codon optimized for expression in humans.

[0359] 9. The nucleic acid of embodiment 8, wherein the sequence encoding the DR-18 polypeptide comprises a DR-18 encoding sequence selected from SEQ ID NOs: 67-84.

[0360] 10. The nucleic acid of any one of the preceding embodiments, wherein the nucleic acid is configured for in vivo expression of the DR-18 polypeptide and CAR.

[0361] 11. The nucleic acid of any one of the preceding embodiments, further comprising a sequence encoding a cytokine selected from the group consisting of: interleukin 2, interleukin

[0362] 21, interleukin 23, interleukin 27, interleukin 33, tumor necrosis factor (TNF), TNF ligand superfamily member 15, interferon (IFN) alpha, IFN beta, and IFN gamma.

[0363] 12. The nucleic acid of any one of the preceding embodiments, comprising a plurality of CAR sequences encoding a plurality of different CARs that bind different antigens.

[0364] 13. The nucleic acid of embodiment 12, wherein the plurality of CAR sequences encode a plurality of different CARs that bind to a combination of antigens selected from the group consisting of: CD19 and CD20, CD19 and CD22, BCMA and CD19, CD33 and CLL-1, CD19 and CD276, CD19 and CD20 and CD22, CD19 and IL15R, IL15R and MUC16, IL-15Rα and NKG2D, GPC3 and IL15R, CD16a and CD19 and IL-15Rα, BCMA and CD16a and NKp46, CD123 and CD33, CLDN18.2 and NKG2D, CD20 and CD22, BCMA and SLAMF7, CD19 and CD70, CD19 and IL18R1, GD2 and IL15R, CD70 and IL15R, CD19 and CD8, CD20 and CD79A, CD19 and CD22 and CD8, CD19 and EBV Protein, CD19 and CD7, BCMA and CD38, IL-12 and MUC16, CD33 and FLT3, CD19 and IL-2RP, BCMA and CD70, EGFR and IL-13Rα2, HER2 and IL15R, MICA and MICB, BCMA and TACI, GPC3 and IL-2RP, CD19 and DR5, CD123 and CD33 and CLL-1, CD3 and CD7, CEACAM5 and CEACAM6, CD20 and CD22 and CD38, CD19 and CLDN18.2, AGRE2 and CLL-1, CD123 and TIM3, CLDN18.2 and PDL1, CTLA4 and MSLN and PD-1, STEAP2 and TGFBR2, CEA and IL-15Rα and IL-21R, CD123 and CD33 and CD38 and CLL-1, CD133 and EGFR, BCMA and HPK1, CD8A and NY-ESO-1, CD38 and DR5, CD19 and EGFR, BCMA and CD19 and HER2 and Trop-2, CD19 and MUC1, PDL1 and VEGFR1, CD123 and CD33 and CD38 and CD56 and CLL-1 and MUC1, BCMA and CD138 and CD19 and CD38, BCMA and CD16a and IL-15Rα, NCR2 and NKG2D, CD38 and CLL-1, and CD276 and FGFR4.

[0365] 14. An expression vector comprising the nucleic acid of any one of the preceding embodiments.

[0366] 15. The expression vector of embodiment 14, wherein the expression vector comprises, in operable linkage to the sequence encoding the DR-18 polypeptide, a sequence encoding a signal peptide selected from the group consisting of: Human OSM signal peptide, VSV-G signal peptide, Mouse Ig Kappa signal peptide, Mouse Ig Heavy signal peptide, BM40 signal peptide, Secrecon signal peptide, Human IgKVIII signal peptide, CD33 signal peptide, tPA signal peptide, Human Chymotrypsinogen signal peptide, Human trypsinogen-2 signal peptide, Human IL-2 signal peptide, Gaussia luc signal peptide, Albumin (HSA) signal peptide, Influenza Haemagglutinin signal peptide, Human insulin signal peptide, Silkworm Fibroin LC signal peptide, Interleukin-18-binding protein signal peptide, Interleukin-12 subunit beta signal peptide, Interleukin-12 subunit alpha signal peptide, Interleukin-18 signal peptide, and CD8 signal peptide.

[0367] 16. The expression vector of embodiment 14 or 15, wherein the expression vector comprises, in operable linkage to the sequence encoding the DR-18 polypeptide, a promoter sequence selected from the group consisting of: cytomegalovirus (CMV) promoter, elongation factor 1 alpha (EF1a) promoter, simian virus 40 (SV40) promoter, phosphoglycerate kinase (PGK) ubiquitin C (UBC) promoter, human beta actin promoter, CAG promoter, LC immunoglobulin promoter, HC immunoglobulin promoter, herpes simplex virus thymidine kinase promoter, and spleen focus-forming virus (SFFV) promoter.

[0368] 17. The expression vector of embodiment 16, wherein the expression vector comprises, in operable linkage to the sequence encoding the DR-18 polypeptide, a human EF1a promoter.

[0369] 18. The expression vector of any one of embodiments 14-17, wherein the sequence encoding the DR-18 polypeptide and the sequence encoding the CAR are operably linked by a multicistronic element.

[0370] 19. The expression vector of embodiment 18, wherein the multicistronic element is an internal ribosome entry site.

[0371] 20. The expression vector of embodiment 18, wherein the multicistronic element is a sequence encoding 2A peptide selected from the group consisting of: T2A, P2A, E2A, and F2A.

[0372] 21. A cell comprising the nucleic acid or expression vector of any one of the preceding embodiments.

[0373] 22. The cell of embodiment 21, wherein the cell is selected from the group consisting of: a T cell, a Naïve T cell (TN), stem cell memory T cell (T_{SCM}), a central memory T cell (T_{CM}), a resident memory T cell (TRM), an effector T cell (TEFF), an effector memory cell (TEM), an alpha-beta ($\alpha\beta$) T cell, a gamma-delta ($\gamma\alpha$) T cell, a Natural Killer (NK) cell, a CD4+ T cell, a CD4+Th1 cell, a CD4+Th2 cell, a CD4+Th9 cell, a CD4+Th17 cell, a CD4+Th22 cell, a CD4+T Regulatory Cell (Treg), a CD4+T follicular helper cell (Tfh), a resting Treg cell, an effector Treg cell, an effector memory Treg cell, a CD8+ T cell, a CD8+Tc1 cell, a CD8+Tc2 cell, a CD8+Tc9 cell, a CD8+Th17 cell, a CD8+Treg cell, a macrophage, a B Cell, a dendritic cell, a stem cell, a hematopoietic stem cell (HSC), and an induced pluripotent stem cell (iPS cell).

[0374] 23. A method of treating a subject for cancer, the method comprising administering to the subject an effective amount of a nucleic acid, an expression vector, or a cell of any one of the preceding embodiments.

[0375] 24. The method of embodiment 23, wherein the cancer is selected from the group consisting of: Acute Lymphoblastic Leukemia (ALL), Acute Myeloid Leukemia (AML), Adrenocortical Carcinoma, an AIDS-Related Cancer, Anal Cancer, Appendix Cancer, Astrocytomas, Atypical Teratoid/Rhabdoid Tumor, Basal Cell Carcinoma, Bile Duct Cancer, Bladder Cancer, Bone Cancer, Brain Stem Glioma, Brain Tumor, Breast Cancer, Bronchial Tumors, Burkitt Lymphoma, Carcinoid Tumor, Carcinoma of Unknown Primary, Cardiac (Heart) Tumors, Central Nervous System Tumor, Cervical Cancer, Childhood Cancers, Chordoma, Chronic Lymphocytic Leukemia (CLL), Chronic Myelogenous Leukemia (CML), Chronic Myeloproliferative Neoplasms, Colon Cancer, Colorectal Cancer, Craniopharyngioma, Cutaneous T-Cell Lymphoma, Duct Cancer, Ductal Carcinoma In Situ (DCIS), Embryonal Tumors, Endometrial Cancer, Ependymoma, Esophageal Cancer, Esthesioneuroblastoma, Ewing Sarcoma, Extracranial Germ Cell Tumor, Extragonadal Germ Cell Tumor, Extrahepatic Bile Duct Cancer, Eye Cancer, Fibrous Histiocytoma of Bone, Gallbladder Cancer, Gastric (Stomach) Cancer, Gastrointestinal Carcinoid Tumor, Gastrointestinal Stromal Tumors (GIST), Germ Cell Tumor, Gestational Trophoblastic Disease, Glioma, Hairy Cell Leukemia, Head and Neck Cancer, Heart Cancer, Hepatocellular (Liver) Cancer, Histiocytosis, Hodgkin Lymphoma, Hypopharyngeal Cancer, Intraocular Melanoma, Islet Cell Tumors, Kaposi Sarcoma, Kidney Cancer, Langerhans Cell Histiocytosis, Laryngeal Cancer, Leukemia, Lip and Oral Cavity Cancer, Liver Cancer (Primary), Lobular Carcinoma In Situ (LCIS), Lung Cancer, Lymphoma, Macroglobulinemia, Male Breast Cancer, Malignant Fibrous Histiocytoma of Bone and Osteosarcoma, Melanoma, Merkel Cell Carcinoma, Mesothelioma, Metastatic Squamous Neck Cancer with Occult Primary, Midline Tract Carcinoma Involving NUT Gene, Mouth Cancer, Multiple Endocrine Neoplasia Syndromes, Multiple Myeloma/Plasma Cell Neoplasm, Mycosis Fungoides, Myelodysplastic Syndromes, Myelodysplastic/Myeloproliferative Neoplasms, Myelogenous Leukemia, Myeloid Leukemia, Myeloproliferative Neoplasms, Nasal Cavity and Paranasal Sinus Cancer, Nasopharyngeal Cancer, Neuroblastoma, Non-Hodgkin Lymphoma, Non-Small Cell Lung Cancer, Oral Cancer, Oral Cavity Cancer, Oropharyngeal Cancer, Osteosarcoma and Malignant Fibrous Histiocytoma of Bone, Ovarian Cancer, Pancreatic Cancer, Pancreatic Neuroendocrine Tumors, Papillomatosis, Paraganglioma, Paranasal Sinus and Nasal Cavity Cancer, Parathyroid Cancer, Penile Cancer, Pharyngeal Cancer, Pheochromocytoma, Pituitary Tumor, Pleuropulmonary Blastoma, Primary Central Nervous System (CNS) Lymphoma, Prostate Cancer, Rectal Cancer, Renal Cell (Kidney) Cancer, Renal Pelvis and Ureter, Transitional Cell Cancer, Retinoblastoma, Rhabdomyosarcoma, Salivary Gland Cancer, Sarcoma, Sezary Syndrome, Skin Cancer, Small Cell Lung Cancer, Small Intestine Cancer, Soft Tissue Sarcoma, Squamous Cell Carcinoma, Squamous Neck Cancer, Stomach (Gastric) Cancer, T-Cell Lymphoma, Testicular Cancer, Throat Cancer, Thymoma and Thymic Carcinoma, Thyroid Cancer, Transitional Cell Cancer of the Renal Pelvis and Ureter, Ureter and Renal Pelvis Cancer, Urethral Cancer, Uterine Cancer, Uterine Sarcoma, Vaginal Cancer, Vulvar Cancer, Waldenstrom Macroglobulinemia, and Wilms Tumor.

[0376] 25. The method of embodiment 23 or 24, wherein the subject does not undergo lymphodepletion during or prior to the administering.

[0377] 26. The method of any one of embodiments 23-25, wherein the amount of the nucleic acid, the expression vector, or the cell administered to the subject is less than the amount of a corresponding nucleic acid, expression vector, or cell comprising the sequence encoding the CAR administered to the subject to treat the cancer without the sequence encoding the DR-18 polypeptide.

[0378] 27. The method of any one of embodiments 23-26, wherein the administering comprises delivering multiple doses of the nucleic acid, the expression vector, or the cell to the subject and the dosing frequency is less than the dosing frequency of a corresponding nucleic acid, expression vector, or cell comprising the sequence encoding the CAR administered to the subject to treat the cancer without the sequence encoding the DR-18 polypeptide.

[0379] 28. A method, the method comprising contacting a human cell with the nucleic acid or expression vector of any one of embodiments 1-20 under conditions sufficient for delivery of the nucleic acid or expression vector into the human cell.

[0380] 29. The method of embodiment 28, wherein the contacting is performed under conditions sufficient for the expression by the human cell of the CAR from the CAR encoding sequence.

[0381] 30. The method of embodiment 28 or 29, wherein the contacting is performed under conditions sufficient for the expression by the human cell of the CAR from the CAR coding sequence, the DR-18 polypeptide from the DR-18 polypeptide coding sequence, or both.

[0382] 31. A nucleic acid, or expression vector, comprising or consisting of the nucleic acid, comprising a sequence encoding a decoy-resistant interleukin 18 (DR-18) polypeptide, wherein the sequence encoding the DR-18 polypeptide is codon optimized for expression in human cells.

[0383] 32. The nucleic acid or expression vector of embodiment 31, wherein the nucleic acid is configured for in vivo expression of the DR-18 polypeptide.

[0384] 33. The nucleic acid or expression vector of embodiment 31 or 32, wherein the DR-18 polypeptide comprises the following amino acid sequence: XFGKXESXLSVIRNLNDQVLFIDQGNR-PLFEDMTDSDXRDNAPRTI-FIISXYXDXXXRXXAVTISV KXEKISTLSXXNKI-ISFKEMNPPDNIKDTKSDIIFFXRXVPGHXXKXQ FESSSYEGYFLAXEKERD LFKLILKKEDELGDR-SIMFTXQXED (SEQ ID NO: 66), wherein the X at pos. 1 is Y, R or H; the X at pos. 5 is L, H, I or Y; the X at pos. 8 is K, Q or R; the X at pos. 38 is C or S; the X at pos. 51 is M, T, K, D, N, E or R; the X at pos. 53 is K, R, G, S or T; the X at pos. 55 is S, K or R; the X at pos. 56 is Q, E, A, R, V, G, K, L or R; the X at pos. 57 is P, L, G, A or K; the X at pos. 59 is G, A or T; the X at pos. 60 is M, K, Q, R or L; the X at pos. 68 is C, S, G, A, V, D, E or N; the X at pos. 76 is C or S; the X at pos. 77 is E or D; the X at pos. 103 is Q, E, K, P, A or R; the X at pos. 105 is S, D, N, R, K or A; the X at pos. 110 is D, K, H, N, Q, E, S or G; the X at pos. 111 is N, H, Y, D, R, S or G; the X at pos. 113 is M, V, R, T or K; the X at pos. 127 is C or S; the X at pos. 153 is V, I, T or A; and the X at pos. 155 is N, K or H. 34. The nucleic acid or expression vector of any one of embodiments 31-33, wherein the sequence encoding the DR-18 polypeptide comprises a DR-18 encoding sequence selected from SEQ ID NOs: 67-84.

[0385] 35. The nucleic acid or expression vector of any one of embodiments 31-34, further comprising a sequence encoding a cytokine selected from the group consisting of: interleukin 2, interleukin 21, interleukin 23, interleukin 27, interleukin 33, tumor necrosis factor (TNF), TNF ligand superfamily member 15, interferon (IFN) alpha, IFN beta, and IFN gamma.

[0386] 36. The nucleic acid or expression vector of any one of embodiments 31-35, further comprising, in operable linkage to the sequence encoding the DR-18 polypeptide, a sequence encoding a signal peptide selected from the group consisting of: Human OSM signal peptide, VSV-G signal peptide, Mouse Ig Kappa signal peptide, Mouse Ig Heavy signal peptide, BM40 signal peptide, Secrecon signal peptide, Human IgKVIII signal peptide, CD33 signal peptide, tPA signal peptide, Human Chymotrypsinogen signal peptide, Human trypsinogen-2 signal peptide, Human IL-2 signal peptide, Gaussia luc signal peptide, Albumin (HSA) signal peptide, Influenza Haemagglutinin signal peptide, Human insulin signal peptide, Silkworm Fibroin LC signal peptide, Interleukin-18-binding protein signal peptide, Interleukin-12 subunit beta signal peptide, Interleukin-12 subunit alpha signal peptide, Interleukin-18 signal peptide, and CD8 signal peptide.

[0387] 37. The nucleic acid or expression vector of any one of embodiments 31-36, further comprising, in operable linkage to the sequence encoding the DR-18 polypeptide, a promoter sequence selected from the group consisting of: cytomegalovirus (CMV) promoter, elongation factor 1 alpha (EF1a) promoter, simian virus 40 (SV40) promoter, phosphoglycerate kinase (PGK) ubiquitin C (UBC) promoter, human beta actin promoter, CAG promoter, LC immunoglobulin promoter, HC immunoglobulin promoter, herpes simplex virus thymidine kinase promoter, and spleen focus-forming virus (SFFV) promoter.

[0388] 38. The nucleic acid or expression vector of any one of embodiments 31-37, further comprising a chimeric antigen receptor (CAR) sequence encoding a CAR, wherein the CAR is selected from the group consisting of:

[0389] i) an anti-5T4 CAR comprising an antigen binding domain (ABD) that specifically binds 5T4; ii) an anti-AFP CAR comprising an ABD that specifically binds AFP;

[0390] iii) an anti-ALPP CAR comprising an ABD that specifically binds ALPP;

[0391] iv) an anti-ALPPL2 CAR comprising an ABD that specifically binds ALPPL2;

[0392] v) an anti-APN (CD13) CAR comprising an ABD that specifically binds APN (CD13);

[0393] vi) an anti-APRIL CAR comprising an ABD that specifically binds APRIL;

[0394] vii) an anti-AXL CAR comprising an ABD that specifically binds AXL;

[0395] viii) an anti-B7-H3 CAR comprising an ABD that specifically binds B7-H3;

[0396] ix) an anti-B7-H4 CAR comprising an ABD that specifically binds B7-H4;

[0397] x) an anti-BAFF-R CAR comprising an ABD that specifically binds BAFF-R;

[0398] xi) an anti-BCMA CAR comprising an ABD that specifically binds BCMA;

[0399] xii) an anti-BSG CAR comprising an ABD that specifically binds BSG;

[0400] xiii) an anti-CAIX CAR comprising an ABD that specifically binds CAIX;

[0401] xiv) an anti-CD123 CAR comprising an ABD that specifically binds CD123;

[0402] xv) an anti-CD133 CAR comprising an ABD that specifically binds CD133;

[0403] xvi) an anti-CD138 CAR comprising an ABD that specifically binds CD138;

[0404] xvii) an anti-CD19 CAR comprising an ABD that specifically binds CD19;

[0405] xviii) an anti-CD1A CAR comprising an ABD that specifically binds CD1A;

[0406] xix) an anti-CD2 CAR comprising an ABD that specifically binds CD2;

[0407] xx) an anti-CD20 CAR comprising an ABD that specifically binds CD20;
[0408] xxi) an anti-CD22 CAR comprising an ABD that specifically binds CD22;
[0409] xxii) an anti-CD24 CAR comprising an ABD that specifically binds CD24;
[0410] xxiii) an anti-CD276 CAR comprising an ABD that specifically binds CD276;
[0411] xxiv) an anti-CD3 CAR comprising an ABD that specifically binds CD3;
[0412] xxv) an anti-CD30 CAR comprising an ABD that specifically binds CD30;
[0413] xxvi) an anti-CD33 CAR comprising an ABD that specifically binds CD33;
[0414] xxvii) an anti-CD38 CAR comprising an ABD that specifically binds CD38;
[0415] xxviii) an anti-CD4 CAR comprising an ABD that specifically binds CD4;
[0416] xxix) an anti-CD43 CAR comprising an ABD that specifically binds CD43;
[0417] xxx) an anti-CD44 CAR comprising an ABD that specifically binds CD44;
[0418] xxxi) an anti-CD44v6 CAR comprising an ABD that specifically binds CD44v6;
[0419] xxxii) an anti-CD45 CAR comprising an ABD that specifically binds CD45;
[0420] xxxiii) an anti-CD5 CAR comprising an ABD that specifically binds CD5;
[0421] xxxiv) an anti-CD56 CAR comprising an ABD that specifically binds CD56;
[0422] xxxv) an anti-CD7 CAR comprising an ABD that specifically binds CD7;
[0423] xxxvi) an anti-CD70 CAR comprising an ABD that specifically binds CD70;
[0424] xxxvii) an anti-CD72 CAR comprising an ABD that specifically binds CD72;
[0425] xxxviii) an anti-CD83 CAR comprising an ABD that specifically binds CD83;
[0426] xxxix) an anti-CD84 CAR comprising an ABD that specifically binds CD84;
[0427] xl) an anti-CD99 CAR comprising an ABD that specifically binds CD99;
[0428] xli) an anti-CDCP1 CAR comprising an ABD that specifically binds CDCP1;
[0429] xlii) an anti-CDH17 CAR comprising an ABD that specifically binds CDH17;
[0430] xliii) an anti-CEA CAR comprising an ABD that specifically binds CEA;
[0431] xliv) an anti-CEACAM5 CAR comprising an ABD that specifically binds CEACAM5;
[0432] xlv) an anti-CLDN18.2 CAR comprising an ABD that specifically binds CLDN18.2;
[0433] xlvi) an anti-CLDN6 CAR comprising an ABD that specifically binds CLDN6;
[0434] xlvii) an anti-CLEC4K (CD207) CAR comprising an ABD that specifically binds CLEC4K (CD207);
[0435] xlviii) an anti-CLL-1 CAR comprising an ABD that specifically binds CLL-1;
[0436] xlix) an anti-CSPG4 CAR comprising an ABD that specifically binds CSPG4;
[0437] l) an anti-DLL3 CAR comprising an ABD that specifically binds DLL3;
[0438] li) an anti-DR5 CAR comprising an ABD that specifically binds DR5;
[0439] lii) an anti-EBV Protein CAR comprising an ABD that specifically binds EBV Protein;
[0440] liii) an anti-EGFR CAR comprising an ABD that specifically binds EGFR;
[0441] liv) an anti-EGFRvIII CAR comprising an ABD that specifically binds EGFRvII;
[0442] lv) an anti-EMR1 CAR comprising an ABD that specifically binds EMR1;
[0443] lvi) an anti-enkephalinase (CD10) CAR comprising an ABD that specifically binds enkephalinase (CD10);
[0444] lvii) an anti-EpCAM CAR comprising an ABD that specifically binds EpCAM;
[0445] lviii) an anti-EphA2 CAR comprising an ABD that specifically binds EphA2;
[0446] lix) an anti-EphA3 CAR comprising an ABD that specifically binds EphA3;
[0447] lx) an anti-FAP CAR comprising an ABD that specifically binds FAP;
[0448] lxi) an anti-FGFR4 CAR comprising an ABD that specifically binds FGFR4;
[0449] lxii) an anti-FLT3 CAR comprising an ABD that specifically binds FLT3;
[0450] lxiii) an anti-FOLR1 CAR comprising an ABD that specifically binds FOLR1;
[0451] lxiv) an anti-FSHR CAR comprising an ABD that specifically binds FSHR;
[0452] lxv) an anti-GC-C CAR comprising an ABD that specifically binds GC-C;
[0453] lxvi) an anti-GD2 CAR comprising an ABD that specifically binds GD2;
[0454] lxvii) an anti-GFRA4 CAR comprising an ABD that specifically binds GFRA4;
[0455] lxviii) an anti-Globo H CAR comprising an ABD that specifically binds Globo H;
[0456] lxix) an anti-GM2 (ganglioside M2) CAR comprising an ABD that specifically binds GM2 (ganglioside M2);
[0457] lxx) an anti-gp100 CAR comprising an ABD that specifically binds gp100;
[0458] lxxi) an anti-GPC1 CAR comprising an ABD that specifically binds GPC1;
[0459] lxxii) an anti-GPC2 CAR comprising an ABD that specifically binds GPC2;
[0460] lxxiii) an anti-GPC3 CAR comprising an ABD that specifically binds GPC3;
[0461] lxxiv) an anti-HAAH (ASPH) CAR comprising an ABD that specifically binds HAAH (ASPH);
[0462] lxxv) an anti-HER2 CAR comprising an ABD that specifically binds HER2;
[0463] lxxvi) an anti-HGF CAR comprising an ABD that specifically binds HGF;
[0464] lxxvii) an anti-HIV envelope protein gpl20 CAR comprising an ABD that specifically binds HIV envelope protein gpl20;
[0465] lxxviii) an anti-HIV-1 pol CAR comprising an ABD that specifically binds HIV-1 pol;
[0466] lxxix) an anti-HLA-A2 CAR comprising an ABD that specifically binds HLA-A2;
[0467] lxxx) an anti-HLA-G CAR comprising an ABD that specifically binds HLA-G;
[0468] lxxxi) an anti-HSP70 heat-shock protein CAR comprising an ABD that specifically binds HSP70 heat-shock protein;

[0469] lxxxii) an anti-ICAM-1 CAR comprising an ABD that specifically binds ICAM-1;

[0470] lxxxiii) an anti-IL-10R CAR comprising an ABD that specifically binds IL-10R;

[0471] lxxxiv) an anti-IL-13Rα2 CAR comprising an ABD that specifically binds IL-13Rα2;

[0472] lxxxv) an anti-integrin αvβ6 CAR comprising an ABD that specifically binds integrin αvβ6;

[0473] lxxxvi) an anti-ITGB7 CAR comprising an ABD that specifically binds ITGB7;

[0474] lxxxvii) an anti-KKLC1 CAR comprising an ABD that specifically binds KKLC1;

[0475] lxxxviii) an anti-KMA CAR comprising an ABD that specifically binds KMA;

[0476] lxxxix) an anti-L1CAM CAR comprising an ABD that specifically binds L1CAM;

[0477] xc) an anti-Lewis-Y antigen CAR comprising an ABD that specifically binds Lewis-Y antigen;

[0478] xci) an anti-LGR5 CAR comprising an ABD that specifically binds LGR5;

[0479] xcii) an anti-LILRB4 CAR comprising an ABD that specifically binds LILRB4;

[0480] xciii) an anti-LMP1 CAR comprising an ABD that specifically binds LMP1;

[0481] xciv) an anti-MAGEA3 CAR comprising an ABD that specifically binds MAGEA3;

[0482] xcv) an anti-M-CSF CAR comprising an ABD that specifically binds M-CSF;

[0483] xcvi) an anti-MG7 CAR comprising an ABD that specifically binds MG7;

[0484] xcvii) an anti-MICA CAR comprising an ABD that specifically binds MICA;

[0485] xcviii) an anti-MMP2 CAR comprising an ABD that specifically binds MMP2;

[0486] xcix) an anti-MSLN CAR comprising an ABD that specifically binds MSLN;

[0487] c) an anti-MUC1/MUC16 CAR comprising an ABD that specifically binds MUC1/MUC16; ci) an anti-NCR3LG1 CAR comprising an ABD that specifically binds NCR3LG1; cii) an anti-NECTIN2 CAR comprising an ABD that specifically binds NECTIN2; ciii) an anti-nectin-4 CAR comprising an ABD that specifically binds nectin-4; civ) an anti-NKG2D CAR comprising an ABD that specifically binds NKG2D;

[0488] cv) an anti-NKG2DL CAR comprising an ABD that specifically binds NKG2DL;

[0489] cvi) an anti-NR2F6 CAR comprising an ABD that specifically binds NR2F6;

[0490] cvii) an anti-NY-ESO-1 CAR comprising an ABD that specifically binds NY-ESO-1;

[0491] cviii) an anti-OPCML CAR comprising an ABD that specifically binds OPCML;

[0492] cix) an anti-PD-1 CAR comprising an ABD that specifically binds PD-1;

[0493] cx) an anti-PDGFRα CAR comprising an ABD that specifically binds PDGFRα;

[0494] cxi) an anti-PDL1 CAR comprising an ABD that specifically binds PDL1;

[0495] cxii) an anti-PLA2R CAR comprising an ABD that specifically binds PLA2R;

[0496] cxiii) an anti-PR1 CAR comprising an ABD that specifically binds PR1;

[0497] cxiv) an anti-PSCA CAR comprising an ABD that specifically binds PSCA;

[0498] cxv) an anti-PTK7 CAR comprising an ABD that specifically binds PTK7;

[0499] cxvi) an anti-PVR CAR comprising an ABD that specifically binds PVR;

[0500] cxvii) an anti-ROBO1 CAR comprising an ABD that specifically binds ROBO1;

[0501] cxviii) an anti-ROR1 CAR comprising an ABD that specifically binds ROR1;

[0502] cxix) an anti-ROR2 CAR comprising an ABD that specifically binds ROR2;

[0503] cxx) an anti-SEMA4A CAR comprising an ABD that specifically binds SEMA4A;

[0504] cxxi) an anti-SLAMF7 CAR comprising an ABD that specifically binds SLAMF7;

[0505] cxxii) an anti-TAG-72 CAR comprising an ABD that specifically binds TAG-72;

[0506] cxxiii) an anti-TGF-β CAR comprising an ABD that specifically binds TGF-β;

[0507] cxxiv) an anti-TM4SF1 CAR comprising an ABD that specifically binds TM4SF1;

[0508] cxxv) an anti-TRAIL CAR comprising an ABD that specifically binds TRAIL;

[0509] cxxvi) an anti-TRBC1 CAR comprising an ABD that specifically binds TRBC1;

[0510] cxxvii) an anti-TRBC2 CAR comprising an ABD that specifically binds TRBC2;

[0511] cxxviii) an anti-Trop-2 CAR comprising an ABD that specifically binds Trop-2;

[0512] cxxix) an anti-TSHR CAR comprising an ABD that specifically binds TSHR;

[0513] cxxx) an anti-TSLPR CAR comprising an ABD that specifically binds TSLPR;

[0514] cxxxi) an anti-ULBP1 CAR comprising an ABD that specifically binds ULBP1;

[0515] cxxxii) an anti-αvβ3 integrin CAR comprising an ABD that specifically binds αvβ3 integrin; and

[0516] cxxxiii) an anti-K light chain of human immunoglobulin CAR comprising an ABD that specifically binds K light chain of human immunoglobulin.

[0517] 39. The nucleic acid or expression vector of embodiment 38, wherein the CAR is selected from the group consisting of: a) inaticabtagene autoleucel; b) actalycabtagene autoleucel; c) relmacabtagene autoleucel; d) lisocabtagene maraleucel; e) brexucabtagene autoleucel; f) axicabtagene ciloleucel; g) tisagenlecleucel; h) obecabatagene autoleucel; i) azercabtagene zapreleucel; j) rapcabtagene autoleucel; and k) idecabtagene vicleucel.

[0518] 40. The nucleic acid or expression vector of any one of embodiments 31-39, wherein the sequence encoding the CAR and the sequence encoding the DR-18 polypeptide are operably linked by a multicistronic element.

[0519] 41. The nucleic acid or expression vector of embodiment 40, wherein the multicistronic element is an internal ribosome entry site.

[0520] 42. The nucleic acid or expression vector of embodiment 40, wherein the multicistronic element is a sequence encoding 2A peptide selected from the group consisting of: T2A, P2A, E2A, and F2A.

[0521] 43. The nucleic acid or expression vector of any one of embodiments 38-42, wherein the CAR is an anti-EGFRvIII CAR comprising a means for specifically binding EGFRvIII.

[0522] 44. The nucleic acid or expression vector of embodiment 43, wherein the anti-EGFRvIII CAR comprises an antigen binding domain comprising one or more of light chain complementary determining region 1 (LC CDR1), light chain complementary determining region 2 (LC CDR2), and light chain complementary determining region 3 (LC CDR3) of SEQ ID NO: 110, and one or more of heavy chain complementary determining region 1 (HC CDR1), heavy chain complementary determining region 2 (HC CDR2), and heavy chain complementary determining region 3 (HC CDR3) of SEQ ID NO:111.

[0523] 45. The nucleic acid or expression vector of embodiment 44, wherein the antigen binding domain comprises LC CDR1 (SEQ ID NO:112), LC CDR2 (SEQ ID NO:113), and LC CDR3 (SEQ ID NO:114); and HC CDR1 (SEQ ID NO:115), HC CDR2 (SEQ ID NO:116), and HC CDR3 (SEQ ID NO:117).

[0524] 46. The nucleic acid or expression vector of embodiment 45, wherein the antigen binding domain comprises a light chain variable region comprising sequence SEQ ID NO: 110 and a heavy chain variable region comprising sequence SEQ ID NO:111.

[0525] 47. The nucleic acid or expression vector of any one of embodiments 31-46, comprising a plurality of CAR sequences encoding a plurality of different CARs that bind different antigens.

[0526] 48. The nucleic acid or expression vector of embodiment 47, wherein the plurality of CAR sequences encode a plurality of different CARs that bind to a combination of antigens selected from the group consisting of: CD19 and CD20, CD19 and CD22, BCMA and CD19, CD33 and CLL-1, CD19 and CD276, CD19 and CD20 and CD22, CD19 and IL15R, IL15R and MUC16, IL-15Rα and NKG2D, GPC3 and IL15R, CD16a and CD19 and IL-15Rα, BCMA and CD16a and NKp46, CD123 and CD33, CLDN18.2 and NKG2D, CD20 and CD22, BCMA and SLAMF7, CD19 and CD70, CD19 and IL18R1, GD2 and IL15R, CD70 and IL15R, CD19 and CD8, CD20 and CD79A, CD19 and CD22 and CD8, CD19 and EBV Protein, CD19 and CD7, BCMA and CD38, IL-12 and MUC16, CD33 and FLT3, CD19 and IL-2RP, BCMA and CD70, EGFR and IL-13Rα2, HER2 and IL15R, MICA and MICB, BCMA and TACI, GPC3 and IL-2RP, CD19 and DR5, CD123 and CD33 and CLL-1, CD3 and CD7, CEACAM5 and CEACAM6, CD20 and CD22 and CD38, CD19 and CLDN18.2, AGRE2 and CLL-1, CD123 and TIM3, CLDN18.2 and PDL1, CTLA4 and MSLN and PD-1, STEAP2 and TGFBR2, CEA and IL-15Rα and IL-21R, CD123 and CD33 and CD38 and CLL-1, CD133 and EGFR, BCMA and HPK1, CD8A and NY-ESO-1, CD38 and DR5, CD19 and EGFR, BCMA and CD19 and HER2 and Trop-2, CD19 and MUC1, PDL1 and VEGFR1, CD123 and CD33 and CD38 and CD56 and CLL-1 and MUC1, BCMA and CD138 and CD19 and CD38, BCMA and CD16a and IL-15Rα, NCR2 and NKG2D, CD38 and CLL-1, and CD276 and FGFR4.

[0527] 49. A cell comprising the nucleic acid or expression vector of any one of embodiments 31-48.

[0528] 50. The cell of embodiment 49, wherein the cell is selected from the group consisting of: a T cell, a Naïve T cell (TN), stem cell memory T cell (T_{SCM}), a central memory T cell (T_{CM}), a resident memory T cell (TRM), an effector T cell (TEFF), an effector memory cell (TEM), an alpha-beta (αβ) T cell, a gamma-delta (γα) T cell, a Natural Killer (NK) cell, a CD4+ T cell, a CD4+Th1 cell, a CD4+Th2 cell, a CD4+Th9 cell, a CD4+Th17 cell, a CD4+Th22 cell, a CD4+T Regulatory Cell (Treg), a CD4+T follicular helper cell (Tfh), a resting Treg cell, an effector Treg cell, an effector memory Treg cell, a CD8+ T cell, a CD8+Tc1 cell, a CD8+Tc2 cell, a CD8+Tc9 cell, a CD8+Th17 cell, a CD8+Treg cell, a macrophage, a B Cell, a dendritic cell, a stem cell, a hematopoietic stem cell (HSC), and an induced pluripotent stem cell (iPS cell).

[0529] 51. A method of treating a subject for cancer, the method comprising administering to the subject an effective amount of a nucleic acid, an expression vector, or a cell of any one of embodiments 31-50.

[0530] 52. The method of embodiment 51, wherein the cancer is selected from the group consisting of: Acute Lymphoblastic Leukemia (ALL), Acute Myeloid Leukemia (AML), Adrenocortical Carcinoma, an AIDS-Related Cancer, Anal Cancer, Appendix Cancer, Astrocytomas, Atypical Teratoid/Rhabdoid Tumor, Basal Cell Carcinoma, Bile Duct Cancer, Bladder Cancer, Bone Cancer, Brain Stem Glioma, Brain Tumor, Breast Cancer, Bronchial Tumors, Burkitt Lymphoma, Carcinoid Tumor, Carcinoma of Unknown Primary, Cardiac (Heart) Tumors, Central Nervous System Tumor, Cervical Cancer, Childhood Cancers, Chordoma, Chronic Lymphocytic Leukemia (CLL), Chronic Myelogenous Leukemia (CML), Chronic Myeloproliferative Neoplasms, Colon Cancer, Colorectal Cancer, Craniopharyngioma, Cutaneous T-Cell Lymphoma, Duct Cancer, Ductal Carcinoma In Situ (DCIS), Embryonal Tumors, Endometrial Cancer, Ependymoma, Esophageal Cancer, Esthesioneuroblastoma, Ewing Sarcoma, Extracranial Germ Cell Tumor, Extragonadal Germ Cell Tumor, Extrahepatic Bile Duct Cancer, Eye Cancer, Fibrous Histiocytoma of Bone, Gallbladder Cancer, Gastric (Stomach) Cancer, Gastrointestinal Carcinoid Tumor, Gastrointestinal Stromal Tumors (GIST), Germ Cell Tumor, Gestational Trophoblastic Disease, Glioma, Hairy Cell Leukemia, Head and Neck Cancer, Heart Cancer, Hepatocellular (Liver) Cancer, Histiocytosis, Hodgkin Lymphoma, Hypopharyngeal Cancer, Intraocular Melanoma, Islet Cell Tumors, Kaposi Sarcoma, Kidney Cancer, Langerhans Cell Histiocytosis, Laryngeal Cancer, Leukemia, Lip and Oral Cavity Cancer, Liver Cancer (Primary), Lobular Carcinoma In Situ (LCIS), Lung Cancer, Lymphoma, Macroglobulinemia, Male Breast Cancer, Malignant Fibrous Histiocytoma of Bone and Osteosarcoma, Melanoma, Merkel Cell Carcinoma, Mesothelioma, Metastatic Squamous Neck Cancer with Occult Primary, Midline Tract Carcinoma Involving NUT Gene, Mouth Cancer, Multiple Endocrine Neoplasia Syndromes, Multiple Myeloma/Plasma Cell Neoplasm, Mycosis Fungoides, Myelodysplastic Syndromes, Myelodysplastic/Myeloproliferative Neoplasms, Myelogenous Leukemia, Myeloid Leukemia, Myeloproliferative Neoplasms, Nasal Cavity and Paranasal Sinus Cancer, Nasopharyngeal Cancer, Neuroblastoma, Non-Hodgkin Lymphoma, Non-Small Cell Lung Cancer, Oral Cancer, Oral Cavity Cancer, Oropharyngeal Cancer, Osteosarcoma and Malignant Fibrous Histiocytoma of Bone, Ovarian Cancer, Pancreatic Cancer, Pancreatic Neuroendocrine Tumors, Papillomatosis, Paraganglioma, Paranasal Sinus and Nasal Cavity Cancer, Parathyroid Cancer, Penile Cancer, Pharyngeal Cancer, Pheochromocytoma, Pituitary Tumor, Pleuropulmonary Blastoma, Primary Central Nervous System (CNS) Lymphoma, Prostate Cancer, Rectal Cancer, Renal Cell (Kidney) Cancer, Renal Pelvis and Ureter, Transitional Cell Cancer, Retinoblastoma, Rhabdomyosarcoma, Salivary Gland Cancer, Sarcoma, Sezary Syndrome, Skin Cancer, Small Cell Lung Cancer, Small Intestine Cancer, Soft Tissue Sarcoma, Squamous Cell Carcinoma, Squamous Neck Cancer, Stomach (Gastric) Cancer, T-Cell Lymphoma, Testicular Cancer, Throat Cancer, Thymoma and Thymic Carcinoma, Thyroid Cancer, Transitional Cell Cancer of the Renal Pelvis and Ureter, Ureter and Renal Pelvis Cancer, Urethral Cancer, Uterine Cancer, Uterine Sarcoma, Vaginal Cancer, Vulvar Cancer, Waldenstrom Macroglobulinemia, and Wilms Tumor.

[0531] 53. The method of embodiment 52, wherein the subject does not undergo lymphodepletion during or prior to the administering.

[0532] 54. The method of embodiment 52 or 53, wherein the amount of the nucleic acid, the expression vector, or the cell administered to the subject is less than the amount of a corresponding nucleic acid, expression vector, or cell comprising the sequence encoding the CAR administered to the subject to treat the cancer without the sequence encoding the DR-18 polypeptide.

[0533] 55. The method of any one of embodiments 52-54, wherein the administering comprises delivering multiple doses of the nucleic acid, the expression vector, or the cell to the subject and the dosing frequency is less than the dosing frequency of a corresponding nucleic acid, expression vector, or cell comprising the sequence encoding the CAR administered to the subject to treat the cancer without the sequence encoding the DR-18 polypeptide.

[0534] 56. A method of making a population of therapeutic cells, the method comprising:

[0535] a) contacting a human cell with a nucleic acid comprising a chimeric antigen receptor (CAR) sequence encoding a CAR and a sequence encoding a decoy-resistant interleukin 18 (DR-18) polypeptide; wherein the nucleic acid is configured to result in expression and secretion of the DR-18 by the human cell ex vivo;

[0536] b) culturing the contacted human cell ex vivo in the presence of the secreted DR-18 polypeptide, thereby producing a population of therapeutic cells, wherein the population of therapeutic cells is enhanced as compared to a corresponding population of cells generated without the sequence encoding the DR-18 polypeptide.

[0537] 57. The method of embodiment 56, wherein the population of therapeutic cells demonstrates enhanced ex vivo proliferation as compared to the corresponding population of cells.

[0538] 58. The method of embodiment 56 or 57, wherein the population of therapeutic cells demonstrates enhanced in vivo proliferation as compared to the corresponding population of cells.

[0539] 59. The method of any one of embodiments 56-58, wherein the population of therapeutic cells demonstrates enhanced in vivo persistence as compared to the corresponding population of cells.

[0540] 60. The method of any one of embodiments 56-59, wherein the population of therapeutic cells demonstrates enhanced tumor cell killing as compared to the corresponding population of cells.

EXAMPLES

[0541] The following examples are put forth so as to provide those of ordinary skill in the art with a complete disclosure and description of how to make and use the present invention, and are not intended to limit the scope of what the inventors regard as their invention nor are they intended to represent that the experiments below are all or the only experiments performed. Efforts have been made to ensure accuracy with respect to numbers used (e.g. amounts, temperature, etc.) but some experimental errors and deviations should be accounted for. Unless indicated otherwise, parts are parts by weight, molecular weight is weight average molecular weight, temperature is in degrees Centigrade, and pressure is at or near atmospheric.

Example 1—DR-18 Armored Anti-EGFRvIII CAR T Cells

[0542] The epidermal growth factor receptor (EGFR) is overexpressed in a variety of human epithelial tumors, often as a consequence of gene amplification. A deletion of exons 2-7 of the EGFR gene is the most common extracellular domain mutation of EGFR and is known as epidermal growth factor receptor variant Ill (EGFRvIII). Aberrant EGFRvII signaling has been shown to be important in driving tumor progression and often correlates with poor prognosis. Studies have indicated that chimeric antigen receptor (CAR) T cell therapy directed to EGFRvIII antigen is safe and that CART-EGFRvIII cells engraft and traffic to tumor sites (see e.g., O'Rourke et al. *Sci Transl Med.* 2017; 9(399): eaaa0984). However, while initial studies observed a specific loss or decreased expression of EGFRvII antigen in tumors that correlated with CART-EGFRvIII treatment, tumor regression from CART-EGFRvIII monotherapy alone was not observed and clinical benefit could not be confirmed in the 10 patients tested (Id.).

[0543] Bicistronic DR-18 armored anti-EGFRvIII CAR constructs were designed to encode human DR-18 polypeptide (hDR18) (SEQ ID NO: 22) with a heterologous human IL2 variant signal peptide (SEQ ID NO:141) and anti-EGFRvIII CAR (SEQ ID NO: 109) in various configurations. EGFRvII may be referred to hereafter in this Example as EGFR for brevity. In some instances, coding sequences were separated by a sequence encoding a 2A peptide sequence and expression of both components was driven by a human elongation factor-1 alpha (EF1a) promoter. In other instances, expression of the anti-EGFR CAR was driven by the EF1a promoter and expression of the hDR18 was driven by a nuclear factor of activated T cells (NFAT) responsive promoter that contained a minimal interleukin 2 promoter (mIL2p) operably linked to a six NFAT response elements (6×NFAT-Bs). All constructs further included woodchuck hepatitis virus post-transcriptional regulatory element

(WPRE). Corresponding control constructs were also designed, including bicistronic constructs wherein the hDR-18 coding sequence was replaced sequence encoding a luciferase (Luc) reporter and a monocistronic control containing an anti-EGFR CAR coding sequence without a hDR18 coding sequence. Bicistronic constructs where hDR18 was replaced with sequence encoding wild-type (wt) human interleukin 18 (wt-IL18) (SEQ ID NO: 1) were also designed and created for comparative purposes.

[0544] The general architecture of constructs employed in this example, including "EGFR CAR" (i.e., "CAR-ONLY"), "EGFR CAR-2A-hDR18" and "EGFR CAR-2A-wt-IL18" (i.e., "CAR-2A-IL18"), "EGFR CAR-NFAT-hDR18" and "EGFR CAR-NFAT-wt-IL18" (i.e., "CAR-NFAT-IL18"), "EGFR CAR-2A-Luc" (i.e., CAR-2A-lc"), and "EGFR CAR-NFAT-Luc" (i.e., "CAR-NFAT-luc"), are as shown in FIG. **1**.

[0545] Cells are cultured and CAR lentivirus with the above constructs are packaged, harvested, concentrated, stored, and titred according to standard procedures. To determine expression from titred lentivirus particles, cells are seeded and cultured in the presence of concentrated lentivirus solutions, then CAR expression is measured by flow cytometry following incubation. Once expression from the designed constructs is determined, the produced lentivirus are used to prepare CAR-T cells from activated human pan T cells. CAR T cells transduced with lentivirus carrying each construct are expanded, subcultured, and assessed for transfection efficiency, phenotype (FACS), and IL-18/hDR18 expression (ELISA).

[0546] The produced CAR T cells are tested, and activity verified, in an in vitro cytotoxicity assay. In brief, CAR-T cell cytotoxicity is measured by detecting apoptosis of target tumor cells through flow cytometry after co-culture. Cytotoxicity is tested in various effector-to-target cell ratios. In vitro cytotoxicity is tested with and without supplemental cytokines.

[0547] In vivo mouse xenograft experiments will be performed. Briefly, EGFR-expressing human tumors are engrafted by subcutaneous injection of five million EGFR-expressing tumor cells into the right flank of immunodeficient (NOG) host mice under conditions that will allow for the tumor to establish and grow within the mouse model. Tumor size, as well as animal health and bodyweight, will be monitored throughout the study. Expression of the EGFR target antigen will be verified in the established tumors and EGFR+tumor bearing mice will be administered CAR-NFAT-luc T cells to validate the inducibility of the transgene by CAR-T cell activation. Appropriate control groups, such as mock T cell treated mice and mice treated with CAR-2A-luc T cells will be used to verify and quantify induced luciferase expression.

[0548] After induced expression of luciferase is verified in the EGFR+model, EGFR tumor bearing mice will be intravenously administered mock T cell treatment, CAR-only T cells with cytokine supplementation, CAR-only T cells (without cytokine supplementation), CAR-2A-hDR18 T cells, and CAR-NFAT-hDR18 T cells to assess the relative anti-tumor activity of the various constructs. Corresponding CAR-2A-wt-IL18 and CAR-NFAT-wt-IL18 T cell groups may or may not be further included, e.g., to compare functionality of hDR18 to wt-IL18 in these contexts. In addition to monitoring tumor burden, animal health, and animal bodyweight, blood samples will also be periodically collected for analysis (phenotype by FACS; luciferase, wt-IL18, and hDR18 quantification by ELISA, etc.). Presence, proliferation, and other parameters of the administered cells will also be assessed via samples collected at appropriate timepoints, such as e.g., one day, two days, and one week following administration of the cells. Further analyses may be performed including a cytokine panel analysis, gene expression analyses, cytology, histology, and the like via appropriate collected samples, such as mouse blood, serum, plasma, tumor tissue, and non-tumor tissue.

[0549] Constitutive and inducible hDR18 armored anti-EGFR CAR T cell treated mice is expected to show a greater reduction in tumor growth as compared to the mice treated without hDR18 armoring (i.e., the constructs carrying luciferase in place of hDR18 as well as anti-EGFR CAR only T cell control treated animals). Other beneficial characteristics such as increased proliferation and/or enhanced persistence of the administered cells are expected to be observed in samples collected from the hDR18 armored anti-EGFR CAR T cohorts, e.g., as compared to groups without hDR18 armoring. Beneficial in vitro characteristics, such as enhanced in vitro proliferation as compared to non-hDR18-armored CAR T cells, are also expected to be observed. In some instances, certain beneficial in vivo characteristics, such as e.g., enhanced persistence of the administered cells or reduced tumor growth, may be independent of (or observed in the absence of) increased proliferation of the cells in vitro and/or ex vivo.

[0550] In at least some of the previously described embodiments, one or more elements used in an embodiment can interchangeably be used in another embodiment unless such a replacement is not technically feasible. It will be appreciated by those skilled in the art that various other omissions, additions and modifications may be made to the methods and structures described above without departing from the scope of the claimed subject matter. All such modifications and changes are intended to fall within the scope of the subject matter, as defined by the appended claims.

[0551] It will be understood by those within the art that, in general, terms used herein, and especially in the appended claims (e.g., bodies of the appended claims) are generally intended as "open" terms (e.g., the term "including" should be interpreted as "including but not limited to," the term "having" should be interpreted as "having at least," the term "includes" should be interpreted as "includes but is not limited to," etc.). It will be further understood by those within the art that if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases "at least one" and "one or more" to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles "a" or "an" limits any particular claim containing such introduced claim recitation to embodiments containing only one such recitation, even when the same claim includes the introductory phrases "one or more" or "at least one" and indefinite articles such as "a" or "an" (e.g., "a" and/or "an" should be interpreted to mean "at least one" or "one or more"); the same holds true for the use of definite articles used to introduce claim recitations. In addition, even if a specific number of an introduced claim recitation is explicitly recited, those skilled in the art will recognize that such recitation should be interpreted to mean at least the recited number (e.g., the bare recitation of "two recitations," without other modifiers, means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to "at least one of A, B, and C, etc." is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., "a system having at least one of A, B, and C" would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). In those instances where a convention analogous to "at least one of A, B, or C, etc." is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., "a system having at least one of A, B, or C" would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). It will be further understood by those within the art that virtually any disjunctive word and/or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms. For example, the phrase "A or B" will be understood to include the possibilities of "A" or "B" or "A and B."

[0552] In addition, where features or aspects of the disclosure are described in terms of Markush groups, those skilled in the art will recognize that the disclosure is also thereby described in terms of any individual member or subgroup of members of the Markush group.

[0553] As will be understood by one skilled in the art, for any and all purposes, such as in terms of providing a written description, all ranges disclosed herein also encompass any and all possible sub-ranges and combinations of sub-ranges thereof. Any listed range can be easily recognized as sufficiently describing and enabling the same range being broken down into at least equal halves, thirds, quarters, fifths, tenths, etc. As a non-limiting example, each range discussed herein can be readily broken down into a lower third, middle third and upper third, etc. As will also be understood by one skilled in the art all language such as "up to," "at least," "greater than," "less than," and the like include the number recited and refer to ranges which can be subsequently broken down into sub-ranges as discussed above. Finally, as will be understood by one skilled in the art, a range includes each individual member. Thus, for example, a group having 1-3 articles refers to groups having 1, 2, or 3 articles. Similarly, a group having 1-5 articles refers to groups having 1, 2, 3, 4, or 5 articles, and so forth.

[0554] Although the foregoing invention has been described in some detail by way of illustration and example for purposes of clarity of understanding, it is readily apparent to those of ordinary skill in the art in light of the teachings of this invention that certain changes and modifications may be made thereto without departing from the spirit or scope of the appended claims.

[0555] Accordingly, the preceding merely illustrates the principles of the invention. It will be appreciated that those skilled in the art will be able to devise various arrangements which, although not explicitly described or shown herein, embody the principles of the invention and are included within its spirit and scope. Furthermore, all examples and conditional language recited herein are principally intended to aid the reader in understanding the principles of the invention and the concepts contributed by the inventors to furthering the art, and are to be construed as being without limitation to such specifically recited examples and conditions. Moreover, all statements herein reciting principles, aspects, and embodiments of the invention as well as specific examples thereof, are intended to encompass both structural and functional equivalents thereof. Additionally, it is intended that such equivalents include both currently known equivalents and equivalents developed in the future, i.e., any elements developed that perform the same function, regardless of structure. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims.

[0556] The scope of the present invention, therefore, is not intended to be limited to the exemplary embodiments shown and described herein. Rather, the scope and spirit of present invention is embodied by the appended claims. In the claims, 35 U.S.C. § 112(f) or 35 U.S.C. § 112(6) is expressly defined as being invoked for a limitation in the claim only when the exact phrase "means for" or the exact phrase "step for" is recited at the beginning of such limitation in the claim; if such exact phrase is not used in a limitation in the claim, then 35 U.S.C. § 112 (f) or 35 U.S.C. § 112(6) is not invoked.

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                        organism = synthetic construct
SEQUENCE: 15
YFGKLESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDSRD NAPRTIFIIS KYSDSLARGL  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFQRDVPGHS RKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQNED                          157

SEQ ID NO: 16           moltype = AA  length = 157
FEATURE                 Location/Qualifiers
source                  1..157
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 16
YFGKLESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS KYSDSLARGL  60
AVTISVKSEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFQRDVPGHS RKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQNED                          157

SEQ ID NO: 17           moltype = AA  length = 157
FEATURE                 Location/Qualifiers
source                  1..157
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 17
YFGKLESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS KYSDSLARGL  60
AVTISVKCEK ISTLSSENKI ISFKEMNPPD NIKDTKSDII FFQRDVPGHS RKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQNED                          157
```

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SEQ ID NO: 18          moltype = AA  length = 157
FEATURE                Location/Qualifiers
source                 1..157
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 18
YFGKLESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS KYSDSLARGL  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFQRDVPGHS RKMQFESSSY  120
EGYFLASEKE RDLFKLILKK EDELGDRSIM FTVQNED                           157

SEQ ID NO: 19          moltype = AA  length = 157
FEATURE                Location/Qualifiers
source                 1..157
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 19
YFGKLESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDSRD NAPRTIFIIS KYSDSLARGL  60
AVTISVKGEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFQRDVPGHS RKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQNED                           157

SEQ ID NO: 20          moltype = AA  length = 157
FEATURE                Location/Qualifiers
source                 1..157
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 20
YFGKLESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDSRD NAPRTIFIIS KYSDSLARGL  60
AVTISVKAEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFQRDVPGHS RKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQNED                           157

SEQ ID NO: 21          moltype = AA  length = 157
FEATURE                Location/Qualifiers
source                 1..157
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 21
YFGKLESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDSRD NAPRTIFIIS KYSDSLARGL  60
AVTISVKVEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFQRDVPGHS RKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQNED                           157

SEQ ID NO: 22          moltype = AA  length = 157
FEATURE                Location/Qualifiers
source                 1..157
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 22
YFGKLESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDSRD NAPRTIFIIS KYSDSLARGL  60
AVTISVKDEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFQRDVPGHS RKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQNED                           157

SEQ ID NO: 23          moltype = AA  length = 157
FEATURE                Location/Qualifiers
source                 1..157
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 23
YFGKLESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDSRD NAPRTIFIIS KYSDSLARGL  60
AVTISVKEEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFQRDVPGHS RKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQNED                           157

SEQ ID NO: 24          moltype = AA  length = 157
FEATURE                Location/Qualifiers
source                 1..157
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 24
YFGKLESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDSRD NAPRTIFIIS KYSDSLARGL  60
AVTISVKNEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFQRDVPGHS RKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQNED                           157

SEQ ID NO: 25          moltype = AA  length = 157
FEATURE                Location/Qualifiers
source                 1..157
                       mol_type = protein
                       organism = synthetic construct
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SEQUENCE: 25
YFGKLESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS TYKDSQPRGK  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFQRDVPGHK HKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQNED                           157

SEQ ID NO: 26          moltype = AA  length = 157
FEATURE                Location/Qualifiers
source                 1..157
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 26
YFGKLESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS TYKDKQPRAK  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFQRDVPGHK HKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTIQNED                           157

SEQ ID NO: 27          moltype = AA  length = 157
FEATURE                Location/Qualifiers
source                 1..157
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 27
RFGKLESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS TYKDSQPRGK  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFQRDVPGHK HKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQNED                           157

SEQ ID NO: 28          moltype = AA  length = 157
FEATURE                Location/Qualifiers
source                 1..157
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 28
RFGKLESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS TYRDSQPRGK  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFQRNVPGHK YKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQNED                           157

SEQ ID NO: 29          moltype = AA  length = 157
FEATURE                Location/Qualifiers
source                 1..157
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 29
YFGKLESQLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS TYKDKQPRTK  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFQRRVPGHH NKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQKED                           157

SEQ ID NO: 30          moltype = AA  length = 157
FEATURE                Location/Qualifiers
source                 1..157
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 30
YFGKLESRLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS KYKDKQPRAQ  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFQRDVPGHK HKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTIQNED                           157

SEQ ID NO: 31          moltype = AA  length = 157
FEATURE                Location/Qualifiers
source                 1..157
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 31
YFGKLESRLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS DYKDKQPRAX  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFQRDVPGHK HKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTIQNED                           157

SEQ ID NO: 32          moltype = AA  length = 157
FEATURE                Location/Qualifiers
source                 1..157
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 32
YFGKHESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS TYRDSQPRGK  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFQRDVPGHN NKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTTQNED                           157

SEQ ID NO: 33          moltype = AA  length = 157
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FEATURE                  Location/Qualifiers
source                   1..157
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 33
YFGKIESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS KYKDKQPRAQ  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFQRKVPGHQ HKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQKED                           157

SEQ ID NO: 34            moltype = AA  length = 157
FEATURE                  Location/Qualifiers
source                   1..157
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 34
YFGKIESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS TYKDRQPRGK  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFERDVPGHH HKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTIQNED                           157

SEQ ID NO: 35            moltype = AA  length = 157
FEATURE                  Location/Qualifiers
source                   1..157
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 35
YFGKIESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS TYKDKQPRGK  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFQRDVPGHK HKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTTQHED                           157

SEQ ID NO: 36            moltype = AA  length = 157
FEATURE                  Location/Qualifiers
source                   1..157
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 36
YFGKIESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS TYKDKQPRAK  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFQRRVPGHH HKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTIQKED                           157

SEQ ID NO: 37            moltype = AA  length = 157
FEATURE                  Location/Qualifiers
source                   1..157
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 37
YFGKIESRLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS TYKDKQPRGK  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFQRDVPGHD YKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTIQKED                           157

SEQ ID NO: 38            moltype = AA  length = 157
FEATURE                  Location/Qualifiers
source                   1..157
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 38
YFGKYESRLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS TYRDSQPRGK  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFQRDVPGHE HKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQKED                           157

SEQ ID NO: 39            moltype = AA  length = 157
FEATURE                  Location/Qualifiers
source                   1..157
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 39
HFGKYESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS TYRDSQPRGK  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFQRDVPGHH NKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQKED                           157

SEQ ID NO: 40            moltype = AA  length = 157
FEATURE                  Location/Qualifiers
source                   1..157
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 40
RFGKLESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS TYRDSQPRAK  60
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AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFQRDVPGHQ HKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTAQKED                           157

SEQ ID NO: 41           moltype = AA  length = 157
FEATURE                 Location/Qualifiers
source                  1..157
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 41
RFGKLESRLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS DYRDSQPRGR  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFKRNVPGHK YKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQHED                           157

SEQ ID NO: 42           moltype = AA  length = 157
FEATURE                 Location/Qualifiers
source                  1..157
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 42
RFGKLESRLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS NYRDSQPRGQ  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFKRRVPGHN HKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQKED                           157

SEQ ID NO: 43           moltype = AA  length = 157
FEATURE                 Location/Qualifiers
source                  1..157
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 43
RFGKLESRLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS TYKDSQPRGK  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFQRDVPGHK HKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQNED                           157

SEQ ID NO: 44           moltype = AA  length = 157
FEATURE                 Location/Qualifiers
source                  1..157
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 44
RFGKHESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS TYRDSQPRGK  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFERNVPGHK YKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQNED                           157

SEQ ID NO: 45           moltype = AA  length = 157
FEATURE                 Location/Qualifiers
source                  1..157
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 45
RFGKLESRLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS TYRDSQPRAK  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFERDVPGHQ HKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTIQXED                           157

SEQ ID NO: 46           moltype = AA  length = 157
FEATURE                 Location/Qualifiers
source                  1..157
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 46
RFGKLESRLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS TYRDSQPRTK  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFQRNVPGHH DKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQHED                           157

SEQ ID NO: 47           moltype = AA  length = 157
FEATURE                 Location/Qualifiers
source                  1..157
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 47
RFGKLESRLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS TYKDSQPRAK  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFERDVPGHQ HKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTIQKED                           157

SEQ ID NO: 48           moltype = AA  length = 157
FEATURE                 Location/Qualifiers
source                  1..157
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                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 48
RFGKYESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS TYKDSQPRTK  60
AVTISVKCEK ISTLSCDNKI ISFKEMNPPD NIKDTKSDII FFQRDVPGHK HKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQNED                          157

SEQ ID NO: 49           moltype = AA  length = 157
FEATURE                 Location/Qualifiers
source                  1..157
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 49
RFGKLESRLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS TYRDSQPRTK  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFQRKVPGHN HKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQKED                          157

SEQ ID NO: 50           moltype = AA  length = 157
FEATURE                 Location/Qualifiers
source                  1..157
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 50
YFGKLESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS EYKDSELRGR  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFPRAVPGHN RKVQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQNED                          157

SEQ ID NO: 51           moltype = AA  length = 157
FEATURE                 Location/Qualifiers
source                  1..157
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 51
YFGKLESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS KYKDSAGRGL  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFERDVPGHS NKVQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQNED                          157

SEQ ID NO: 52           moltype = AA  length = 157
FEATURE                 Location/Qualifiers
source                  1..157
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 52
YFGKLESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS KYGDSAARGL  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFQRSVPGHK RKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQNED                          157

SEQ ID NO: 53           moltype = AA  length = 157
FEATURE                 Location/Qualifiers
source                  1..157
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 53
YFGKLESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS KYGDSRGRGL  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFERDVPGHN SKRQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQNED                          157

SEQ ID NO: 54           moltype = AA  length = 157
FEATURE                 Location/Qualifiers
source                  1..157
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 54
YFGKLESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS KYGDSVPRGL  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFARAVPGHS RKTQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQNED                          157

SEQ ID NO: 55           moltype = AA  length = 157
FEATURE                 Location/Qualifiers
source                  1..157
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 55
YFGKLESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS KYSDSGARGL  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFARAVPGHG RKTQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQNED                          157
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SEQ ID NO: 56          moltype = AA  length = 157
FEATURE                Location/Qualifiers
source                 1..157
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 56
YFGKLESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS KYSDSKARGM  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFARDVPGHS SKRQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQNED                           157

SEQ ID NO: 57          moltype = AA  length = 157
FEATURE                Location/Qualifiers
source                 1..157
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 57
YFGKLESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS KYSDSLARGL  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFQRDVPGHS RKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQNED                           157

SEQ ID NO: 58          moltype = AA  length = 157
FEATURE                Location/Qualifiers
source                 1..157
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 58
YFGKLESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS KYSDSRARGL  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFQRNVPGHG RKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQNED                           157

SEQ ID NO: 59          moltype = AA  length = 157
FEATURE                Location/Qualifiers
source                 1..157
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 59
YFGKLESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS KYSDSRARGL  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFARSVPGHG RKTQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQNED                           157

SEQ ID NO: 60          moltype = AA  length = 157
FEATURE                Location/Qualifiers
source                 1..157
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 60
YFGKLESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS KYSDSRARGL  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFARDVPGHS GKRQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQNED                           157

SEQ ID NO: 61          moltype = AA  length = 157
FEATURE                Location/Qualifiers
source                 1..157
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 61
YFGKLESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS KYTDSRPRGL  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFERDVPGHS SKKQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQNED                           157

SEQ ID NO: 62          moltype = AA  length = 157
FEATURE                Location/Qualifiers
source                 1..157
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 62
YFGKLESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS KYTDSRARGM  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFERDVPGHN DKRQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQNED                           157

SEQ ID NO: 63          moltype = AA  length = 157
FEATURE                Location/Qualifiers
source                 1..157
                       mol_type = protein
                       organism = synthetic construct
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SEQUENCE: 63
YFGKLESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS RYKDSGKRGL  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFRRSVPGHS RKVQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQNED                           157

SEQ ID NO: 64          moltype = AA  length = 157
FEATURE                Location/Qualifiers
source                 1..157
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 64
YFGKLESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS KYGDSGARGL  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFERDVPGHS GKVQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQNED                           157

SEQ ID NO: 65          moltype = AA  length = 157
FEATURE                Location/Qualifiers
source                 1..157
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 65
YFGKLESKLS VIRNLNDQVL FIDQGNRPLF EDMTDSDCRD NAPRTIFIIS KYGDSRPRGM  60
AVTISVKCEK ISTLSCENKI ISFKEMNPPD NIKDTKSDII FFQRAVPGHN RKMQFESSSY  120
EGYFLACEKE RDLFKLILKK EDELGDRSIM FTVQNED                           157

SEQ ID NO: 66          moltype = AA  length = 157
FEATURE                Location/Qualifiers
source                 1..157
                       mol_type = protein
                       organism = synthetic construct
VARIANT                1
                       note = X at pos. 1 is Y, R or H
VARIANT                5
                       note = X at pos. 5 is L, H, I or Y
VARIANT                8
                       note = X at pos. 8 is K, Q or R
VARIANT                38
                       note = X at pos. 38 is C or S
VARIANT                51
                       note = X at pos. 51 is M, T, K, D, N, E or R
VARIANT                53
                       note = X at pos. 53 is K, R, G, S or T
VARIANT                55
                       note = X at pos. 55 is S, K or R
VARIANT                56
                       note = X at pos. 56 is Q, E, A, R, V, G, K, L or R
VARIANT                57
                       note = X at pos. 57 is P, L, G, A or K
VARIANT                59
                       note = X at pos. 59 is G, A or T
VARIANT                60
                       note = X at pos. 60 is M, K, Q, R or L
VARIANT                68
                       note = X at pos. 68 is C, S, G, A, V, D, E or N
VARIANT                76
                       note = X at pos. 76 is C or S
VARIANT                77
                       note = X at pos. 77 is E or D
VARIANT                103
                       note = X at pos. 103 is Q, E, K, P, A or R
VARIANT                105
                       note = X at pos. 105 is S, D, N, R, K or A
VARIANT                110
                       note = X at pos. 110 is D, K, H, N, Q, E, S or G
VARIANT                111
                       note = X at pos. 111 is N, H, Y, D, R, S or G
VARIANT                113
                       note = X at pos. 113 is M, V, R, T or K
VARIANT                127
                       note = X at pos. 127 is C or S
VARIANT                153
                       note = X at pos. 153 is V, I, T or A
VARIANT                155
                       note = X at pos. 155 is N, K or H
SEQUENCE: 66
XFGKXESXLS VIRNLNDQVL FIDQGNRPLF EDMTDSDXRD NAPRTIFIIS XYXDXXXRXX  60
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AVTISVKXEK ISTLSXXNKI ISFKEMNPPD NIKDTKSDII FFXRXVPGHX XKXQFESSSY  120
EGYFLAXEKE RDLFKLILKK EDELGDRSIM FTXQXED                           157

SEQ ID NO: 67          moltype = DNA  length = 471
FEATURE                Location/Qualifiers
source                 1..471
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 67
tatttcggca aactggaaag caagctcagc gtgatccgga acctgaacga ccaagtgctg  60
ttcatcgacc agggcaaccg gcctctgttt gaggacatga cagacagcga ctgcagagat  120
aatgcccctc ggaccatctt catcatctct acctacaagg attctcagcc tagaggcaag  180
gccgtgacca tctctgtcaa gtgcgagaag atcagcacac tgagctgcga gaacaagatc  240
attagcttca aagaaatgaa cccccccgac aatatcaagg acaccaagtc cgacatcatc  300
tttttccaga gagatgtgcc aggccacaag cacaaaatgc agttcgagag cagctcctac  360
gagggctact tcctggcttg tgaaaaggaa agagacctgt tcaagctgat cctgaaaaag  420
gaagatgagc tgggagatag aagcatcatg ttcaccgtgc agaacgagga c           471

SEQ ID NO: 68          moltype = DNA  length = 474
FEATURE                Location/Qualifiers
source                 1..474
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 68
tacttcggca aactggagtc caagctgagc gtgatcagga atctgaacga tcaggtgctc  60
ttcatcgatc agggcaatag acctctgttt gaggacatga ctgactccga ctgtcgggac  120
aacgcccccc gaacaatctt tattatcagc acctacaagg acagccagcc caggggggaag  180
gccgtgacta tctccgtgaa gtgcgaaaag atcagcaccc tgtcatgtga gaacaagatt  240
atctccttta aggagatgaa ccccccccgat aacattaagg acaccaagag cgacatcatc  300
ttcttccagc gcgacgtgcc aggacacaag cacaaaatgc agtttgagag cagcagctac  360
gaagggtatt tcctggcctg tgagaaggaa agagatctgt tcaagctgat cctgaagaag  420
gaggacgagc tgggcgaccg gagcatcatg tttaccgtgc agaacgagga ctga        474

SEQ ID NO: 69          moltype = DNA  length = 471
FEATURE                Location/Qualifiers
source                 1..471
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 69
tacttcggca agctggaaag caagctgagc gtgatccgga acctgaacga ccaagtgctg  60
ttcatcgacc agggaaatag acctctgttc gaggacatga ccgacagcga ctgcagagat  120
aatgccccta gaacaatctt catcatcagc acatacaaag ataagcagcc tcgggctaag  180
gccgtgacca tcagcgtcaa gtgcgagaaa atctctaccc tgtcttgtga aaacaagatc  240
atctccttca aggaaatgaa ccccccccgac aacatcaagg acaccaagtc cgacatcatt  300
ttcttccagc gggacgtgcc aggccacaag cacaaaatgc agttcgagag cagctcttat  360
gagggctact ttctggcctg cgagaaggaa agagatctgt ttaagctcat cctgaagaaa  420
gaagatgagc tgggcgacag aagcatcatg ttcaccatcc agaacgagga t           471

SEQ ID NO: 70          moltype = DNA  length = 474
FEATURE                Location/Qualifiers
source                 1..474
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 70
tactttggca agctggagtc caaactgtcc gtgatccgca acctgaatga tcaggtgctg  60
ttcatcgacc agggcaaccg cccccctgttc gaggacatga ccgatagcga ctgcagggat  120
aacgcccctc ggactatctt tatcatcagc acatacaagg acaaacagcc cagagccaag  180
gccgtgacca tcagcgtcaa gtgtgagaag attagcactc tgtcctgtga gaacaagatc  240
attagcttta aggaaatgaa cccacctgat aatatcaagg acaccaagag cgacattatc  300
ttctttcaga gagatgtgcc cggccacaag cacaagatgc agttcgagag ctctagctac  360
gagggctact tcctggcttg cgaaaaggag agagacctgt ttaaactcat tctgaagaag  420
gaggacgaac tgggcgatag atccatcatg ttcaccatcc agaacgagga ttga        474

SEQ ID NO: 71          moltype = DNA  length = 471
FEATURE                Location/Qualifiers
source                 1..471
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 71
cggttcggca aactggaatc taagctgtct gtgatccgga acctgaacga ccaagtgctg  60
ttcatcgacc agggcaatcg cccccctgttt gaggatatga ccgacagcga ctgcagagat  120
aacgccccta gaacaatctt catcatcagc acctacaagg atagccagcc tcggggcaag  180
gccgtgacaa tcagcgtcaa gtgcgaaaaa atctccaccc tgagctgtga aaacaagatc  240
atctctttta aggaaatgaa cccacctgat aatatcaagg acaccaagtc cgacatcatt  300
ttcttccaga gagatgtgcc cggacacaag cacaaaatgc agttcgagag cagcagctac  360
gagggctact tcctggcttg cgagaaggaa agagacctgt tcaagctcat cctgaagaaa  420
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gaggacgagc tgggcgacag aagcatcatg ttcaccgtgc agaacgagga c          471

SEQ ID NO: 72           moltype = DNA  length = 474
FEATURE                 Location/Qualifiers
source                  1..474
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 72
cgcttcggca aactggagtc caagctgagc gtgatcagga atctgaacga tcaggtgctc  60
ttcatcgatc agggcaatag acctctgttt gaggacatga ctgactccga ctgtcgggac  120
aacgcccccc gaacaatctt tattatcagc acctacaagg acagccagcc caggggggaag  180
gccgtgacta tctccgtgaa gtgcgaaaag atcagcaccc tgtcatgtga gaacaagatt  240
atctccttta aggagatgaa cccccccgat aacattaagg acaccaagag cgacatcatc  300
ttcttccagc gcgacgtgcc aggacacaag cacaaaatgc agtttgagag cagcagctac  360
gaagggtatt tcctggcctg tgagaaggaa agagatctgt tcaagctgat cctgaagaag  420
gaggacgagc tgggcgaccg gagcatcatg tttaccgtgc agaacgagga ctga        474

SEQ ID NO: 73           moltype = DNA  length = 471
FEATURE                 Location/Qualifiers
source                  1..471
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 73
aggttcggca aactggaaag caagctctcc gtgatccgga acctgaacga ccaggtgctg  60
tttatcgacc agggaaatag acctctgttc gaggacatga cagatagcga ctgcagagat  120
aatgcccctc ggaccatctt catcatctct acctaccggg acagccagcc tagaggcaag  180
gccgtgacca tctctgtcaa gtgcgagaag atcagcaccc tgagctgcga aaacaagatc  240
atcagcttca aggaaatgaa ccccccagac aacatcaagg acacaaagtc cgacatcatt  300
ttcttccaga gaaacgtgcc cggccacaag tacaagatgc aatttgagtc tagcagctac  360
gagggctatt tcctggcttg tgaaaaggaa agagatctgt tcaaactgat cctgaagaaa  420
gaggacgagc tgggcgatag aagcatcatg ttcaccgtgc agaacgagga t          471

SEQ ID NO: 74           moltype = DNA  length = 474
FEATURE                 Location/Qualifiers
source                  1..474
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 74
cgcttcggca aactggagtc caagctgagc gtgatcagga atctgaacga tcaggtgctc  60
ttcatcgatc agggcaatag acctctgttt gaggacatga ctgactccga ctgtcgggac  120
aacgcccccc gaacaatctt tattatcagc acctaccggg acagccagcc caggggggaag  180
gccgtgacta tctccgtgaa gtgcgaaaag atcagcaccc tgtcatgtga gaacaagatt  240
atctccttta aggagatgaa cccccccgat aacattaagg acaccaagag cgacatcatc  300
ttcttccagc gcaacgtgcc aggacacaag tacaaaatgc agtttgagag cagcagctac  360
gaagggtatt tcctggcctg tgagaaggaa agagatctgt tcaagctgat cctgaagaag  420
gaggacgagc tgggcgaccg gagcatcatg tttaccgtgc agaacgagga ctga        474

SEQ ID NO: 75           moltype = DNA  length = 471
FEATURE                 Location/Qualifiers
source                  1..471
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 75
tatttcggca agctggaaag caagctctct gtgatccgga acctgaacga ccaagtgctg  60
ttcatcgacc agggcaaccg gcctctgttc gaggatatga ccgacagcga ctgcagagat  120
aatgcccctc gcacaatctt catcatttct aaatacagcg atagcctggc cagaggcctg  180
gctgtgacca tcagcgtcaa gtgcgagaaa atctccaccc tgagctgcga gaacaagatc  240
atctctttca aggaaatgaa ccccccctgac aatatcaagg acacaaagtc cgacatcatc  300
tttttccaga gagatgtgcc aggacacagc agaaagatgc agttcgagag cagctcctac  360
gagggctact tcctggcctg tgaaaaggaa cgggacctgt ttaagctgat cctgaagaaa  420
gaagatgagc tgggcgacag aagcatcatg ttcaccgtgc agaacgagga c          471

SEQ ID NO: 76           moltype = DNA  length = 474
FEATURE                 Location/Qualifiers
source                  1..474
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 76
tatttcggca aactggagag caagctgagc gtgattcgga acctgaatga ccaggtgctg  60
ttcattgacc agggcaatcg ccctctgttc gaagacatga ccgactccga ttgtagagac  120
aacgcccctc ggaccatctt cattatcagc aagtacagcg acagcctggc ccgcggcctg  180
gctgtgacaa tcagcgtgaa gtgcgagaag atcagcacac tgtcctgcga gaacaaaatc  240
atcagcttta aagagatgaa tccccctgac aatatcaaag atactaaatc tgacatcatc  300
ttcttccagc gggacgtgcc tggccactct agaaaaatgc agttcgagtc ctccagttac  360
gaggggtact tcctggcctg cgagaaggag cgggacctgt tcaaactgat cctgaaaaaa  420
gaagacgagc tgggcgatcg gagcatcatg ttcactgtgc agaacgagga ttga        474
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SEQ ID NO: 77          moltype = DNA  length = 471
FEATURE                Location/Qualifiers
source                 1..471
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 77
tacttcggca agctggaaag caagctcagc gtcatccgga acctgaacga ccaggtgctg  60
ttcatcgacc aaggcaaccg gcctctgttt gaggacatga ccgactctga ttctagagat  120
aatgccccta gaaccatctt catcatctcc aagtacagcg atagcctggc cagaggcctg  180
gctgtgacaa tctctgtgaa gagcgagaaa atcagcaccc tgtcttgtga aaacaagatc  240
atcagcttca aggaaatgaa ccctccagac aatatcaagg acacaaagtc cgacatcatt  300
ttcttccaga gagatgtgcc cggccacagc aggaagatgc agttcgagtc cagcagctac  360
gaaggatatt ttctggcctg cgagaaggaa cgggacctgt tcaaactgat cctgaagaaa  420
gaggacgagc tgggcgacag aagcatcatg ttcaccgtgc agaacgagga t           471

SEQ ID NO: 78          moltype = DNA  length = 474
FEATURE                Location/Qualifiers
source                 1..474
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 78
tattttggga agctggagag caaactgagc gtgatccgga atctgaatga ccaggtgctg  60
ttcattgacc agggcaatcg accactgttc gaagatatga ccgatagcga cagcagggac  120
aatgcccccca ggaccatctt tatcatcagc aagtacagcg atagcctggc caggggactg  180
gccgtgacaa tcagcgtgaa gtccgagaag atcagcaccc tgtcctgcga gaacaagatc  240
atttctttca aggagatgaa cccccctgac aacatcaagg acacaaagag cgacatcatc  300
ttctttcaga gagacgtgcc cggacacagc cgcaagatgc agtttgaatc ttccagctac  360
gaagggtact tcctggcctg tgaaaaagag cgcgatctgt tcaaactgat cctcaagaag  420
gaggacgagc tgggcgatcg ctctatcatg ttcacagtgc agaacgagga ctga        474

SEQ ID NO: 79          moltype = DNA  length = 471
FEATURE                Location/Qualifiers
source                 1..471
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 79
tatttcggca agctggaaag caagctctct gtgatccgga acctgaacga ccaagtgctg  60
tttatcgacc agggaaatag acctctgttc gaggacatga ccgacagcga cagccgggat  120
aacgcccctc gcaccatctt catcatctcc aagtacagcg atagcctggc tagaggcctg  180
gccgtgacca tcagcgtcaa gggcgagaag atctctacac tgagctgtga aaacaagatc  240
atcagcttca aggaaatgaa cccccccgac aatatcaagg acacaaagtc cgacatcatt  300
ttcttccaga gagatgtgcc tggccacagc agaaagatgc agttcgagtc cagcagctac  360
gagggctact ttctggcctg cgagaaagaa cgggacctgt tcaaactgat cctgaaaaag  420
gaagacgagc tgggcgatag atctatcatg ttcaccgtgc agaacgagga t           471

SEQ ID NO: 80          moltype = DNA  length = 474
FEATURE                Location/Qualifiers
source                 1..474
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 80
tattttggga agctggagag caaactgagc gtgatccgga atctgaatga ccaggtgctg  60
ttcattgacc agggcaatcg accactgttc gaagatatga ccgatagcga cagcagggac  120
aatgcccccca ggaccatctt tatcatcagc aagtacagcg atagcctggc caggggactg  180
gccgtgacaa tcagcgtgaa gggcgagaag atcagcaccc tgtcctgcga gaacaagatc  240
atttctttca aggagatgaa cccccctgac aacatcaagg acacaaagag cgacatcatc  300
ttctttcaga gagacgtgcc cggacacagc cgcaagatgc agtttgaatc ttccagctac  360
gaagggtact tcctggcctg tgaaaaagag cgcgatctgt tcaaactgat cctcaagaag  420
gaggacgagc tgggcgatcg ctctatcatg ttcacagtgc agaacgagga ctga        474

SEQ ID NO: 81          moltype = DNA  length = 471
FEATURE                Location/Qualifiers
source                 1..471
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 81
tatttcggca agctggaaag caagctgagc gtcatccgga acctcaacga ccaagtgctg  60
tttatcgacc agggcaaccg gcctctgttc gaggacatga cagacagcga cagcagagat  120
aatgccccta gaaccatctt catcatttcc aagtacagcg attctctggc cagaggcctg  180
gccgtgacca tctctgtgaa ggccgagaaa atctccaccc tgagctgcga aaacaagatc  240
atcagcttca aggagatgaa ccctccagac aatatcaagg acaccaagag cgacatcatc  300
tttttccagc gggacgtgcc cggccactct cgcaagatgc agttcgagtc cagcagctac  360
gagggctact tcctggcttg tgaaaaggaa agagatctgt tcaaactgat cctgaaaaag  420
gaagatgagc tgggagatag aagcatcatg ttcacagtgc agaacgagga c           471
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SEQ ID NO: 82          moltype = DNA  length = 474
FEATURE                Location/Qualifiers
source                 1..474
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 82
tattttggga agctggagag caaactgagc gtgatccgga atctgaatga ccaggtgctg  60
ttcattgacc agggcaatcg accactgttc gaagatatga ccgatagcga cagcagggac  120
aatgcccccca ggaccatctt tatcatcagc aagtacagcg atagcctggc caggggactg  180
gccgtgacaa tcagcgtgaa ggccgagaag atcagcaccc tgtcctgcga gaacaagatc  240
atttctttca aggagatgaa ccccccctgac aacatcaagg acacaaagag cgacatcatc  300
ttctttcaga gagacgtgcc cggacacagc cgcaagatgc agtttgaatc ttccagctac  360
gaagggtact tcctggcctg tgaaaaagag cgcgatctgt tcaaactgat cctcaagaag  420
gaggacgagc tgggcgatcg ctctatcatg ttcacagtgc agaacgagga ctga        474

SEQ ID NO: 83          moltype = DNA  length = 471
FEATURE                Location/Qualifiers
source                 1..471
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 83
tatttcggca agctggaatc taaactctcc gtgatccgga acctgaacga ccaagtgctg  60
ttcatcgacc agggaaatag acctctgttc gaggatatga cagacagcga tagccgcgat  120
aatgccccta gaaccatctt tatcattagc aagtacagcg acagcctggc cagaggcctg  180
gccgtgacaa tcagcgtcaa ggacgagaaa atctccaccc tgagctgcga gaacaagatc  240
atcagcttca aggaaatgaa ccctccagac aacatcaagg acaccaagtc cgatatcatc  300
ttcttccagc gggacgtgcc cggccactct agaaagatgc agttcgagag ctctagctac  360
gagggctact tcctggcttg tgaaaaagaa cgggacctgt ttaagctgat cctgaagaag  420
gaagatgagc tgggcgacag aagcatcatg ttcaccgtgc agaacgagga c           471

SEQ ID NO: 84          moltype = DNA  length = 474
FEATURE                Location/Qualifiers
source                 1..474
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 84
tattttggga agctggagag caaactgagc gtgatccgga atctgaatga ccaggtgctg  60
ttcattgacc agggcaatcg accactgttc gaagatatga ccgatagcga cagcagggac  120
aatgccccca ggaccatctt tatcatcagc aagtacagcg atagcctggc caggggactg  180
gccgtgacaa tcagcgtgaa ggacgagaag atcagcaccc tgtcctgcga gaacaagatc  240
atttctttca aggagatgaa ccccccctgac aacatcaagg acacaaagag cgacatcatc  300
ttctttcaga gagacgtgcc cggacacagc cgcaagatgc agtttgaatc ttccagctac  360
gaagggtact tcctggcctg tgaaaaagag cgcgatctgt tcaaactgat cctcaagaag  420
gaggacgagc tgggcgatcg ctctatcatg ttcacagtgc agaacgagga ctga        474

SEQ ID NO: 85          moltype = DNA  length = 471
FEATURE                Location/Qualifiers
source                 1..471
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 85
tatttcggca agctggaaag caagctctcc gtgatcagaa acctgaacga ccaagtgctg  60
ttcatcgacc agggcaacag acctctgttt gaggatatga ccgactctga ttgcagagat  120
aatgccccta gaaccatctt tatcatcagc atgtacaagg acagccagcc tcggggaatg  180
gccgtgacca tcagcgtcaa gtgcgagaag atctctaccc tgagctgcga aaacaaaatc  240
atctccttca aggaaatgaa ccccccagac aatatcaagg atacaaagag cgacatcatt  300
ttcttccagc ggagcgtgcc cggccacgac aacaagatgc agttcgagag ctctagctac  360
gagggctact tcctggcttg tgaaaaggaa cgggacctgt tcaaactgat cctgaagaaa  420
gaggacgagc tgggcgatag aagcatcatg ttcacagtgc agaacgagga c           471

SEQ ID NO: 86          moltype = DNA  length = 474
FEATURE                Location/Qualifiers
source                 1..474
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 86
tacttcggca aactggagtc caagctgagc gtgatcagga atctgaacga tcaggtgctc  60
ttcatcgatc agggcaatag acctctgttt gaggacatga ctgactccga ctgtcgggac  120
aacgcccccc gaacaatctt tattatcagc atgtacaagg acagccagcc cagggggatg  180
gccgtgacta tctccgtgaa gtgcgaaaag atcagcaccc tgtcatgtga gaacaagatt  240
atctccttta aggagatgaa cccccccgat aacattaagg acaccaagag cgacatcatc  300
ttcttccagc gctccgtgcc aggacacgat aacaaaatgc agtttgagag cagcagctac  360
gaagggtatt tcctggcctg tgagaaggaa agagatctgt tcaagctgat cctgaagaag  420
gaggacgagc tgggcgaccg gagcatcatg tttaccgtgc agaacgagga ctga        474

SEQ ID NO: 87          moltype = DNA  length = 1184
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FEATURE                 Location/Qualifiers
source                  1..1184
                        mol_type = other DNA
                        organism = Homo sapiens
SEQUENCE: 87
cgtgaggctc cggtgcccgt cagtgggcag agcgcacatc gcccacagtc cccgagaagt  60
tggggggagg ggtcggcaat tgaaccggtg cctagagaag gtggcgcggg gtaaactggg  120
aaagtgatgt cgtgtactgg ctccgccttt ttcccgaggg tgggggagaa ccgtatataa  180
gtgcagtagt cgccgtgaac gttctttttc gcaacgggtt tgccgccaga acacaggtaa  240
gtgccgtgtg tggttcccgc gggcctggcc tctttacggg ttatggccct tgcgtgcctt  300
gaattacttc cacctggctg cagtacgtga ttcttgatcc cgagcttcgg gttggaagtg  360
ggtgggagag ttcgaggcct tgcgcttaag gagccccttc gcctcgtgct tgagttgagg  420
cctggcctgg gcgctggggc cgccgcgtgc gaatctggtg gcaccttcgc gcctgtctcg  480
ctgctttcga taagtctcta gccatttaaa atttttgatg acctgctgcg acgctttttt  540
tctggcaaga tagtcttgta aatgcgggcc aagatctgca cactggtatt tcggtttttg  600
gggccgcggg cggcgacggg gcccgtgcgt cccagcgcac atgttcggcg aggcggggcc  660
tgcgagcgcg gccaccgaga atcggacggg ggtagtctca agctggccgg cctgctctgg  720
tgcctggcct cgcgccgccg tgtatcgccc cgccctgggc ggcaaggctg gcccggtcgg  780
caccagttgc gtgagcggaa agatggccgc ttcccggccc tgctgcaggg agctcaaaat  840
ggaggacgcg gcgctcggga gagcgggcgg gtgagtcacc cacacaaagg aaaagggcct  900
ttccgtcctc agccgtcgct tcatgtgact ccacggagta ccgggcgccg tccaggcacc  960
tcgattagtt ctcgagcttt tggagtacgt cgtctttagg ttggggggag gggttttatg  1020
cgatggagtt tccccacact gagtgggtgg agactgaagt taggccagct tggcacttga  1080
tgtaattctc cttggaattt gccctttttg agtttggatc ttggttcatt ctcaagcctc  1140
agacagtggt tcaaagtttt tttcttccat ttcaggtgtc gtga                   1184

SEQ ID NO: 88           moltype = AA  length = 219
FEATURE                 Location/Qualifiers
source                  1..219
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 88
MCPARSLLLV ATLVLLDHLS LARNLPVATP DPGMFPCLHH SQNLLRAVSN MLQKARQTLE  60
FYPCTSEEID HEDITKDKTS TVEACLPLEL TKNESCLNSR ETSFITNGSC LASRKTSFMM  120
ALCLSSIYED LKMYQVEFKT MNAKLLMDPK RQIFLDQNML AVIDELMQAL NFNSETVPQK  180
SSLEEPDFYK TKIKLCILLH AFRIRAVTID RVMSYLNAS                         219

SEQ ID NO: 89           moltype = AA  length = 197
FEATURE                 Location/Qualifiers
source                  1..197
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 89
RNLPVATPDP GMFPCLHHSQ NLLRAVSNML QKARQTLEFY PCTSEEIDHE DITKDKTSTV  60
EACLPLELTK NESCLNSRET SFITNGSCLA SRKTSFMMAL CLSSIYEDLK MYQVEFKTMN  120
AKLLMDPKRQ IFLDQNMLAV IDELMQALNF NSETVPQKSS LEEPDFYKTK IKLCILLHAF  180
RIRAVTIDRV MSYLNAS                                                 197

SEQ ID NO: 90           moltype = AA  length = 328
FEATURE                 Location/Qualifiers
source                  1..328
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 90
MCHQQLVISW FSLVFLASPL VAIWELKKDV YVVELDWYPD APGEMVVLTC DTPEEDGITW  60
TLDQSSEVLG SGKTLTIQVK EFGDAGQYTC HKGGEVLSHS LLLLHKKEDG IWSTDILKDQ  120
KEPKNKTFLR CEAKNYSGRF TCWWLTTIST DLTFSVKSSR GSSDPQGVTC GAATLSAERV  180
RGDNKEYEYS VECQEDSACP AAEESLPIEV MVDAVHKLKY ENYTSSFFIR DIIKPDPPKN  240
LQLKPLKNSR QVEVSWEYPD TWSTPHSYFS LTFCVQVQGK SKREKKDRVF TDKTSATVIC  300
RKNASISVRA QDRYYSSSWS EWASVPCS                                     328

SEQ ID NO: 91           moltype = AA  length = 306
FEATURE                 Location/Qualifiers
source                  1..306
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 91
IWELKKDVYV VELDWYPDAP GEMVVLTCDT PEEDGITWTL DQSSEVLGSG KTLTIQVKEF  60
GDAGQYTCHK GGEVLSHSLL LLHKKEDGIW STDILKDQKE PKNKTFLRCE AKNYSGRFTC  120
WWLTTISTDL TFSVKSSRGS SDPQGVTCGA ATLSAERVRG DNKEYEYSVE CQEDSACPAA  180
EESLPIEVMV DAVHKLKYEN YTSSFFIRDI IKPDPPKNLQ LKPLKNSRQV EVSWEYPDTW  240
STPHSYFSLT FCVQVQGKSK REKKDRVFTD KTSATVICRK NASISVRAQD RYYSSSWSEW  300
ASVPCS                                                             306

SEQ ID NO: 92           moltype = AA  length = 328
FEATURE                 Location/Qualifiers
```

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```
source                  1..328
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 92
MCHQQLVISW FSLVFLASPL VAIWELKKDV YVVELDWYPD APGEMVVLTC DTPEEDGITW  60
TLDQSSEVLG SGKTLTIQVK EFGDAGQYTC HKGGEVLSHS LLLLHKKEDG IWSTDILKDQ  120
KEPKNKTFLR CEAKNYSGRF TCWWLTTIST DLTFSVKSSR GSSDPQGVTC GAATLSAERV  180
RGDNKEYEYS VECQEDSACP AAEESLPIEV MVDAVAKLKY ENYTSSFFIR DIIKPDPPKN  240
LQLKPLKNSR QVEVSWEYPD TWSTPHSYFS LTFCVQVQGK SKREKKDRVF TDKTSATVIC  300
RKNASISVRA QDRYYSSSWS EWASVPCS                                   328

SEQ ID NO: 93           moltype = AA  length = 328
FEATURE                 Location/Qualifiers
source                  1..328
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 93
MCHQQLVISW FSLVFLASPL VAIWELKKDV YVVELDWYPD APGEMVVLTC DTPEEDGITW  60
TLDQSSEVLG SGKTLTIQVK EFGDAGQYTC HKGGEVLSHS LLLLHKKEDG IWSTDILKDQ  120
KEPKNKTFLR CEAKNYSGRF TCWWLTTIST DLTFSVKSSR GSSDPQGVTC GAATLSAERV  180
RGDNKEYEYS VECQEDSACP AAEESLPIEV MVDAVHALKY ENYTSSFFIR DIIKPDPPKN  240
LQLKPLKNSR QVEVSWEYPD TWSTPHSYFS LTFCVQVQGK SKREKKDRVF TDKTSATVIC  300
RKNASISVRA QDRYYSSSWS EWASVPCS                                   328

SEQ ID NO: 94           moltype = AA  length = 328
FEATURE                 Location/Qualifiers
source                  1..328
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 94
MCHQQLVISW FSLVFLASPL VAIWELKKDV YVVELDWYPD APGEMVVLTC DTPEEDGITW  60
TLDQSSEVLG SGKTLTIQVK EFGDAGQYTC HKGGEVLSHS LLLLHKKEDG IWSTDILKDQ  120
KEPKNKTFLR CEAKNYSGRF TCWWLTTIST DLTFSVKSSR GSSDPQGVTC GAATLSAERV  180
RGDNKEYEYS VECQEDSACP AAEESLPIEV MVDAVHKLAY ENYTSSFFIR DIIKPDPPKN  240
LQLKPLKNSR QVEVSWEYPD TWSTPHSYFS LTFCVQVQGK SKREKKDRVF TDKTSATVIC  300
RKNASISVRA QDRYYSSSWS EWASVPCS                                   328

SEQ ID NO: 95           moltype = AA  length = 328
FEATURE                 Location/Qualifiers
source                  1..328
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 95
MCHQQLVISW FSLVFLASPL VAIWELKKDV YVVELDWYPD APGEMVVLTC DTPEEDGITW  60
TLDQSSEVLG SGKTLTIQVK EFGDAGQYTC HKGGEVLSHS LLLLHKKEDG IWSTDILKDQ  120
KEPKNKTFLR CEAKNYSGRF TCWWLTTIST DLTFSVKSSR GSSDPQGVTC GAATLSAERV  180
RGDNKEYEYS VECQEDSACP AAEESLPIEV MVDAVAALKY ENYTSSFFIR DIIKPDPPKN  240
LQLKPLKNSR QVEVSWEYPD TWSTPHSYFS LTFCVQVQGK SKREKKDRVF TDKTSATVIC  300
RKNASISVRA QDRYYSSSWS EWASVPCS                                   328

SEQ ID NO: 96           moltype = AA  length = 328
FEATURE                 Location/Qualifiers
source                  1..328
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 96
MCHQQLVISW FSLVFLASPL VAIWELKKDV YVVELDWYPD APGEMVVLTC DTPEEDGITW  60
TLDQSSEVLG SGKTLTIQVK EFGDAGQYTC HKGGEVLSHS LLLLHKKEDG IWSTDILKDQ  120
KEPKNKTFLR CEAKNYSGRF TCWWLTTIST DLTFSVKSSR GSSDPQGVTC GAATLSAERV  180
RGDNKEYEYS VECQEDSACP AAEESLPIEV MVDAVAKLAY ENYTSSFFIR DIIKPDPPKN  240
LQLKPLKNSR QVEVSWEYPD TWSTPHSYFS LTFCVQVQGK SKREKKDRVF TDKTSATVIC  300
RKNASISVRA QDRYYSSSWS EWASVPCS                                   328

SEQ ID NO: 97           moltype = AA  length = 328
FEATURE                 Location/Qualifiers
source                  1..328
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 97
MCHQQLVISW FSLVFLASPL VAIWELKKDV YVVELDWYPD APGEMVVLTC DTPEEDGITW  60
TLDQSSEVLG SGKTLTIQVK EFGDAGQYTC HKGGEVLSHS LLLLHKKEDG IWSTDILKDQ  120
KEPKNKTFLR CEAKNYSGRF TCWWLTTIST DLTFSVKSSR GSSDPQGVTC GAATLSAERV  180
RGDNKEYEYS VECQEDSACP AAEESLPIEV MVDAVHALAY ENYTSSFFIR DIIKPDPPKN  240
LQLKPLKNSR QVEVSWEYPD TWSTPHSYFS LTFCVQVQGK SKREKKDRVF TDKTSATVIC  300
RKNASISVRA QDRYYSSSWS EWASVPCS                                   328
```

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SEQ ID NO: 98          moltype = AA  length = 328
FEATURE                Location/Qualifiers
source                 1..328
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 98
MCHQQLVISW FSLVFLASPL VAIWELKKDV YVVELDWYPD APGEMVVLTC DTPEEDGITW  60
TLDQSSEVLG SGKTLTIQVK EFGDAGQYTC HKGGEVLSHS LLLLHKKEDG IWSTDILKDQ  120
KEPKNKTFLR CEAKNYSGRF TCWWLTTIST DLTFSVKSSR GSSDPQGVTC GAATLSAERV  180
RGDNKEYEYS VECQEDSACP AAEESLPIEV MVDAVAALAY ENYTSSFFIR DIIKPDPPKN  240
LQLKPLKNSR QVEVSWEYPD TWSTPHSYFS LTFCVQVQGK SKREKKDRVF TDKTSATVIC  300
RKNASISVRA QDRYYSSSWS EWASVPCS                                     328

SEQ ID NO: 99          moltype = AA  length = 328
FEATURE                Location/Qualifiers
source                 1..328
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 99
MCHQQLVISW FSLVFLASPL VAIWELKKDV YVVELDWYPD APGEMVVLTC DTPEEDGITW  60
TLDQSSEVLG SGKTLTIQVK AFGDAGQYTC HKGGEVLSHS LLLLHKKEDG IWSTDILKDQ  120
KEPKNKTFLR CEAKNYSGRF TCWWLTTIST DLTFSVKSSR GSSDPQGVTC GAATLSAERV  180
RGDNKEYEYS VECQEDSACP AAEESLPIEV MVDAVHKLKY ENYTSSFFIR DIIKPDPPKN  240
LQLKPLKNSR QVEVSWEYPD TWSTPHSYFS LTFCVQVQGK SKREKKDRVF TDKTSATVIC  300
RKNASISVRA QDRYYSSSWS EWASVPCS                                     328

SEQ ID NO: 100         moltype = AA  length = 328
FEATURE                Location/Qualifiers
source                 1..328
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 100
MCHQQLVISW FSLVFLASPL VAIWELKKDV YVVELDWYPD APGEMVVLTC DTPEEDGITW  60
TLDQSSEVLG SGKTLTIQVK EAGDAGQYTC HKGGEVLSHS LLLLHKKEDG IWSTDILKDQ  120
KEPKNKTFLR CEAKNYSGRF TCWWLTTIST DLTFSVKSSR GSSDPQGVTC GAATLSAERV  180
RGDNKEYEYS VECQEDSACP AAEESLPIEV MVDAVHKLKY ENYTSSFFIR DIIKPDPPKN  240
LQLKPLKNSR QVEVSWEYPD TWSTPHSYFS LTFCVQVQGK SKREKKDRVF TDKTSATVIC  300
RKNASISVRA QDRYYSSSWS EWASVPCS                                     328

SEQ ID NO: 101         moltype = AA  length = 328
FEATURE                Location/Qualifiers
source                 1..328
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 101
MCHQQLVISW FSLVFLASPL VAIWELKKDV YVVELDWYAA APGEMVVLTC DTPEEDGITW  60
TLDQSSEVLG SGKTLTIQVK AAGDAGQYTC HKGGEVLSHS LLLLHKKEDG IWSTDILKDQ  120
KEPKNKTFLR CEAKNYSGRF TCWWLTTIST DLTFSVKSSR GSSDPQGVTC GAATLSAERV  180
RGDNKEYEYS VECQEDSACP AAEESLPIEV MVDAVHKLKY ENYTSSFFIR DIIKPDPPKN  240
LQLKPLKNSR QVEVSWEYPD TWSTPHSYFS LTFCVQVQGK SKREKKDRVF TDKTSATVIC  300
RKNASISVRA QDRYYSSSWS EWASVPCS                                     328

SEQ ID NO: 102         moltype = AA  length = 328
FEATURE                Location/Qualifiers
source                 1..328
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 102
MCHQQLVISW FSLVFLASPL VAIWELKKDV YVVELDWYPD APGEMVVLTC DTPEEDGITW  60
TLDQSSEVLG SGKTLTIQVK EFGDAGQYTC HKGGEVLSHS LLLLHAKEDG IWSTDILKDQ  120
KEPKNKTFLR CEAKNYSGRF TCWWLTTIST DLTFSVKSSR GSSDPQGVTC GAATLSAERV  180
RGDNKEYEYS VECQEDSACP AAEESLPIEV MVDAVHKLKY ENYTSSFFIR DIIKPDPPKN  240
LQLKPLKNSR QVEVSWEYPD TWSTPHSYFS LTFCVQVQGK SKREKKDRVF TDKTSATVIC  300
RKNASISVRA QDRYYSSSWS EWASVPCS                                     328

SEQ ID NO: 103         moltype = AA  length = 328
FEATURE                Location/Qualifiers
source                 1..328
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 103
MCHQQLVISW FSLVFLASPL VAIWELKKDV YVVELDWYPD APGEMVVLTC DTPEEDGITW  60
TLDQSSEVLG SGKTLTIQVK EFGDAGQYTC HKGGEVLSHS LLLLHKKEDG IWSTDILKDQ  120
KEPKNKTFLR CEAKNYSGRF TCWWLTTIST DLTFSVKSSR GSSDPQGVTC GAATLSAERV  180
RGDNKEYEYS VECQEDSACP AAEESLPIEV MVDAVHALKY ENYTSSFFIR DIIKPDPPKN  240
LQLKPLKNSR QVEVSWEYPD TWSTPHSYFS LTFCVQVQGK SKREKKDRVF TDKTSATVIC  300
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RKNASISVRA QDRYYSSSWS EWASVPCS                                 328

SEQ ID NO: 104          moltype = AA  length = 328
FEATURE                 Location/Qualifiers
source                  1..328
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 104
MCHQQLVISW FSLVFLASPL VAIWELKKDV YVVELDWYPD APGEMVVLTC DTPEEDGITW  60
TLDQSSEVLG SGKTLTIQVK EFGDAGQYTC HKGGEVLSHS LLLLHKKEDG IWSTDILKDQ  120
KEPKNKTFLR CEAKNYSGRF TCWWLTTIST DLTFSVKSSR GSSDPQGVTC GAATLSAERV  180
RGDNKEYEYS VECQEDSACP AAEESLPIEV MVDAVHKLAY ENYTSSFFIR DIIKPDPPKN  240
LQLKPLKNSR QVEVSWEYPD TWSTPHSYFS LTFCVQVQGK SKREKKDRVF TDKTSATVIC  300
RKNASISVRA QDRYYSSSWS EWASVPCS                                 328

SEQ ID NO: 105          moltype = AA  length = 328
FEATURE                 Location/Qualifiers
source                  1..328
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 105
MCHQQLVISW FSLVFLASPL VAIWELKKDV YVVELDWYPD APGEMVVLTC DTPEEDGITW  60
TLDQSSEVLG SGKTLTIQVK AAGDAGQYTC HKGGEVLSHS LLLLHKKEDG IWSTDILKDQ  120
KEPKNKTFLR CEAKNYSGRF TCWWLTTIST DLTFSVKSSR GSSDPQGVTC GAATLSAERV  180
RGDNKEYEYS VECQEDSACP AAEESLPIEV MVDAVHKLKY ENYTSSFFIR DIIKPDPPKN  240
LQLKPLKNSR QVEVSWEYPD TWSTPHSYFS LTFCVQVQGK SKREKKDRVF TDKTSATVIC  300
RKNASISVRA QDRYYSSSWS EWASVPCS                                 328

SEQ ID NO: 106          moltype = AA  length = 328
FEATURE                 Location/Qualifiers
source                  1..328
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 106
MCHQQLVISW FSLVFLASPL VAIWELKKDV YVVELDWYPD APGEMVVLTC DTPEEDGITW  60
TLDQSSEVLG SGKTLTIQVK AAGDAGQYTC HKGGEVLSHS LLLLHAKEDG IWSTDILKDQ  120
KEPKNKTFLR CEAKNYSGRF TCWWLTTIST DLTFSVKSSR GSSDPQGVTC GAATLSAERV  180
RGDNKEYEYS VECQEDSACP AAEESLPIEV MVDAVHKLKY ENYTSSFFIR DIIKPDPPKN  240
LQLKPLKNSR QVEVSWEYPD TWSTPHSYFS LTFCVQVQGK SKREKKDRVF TDKTSATVIC  300
RKNASISVRA QDRYYSSSWS EWASVPCS                                 328

SEQ ID NO: 107          moltype = AA  length = 328
FEATURE                 Location/Qualifiers
source                  1..328
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 107
MCHQQLVISW FSLVFLASPL VAIWELKKDV YVVELDAYPD APGEMVVLTC DTPEEDGITW  60
TLDQSSEVLG SGKTLTIQVK AAGDAGQYTC HKGGEVLSHS LLLLHKKEDG IWSTDILKDQ  120
KEPKNKTFLR CEAKNYSGRF TCWWLTTIST DLTFSVKSSR GSSDPQGVTC GAATLSAERV  180
RGDNKEYEYS VECQEDSACP AAEESLPIEV MVDAVHKLKY ENYTSSFFIR DIIKPDPPKN  240
LQLKPLKNSR QVEVSWEYPD TWSTPHSYFS LTFCVQVQGK SKREKKDRVF TDKTSATVIC  300
RKNASISVRA QDRYYSSSWS EWASVPCS                                 328

SEQ ID NO: 108          moltype = AA  length = 328
FEATURE                 Location/Qualifiers
source                  1..328
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 108
MCHQQLVISW FSLVFLASPL VAIWELKKDV YVVELDWYPD APGEMVVLTC DTPEEDGITW  60
TLDQSSEVLG SGKTLTIQVK AAGDAGQYTC HKGGEVLSHS LLLLHAKADG IWSTAILKDQ  120
KEPKNKTFLR CEAKNYSGRF TCWWLTTIST DLTFSVKSSR GSSDPQGVTC GAATLSAERV  180
RGDNKEYEYS VECQEDSACP AAEESLPIEV MVDAVHKLKY ENYTSSFFIR DIIKPDPPKN  240
LQLKPLKNSR QVEVSWEYPD TWSTPHSYFS LTFCVQVQGK SKREKKDRVF TDKTSATVIC  300
RKNASISVRA QDRYYSSSWS EWASVPCS                                 328

SEQ ID NO: 109          moltype = AA  length = 481
FEATURE                 Location/Qualifiers
source                  1..481
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 109
MALPVTALLL PLALLLHAAR PDVVMTQSPD SLAVSLGERA TINCKSSQSL LDSDGKTYLN  60
WLQQKPGQPP KRLISLVSKL DSGVPDRFSG SGSGTDFTLT ISSLQAEDVA VYYCWQGTHF  120
PGTFGGGTKV EIKGGGGSGG GGSGGGGSEI QLVQSGAEVK KPGESLRISC KGSGFNIEDY  180
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YIHWVRQMPG KGLEWMGRID PENDETKYGP IFQGHVTISA DTSINTVYLQ WSSLKASDTA  240
MYYCAFRGGV YWGQGTTVTV SSTTTPAPRP PTPAPTIASQ PLSLRPEACR PAAGGAVHTR  300
GLDFACDIYI WAPLAGTCGV LLLSLVITRS KRSRLLHSDY MNMTPRRPGP TRKHYQPYAP  360
PRDFAAYRSR VKFSRSADAP AYQQGQNQLY NELNLGRREE YDVLDKRRGR DPEMGGKPRR  420
KNPQEGLYNE LQKDKMAEAY SEIGMKGERR RGKGHDGLYQ GLSTATKDTY DALHMQALPP  480
R                                                                  481

SEQ ID NO: 110          moltype = AA  length = 112
FEATURE                 Location/Qualifiers
source                  1..112
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 110
DVVMTQSPDS LAVSLGERAT INCKSSQSLL DSDGKTYLNW LQQKPGQPPK RLISLVSKLD  60
SGVPDRFSGS GSGTDFTLTI SSLQAEDVAV YYCWQGTHFP GTFGGGTKVE IK          112

SEQ ID NO: 111          moltype = AA  length = 114
FEATURE                 Location/Qualifiers
source                  1..114
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 111
EIQLVQSGAE VKKPGESLRI SCKGSGFNIE DYYIHWVRQM PGKGLEWMGR IDPENDETKY  60
GPIFQGHVTI SADTSINTVY LQWSSLKASD TAMYYCAFRG GVYWGQGTTV TVSS        114

SEQ ID NO: 112          moltype = AA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 112
KSSQSLLDSD GKTYLN                                                  16

SEQ ID NO: 113          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 113
LVSKLDS                                                            7

SEQ ID NO: 114          moltype = AA  length = 9
FEATURE                 Location/Qualifiers
source                  1..9
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 114
WQGTHFPGT                                                          9

SEQ ID NO: 115          moltype = AA  length = 5
FEATURE                 Location/Qualifiers
source                  1..5
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 115
DYYIH                                                              5

SEQ ID NO: 116          moltype = AA  length = 17
FEATURE                 Location/Qualifiers
source                  1..17
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 116
RIDPENDETK YGPIFQG                                                 17

SEQ ID NO: 117          moltype = AA  length = 5
FEATURE                 Location/Qualifiers
source                  1..5
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 117
RGGVY                                                              5

SEQ ID NO: 118          moltype = AA  length = 25
FEATURE                 Location/Qualifiers
source                  1..25
```

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                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 118
MGVLLTQRTL LSLVLALLFP SMASM                                        25

SEQ ID NO: 119          moltype = AA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 119
MKCLLYLAFL FIGVNC                                                  16

SEQ ID NO: 120          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 120
METDTLLLWV LLLWVPGSTG D                                            21

SEQ ID NO: 121          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 121
MGWSCIILFL VATATGVHS                                               19

SEQ ID NO: 122          moltype = AA  length = 17
FEATURE                 Location/Qualifiers
source                  1..17
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 122
MRAWIFFLLC LAGRALA                                                 17

SEQ ID NO: 123          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 123
MWWRLWWLLL LLLLLWPMVW A                                            21

SEQ ID NO: 124          moltype = AA  length = 22
FEATURE                 Location/Qualifiers
source                  1..22
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 124
MDMRVPAQLL GLLLLWLRGA RC                                           22

SEQ ID NO: 125          moltype = AA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 125
MPLLLLLPLL WAGALA                                                  16

SEQ ID NO: 126          moltype = AA  length = 23
FEATURE                 Location/Qualifiers
source                  1..23
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 126
MDAMKRGLCC VLLLCGAVFV SPS                                          23

SEQ ID NO: 127          moltype = AA  length = 18
FEATURE                 Location/Qualifiers
source                  1..18
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 127
MAFLWLLSCW ALLGTTFG                                                18
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SEQ ID NO: 128          moltype = AA  length = 15
FEATURE                 Location/Qualifiers
source                  1..15
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 128
MNLLLILTFV AAAVA                                                   15

SEQ ID NO: 129          moltype = AA  length = 20
FEATURE                 Location/Qualifiers
source                  1..20
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 129
MYRMQLLSCI ALSLALVTNS                                              20

SEQ ID NO: 130          moltype = AA  length = 17
FEATURE                 Location/Qualifiers
source                  1..17
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 130
MGVKVLFALI CIAVAEA                                                 17

SEQ ID NO: 131          moltype = AA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 131
MKWVTFISLL FSSAYS                                                  16

SEQ ID NO: 132          moltype = AA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 132
MKTIIALSYI FCLVLG                                                  16

SEQ ID NO: 133          moltype = AA  length = 24
FEATURE                 Location/Qualifiers
source                  1..24
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 133
MALWMRLLPL LALLALWGPD PAAA                                         24

SEQ ID NO: 134          moltype = AA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 134
MKPIFLVLLV VTSAYA                                                  16

SEQ ID NO: 135          moltype = AA  length = 28
FEATURE                 Location/Qualifiers
source                  1..28
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 135
MRHNWTPDLS PLWVLLLCAH VVTLLVRA                                     28

SEQ ID NO: 136          moltype = AA  length = 22
FEATURE                 Location/Qualifiers
source                  1..22
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 136
MCHQQLVISW FSLVFLASPL VA                                           22

SEQ ID NO: 137          moltype = AA  length = 22
FEATURE                 Location/Qualifiers
source                  1..22
```

-continued

```
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 137
MCPARSLLLV ATLVLLDHLS LA                                   22

SEQ ID NO: 138           moltype = AA  length = 36
FEATURE                  Location/Qualifiers
source                   1..36
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 138
MAAEPVEDNC INFVAMKFID NTLYFIAEDD ENLESD                    36

SEQ ID NO: 139           moltype = AA  length = 21
FEATURE                  Location/Qualifiers
source                   1..21
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 139
MALPVTALLL PLALLLHAAR P                                    21

SEQ ID NO: 140           moltype = AA  length = 19
FEATURE                  Location/Qualifiers
source                   1..19
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 140
MEFGLSWVFL VALFRGVQS                                       19

SEQ ID NO: 141           moltype = AA  length = 22
FEATURE                  Location/Qualifiers
source                   1..22
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 141
MGYRMQLLSC IALSLALVTN SG                                   22

SEQ ID NO: 142           moltype = DNA  length = 553
FEATURE                  Location/Qualifiers
source                   1..553
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 142
acgttactgg ccgaagccgc ttggaataag gccggtgtgc gtttgtctat atgttatttt  60
ccaccatatt gccgtctttt ggcaatgtga gggcccggaa acctggccct gtcttcttga  120
cgagcattcc taggggtctt tcccctctcg ccaaaggaat gcaaggtctg ttgaatgtcg  180
tgaaggaagc agttcctctg gaagcttctt gaagacaaac aacgtctgta gcgacccttt  240
gcaggcagcg gaacccccca cctggcgaca ggtgcctctg cggccaaaag ccacgtgtat  300
aagatacacc tgcaaaggcg gcacaacccc agtgccacgt tgtgagttgg atagttgtgg  360
aaagagtcaa atggctctcc tcaagcgtat tcaacaaggg gctgaaggat gcccagaagg  420
taccccattg tatgggatct gatctggggc ctcggtgcac atgctttaca tgtgtttagt  480
cgaggttaaa aaaacgtcta ggccccccga accacgggga cgtggttttc ctttgaaaaa  540
cacgatgata ata                                                     553

SEQ ID NO: 143           moltype = AA  length = 21
FEATURE                  Location/Qualifiers
source                   1..21
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 143
GSGEGRGSLL TCGDVEENPG P                                    21

SEQ ID NO: 144           moltype = AA  length = 22
FEATURE                  Location/Qualifiers
source                   1..22
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 144
GSGATNFSLL KQAGDVEENP GP                                   22

SEQ ID NO: 145           moltype = AA  length = 23
FEATURE                  Location/Qualifiers
source                   1..23
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 145
```

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GSGQCTNYAL LKLAGDVESN PGP                                        23

SEQ ID NO: 146          moltype = AA  length = 25
FEATURE                 Location/Qualifiers
source                  1..25
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 146
GSGVKQTLNF DLLKLAGDVE SNPGP                                      25

SEQ ID NO: 147          moltype = AA  length = 121
FEATURE                 Location/Qualifiers
source                  1..121
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 147
QVQLVQSGAE VKKPGASVKV SCKASGYIFT SSIMHWVRQA PGQGLEWIGY IKPYNDGTKY  60
NEKFKGRATL TSDRSTSTAY MELSSLRSED TAVYYCAREG GNDYYDTMDY WGQGTLVTVS  120
S                                                                121

SEQ ID NO: 148          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 148
EIQMTQSPSS LSASVGDRVT ITCRASQDIN SYLSWFQQKP GKAPKTLIYR VNRLVDGVPS  60
RFSGSGSGND YTLTISSLQP EDFATYYCLQ YDAFPYTFGQ GTKVEIK              107

SEQ ID NO: 149          moltype = AA  length = 120
FEATURE                 Location/Qualifiers
source                  1..120
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 149
EVQLVQSGAE VKKPGESLKI SCKGSGYSFT DYYMKWARQM PGKGLEWMGD IIPSNGATFY  60
NQKFKGQVTI SADKSISTTY LQWSSLKASD TAMYYCARSH LLRASWFAYW GQGTMVTVSS  120

SEQ ID NO: 150          moltype = AA  length = 113
FEATURE                 Location/Qualifiers
source                  1..113
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 150
DIVMTQSPDS LAVSLGERAT INCESSQSLL NSGNQKNYLT WYQQKPGQPP KPLIYWASTR  60
ESGVPDRFSG SGSGTDFTLT ISSLQAEDVA VYYCQNDYSY PYTFGQGTKL EIK        113

SEQ ID NO: 151          moltype = AA  length = 120
FEATURE                 Location/Qualifiers
source                  1..120
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 151
QVQLQQSGAE VKKPGASVKV SCKASGYTFT DYYMKWVKQS HGKSLEWMGD IIPSNGATFY  60
NQKFKGKATL TVDRSTSTAY MELSSLRSED TAVYYCARSH LLRASWFAYW GQGTLVTVSS  120

SEQ ID NO: 152          moltype = AA  length = 113
FEATURE                 Location/Qualifiers
source                  1..113
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 152
DFVMTQSPDS LAVSLGERAT INCKSSQSLL NTGNQKNYLT WYQQKPGQPP KLLIYWASTR  60
ESGVPDRFTG SGSGTDFTLT ISSLQAEDVA VYYCQNDYSY PYTFGGGTKL EIK        113

SEQ ID NO: 153          moltype = AA  length = 120
FEATURE                 Location/Qualifiers
source                  1..120
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 153
QVQLEQPGAE VVKPGASVKV SCKTSGYTFT SNWMHWVKQT PGKGLEWIGE IDPSDSYTNY  60
NQKFDGKAKL TVDKSSSTAY MEVSDLTAED SATYYCARGS NPYYYAMDYW GQGTSVTVSS  120

SEQ ID NO: 154          moltype = AA  length = 104
FEATURE                 Location/Qualifiers
```

-continued

```
source                  1..104
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 154
QIVLTQSPAT LSASPGEKAT MTCSASSGVN YMHWYQQKPG TSPKRWIYDT DKTASGVPAR  60
FSGSGSGTSY SLTISSMEAE DAATYYCHQR GSYTFGGGTK LEIK                  104

SEQ ID NO: 155          moltype = AA  length = 120
FEATURE                 Location/Qualifiers
source                  1..120
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 155
QVQLVQPGAE VVKPGASVKL SCKTSGYTFT SNWMHWVKQA PGQGLEWIGE IDPSDSYTNY  60
NQNFQGKAKL TVDKSTSTAY MEVSSLRSDD TAVYYCARGS NPYYYAMDYW GQGTSVTVSS  120

SEQ ID NO: 156          moltype = AA  length = 104
FEATURE                 Location/Qualifiers
source                  1..104
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 156
EIVLTQSPAI MSASPGERVT MTCSASSGVN YMHWYQQKPG TSPRRWIYDT SKLASGVPAR  60
FSGSGSGTDY SLTISSMEPE DAATYYCHQR GSYTFGGGTK LEIK                  104

SEQ ID NO: 157          moltype = AA  length = 120
FEATURE                 Location/Qualifiers
source                  1..120
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 157
QVQLQESGPG LVKPSQTLSL TCTVSGGSIS TSGMGVGWIR QHPGKGLEWI GHIWWDDDKR  60
YNPALKSRVT ISVDTSKNQF SLKLSSVTAA DTAVYYCARM ELWSYYFDYW GQGTLVTVSS  120

SEQ ID NO: 158          moltype = AA  length = 106
FEATURE                 Location/Qualifiers
source                  1..106
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 158
EIVLTQSPAT LSLSPGERAT LSCSASSSVS YMHWYQQKPG QAPRLLIYDT SKLASGIPAR  60
FSGSGSGTDF TLTISSLEPE DVAVYYCFQG SVYPFTFGQG TKLEIK                106

SEQ ID NO: 159          moltype = AA  length = 121
FEATURE                 Location/Qualifiers
source                  1..121
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 159
EVQLVESGGG LVQPGGSLRL SCAASGFTFS SSWMNWVRQA PGKGLEWVGR IYPGDGDTNY  60
NVKFKGRFTI SRDDSKNSLY LQMNSLKTED TAVYYCARSG FITTVRDFDY WGQGTLVTVS  120
S                                                                 121

SEQ ID NO: 160          moltype = AA  length = 111
FEATURE                 Location/Qualifiers
source                  1..111
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 160
EIVLTQSPDF QSVTPKEKVT ITCRASESVD TFGISFMNWF QQKPDQSPKL LIHEASNQGS  60
GVPSRFSGSG SGTDFTLTIN SLEAEDAATY YCQQSKEVPF TFGGGTKVEI K          111

SEQ ID NO: 161          moltype = AA  length = 122
FEATURE                 Location/Qualifiers
source                  1..122
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 161
EVQLVESGGG LVKPGGSLRL SCAASGYTFS NYWIGWVRQA PGKGLEWIGD IYPGGNYIRN  60
NEKFKDKTTL SADTSKNTAY LQMNSLKTED TAVYYCGSSF GSNYVFAWFT YWGQGTLVTV  120
SS                                                                122

SEQ ID NO: 162          moltype = AA  length = 112
FEATURE                 Location/Qualifiers
source                  1..112
                        mol_type = protein
```

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                        organism = sythetic construct
SEQUENCE: 162
DIVMTQSPLS LPVTPGEPAS ISCRSSQRLL SSYGHTYLHW YLQKPGQSPQ LLIYEVSNRF  60
SGVPDRFSGS GSGTDFTLKI SRVEAEDVGV YYCSQSTHVP LTFGQGTKVE IK          112

SEQ ID NO: 163          moltype = AA  length = 120
FEATURE                 Location/Qualifiers
source                  1..120
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 163
QVQLVQPGAE VVKPGASVKL SCKTSGYTFT SNWMHWVKQA PGQGLEWIGE IDPSDSYTNY  60
NQNFQGKAKL TVDKSTSTAY MEVSSLRSDD TAVYYCARGS NPYYYAMDYW GQGTSVTVSS  120

SEQ ID NO: 164          moltype = AA  length = 104
FEATURE                 Location/Qualifiers
source                  1..104
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 164
EIVLTQSPAI MSASPGERVT MTCSASSGVN YMHWYQQKPG TSPRRWIYDT SKLASGVPAR  60
FSGSGSGTSY SLTISSMEPE DAATYYCHQR GSYTFGGGTK LEIK                  104

SEQ ID NO: 165          moltype = AA  length = 119
FEATURE                 Location/Qualifiers
source                  1..119
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 165
EVQLVESGGD LVQPGRSLRL SCAASGFIFS NYGMSWVRQA PGKGLEWVAT ISSASTYSYY  60
PDSVKGRFTI SRDNAKNSLY LQMNSLRVED TALYYCGRHS DGNFAFGYWG QGTLVTVSS   119

SEQ ID NO: 166          moltype = AA  length = 112
FEATURE                 Location/Qualifiers
source                  1..112
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 166
DVLMTQSPLS LPVTPGEPAS ISCRSSRNIV HINGDTYLEW YLQKPGQSPQ LLIYKVSNRF  60
SGVPDRFSGS GSGTDFTLKI SRVEAEDVGV YYCFQGSLLP WTFGQGTKVE IK          112

SEQ ID NO: 167          moltype = AA  length = 121
FEATURE                 Location/Qualifiers
source                  1..121
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 167
EVQLVESGGG LVKPGGSLKL SCAASGYTFT SYVMHWVRQA PGKGLEWIGY INPYNDGTKY  60
NEKFQGRVTI SSDKSISTAY MELSSLRSED TAMYYCARGT YYYGTRVFDY WGQGTLVTVS  120
S                                                                  121

SEQ ID NO: 168          moltype = AA  length = 112
FEATURE                 Location/Qualifiers
source                  1..112
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 168
DIVMTQSPAT LSLSPGERAT LSCRSSKSLQ NVNGNTYLYW FQQKPGQSPQ LLIYRMSNLN  60
SGVPDRFSGS GSGTEFTLTI SSLEPEDFAV YYCMQHLEYP ITFGAGTKLE IK          112

SEQ ID NO: 169          moltype = AA  length = 117
FEATURE                 Location/Qualifiers
source                  1..117
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 169
EVQLVESGGG LVKPGGSLRL SCAASGFTFS AYAMNWVRQA PGKGLEWVGR IRTKNNNYAT  60
YYADSVKDRF TISRDDSKNT LYLQMNSLKT EDTAVYYCTT FYGNGVWGQG TLVTVSS     117

SEQ ID NO: 170          moltype = AA  length = 112
FEATURE                 Location/Qualifiers
source                  1..112
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 170
DVVMTQSPLS LPVTLGQPAS ISCKSSQSLL DSDGKTFLNW FQQRPGQSPR RLIYLVSKLD  60
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SGVPDRFSGS GSGTDFTLKI SRVEAEDVGV YYCWQGTHFP YTFGQGTRLE IK          112

SEQ ID NO: 171          moltype = AA  length = 121
FEATURE                 Location/Qualifiers
source                  1..121
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 171
EVQLVESGGG LVKPGGSLKL SCAASGYTFT SYVMHWVRQA PGKGLEWIGY INPYNDGTKY  60
NEKFQGRVTI SSDKSISTAY MELSSLRSED TAMYYCARGT YYYGTRVFDY WGQGTLVTVS  120
S                                                                   121

SEQ ID NO: 172          moltype = AA  length = 112
FEATURE                 Location/Qualifiers
source                  1..112
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 172
DIVMTQSPAT LSLSPGERAT LSCRSSKSLQ NVNGNTYLYW FQQKPGQSPQ LLIYRMSNLN  60
SGVPDRFSGS GSGTEFTLTI SSLEPEDFAV YYCMQHLEYP ITFGAGTKLE IK          112

SEQ ID NO: 173          moltype = AA  length = 121
FEATURE                 Location/Qualifiers
source                  1..121
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 173
EVQLVQPGAE VVKPGASVKV SCKASGYTFT SYNMHWVRQA PGRGLEWMGA IYPGNGDTSY  60
NQKFKGRVTM TRDKSTSTVY MELSSLRSED TAVYYCARST YYGGDWYFNV WGQGTLVTVS  120
S                                                                   121

SEQ ID NO: 174          moltype = AA  length = 106
FEATURE                 Location/Qualifiers
source                  1..106
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 174
QIVLSQSPAI LSASPGERVT LTCRASSSVS YIHWFQQKPG KAPKPLIYAT SNLASGVPSR  60
FSGSGSGTDF SLTISRVEPE DFAVYYCQQW TSNPPTFGGG TKVEIK                  106

SEQ ID NO: 175          moltype = AA  length = 122
FEATURE                 Location/Qualifiers
source                  1..122
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 175
QAYLQQSGAE LVRPGASVKM SCKASGYTFT SYNMHWVKQT PRQGLEWIGA IYPGNGDTSY  60
NQKFKGKATL TVDKSSSTAY MQLSSLTSED SAVYFCARVV YYSNSYWYFD VWGTGTTVTV  120
SA                                                                  122

SEQ ID NO: 176          moltype = AA  length = 106
FEATURE                 Location/Qualifiers
source                  1..106
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 176
QIVLSQSPAI LSASPGEKVT MTCRASSSVS YMHWYQQKPG SSPKPWIYAP SNLASGVPAR  60
FSGSGSGTSY SLTISRVEAE DAATYYCQQW SFNPPTFGAG TKLELK                  106

SEQ ID NO: 177          moltype = AA  length = 121
FEATURE                 Location/Qualifiers
source                  1..121
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 177
EVQLVQSGAE VKKPGESLKI SCKGSGRTFT SYNMHWVRQM PGKGLEWMGA IYPLTGDTSY  60
NQKSKLQVTI SADKSISTAY LQWSSLKASD TAMYYCARST YVGGDWQFDV WGKGTTVTVS  120
S                                                                   121

SEQ ID NO: 178          moltype = AA  length = 106
FEATURE                 Location/Qualifiers
source                  1..106
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 178
EIVLTQSPGT LSLSPGERAT LSCRASSSVP YIHWYQQKPG QAPRLLIYAT SALASGIPDR  60
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FSGSGSGTDF TLTISRLEPE DFAVYYCQQW LSNPPTFGQG TKLEIK                106

SEQ ID NO: 179          moltype = AA  length = 122
FEATURE                 Location/Qualifiers
source                  1..122
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 179
EVQLVESGGG LVQPGGSLRL SCAASGYTFT SYNMHWVRQA PGKGLEWVGA IYPGNGDTSY  60
NQKFKGRFTI SVDKSKNTLY LQMNSLRAED TAVYYCARVV YYSNSYWYFD VWGQGTLVTV  120
SS                                                                 122

SEQ ID NO: 180          moltype = AA  length = 106
FEATURE                 Location/Qualifiers
source                  1..106
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 180
DIQMTQSPSS LSASVGDRVT ITCRASSSVS YMHWYQQKPG KAPKPLIYAP SNLASGVPSR  60
FSGSGSGTDF TLTISSLQPE DFATYYCQQW SFNPPTFGQG TKVEIK                106

SEQ ID NO: 181          moltype = AA  length = 122
FEATURE                 Location/Qualifiers
source                  1..122
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 181
EVQLVESGGG LVQPGRSLRL SCAASGFTFN DYAMHWVRQA PGKGLEWVST ISWNSGSIGY  60
ADSVKGRFTI SRDNAKKSLY LQMNSLRAED TALYYCAKDI QYGNYYYGMD VWGQGTTVTV  120
SS                                                                 122

SEQ ID NO: 182          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 182
EIVLTQSPAT LSLSPGERAT LSCRASQSVS SYLAWYQQKP GQAPRLLIYD ASNRATGIPA  60
RFSGSGSGTD FTLTISSLEP EDFAVYYCQQ RSNWPITFGQ GTRLEIK               107

SEQ ID NO: 183          moltype = AA  length = 121
FEATURE                 Location/Qualifiers
source                  1..121
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 183
QVQLQQPGAE LVKPGASVKM SCKASGYTFT SYNMHWVKQT PGRGLEWIGA IYPGNGDTSY  60
NQKFKGKATL TADKSSSTAY MQLSSLTSED SAVYYCARST YYGGDWYFNV WGAGTTVTVS  120
A                                                                  121

SEQ ID NO: 184          moltype = AA  length = 106
FEATURE                 Location/Qualifiers
source                  1..106
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 184
QIVLSQSPAI LSASPGEKVT MTCRASSSVS YIHWFQQKPG SSPKPWIYAT SNLASGVPVR  60
FSGSGSGTSY SLTISRVEAE DAATYYCQQW TSNPPTFGGG TKLEIK                106

SEQ ID NO: 185          moltype = AA  length = 121
FEATURE                 Location/Qualifiers
source                  1..121
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 185
QVQLQQPGAE LVKPGASVKM SCKASGYTFT SYNMHWVKQT PGRGLEWIGA IYPGNGDTSY  60
NQKFKGKATL TADKSSSTAY MQLSSLTSED SAVYYCARST YYGGDWYFNV WGAGTTVTVS  120
A                                                                  121

SEQ ID NO: 186          moltype = AA  length = 106
FEATURE                 Location/Qualifiers
source                  1..106
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 186
QIVLSQSPAI LSASPGEKVT MTCRASSSVS YIHWFQQKPG SSPKPWIYAT SNLASGVPVR  60
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FSGSGSGTSY SLTISRVEAE DAATYYCQQW TSNPPTFGGG TKLEIK                106

SEQ ID NO: 187          moltype = AA  length = 118
FEATURE                 Location/Qualifiers
source                  1..118
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 187
QAYLQQSGAE LVRPGASVKM SCKASGYTFT SYNMHWVKQT PRQGLEWIGG IYPGNGDTSY  60
NQKFKGKATL TVGKSSSTAY MQLSSLTSED SAVYFCARYD YNYAMDYWGQ GTSVTVSS    118

SEQ ID NO: 188          moltype = AA  length = 106
FEATURE                 Location/Qualifiers
source                  1..106
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 188
QIVLSQSPAI LSASPGEKVT MTCRASSSVS YMHWYQQKPG SSPKPWIYAT SNLASGVPAR  60
FSGSGSGTSY SFTISRVEAE DAATYYCQQW TFNPPTFGGG TRLEIK                106

SEQ ID NO: 189          moltype = AA  length = 121
FEATURE                 Location/Qualifiers
source                  1..121
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 189
QVQLQQSGAE VKKPGSSVKV SCKASGYTFT SYNMHWVKQA PGQGLEWIGA IYPGMGDTSY  60
NQKFKGKATL TADESTNTAY MELSSLRSED TAFYYCARST YYGGDWYFDV WGQGTTVTVS  120
S                                                                 121

SEQ ID NO: 190          moltype = AA  length = 106
FEATURE                 Location/Qualifiers
source                  1..106
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 190
DIQLTQSPSS LSASVGDRVT MTCRASSSVS YIHWFQQKPG KAPKPWIYAT SNLASGVPVR  60
FSGSGSGTDY TFTISSLQPE DIATYYCQQW TSNPPTFGGG TKLEIK                106

SEQ ID NO: 191          moltype = AA  length = 122
FEATURE                 Location/Qualifiers
source                  1..122
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 191
EVQLQQSGAE LVRPGASVKM SCKASGYTFT SYNMHWVKQT PRQGLEWIGA IYPGNGDTSY  60
NQKFKGKATL TVDKSSSTAY MQLSSLTSED SAVYFCARVV YYSNSYWYFD VWGTGTTVTV  120
SS                                                                122

SEQ ID NO: 192          moltype = AA  length = 106
FEATURE                 Location/Qualifiers
source                  1..106
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 192
DIELSQSPAI LSASPGEKVT MTCRASSSVS YMHWYQQKPG SSPKPWIYAP SNLASGVPAR  60
FSGSGSGTSY SLTISRVEAE DAATYYCQQW SFNPPTFGAG TKLEIK                106

SEQ ID NO: 193          moltype = AA  length = 116
FEATURE                 Location/Qualifiers
source                  1..116
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 193
QVQLVQSGAE VKKPGSSVKV SCKASGYTFT SYWLHWVRQA PGQGLEWIGY INPRNDYTEY  60
NQNFKDKATI TADESTNTAY MELSSLRSED TAFYFCARRD ITTFYWGQGT TVTVSS      116

SEQ ID NO: 194          moltype = AA  length = 112
FEATURE                 Location/Qualifiers
source                  1..112
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 194
DIQLTQSPSS LSASVGDRVT MSCKSSQSVL YSANHKNYLA WYQQKPGKAP KLLIYWASTR  60
ESGVPSRFSG SGSGTDFTFT ISSLQPEDIA TYYCHQYLSS WTFGGGTKLE IK          112
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SEQ ID NO: 195          moltype = AA  length = 121
FEATURE                 Location/Qualifiers
source                  1..121
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 195
EVQLVQSGAE VKKPGASVKV SCKASGYRFT NYWIHWVRQA PGQGLEWIGG INPGNNYATY  60
RRKFQGRVTM TADTSTSTVY MELSSLRSED TAVYYCTREG YGNYGAWFAY WGQGTLVTVS  120
S                                                                  121

SEQ ID NO: 196          moltype = AA  length = 112
FEATURE                 Location/Qualifiers
source                  1..112
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 196
DVQVTQSPSS LSASVGDRVT ITCRSSQSLA NSYGNTFLSW YLHKPGKAPQ LLIYGISNRF  60
SGVPDRFSGS GSGTDFTLTI SSLQPEDFAT YYCLQGTHQP YTFGQGTKVE IK          112

SEQ ID NO: 197          moltype = AA  length = 120
FEATURE                 Location/Qualifiers
source                  1..120
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 197
EVQLVESGGG LVQPGGSLRL SCAASGYEFS RSWMNWVRQA PGKGLEWVGR IYPGDGDTNY  60
SGKFKGRFTI SADTSKNTAY LQMNSLRAED TAVYYCARDG SSWDWYFDVW GQGTLVTVSS  120

SEQ ID NO: 198          moltype = AA  length = 112
FEATURE                 Location/Qualifiers
source                  1..112
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 198
DIQMTQSPSS LSASVGDRVT ITCRSSQSIV HSVGNTFLEW YQQKPGKAPK LLIYKVSNRF  60
SGVPSRFSGS GSGTDFTLTI SSLQPEDFAT YYCFQGSQFP YTFGQGTKVE IK          112

SEQ ID NO: 199          moltype = AA  length = 123
FEATURE                 Location/Qualifiers
source                  1..123
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 199
QVQLQESGGG LVKPGGSLKL SCAASGFAFS IYDMSWVRQT PEKRLEWVAY ISSGGGTTYY  60
PDTVKGRFTI SRDNAKNTLY LQMSSLKSED TAMYYCARHS GYGSSYGVLF AYWGQGTLVT  120
VSS                                                                123

SEQ ID NO: 200          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 200
DIQLTQTTSS LSASLGDRVT ISCRASQDIS NYLNWYQQKP DGTVKLLIYY TSILHSGVPS  60
RFSGSGSGTD YSLTISNLEQ EDFATYFCQQ GNTLPWTFGG GTKLEIK                107

SEQ ID NO: 201          moltype = AA  length = 117
FEATURE                 Location/Qualifiers
source                  1..117
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 201
QIQLQQSGPE VVKPGASVKI SCKASGYTFT DYYITWVKQK PGQGLEWIGW IYPGSGNTKY  60
NEKFKGKATL TVDTSSSTAF MQLSSLTSED TAVYFCANYG NYWFAYWGQG TQVTVSA     117

SEQ ID NO: 202          moltype = AA  length = 111
FEATURE                 Location/Qualifiers
source                  1..111
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 202
DIVLTQSPAS LAVSLGQRAT ISCKASQSVD FDGDSYMNWY QQKPGQPPKV LIYAASNLES  60
GIPARFSGSG SGTDFTLNIH PVEEEDAATY YCQQSNEDPW TFGGGTKLEI K           111

SEQ ID NO: 203          moltype = AA  length = 112
FEATURE                 Location/Qualifiers
```

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```
source                 1..112
                       mol_type = protein
                       organism = sythetic construct
SEQUENCE: 203
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS AYYWSWIRQP PGKGLEWIGD INHGGGTNYN  60
PSLKSRVTIS VDTSKNQFSL KLNSVTAADT AVYYCASLTA YWGQGSLVTV SS          112

SEQ ID NO: 204         moltype = AA  length = 107
FEATURE                Location/Qualifiers
source                 1..107
                       mol_type = protein
                       organism = sythetic construct
SEQUENCE: 204
DIQMTQSPTS LSASVGDRVT ITCRASQGIS SWLTWYQQKP EKAPKSLIYA ASSLQSGVPS  60
RFSGSGSGTD FTLTISSLQP EDFATYYCQQ YDSYPITFGQ GTRLEIK                107

SEQ ID NO: 205         moltype = AA  length = 116
FEATURE                Location/Qualifiers
source                 1..116
                       mol_type = protein
                       organism = sythetic construct
SEQUENCE: 205
EVQLVQSGAE VKKPGSSVKV SCKASGYTIT DSNIHWVRQA PGQSLEWIGY IYPYNGGTDY  60
NQKFKNRATL TVDNPTNTAY MELSSLRSED TAFYYCVNGN PWLAYWGQGT LVTVSS      116

SEQ ID NO: 206         moltype = AA  length = 111
FEATURE                Location/Qualifiers
source                 1..111
                       mol_type = protein
                       organism = sythetic construct
SEQUENCE: 206
DIQLTQSPST LSASVGDRVT ITCRASESLD NYGIRFLTWF QQKPGKAPKL LMYAASNQGS  60
GVPSRFSGSG SGTEFTLTIS SLQPDDFATY YCQQTKEVPW SFGQGTKVEV K           111

SEQ ID NO: 207         moltype = AA  length = 116
FEATURE                Location/Qualifiers
source                 1..116
                       mol_type = protein
                       organism = sythetic construct
SEQUENCE: 207
QVQLVQSGAE VKKPGSSVKV SCKASGYTFT DYNMHWVRQA PGQGLEWIGY IYPYNGGTGY  60
NQKFKSKATI TADESTNTAY MELSSLRSED TAVYYCARGR PAMDYWGQGT LVTVSS      116

SEQ ID NO: 208         moltype = AA  length = 111
FEATURE                Location/Qualifiers
source                 1..111
                       mol_type = protein
                       organism = sythetic construct
SEQUENCE: 208
DIQMTQSPSS LSASVGDRVT ITCRASESVD NYGISFMNWF QQKPGKAPKL LIYAASNQGS  60
GVPSRFSGSG SGTDFTLTIS SLQPDDFATY YCQQSKEVPW TFGQGTKVEI K           111

SEQ ID NO: 209         moltype = AA  length = 117
FEATURE                Location/Qualifiers
source                 1..117
                       mol_type = protein
                       organism = sythetic construct
SEQUENCE: 209
QVQLVQSGAE VKKPGASVKV SCKASGYTFT NYDINWVRQA PGQGLEWIGW IYPGDGSTKY  60
NEKFKAKATL TADTSTSTAY MELRSLRSDD TAVYYCASGY EDAMDYWGQG TTVTVSS     117

SEQ ID NO: 210         moltype = AA  length = 107
FEATURE                Location/Qualifiers
source                 1..107
                       mol_type = protein
                       organism = sythetic construct
SEQUENCE: 210
DIQMTQSPSS LSASVGDRVT INCKASQDIN SYLSWFQQKP GKAPKTLIYR ANRLVDGVPS  60
RFSGSGSGQD YTLTISSLQP EDFATYYCLQ YDEFPLTFGG GTKVEIK                107

SEQ ID NO: 211         moltype = AA  length = 122
FEATURE                Location/Qualifiers
source                 1..122
                       mol_type = protein
                       organism = sythetic construct
SEQUENCE: 211
```

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EVQLLESGGG LVQPGGSLRL SCAVSGFTFN SFAMSWVRQA PGKGLEWVSA ISGSGGGTYY  60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYFCAKDK ILWFGEPVFD YWGQGTLVTV  120
SS                                                                122

SEQ ID NO: 212          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 212
EIVLTQSPAT LSLSPGERAT LSCRASQSVS SYLAWYQQKP GQAPRLLIYD ASNRATGIPA  60
RFSGSGSGTD FTLTISSLEP EDFAVYYCQQ RSNWPPTFGQ GTKVEIK               107

SEQ ID NO: 213          moltype = AA  length = 121
FEATURE                 Location/Qualifiers
source                  1..121
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 213
QVQLVQSGAE VKKPGSSVKV SCKAFGGTFS SYAISWVRQA PGQGLEWMGR IIRFLGIANY  60
AQKFQGRVTL IADKSTNTAY MELSSLRSED TAVYYCAGEP GERDPDAVDI WGQGTMVTVS  120
S                                                                 121

SEQ ID NO: 214          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 214
DIQMTQSPSS LSASVGDRVT ITCRASQGIR SWLAWYQQKP EKAPKSLIYA ASSLQSGVPS  60
RFSGSGSGTD FTLTISSLQP EDFATYYCQQ YNSYPLTFGG GTKVEIK               107

SEQ ID NO: 215          moltype = AA  length = 120
FEATURE                 Location/Qualifiers
source                  1..120
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 215
QVQLVESGGG LVQPGGSLRL SCAASGFTFS SYYMNWVRQA PGKGLEWVSG ISGDPSNTYY  60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCARDL PLVYTGFAYW GQGTLVTVSS  120

SEQ ID NO: 216          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 216
DIELTQPPSV SVAPGQTARI SCSGDNLRHY YVYWYQQKPG QAPVLVIYGD SKRPSGIPER  60
FSGSNSGNTA TLTISGTQAE DEADYYCQTY TGGASLVFGG GTKLTVL               107

SEQ ID NO: 217          moltype = AA  length = 120
FEATURE                 Location/Qualifiers
source                  1..120
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 217
QVQLVQSGAE VAKPGTSVKL SCKASGYTFT DYWMQWVKQR PGQGLEWIGT IYPGDGDTGY  60
AQKFQGKATL TADKSSKTVY MHLSSLASED SAVYYCARGD YYGSNSLDYW GQGTSVTVSS  120

SEQ ID NO: 218          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 218
DIVMTQSHLS MSTSLGDPVS ITCKASQDVS TVVAWYQQKP GQSPRRLIYS ASYRYIGVPD  60
RFTGSGAGTD FTFTISSVQA EDLAVYYCQQ HYSPPYTFGG GTKLEIK               107

SEQ ID NO: 219          moltype = AA  length = 119
FEATURE                 Location/Qualifiers
source                  1..119
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 219
QVQLQESGPG LVKPSETLSL TCTVSGFSLT SYGIHWLRQP PGKGLEWIGV IWRGGSTDYN  60
PSLKSRVTIS KDTSKSQVSL KLSSVTAADT AVYYCAKGKV TTGFYFDFWG QGTLVTVSS   119
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SEQ ID NO: 220          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 220
EIVMTQSPPT LSLSPGERVT LSCRASEDIY NRLVWYQQKP GQAPRLLISG VTSLETSIPA  60
RFSGSGSGTD YTLTISSLQP EDFAVYYCQQ YWSTPYTFGQ GTKVEIK               107

SEQ ID NO: 221          moltype = AA  length = 123
FEATURE                 Location/Qualifiers
source                  1..123
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 221
EVQLLESGGG LVQPGGSLRL SCAASGFTFD DYGMSWVRQA PGKGLEWVSD ISWNGGKTHY  60
VDSVKGQFTI SRDNSKNTLY LQMNSLRAED TAVYYCARGS LFHDSSGFYF GHWGQGTLVT  120
VSS                                                               123

SEQ ID NO: 222          moltype = AA  length = 110
FEATURE                 Location/Qualifiers
source                  1..110
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 222
QSVLTQPPSA SGTPGQRVTI SCSGSSSNIG DNYVSWYQQL PGTAPKLLIY RDSQRPSGVP  60
DRFSGSKSGT SASLAISGLR SEDEADYYCQ SYDSSLSGSV FGGGTKLTVL            110

SEQ ID NO: 223          moltype = AA  length = 120
FEATURE                 Location/Qualifiers
source                  1..120
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 223
EVQLVQSGAE VKKPGATVKI SCKVSGYTFT DSVMNWVQQA PGKGLEWMGW IDPEYGRTDV  60
AEKFQGRVTI TADTSTDTAY MELSSLRSED TAVYYCARTK YNSGYGFPYW GQGTTVTVSS  120

SEQ ID NO: 224          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 224
DIQMTQSPSS LSASVGDRVT ITCKASQNVD SDVDWYQQKP GKAPKLLIYK ASNDYTGVPS  60
RFSGSGSGTD FTFTISSLQP EDIATYYCMQ SNTHPRTFGG GTKVEIK               107

SEQ ID NO: 225          moltype = AA  length = 122
FEATURE                 Location/Qualifiers
source                  1..122
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 225
EVQLVESGGG LVQPGGSLRL SCAASGFTFS VYYMNWVRQA PGKGLEWVSD INNEGGTTYY  60
ADSVKGRFTI SRDNSKNSLY LQMNSLRAED TAVYYCARDA GYSNHVPIFD SWGQGTLVTV  120
SS                                                                122

SEQ ID NO: 226          moltype = AA  length = 110
FEATURE                 Location/Qualifiers
source                  1..110
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 226
QAVVTQEPSL TVSPGGTVTL TCGLKSGSVT SDNFPTWYQQ TPGQAPRLLI YNTNTRHSGV  60
PDRFSGSILG NKAALTITGA QADDEAEYFC ALFISNPSVE FGGGTQLTVL            110

SEQ ID NO: 227          moltype = AA  length = 118
FEATURE                 Location/Qualifiers
source                  1..118
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 227
QVQLVQSGAE VKKPGASVKV SCKASGYTFT NYGMNWVRQA PGQGLKWMGW INTYTGEPTY  60
ADAFKGRVTM TRDTSISTAY MELSRLRSDD TAVYYCARDY GDYGMDYWGQ GTTVTVSS    118

SEQ ID NO: 228          moltype = AA  length = 111
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FEATURE                 Location/Qualifiers
source                  1..111
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 228
DIVMTQSPDS LAVSLGERAT INCRASKSVS TSGYSFMHWY QQKPGQPPKL LIYLASNLES  60
GVPDRFSGSG SGTDFTLTIS SLQAEDVAVY YCQHSREVPW TFGQGTKVEI K           111

SEQ ID NO: 229          moltype = AA  length = 118
FEATURE                 Location/Qualifiers
source                  1..118
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 229
QVQLVQSGAE VKKPGASVKV SCKASGYTFT GYNMNWVRQA PGQGLEWMGN IDPYYGGTSY  60
NQKFKGRVTM TIDKSTSTVY MELSSLRSED TAVYYCARMY HGNAFDYWGQ GTTVTVSS    118

SEQ ID NO: 230          moltype = AA  length = 113
FEATURE                 Location/Qualifiers
source                  1..113
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 230
DIVMTQSPDS LAVSLGERAT INCKSSQSLL NSGNLKNYLT WYQQKPGQPP KLLIYWASTR  60
KSGVPDRFSG SGSGTDFTLT ISSLQAEDVA VYYCQNDYSY PLTFGGGTKV EIK         113

SEQ ID NO: 231          moltype = AA  length = 118
FEATURE                 Location/Qualifiers
source                  1..118
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 231
QVQLQQPGAE LVRPGASVKL SCKASGYTFT SYWINWVKQR PGQGLEWIGN IYPSDSYTNY  60
NQKFKDKATL TVDKSSSTAY MQLSSPTSED SAVYYCTRSW RGNSFDYWGQ GTTLTVSS    118

SEQ ID NO: 232          moltype = AA  length = 113
FEATURE                 Location/Qualifiers
source                  1..113
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 232
DIVMTQSPSS LTVTAGEKVT MSCKSSQSLL NSGNQKNYLT WYQQKPGQPP KLLIYWASTR  60
ESGVPDRFTG SGSGTDFTLT ISSVQAEDLA VYYCQNDYSY PFTFGSGTKL EIK         113

SEQ ID NO: 233          moltype = AA  length = 118
FEATURE                 Location/Qualifiers
source                  1..118
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 233
QVQLVQSGAE VKKPGASVKV SCKASGYTFT SYYMHWVRQA PGQGLEWMGI INPSGGSTSY  60
AQKFQGRVTM TRDTSTSTVY MELSSLRSED TAVYYCAKGT TGDWFDYWGQ GTLVTVSS    118

SEQ ID NO: 234          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 234
DIQMTQSPSS LSASVGDRVT ITCRASQSIS SYLNWYQQKP GKAPKLLIYA ASSLQSGVPS  60
RFSGSGSGTD FTLTISSLQP EDFATYYCQQ SYSTPPTFGQ GTKVEIK                107

SEQ ID NO: 235          moltype = AA  length = 118
FEATURE                 Location/Qualifiers
source                  1..118
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 235
QVQLVQSGAE VKKPGASVKV SCKASGYTFT NYGMNWVRQA PGQGLEWMGW INTYTGEPTY  60
ADDFKGRVTM TTDTSTSTAY MELRSLRSDD TAVYYCARIG DSSPSDYWGQ GTLVTVSS    118

SEQ ID NO: 236          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
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SEQUENCE: 236
EIVMTQSPAT LSVSPGERAT LSCKASQSVS NDVVWYQQKP GQAPRLLIYY ASNRYTGIPA  60
RFSGSGSGTE FTLTISSLQS EDFAVYYCQQ DYTSPWTFGQ GTKLEIK                107

SEQ ID NO: 237          moltype = AA  length = 118
FEATURE                 Location/Qualifiers
source                  1..118
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 237
QVQLQESGPG LVKPSETLSL TCTVSGGSIS SYYWSWIRQP PGKCLEWIGY VYYSGTTNYN  60
PSLKSRVTIS VDTSKNQFSL KLSSVTAADT AVYYCASIAV TGFYFDYWGQ GTLVTVSS    118

SEQ ID NO: 238          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
source                  1..108
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 238
EIVLTQSPGT LSLSPGERVT LSCRASQRVN NNYLAWYQQR PGQAPRLLIY GASSRATGIP  60
DRFSGSGSGT DFTLTISRLE PEDFAVYYCQ QYDRSPLTFG CGTKLEIK               108

SEQ ID NO: 239          moltype = AA  length = 119
FEATURE                 Location/Qualifiers
source                  1..119
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 239
QVQLQESGPG LVKPSETLSL TCTVSGFSLS NYDVHWVRQA PGKGLEWLGV IWSGGNTDYN  60
TPFTSRLTIS VDTSKNQFSL KLSSVTAADT AVYYCARALD YYDYEFAYWG QGTLVTVSS   119

SEQ ID NO: 240          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 240
EIVLTQSPDF QSVTPKEKVT ITCRASQSIG TNIHWYQQKP DQSPKLLIKY ASESISGIPS  60
RFSGSGSGTD FTLTINSLEA EDAATYYCQQ NNEWPTSFGQ GTKLEIK                107

SEQ ID NO: 241          moltype = AA  length = 119
FEATURE                 Location/Qualifiers
source                  1..119
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 241
QVQLKQSGPG LVQPSQSLSI TCTVSGFSLT NYGVHWVRQS PGKGLEWLGV IWSGGNTDYN  60
TPFTSRLSIN KDNSKSQVFF KMNSLQSNDT AIYYCARALT YYDYEFAYWG QGTLVTVSA   119

SEQ ID NO: 242          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 242
DILLTQSPVI LSVSPGERVS FSCRASQSIG TNIHWYQQRT NGSPRLLIKY ASESISGIPS  60
RFSGSGSGTD FTLSINSVES EDIADYYCQQ NNNWPTTFGA GTKLELK                107

SEQ ID NO: 243          moltype = AA  length = 119
FEATURE                 Location/Qualifiers
source                  1..119
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 243
QVQLQESGPG LVKPSETLSL TCTVSGGSVS SGDYYWTWIR QSPGKGLEWI GHIYYSGNTN  60
YNPSLKSRLT ISIDTSKTQF SLKLSSVTAA DTAIYYCVRD RVTGAFDIWG QGTMVTVSS   119

SEQ ID NO: 244          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 244
DIQMTQSPST LSASVGDRVT ITCQASQDIS NYLNWYQQKP GKAPKLLIYD ASNLETGVPS  60
RFSGSGSGTD FTFTISSLQP EDIATYFCQH FDHLPLAFGG GTKVEIK                107
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SEQ ID NO: 245          moltype = AA  length = 123
FEATURE                 Location/Qualifiers
source                  1..123
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 245
EVQLQQPGSE LVRPGASVKL SCKASGYTFT SYWMHWVKQR PGQGLEWIGN IYPGSRSTNY  60
DEKFKSKATL TVDTSSSTAY MQLSSLTSED SAVYYCTRNG DYYVSSGDAM DYWGQGTSVT  120
VSS                                                                123

SEQ ID NO: 246          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 246
DIQMTQTTSS LSASLGDRVT ISCRTSQDIG NYLNWYQQKP DGTVKLLIYY TSRLHSGVPS  60
RFSGSGSGTD FSLTINNVEQ EDVATYFCQH YNTVPPTFGG GTKLEIK                107

SEQ ID NO: 247          moltype = AA  length = 120
FEATURE                 Location/Qualifiers
source                  1..120
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 247
QVQLVQSGAE VKKPGSSVKV SCKASGFTFT DYKIHWVRQA PGQGLEWMGY FNPNSGYSTY  60
AQKFQGRVTI TADKSTSTAY MELSSLRSED TAVYYCARLS PGGYYVMDAW GQGTTVTVSS  120

SEQ ID NO: 248          moltype = AA  length = 106
FEATURE                 Location/Qualifiers
source                  1..106
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 248
DIQMTQSPSS LSASVGDRVT ITCRASQGIN NYLNWYQQKP GKAPKRLIYN TNNLQTGVPS  60
RFSGSGSGTE FTLTISSLQP EDFATYYCLQ HNSFPTFGQG TKLEIK                 106

SEQ ID NO: 249          moltype = AA  length = 121
FEATURE                 Location/Qualifiers
source                  1..121
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 249
QVQLVQSGAE VKKPGASVKV SCKASGYTFT SHWMHWVRQA PGQGLEWIGE FNPSNGRTNY  60
NEKFKSKATM TVDTSTNTAY MELSSLRSED TAVYYCASRD YDYDGRYFDY WGQGTLVTVS  120
S                                                                  121

SEQ ID NO: 250          moltype = AA  length = 106
FEATURE                 Location/Qualifiers
source                  1..106
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 250
DIQMTQSPSS LSASVGDRVT ITCSASSSVT YMYWYQQKPG KAPKLLIYDT SNLASGVPSR  60
FSGSGSGTDY TFTISSLQPE DIATYYCQQW SSHIFTFGQG TKVEIK                 106

SEQ ID NO: 251          moltype = AA  length = 119
FEATURE                 Location/Qualifiers
source                  1..119
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 251
QVQLQQPGAE LVEPGGSVKL SCKASGYTFT SHWMHWVKQR PGQGLEWIGE INPSSGRNNY  60
NEKFKSKATL TVDKSSSTAY MQFSSLTSED SAVYYCVRYY GYDEAMDYWG QGTSVTVSS   119

SEQ ID NO: 252          moltype = AA  length = 112
FEATURE                 Location/Qualifiers
source                  1..112
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 252
DIVMTQAAFS NPVTLGTSAS ISCRSSKSLL HSNGITYLYW YLQKPGQSPQ LLIYQMSNLA  60
SGVPDRFSSS GSGTDFTLRI SRVEAEDVGV YYCAQNLELP YTFGGGTKLE IK          112

SEQ ID NO: 253          moltype = AA  length = 121
FEATURE                 Location/Qualifiers
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```
source                  1..121
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 253
QVQLQESGPG LVKPSQTLSL TCTVSGGSIS SGDYYWSWIR QPPGKGLEWI GYIYYSGSTD  60
YNPSLKSRVT MSVDTSKNQF SLKVNSVTAA DTAVYYCARV SIFGVGTFDY WGQGTLVTVS  120
S                                                                  121

SEQ ID NO: 254          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 254
EIVMTQSPAT LSLSPGERAT LSCRASQSVS SYLAWYQQKP GQAPRLLIYD ASNRATGIPA  60
RFSGSGSGTD FTLTISSLEP EDFAVYYCHQ YGSTPLTFGG GTKAEIK                107

SEQ ID NO: 255          moltype = AA  length = 123
FEATURE                 Location/Qualifiers
source                  1..123
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 255
QVQLQQSGAE VKKPGSSVKV SCKASGYTFT NYYIYWVRQA PGQGLEWIGG INPTSGGSNF  60
NEKFKTRVTI TADESSTTAY MELSSLRSED TAFYFCTRQG LWFDSDGRGF DFWGQGTTVT  120
VSS                                                                123

SEQ ID NO: 256          moltype = AA  length = 112
FEATURE                 Location/Qualifiers
source                  1..112
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 256
DIQMTQSPSS LSASVGDRVT ITCRSSQNIV HSNGNTYLDW YQQTPGKAPK LLIYKVSNRF  60
SGVPSRFSGS GSGTDFTFTI SSLQPEDIAT YYCFQYSHVP WTFGQGTKLQ IT          112

SEQ ID NO: 257          moltype = AA  length = 119
FEATURE                 Location/Qualifiers
source                  1..119
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 257
QVQLQESGPG LVKPSETLSL TCTVSGGSVS SGDYYWTWIR QSPGKGLEWI GHIYYSGNTN  60
YNPSLKSRLT ISIDTSKTQF SLKLSSVTAA DTAIYYCVRD RVTGAFDIWG QGTMVTVSS   119

SEQ ID NO: 258          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 258
DIQMTQSPSS LSASVGDRVT ITCQASQDIS NYLNWYQQKP GKAPKLLIYD ASNLETGVPS  60
RFSGSGSGTD FTFTISSLQP EDIATYFCQH FDHLPLAFGG GTKVEIK                107

SEQ ID NO: 259          moltype = AA  length = 119
FEATURE                 Location/Qualifiers
source                  1..119
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 259
EVQLVESGGG LVQPGGSLRL SCAASGFSLT NYGVHWVRQA PGKGLEWLGV IWSGGNTDYG  60
NEFTSRFTIS RDNAKNSLYL QMNSLRAEDT AVYYCARALD YYDYEFAYWG QGTMVTVSS   119

SEQ ID NO: 260          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 260
EIVLTQSPAT LSLSPGERAT LSCRASQSIG TNIHWYQQKP GQAPRLLIKY ASESISGIPA  60
RFSGSGSGTD FTLTISSLEP EDFAVYYCQQ NNNWPTSFGG GTKVEIK                107

SEQ ID NO: 261          moltype = AA  length = 119
FEATURE                 Location/Qualifiers
source                  1..119
                        mol_type = protein
```

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                        organism = sythetic construct
SEQUENCE: 261
QVQLKQSGPG LVQPSQSLSI TCTVSGFSLT NYGVHWVRQS PGKGLEWLGV IWSGGNTDYN  60
TPFTSRLSIN KDNSKSQVFF KMNSLQSNDT AIYYCARALT YYDYEFAYWG QGTLVTVST   119

SEQ ID NO: 262          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 262
DILLTQSPVI LSVSPGERVS FSCRASQSIG TNIHWYQQRT NGSPRLLIKY ASESISGIPS  60
RFSGSGSGTD FTLSINSVES EDIADYYCQQ NNNWPTTFGA GTKLELK                107

SEQ ID NO: 263          moltype = AA  length = 125
FEATURE                 Location/Qualifiers
source                  1..125
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 263
QVQLVESGGG VVQPGRSLRL SCAASGFTFS TYGMHWVRQA PGKGLEWVAV IWDDGSYKYY  60
GDSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCARDG ITMVRGVMKD YFDYWGQGTL  120
VTVSS                                                              125

SEQ ID NO: 264          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 264
AIQLTQSPSS LSASVGDRVT ITCRASQDIS SALVWYQQKP GKAPKLLIYD ASSLESGVPS  60
RFSGSESGTD FTLTISSLQP EDFATYYCQQ FNSYPLTFGG GTKVEIK                107

SEQ ID NO: 265          moltype = AA  length = 120
FEATURE                 Location/Qualifiers
source                  1..120
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 265
EVQLVESGGG LVQPGGSLRL SCAASGFTIS TNAMSWVRQA PGKGLEWIGV ITGRDITYYA  60
SWAKGRFTIS RDNSKNTLYL QMNSLRAEDT AVYYCARDGG SSAITSNNIW GQGTLVTVSS  120

SEQ ID NO: 266          moltype = AA  length = 112
FEATURE                 Location/Qualifiers
source                  1..112
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 266
DVVMTQSPST LSASVGDRVT INCQASESIS SWLAWYQQKP GKAPKLLIYE ASKLASGVPS  60
RFSGSGSGTE FTLTISSLQP DDFATYYCQG YFYFISRTYV NSFGGGTKVE IK          112

SEQ ID NO: 267          moltype = AA  length = 124
FEATURE                 Location/Qualifiers
source                  1..124
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 267
QVTLKESGPT LVKPTETLTL TCTLSGFSLN NARMGVSWIR QPPGKCLEWL AHIFSNDEKS  60
YSTSLKNRLT ISKDSSKTQV VLTMTNVDPV DTATYYCARI VGYGSGWYGF FDYWGQGTLV  120
TVSS                                                               124

SEQ ID NO: 268          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 268
DIQMTQSPSS LSASVGDRVT ITCRASQGIR NDLGWYQQKP GKAPKRLIYA ASTLQSGVPS  60
RFSGSGSGTE FTLTISSLQP EDFATYYCLQ HNSYPLTFGC GTKVEIK                107

SEQ ID NO: 269          moltype = AA  length = 113
FEATURE                 Location/Qualifiers
source                  1..113
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 269
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EVQLLQSGPE LEKPGASVMI SCKASGSSFT GYNMNWVRQN IGKSLEWIGA IDPYYGGTSY  60
NQKFKGRATL TVDKSSSTAY MHLKSLTSED SAVYYCVSGM EYWGQGTSVT VSS         113

SEQ ID NO: 270          moltype = AA  length = 113
FEATURE                 Location/Qualifiers
source                  1..113
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 270
EIVMTQSPAT LSVSPGERAT LSCRSSQSLV HRNGNTYLHW YLQKPGQSPK LLIHKVSNRF  60
SGVPDRFSGS GSGTDFTLKI SRVEAEDLGV YFCSQSTHVP PLTFGAGTKL ELK         113

SEQ ID NO: 271          moltype = AA  length = 113
FEATURE                 Location/Qualifiers
source                  1..113
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 271
EVQLVQSGAE VEKPGASVKI SCKASGSSFT GYNMNWVRQN IGKSLEWIGA IDPYYGGTSY  60
NQKFKGRATL TVDKSTSTAY MHLKSLRSED TAVYYCVSGM EYWGQGTSVT VSS         113

SEQ ID NO: 272          moltype = AA  length = 113
FEATURE                 Location/Qualifiers
source                  1..113
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 272
DVVMTQTPLS LPVTPGEPAS ISCRSSQSLV HRNGNTYLHW YLQKPGQSPK LLIHKVSNRF  60
SGVPDRFSGS GSGTDFTLKI SRVEAEDLGV YFCSQSTHVP PLTFGAGTKL ELK         113

SEQ ID NO: 273          moltype = AA  length = 119
FEATURE                 Location/Qualifiers
source                  1..119
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 273
QVQLVESGPG VVQPGRSLRI SCAVSGFSVT NYGVHWVRQP PGKGLEWLGV IWAGGITNYN  60
SAFMSRLTIS KDNSKNTVYL QMNSLRAEDT AMYYCASRGG HYGYALDYWG QGTLVTVSS   119

SEQ ID NO: 274          moltype = AA  length = 104
FEATURE                 Location/Qualifiers
source                  1..104
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 274
EIVMTQTPAT LSVSAGERVT ITCKASQSVS NDVTWYQQKP GQAPRLLIYS ASNRYSGVPA  60
RFSGSGYGTE FTFTISSVQS EDFAVYFCQQ DYSSFGQGTK LEIK                   104

SEQ ID NO: 275          moltype = AA  length = 115
FEATURE                 Location/Qualifiers
source                  1..115
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 275
QVQLVQSGAE VKKPGASVKV SCKASGYTFT DYEMHWVRQA PGQGLEWMGA LDPKTGDTAY  60
SQKFKGRVTL TADKSTSTAY MELSSLTSED TAVYYCTRFY SYTYWGQGTL VTVSS       115

SEQ ID NO: 276          moltype = AA  length = 112
FEATURE                 Location/Qualifiers
source                  1..112
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 276
DVVMTQSPLS LPVTPGEPAS ISCRSSQSLV HSNRNTYLHW YLQKPGQSPQ LLIYKVSNRF  60
SGVPDRFSGS GSGTDFTLKI SRVEAEDVGV YYCSQNTHVP PTFGQGTKLE IK          112

SEQ ID NO: 277          moltype = AA  length = 120
FEATURE                 Location/Qualifiers
source                  1..120
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 277
EVQLVESGGG LVQPGGSLRL SCAASGFNIK DTYIHWVRQA PGKGLEWVAR IYPTNGYTRY  60
ADSVKGRFTI SADTSKNTAY LQMNSLRAED TAVYYCSRWG GDGFYAMDYW GQGTLVTVSS  120

SEQ ID NO: 278          moltype = AA  length = 107
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FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 278
DIQMTQSPSS LSASVGDRVT ITCRASQDVN TAVAWYQQKP GKAPKLLIYS ASFLYSGVPS  60
RFSGSRSGTD FTLTISSLQP EDFATYYCQQ HYTTPPTFGQ GTKVEIK               107

SEQ ID NO: 279          moltype = AA  length = 120
FEATURE                 Location/Qualifiers
source                  1..120
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 279
EVQLVESGGG LVQPGGSLRL SCAASGFNIK DTYIHWVRQA PGKGLEWVAR IYPTNGYTRY  60
ADSVKGRFTI SADTSKNTAY LQMNSLRAED TAVYYCSRWG GDGFYAMDYW GQGTLVTVSS  120

SEQ ID NO: 280          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 280
DIQMTQSPSS LSASVGDRVT ITCRASQDVN TAVAWYQQKP GKAPKLLIYS ASFLYSGVPS  60
RFSGSRSGTD FTLTISSLQP EDFATYYCQQ HYTTPPTFGQ GTKVEIK               107

SEQ ID NO: 281          moltype = AA  length = 115
FEATURE                 Location/Qualifiers
source                  1..115
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 281
EVQLVQSGAE VKKPGATVKI SCKVSGYTFT DYYIHWVQQA PGKGLEWMGR VNPDHGDSYY  60
NQKFKDKATI TADKSTDTAY MELSSLRSED TAVYFCARNY LFDHWGQGTL VTVSS       115

SEQ ID NO: 282          moltype = AA  length = 105
FEATURE                 Location/Qualifiers
source                  1..105
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 282
DIQMTQSPSS VSASVGDRVT ITCKASQDVG TAVAWYQQKP GKAPKLLIYW ASIRHTGVPS  60
RFSGSGSGTD FTLTISSLQP EDFATYYCHQ FATYTFGGGT KVEIK                 105

SEQ ID NO: 283          moltype = AA  length = 119
FEATURE                 Location/Qualifiers
source                  1..119
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 283
QVQLVESGGG LVQPGGSLRL SCAASGFTFR SYAMSWVRQA PGKGLEWVSA ISGRGDNTYY  60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCAKMT SNAFAFDYWG QGTLVTVSS   119

SEQ ID NO: 284          moltype = AA  length = 111
FEATURE                 Location/Qualifiers
source                  1..111
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 284
QSVLTQPPSV SGAPGQRVTI SCTGSSSNIG AGYGVHWYQQ LPGTAPKLLI YGNTNRPSGV  60
PDRFSGFKSG TSASLAITGL QAEDEADYYC QSYDSSLSGW VFGGGTKLTV L           111

SEQ ID NO: 285          moltype = AA  length = 120
FEATURE                 Location/Qualifiers
source                  1..120
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 285
QVQLQQSGPE LVKPGASLKL SCTASGFNIK DTYIHWVKQR PEQGLEWIGR IYPTNGYTRY  60
DPKFQDKATI TADTSSNTAY LQVSRLTSED TAVYYCSRWG GDGFYAMDYW GQGASVTVSS  120

SEQ ID NO: 286          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
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SEQUENCE: 286
DIVMTQSHKF MSTSVGDRVS ITCKASQDVN TAVAWYQQKP GHSPKLLIYS ASFRYTGVPD  60
RFTGSRSGTD FTFTISSVQA EDLAVYYCQQ HYTTPPTFGG GTKVEIK               107

SEQ ID NO: 287          moltype = AA  length = 120
FEATURE                 Location/Qualifiers
source                  1..120
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 287
EVQLLESGGG LVQPGGSLRL SCAASGFTFS SYAMSWVRQA PGKGLEWVSA ISGSGGSTYY  60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCAILG ATSLSAFDIW GQGTMVTVSS  120

SEQ ID NO: 288          moltype = AA  length = 112
FEATURE                 Location/Qualifiers
source                  1..112
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 288
QSVLTQPPSV SGAPGQRVTI SCTGSSSNIG AGYDVHWYQQ LPGTAPKLLI YGNSNRPSGV  60
PDRFSGSKSG TSASLAITGL QAEDEADYYC QSYDSSLSGS GVFGGGTKLT VL         112

SEQ ID NO: 289          moltype = AA  length = 119
FEATURE                 Location/Qualifiers
source                  1..119
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 289
EVQLVESGGG LVQPGGSLRL SCAASGFTFT DYTMDWVRQA PGKGLEWVAD VNPNSGGSIY  60
NQRFKGRFTL SVDRSKNTLY LQMNSLRAED TAVYYCARNL GPSFYFDYWG QGTLVTVSS   119

SEQ ID NO: 290          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 290
DIQMTQSPSS LSASVGDRVT ITCKASQDVS IGVAWYQQKP GKAPKLLIYS ASYRYTGVPS  60
RFSGSGSGTD FTLTISSLQP EDFATYYCQQ YYIYPYTFGQ GTKVEIK               107

SEQ ID NO: 291          moltype = AA  length = 120
FEATURE                 Location/Qualifiers
source                  1..120
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 291
EVQLVESGGG LVQPGGSLRL SCAASGFNIK DTYIHWVRQA PGKGLEWVAR IYPTNGYTRY  60
ADSVKGRFTI SADTSKNTAY LQMNSLRAED TAVYYCSRWG GDGFYAMDYW GQGTLVTVSS  120

SEQ ID NO: 292          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 292
DIQMTQSPSS LSASVGDRVT ITCRASQDVN TAVAWYQQKP GKAPKLLIYS ASFLYSGVPS  60
RFSGSRSGTD FTLTISSLQP EDFATYYCQQ HYTTPPTFGQ GTKVEIK               107

SEQ ID NO: 293          moltype = AA  length = 120
FEATURE                 Location/Qualifiers
source                  1..120
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 293
EVQLVESGGG LVQPGGSLRL SCAASGFNIK DTYIHWVRQA PGKGLEWVAR IYPTNGYTRY  60
ADSVKGRFTI SADTSKNTAY LQMNSLRAED TAVYYCSRWG GDGFYAMDYW GQGTLVTVSS  120

SEQ ID NO: 294          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 294
DIQMTQSPSS LSASVGDRVT ITCRASQDVN TAVAWYQQKP GKAPKLLIYS ASFLYSGVPS  60
RFSGSRSGTD FTLTISSLQP EDFATYYCQQ HYTTPPTFGQ GTKVEIK               107
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SEQ ID NO: 295          moltype = AA  length = 119
FEATURE                 Location/Qualifiers
source                  1..119
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 295
QVQLQQSGPE LEKPGASVKI SCKASGYSFT GYTMNWVKQS HGKSLEWIGL ITPYNGASSY  60
NQKFRGKATL TVDKSSSTAY MDLLSLTSED SAVYFCARGG YDGRGFDYWG SGTPVTVSS   119

SEQ ID NO: 296          moltype = AA  length = 106
FEATURE                 Location/Qualifiers
source                  1..106
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 296
DIELTQSPAI MSASPGEKVT MTCSASSSVS YMHWYQQKSG TSPKRWIYDT SKLASGVPGR  60
FSGSGSGNSY SLTISSVEAE DDATYYCQQW SKHPLTFGSG TKVEIK                 106

SEQ ID NO: 297          moltype = AA  length = 120
FEATURE                 Location/Qualifiers
source                  1..120
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 297
QVELVQSGAE VKKPGESLKI SCKGSGYSFT SYWIGWVRQA PGKGLEWMGI IDPGDSRTRY  60
SPSFQGQVTI SADKSISTAY LQWSSLKASD TAMYYCARGQ LYGGTYMDGW GQGTLVTVSS  120

SEQ ID NO: 298          moltype = AA  length = 111
FEATURE                 Location/Qualifiers
source                  1..111
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 298
DIALTQPASV SGSPGQSITI SCTGTSSDIG GYNSVSWYQQ HPGKAPKLMI YGVNNRPSGV  60
SNRFSGSKSG NTASLTISGL QAEDEADYYC SSYDIESATP VFGGGTKLTV L           111

SEQ ID NO: 299          moltype = AA  length = 117
FEATURE                 Location/Qualifiers
source                  1..117
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 299
QVQLVESGGG LVKPGGSLRL SCAASGFTFS DYYMSWIRQA PGKCLEWISY ISSSESIIYY  60
VDAVKGRFTI SRDNAKNSLY LQMNSLRAED TAVYYCARDV GSHFDYWGQG TLVTVSS     117

SEQ ID NO: 300          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 300
DIQMTQSPSS VSASVGDRVT ITCRASQDIS RWLAWYQQKP GKAPKLLISA ASRLQSGVPS  60
RFSGSGSGTD FTLTISSLQP EDFAIYYCQQ AKSFPRTFGC GTKVEIK                107

SEQ ID NO: 301          moltype = AA  length = 119
FEATURE                 Location/Qualifiers
source                  1..119
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 301
QVQLVQSGAE VKKPGETVKI SCKASDYTFT YYGMNWVKQA PGQGLKWMGW IDTTTGEPTY  60
AQKFQGRIAF SLETSASTAY LQIKSLKSED TATYFCARRG PYNWYFDVWG QGTTVTVSS   119

SEQ ID NO: 302          moltype = AA  length = 112
FEATURE                 Location/Qualifiers
source                  1..112
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 302
DIVMTQSPLS VPVTPGEPVS ISCRSSKSLL HSNGNTYLYW FLQRPGQSPQ LLIYRMSNLV  60
SGVPDRFSGS GSGTAFTLRI SRVEAEDVGV YYCLQHLEYP FTFGPGTKLE LK          112

SEQ ID NO: 303          moltype = AA  length = 119
FEATURE                 Location/Qualifiers
source                  1..119
                        mol_type = protein
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                        organism = sythetic construct
SEQUENCE: 303
QVQLQQSGAE VKKPGASVKV SCEASGYTFP SYVLHWVKQA PGQGLEWIGY INPYNDGTQY  60
NEKFKGKATL TRDTSINTAY MELSRLRSDD TAVYYCARGF GGSYGFAYWG QGTLVTVSS   119

SEQ ID NO: 304          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
source                  1..108
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 304
DIQLTQSPSS LSASVGDRVT MTCSASSSVS SSYLYWYQQK PGKAPKLWIY STSNLASGVP  60
ARFSGSGSGT DFTLTISSLQ PEDSASYFCH QWNRYPYTFG GGTRLEIK              108

SEQ ID NO: 305          moltype = AA  length = 117
FEATURE                 Location/Qualifiers
source                  1..117
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 305
EVQLVESGGG LVQPGGSMRL SCVASGFPFS NYWMNWVRQA PGKGLEWVGE IRLKSNNYTT  60
HYAESVKGRF TISRDDSKNS LYLQMNSLKT EDTAVYYCTR HYYFDYWGQG TLVTVSS     117

SEQ ID NO: 306          moltype = AA  length = 112
FEATURE                 Location/Qualifiers
source                  1..112
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 306
DIVMTQSPLS NPVTPGEPAS ISCRSSKSLL HSNGITYFFW YLQKPGQSPQ LLIYQMSNLA  60
SGVPDRFSGS GSGTDFTLRI SRVEAEDVGV YYCAQNLELP PTFGQGTKVE IK         112

SEQ ID NO: 307          moltype = AA  length = 119
FEATURE                 Location/Qualifiers
source                  1..119
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 307
QVKLQESGAE LARPGASVKL SCKASGYTFT NYWMQWVKQR PGQGLDWIGA IYPGDGNTRY  60
THKFKGKATL TADKSSSTAY MQLSSLASED SGVYYCARGE GNYAWFAYWG QGTTVTVSS   119

SEQ ID NO: 308          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 308
DIELTQSPAS LSASVGETVT ITCQASENIY SYLAWHQQKQ GKSPQLLVYN AKTLAGGVSS  60
RFSGSGSGTH FSLKIKSLQP EDFGIYYCQH HYGILPTFGG GTKLEIK                107

SEQ ID NO: 309          moltype = AA  length = 116
FEATURE                 Location/Qualifiers
source                  1..116
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 309
EVQLVESGGG LVQPGGSLRL SCAASGYSIT NDYAWNWVRQ APGKGLEWVG YISYSGYTTY  60
NPSLKSRFTI SRDTSKNTLY LQMNSLRAED TAVYYCARWT SGLDYWGQGT LVTVSS      116

SEQ ID NO: 310          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 310
DIQMTQSPSS LSASVGDRVT ITCKASDLIH NWLAWYQQKP GKAPKLLIYG ATSLETGVPS  60
RFSGSGSGTD FTLTISSLQP EDFATYYCQQ YWTTPFTFGQ GTKVEIK                107

SEQ ID NO: 311          moltype = AA  length = 117
FEATURE                 Location/Qualifiers
source                  1..117
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 311
QVQLVESGGG LVKPGGSLRL SCAASGFTFS NYYMSWVRQA PGKGLEWISY ISGRGSTIFY  60
ADSVKGRITI SRDNAKNSLF LQMNSLRAED TAVYFCVKDR GGYSPYWGQG TLVTVSS     117
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SEQ ID NO: 312          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
source                  1..108
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 312
DIQMTQSPSS LSASVGDRVT ITCRASQSIS TYLNWYQQKP GKAPKLLIYT ASSLQSGVPS  60
RFSGSGSGTD FTLTISSLQP EDFATYYCQQ SYSTPPITFG QGTRLEIK               108

SEQ ID NO: 313          moltype = AA  length = 115
FEATURE                 Location/Qualifiers
source                  1..115
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 313
QVHLQESGPG LVKPSETLSL TCTVSDDSIS SYYWSWIRQP PGKGLEWIGH ISYSGSANYN  60
PSLKSRVTIS VDTSKNQFSL KLSSVTAADT AVYYCANWDD AFNIWGQGTM VTVSS       115

SEQ ID NO: 314          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
source                  1..108
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 314
EIVLTQSPGT LSLSPGERAT LSCRASQSVS SSYLAWYQQK PGQAPRLLIY GASSRATGIP  60
DRFSGSGSGT DFTLTISRLE PEDFAVYYCQ QYGSSPWTFG QGTKVEIK               108

SEQ ID NO: 315          moltype = AA  length = 114
FEATURE                 Location/Qualifiers
source                  1..114
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 315
QVQLVESGGG VVQPGRSLRL SCAASGFTFL RYAMHWVRQA PGKGLEWVAV ISYDGRYKYY  60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCTTTT FDSWGQGTLV TVSS        114

SEQ ID NO: 316          moltype = AA  length = 113
FEATURE                 Location/Qualifiers
source                  1..113
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 316
DIVMTQTPLS LPVTPGEAAS ISCRSSQSLL DSEDGNTYLD WYLQKPGQSP QLLIYTLSHR  60
ASGVPDRFSG SGSGTDFTLE ISRVEAEDVG VYYCMQRRDF PFTFGQGTKV DIK         113

SEQ ID NO: 317          moltype = AA  length = 113
FEATURE                 Location/Qualifiers
source                  1..113
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 317
QVQLVESGGG VVQPGRSLRL SCAASGFTFS SYGMHWVRQA PGKGLEWVAV IWYDGSNKYY  60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCASNG DHWGQGTLVT VSS         113

SEQ ID NO: 318          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 318
EIVMTQSPAT LSVSPGERAT LSCRASQSVS SNLAWYQQKP GQAPRLLIYG ASTRATGIPA  60
RFSGSGSGTE FTLTISSLQS EDFAVYYCQQ YNNWPRTFGQ GTKVEIK                107

SEQ ID NO: 319          moltype = AA  length = 118
FEATURE                 Location/Qualifiers
source                  1..118
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 319
EIQLVQSGAE VKKPGSSVKV SCKASGYTFT HYGMNWVRQA PGQGLEWVGW VNTYTGEPTY  60
ADDFKGRLTF TLDTSTSTAY MELSSLRSED TAVYYCTREG EGLGFGDWGQ GTTVTVSS    118

SEQ ID NO: 320          moltype = AA  length = 112
FEATURE                 Location/Qualifiers
source                  1..112
```

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                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 320
DVVMTQSPLS LPVTPGEPAS ISCRSSQSIV HSHGDTYLEW YLQKPGQSPQ LLIYKVSNRF  60
SGVPDRFSGS GSGTDFTLKI SRVEAEDVGV YYCFQGSHIP VTFGQGTKLE IK          112

SEQ ID NO: 321          moltype = AA  length = 116
FEATURE                 Location/Qualifiers
source                  1..116
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 321
EVQLVESGGG LVQPGGSLRL SCAASGFTFS SYMMSWVRQA PGKGLEWVAT ISGGGANTYY  60
PDSVKGRFTI SRDNAKNSLY LQMNSLRAED TAVYYCARQL YYFDYWGQGT TVTVSS      116

SEQ ID NO: 322          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 322
DIQMTQSPSS LSASVGDRVT ITCLASQTIG TWLTWYQQKP GKAPKLLIYT ATSLADGVPS  60
RFSGSGSGTD FTLTISSLQP EDFATYYCQQ VYSIPWTFGG GTKVEIK                107

SEQ ID NO: 323          moltype = AA  length = 117
FEATURE                 Location/Qualifiers
source                  1..117
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 323
EVQLLESGGV LVQPGGSLRL SCAASGFTFS NFGMTWVRQA PGKGLEWVSG ISGGGRDTYF  60
ADSVKGRFTI SRDNSKNTLY LQMNSLKGED TAVYYCVKWG NIYFDYWGQG TLVTVSS     117

SEQ ID NO: 324          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 324
DIQMTQSPSS LSASVGDSIT ITCRASLSIN TFLNWYQQKP GKAPNLLIYA ASSLHGGVPS  60
RFSGSGSGTD FTLTIRTLQP EDFATYYCQQ SSNTPFTFGP GTVVDFR                107

SEQ ID NO: 325          moltype = AA  length = 123
FEATURE                 Location/Qualifiers
source                  1..123
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 325
QVQLVQSGAE VKKPGSSVKV SCKASGGTFS SYAISWVRQA PGQGLEWMGG IIPIFDTANY  60
AQKFQGRVTI TADESTSTAY MELSSLRSED TAVYYCARPG LAAAYDTGSL DYWGQGTLVT  120
VSS                                                                123

SEQ ID NO: 326          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 326
EIVLTQSPAT LSLSPGERAT LSCRASQSVR SYLAWYQQKP GQAPRLLIYD ASNRATGIPA  60
RFSGSGSGTD FTLTISSLEP EDFAVYYCQQ RNYWPLTFGQ GTKVEIK                107

SEQ ID NO: 327          moltype = AA  length = 116
FEATURE                 Location/Qualifiers
source                  1..116
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 327
EVQLLESGGG LVQPGGSLRL SCAASGFTFS SYDMSWVRQA PGKGLEWVST ISGGGSYTYY  60
QDSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCASPY YAMDYWGQGT TVTVSS      116

SEQ ID NO: 328          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 328
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DIQLTQSPSF LSAYVGDRVT ITCKASQDVG TAVAWYQQKP GKAPKLLIYW ASTLHTGVPS  60
RFSGSGSGTE FTLTISSLQP EDFATYYCQH YSSYPWTFGQ GTKLEIK                107

SEQ ID NO: 329          moltype = AA  length = 113
FEATURE                 Location/Qualifiers
source                  1..113
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 329
QVQLVESGGG VVQPGRSLRL TCKASGLTFS SSGMHWVRQA PGKGLEWVAV IWYDGSKRYY  60
ADSVKGRFTI SRDNSKNTLF LQMNSLRAED TAVYYCATNN DYWGQGTLVT VSS         113

SEQ ID NO: 330          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 330
EIVLTQSPAT LSLSPGERAT LSCRASQSVS SYLAWYQQKP GQAPRLLIYT ASNRATGIPA  60
RFSGSGSGTD FTLTISSLEP EDFAVYYCQQ YSNWPRTFGQ GTKVEIK                107

SEQ ID NO: 331          moltype = AA  length = 120
FEATURE                 Location/Qualifiers
source                  1..120
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 331
EVMLVESGGG LVQPGGSLRL SCTASGFTFS KSAMSWVRQA PGKGLEWVAY ISGGGGDTYY  60
SSSVKGRFTI SRDNAKNSLY LQMNSLRAED TAVYYCARHS NVNYYAMDYW GQGTLVTVSS  120

SEQ ID NO: 332          moltype = AA  length = 111
FEATURE                 Location/Qualifiers
source                  1..111
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 332
EIVLTQSPAT LSLSPGERAT MSCRASENID VSGISFMNWY QQKPGQAPKL LIYVASNQGS  60
GIPARFSGSG SGTDFTLTIS RLEPEDFAVY YCQQSKEVPW TFGQGTKLEI K           111

SEQ ID NO: 333          moltype = AA  length = 118
FEATURE                 Location/Qualifiers
source                  1..118
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 333
EVQLVESGGG LVKPGGSLRL SCAASGFTFS SYGMSWVRQA PGKRLEWVAT ISGGGRDTYY  60
SDSVKGRFTI SRDNAKNNLY LQMNSLRAED TAVYYCSRQY GTVWFFNWGQ GTLVTVSS    118

SEQ ID NO: 334          moltype = AA  length = 111
FEATURE                 Location/Qualifiers
source                  1..111
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 334
EIVLTQSPAT LSLSPGERAT LSCRASESVD SYGNSFMHWY QQKPGQPPRL LIYAASNQGS  60
GVPARFSGSG SGTDFTLTIS SLEPEDFAMY FCQQSKEVPW TFGQGTKVEI K           111

SEQ ID NO: 335          moltype = AA  length = 114
FEATURE                 Location/Qualifiers
source                  1..114
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 335
QIQLVQSGSE LKKPGASVKV SCKASGYTFT NFGMNWVRQA PGQGLKWMGW ISGYTREPTY  60
AADFKGRFVI SLDTSVSTAY LQISSLKAED TAVYYCARDV FDYWGQGTLV TVSS        114

SEQ ID NO: 336          moltype = AA  length = 110
FEATURE                 Location/Qualifiers
source                  1..110
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 336
DIVLTQSPAS LAVSPGQRAT ITCRASESVD NYGYSFMNWF QQKPGQPPKL LIYRASNLES  60
GVPARFSGSG SRTDFTLTIN PVEADDTANY YCQQSNADPT FGQGTKLEIK             110

SEQ ID NO: 337          moltype = AA  length = 119
```

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FEATURE                Location/Qualifiers
source                 1..119
                       mol_type = protein
                       organism = sythetic construct
SEQUENCE: 337
QVQLVQSGAE VKKPGASVKV SCKASGYTFT NYWIHWVRQA PGQGLEWMGE IDPYDSYTNY  60
NQKFKGRVTM TVDKSTSTVY MELSSLRSED TAVYYCARPG FTYGGMDFWG QGTLVTVSS   119

SEQ ID NO: 338         moltype = AA  length = 113
FEATURE                Location/Qualifiers
source                 1..113
                       mol_type = protein
                       organism = sythetic construct
SEQUENCE: 338
DIQMTQSPSS LSASVGDRVT ITCKSSQSLF NSGNQKNYLA WYQQKPGKVP KLLIYGASTR  60
DSGVPYRFSG SGSGTDFTLT ISSLQPEDVA TYYCQNDHYY PYTFGGGTKV EIK         113

SEQ ID NO: 339         moltype = AA  length = 120
FEATURE                Location/Qualifiers
source                 1..120
                       mol_type = protein
                       organism = sythetic construct
SEQUENCE: 339
QVQLVQSGAE VKKPGASVKV SCKASGYTFT SYYMYWVRQA PGQGLEWMGG VNPSNGGTNF  60
NEKFKSRVTI TADKSTSTAY MELSSLRSED TAVYYCARRD YRYDMGFDYW GQGTTVTVSS  120

SEQ ID NO: 340         moltype = AA  length = 111
FEATURE                Location/Qualifiers
source                 1..111
                       mol_type = protein
                       organism = sythetic construct
SEQUENCE: 340
EIVLTQSPAT LSLSPGERAT ISCRASKGVS TSGYSYLHWY QQKPGQAPRL LIYLASYLES  60
GVPARFSGSG SGTDFTLTIS SLEPEDFATY YCQHSRELPL TFGTGTKVEI K           111

SEQ ID NO: 341         moltype = AA  length = 113
FEATURE                Location/Qualifiers
source                 1..113
                       mol_type = protein
                       organism = sythetic construct
SEQUENCE: 341
QVQLVESGGG VVQPGRSLRL DCKASGITFS NSGMHWVRQA PGKGLEWVAV IWYDGSKRYY  60
ADSVKGRFTI SRDNSKNTLF LQMNSLRAED TAVYYCATND DYWGQGTLVT VSS         113

SEQ ID NO: 342         moltype = AA  length = 107
FEATURE                Location/Qualifiers
source                 1..107
                       mol_type = protein
                       organism = sythetic construct
SEQUENCE: 342
EIVLTQSPAT LSLSPGERAT LSCRASQSVS SYLAWYQQKP GQAPRLLIYD ASNRATGIPA  60
RFSGSGSGTD FTLTISSLEP EDFAVYYCQQ SSNWPRTFGQ GTKVEIK                107

SEQ ID NO: 343         moltype = AA  length = 115
FEATURE                Location/Qualifiers
source                 1..115
                       mol_type = protein
                       organism = sythetic construct
SEQUENCE: 343
QVQLVQSGAE VKKPGSSVKV SCKASGFTFT TYYISWVRQA PGQGLEYLGY INMGSGGTNY  60
NEKFKGRVTI TADKSTSTAY MELSSLRSED TAVYYCAIIG YFDYWGQGTM VTVSS       115

SEQ ID NO: 344         moltype = AA  length = 112
FEATURE                Location/Qualifiers
source                 1..112
                       mol_type = protein
                       organism = sythetic construct
SEQUENCE: 344
DVVMTQSPLS LPVTLGQPAS ISCRSSQSLL DSDGGTYLYW FQQRPGQSPR RLIYLVSTLG  60
SGVPDRFSGS GSGTDFTLKI SRVEAEDVGV YYCMQLTHWP YTFGQGTKLE IK          112

SEQ ID NO: 345         moltype = AA  length = 118
FEATURE                Location/Qualifiers
source                 1..118
                       mol_type = protein
                       organism = sythetic construct
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SEQUENCE: 345
EVQLVESGGG LVQPGGSLRL SCAASGFAFS SYDMSWVRQA PGKGLDWVAT ISGGGRYTYY  60
PDSVKGRFTI SRDNSKNNLY LQMNSLRAED TALYYCANRY GEAWFAYWGQ GTLVTVSS    118

SEQ ID NO: 346          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 346
DIQMTQSPSS MSASVGDRVT FTCRASQDIN TYLSWFQQKP GKSPKTLIYR ANRLVSGVPS  60
RFSGSGSGQD YTLTISSLQP EDMATYYCLQ YDEFPLTFGA GTKLELK                107

SEQ ID NO: 347          moltype = AA  length = 119
FEATURE                 Location/Qualifiers
source                  1..119
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 347
QVQLVQSGAE VKKPGASVKV SCKVSGYSLS KYDMSWVRQA PGKGLEWMGI IYTSGYTDYA  60
QKFQGRVTMT EDTSTDTAYM ELSSLRSEDT AVYYCATGNP YYTNGFNSWG QGTLVTVSS   119

SEQ ID NO: 348          moltype = AA  length = 110
FEATURE                 Location/Qualifiers
source                  1..110
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 348
DIQMTQSPSS LSASVGDRVT ITCQASQSPN NLLAWYQQKP GKAPKLLIYG ASDLPSGVPS  60
RFSGSGSGTD FTLTISSLQP EDFATYYCQN NYYVGPVSYA FGGGTKVEIK             110

SEQ ID NO: 349          moltype = AA  length = 117
FEATURE                 Location/Qualifiers
source                  1..117
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 349
QVQLVQSGSE LKKPGASVKI SCKASGYTFT NYGMNWVRQA PGQGLQWMGW INTDSGESTY  60
AEEFKGRFVF SLDTSVNTAY LQITSLTAED TGMYFCVRVG YDALDYWGQG TLVTVSS     117

SEQ ID NO: 350          moltype = AA  length = 106
FEATURE                 Location/Qualifiers
source                  1..106
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 350
EIVLTQSPSS LSASVGDRVT ITCSARSSVS YMHWFQQKPG KAPKLWIYRT SNLASGVPSR  60
FSGSGSGTSY CLTINSLQPE DFATYYCQQR SSFPLTFGGG TKLEIK                 106

SEQ ID NO: 351          moltype = AA  length = 118
FEATURE                 Location/Qualifiers
source                  1..118
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 351
QVQLVQSGAE VKKPGASVKV SCKASGYTFP SYYMHWVRQA PGQGLEWMGI INPEGGSTAY  60
AQKFQGRVTM TRDTSTSTVY MELSSLRSED TAVYYCARGG TYYDYTYWGQ GTLVTVSS    118

SEQ ID NO: 352          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 352
DIQMTQSPST LSASVGDRVT ITCRASQSIS SWLAWYQQKP GKAPKLLIYE ASSLESGVPS  60
RFSGSGSGTE FTLTISSLQP DDFATYYCQQ YNSFPPTFGG GTKVEIK                107

SEQ ID NO: 353          moltype = AA  length = 121
FEATURE                 Location/Qualifiers
source                  1..121
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 353
QVTLKESGPA LVKPTQTLTL TCTFSGFSLS TSGTCVSWIR QPPGKALEWL ATICWEDSKG  60
YNPSLKSRLT ISKDTSKNQA VLTMTNMDPV DTATYYCARR EDSGYFWFPY WGQGTLVTVS  120
S                                                                  121
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SEQ ID NO: 354          moltype = AA  length = 106
FEATURE                 Location/Qualifiers
source                  1..106
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 354
NIQMTQSPSS LSASVGDRVT ITCKAGQNVN NYLAWYQQKP GKAPKVLIFN ANSLQTGVPS  60
RFSGSGSGTD FTLTISSLQP EDFATYYCQQ YNSWTTFGGG TKVEIK                 106

SEQ ID NO: 355          moltype = AA  length = 129
FEATURE                 Location/Qualifiers
source                  1..129
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 355
QVQLVQSGGG LVQPGGSLRL SCAASGFTFS SYWMYWVRQV PGKGLEWVSA IDTGGGRTYY  60
ADSVKGRFAI SRVNAKNTMY LQMNSLRAED TAVYYCARDE GGGTGWGVLK DWPYGLDAWG  120
QGTLVTVSS                                                          129

SEQ ID NO: 356          moltype = AA  length = 106
FEATURE                 Location/Qualifiers
source                  1..106
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 356
QPVLTQPLSV SVALGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVIYRD SNRPSGIPER  60
FSGSNSGNTA TLTISRAQAG DEADYYCQVW DSSTAVFGTG TKLTVL                 106

SEQ ID NO: 357          moltype = AA  length = 118
FEATURE                 Location/Qualifiers
source                  1..118
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 357
EVQLVQSGGG LVQPGGSLKL SCAASGFTFS SYGMSWVRQA PGKGLDWVAT ISGGGRDTYY  60
PDSVKGRFTI SRDNSKNNLY LQMNSLRAED TALYYCARQK GEAWFAYWGQ GTLVTVSS    118

SEQ ID NO: 358          moltype = AA  length = 111
FEATURE                 Location/Qualifiers
source                  1..111
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 358
DIVLTQSPAS LAVSPGQRAT ITCRASESVD NYGISFMNWF QQKPGQPPKL LIYAASNKGT  60
GVPARFSGSG SGTDFTLNIN PMEENDTAMY FCQQSKEVPW TFGGGTKLEI K           111

SEQ ID NO: 359          moltype = AA  length = 119
FEATURE                 Location/Qualifiers
source                  1..119
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 359
QVQLVQSGAE VKKPGASVKV SCKASGYSFT SYWMNWVRQA PGQGLEWIGV IHPSDSETWL  60
DQKFKDRVTI TVDKSTSTAY MELSSLRSED TAVYYCAREH YGTSPFAYWG QGTLVTVSS   119

SEQ ID NO: 360          moltype = AA  length = 111
FEATURE                 Location/Qualifiers
source                  1..111
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 360
EIVLTQSPAT LSLSPGERAT LSCRASESVD NYGMSFMNWF QQKPGQPPKL LIHAASNQGS  60
GVPSRFSGSG SGTDFTLTIS SLEPEDFAVY FCQQSKEVPY TFGGGTKVEI K           111

SEQ ID NO: 361          moltype = AA  length = 121
FEATURE                 Location/Qualifiers
source                  1..121
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 361
QVQLVQSGSE LKKPGASVKV SCKASNYTFT DYSMHWVRQA PGQGLEWMGW INIETYYPTY  60
ADQFKGRFAF SLDTSVSTAY LQISSLKAED TAVYYCARDY YGRFYYAMDY WGQGTTVTVS  120
S                                                                  121

SEQ ID NO: 362          moltype = AA  length = 108
```

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```
FEATURE                Location/Qualifiers
source                 1..108
                       mol_type = protein
                       organism = sythetic construct
SEQUENCE: 362
EIVLTQSPAT LSLSPGERAT LSCTASSSVS SSYFHWYQQK PGQAPRLLIY STSNLASGIP  60
ARFSGSGSGT DFTLTISRLE PEDFAVYYCH QYHRSPLTFG GGTKVEIK               108

SEQ ID NO: 363         moltype = AA  length = 118
FEATURE                Location/Qualifiers
source                 1..118
                       mol_type = protein
                       organism = sythetic construct
SEQUENCE: 363
EVQLVESGGG LVKPGGSLRL SCAASGFTFS SYGMSWVRQT PEKRLEWVAT ISGGGRDTYY  60
PDSVKGRFTI SRDNAKNNLY LQMSSLRSED TALYYCARQK DTSWFVHWGQ GTLVTVSS    118

SEQ ID NO: 364         moltype = AA  length = 111
FEATURE                Location/Qualifiers
source                 1..111
                       mol_type = protein
                       organism = sythetic construct
SEQUENCE: 364
EIVLTQSPAT LAVSPGERAT ISCRASESVD DYGISFMNWF QQKPGQPPKL LIYVASNQGS  60
GVPARFSGSG SGTDFTLNIH PMEEDDTAMY FCQQSKEVPW TFGGGTKLEI K           111

SEQ ID NO: 365         moltype = AA  length = 117
FEATURE                Location/Qualifiers
source                 1..117
                       mol_type = protein
                       organism = sythetic construct
SEQUENCE: 365
QVQLVQSGAE VKKPGASVKV SCKASGYTFT SYWINWVRQA PGQGLEWMGN IYPGSSLTNY  60
NEKFKNRVTM TRDTSTSTVY MELSSLRSED TAVYYCARLS TGTFAYWGQG TLVTVSS     117

SEQ ID NO: 366         moltype = AA  length = 113
FEATURE                Location/Qualifiers
source                 1..113
                       mol_type = protein
                       organism = sythetic construct
SEQUENCE: 366
DIVMTQSPDS LAVSLGERAT INCKSSQSLW DSGNQKNFLT WYQQKPGQPP KLLIYWTSYR  60
ESGVPDRFSG SGSGTDFTLT ISSLQAEDVA VYYCQNDYFY PHTFGGGTKV EIK         113

SEQ ID NO: 367         moltype = AA  length = 116
FEATURE                Location/Qualifiers
source                 1..116
                       mol_type = protein
                       organism = sythetic construct
SEQUENCE: 367
QVQLVESGGG LVKPGGSLRL SCAASGFTFS NYGMSWIRQA PGKGLEWVST ISGGGSNIYY  60
ADSVKGRFTI SRDNAKNSLY LQMNSLRAED TAVYYCVSYY YGIDFWGQGT SVTVSS      116

SEQ ID NO: 368         moltype = AA  length = 107
FEATURE                Location/Qualifiers
source                 1..107
                       mol_type = protein
                       organism = sythetic construct
SEQUENCE: 368
DIQMTQSPSS LSASVGDRVT ITCKASQDVT TAVAWYQQKP GKAPKLLIYW ASTRHTGVPS  60
RFSGSGSGTD FTLTISSLQP EDFATYYCQQ HYTIPWTFGG GTKLEIK                107

SEQ ID NO: 369         moltype = AA  length = 120
FEATURE                Location/Qualifiers
source                 1..120
                       mol_type = protein
                       organism = sythetic construct
SEQUENCE: 369
QVQLVQSGAE VKKPGSSVKV SCKASGGTFS SYAISWVRQA PGQGLEWMGL IIPMFDTAGY  60
AQKFQGRVAI TVDESTSTAY MELSSLRSED TAVYYCARAE HSSTGTFDYW GQGTLVTVSS  120

SEQ ID NO: 370         moltype = AA  length = 107
FEATURE                Location/Qualifiers
source                 1..107
                       mol_type = protein
                       organism = sythetic construct
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SEQUENCE: 370
DIQMTQSPSS VSASVGDRVT ITCRASQGIS SWLAWYQQKP GKAPKLLISA ASSLQSGVPS  60
RFSGSGSGTD FTLTISSLQP EDFATYYCQQ ANHLPFTFGG GTKVEIK                107

SEQ ID NO: 371          moltype = AA  length = 117
FEATURE                 Location/Qualifiers
source                  1..117
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 371
EVQLVQSGAE VKKPGESLRI SCKGSGYTFT TYWMHWVRQA TGQGLEWMGN IYPGTGGSNF  60
DEKFKNRVTI TADKSTSTAY MELSSLRSED TAVYYCTRWT TGTGAYWGQG TTVTVSS     117

SEQ ID NO: 372          moltype = AA  length = 113
FEATURE                 Location/Qualifiers
source                  1..113
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 372
EIVLTQSPAT LSLSPGERAT LSCKSSQSLL DSGNQKNFLT WYQQKPGQAP RLLIYWASTR  60
ESGVPSRFSG SGSGTDFTFT ISSLEAEDAA TYYCQNDYSY PYTFGQGTKV EIK         113

SEQ ID NO: 373          moltype = AA  length = 118
FEATURE                 Location/Qualifiers
source                  1..118
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 373
QVQLQESGPG LVKPSETLSL TCTVSGFSLT SYGVHWIRQP PGKGLEWIGV IYADGSTNYN  60
PSLKSRVTIS KDTSKNQVSL KLSSVTAADT AVYYCARAYG NYWYIDVWGQ GTTVTVSS    118

SEQ ID NO: 374          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 374
DIVMTQSPDS LAVSLGERAT INCKSSESVS NDVAWYQQKP GQPPKLLINY AFHRFTGVPD  60
RFSGSGYGTD FTLTISSLQA EDVAVYYCHQ AYSSPYTFGQ GTKLEIK                107

SEQ ID NO: 375          moltype = AA  length = 125
FEATURE                 Location/Qualifiers
source                  1..125
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 375
QGQLVQSGAE VKKPGASVKV SCKASGYTFT DYEMHWVRQA PIHGLEWIGV IESETGGTAY  60
NQKFKGRVTI TADKSTSTAY MELSSLRSED TAVYYCAREG ITTVATTYYW YFDVWGQGTT  120
VTVSS                                                              125

SEQ ID NO: 376          moltype = AA  length = 112
FEATURE                 Location/Qualifiers
source                  1..112
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 376
DVVMTQSPLS LPVTLGQPAS ISCRSSQSIV HSNGNTYLEW YLQKPGQSPQ LLIYKVSNRF  60
SGVPDRFSGS GSGTDFTLKI SRVEAEDVGV YYCFQGSHVP LTFGQGTKLE IK          112

SEQ ID NO: 377          moltype = AA  length = 120
FEATURE                 Location/Qualifiers
source                  1..120
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 377
EVQLLESGGG LVQPGGSLRL SCAASGFTFS SYDMSWVRQA PGKGLEWVSL ISGGGSQTYY  60
AESVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYFCASPS GHYFYAMDVW GQGTTVTVSS  120

SEQ ID NO: 378          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 378
DIQMTQSPSS VSASVGDRVT ITCRASQGIS NWLAWYQQKP GKAPKLLIFA ASSLQSGVPS  60
RFSGSGSGTD FTLTISSLQP EDFATYYCQQ AESFPHTFGG GTKVEIK                107
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SEQ ID NO: 379          moltype = AA  length = 123
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source                  1..123
                        mol_type = protein
                        organism = sythetic construct
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QLQLQESGPG LVKPSETLTL TCTVSADSIS STTYYWVWIR QPPGKGLEWI GSISYSGSTY  60
YNPSLKSRVT VSVDTSKNQF SLKLNSVAAT DTALYYCARH LGYNGRYLPF DYWGQGTLVT  120
VSS                                                                123

SEQ ID NO: 380          moltype = AA  length = 110
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source                  1..110
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SEQUENCE: 380
QSALTQPASV SGSPGQSITI SCTGTSSDVG FYNYVSWYQQ HPGKAPELMI YDVSNRPSGV  60
SDRFSGSKSG NTASLTISGL QAEDEADYYC SSYTSISTWV FGGGTKLTVL             110

SEQ ID NO: 381          moltype = AA  length = 122
FEATURE                 Location/Qualifiers
source                  1..122
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 381
EVQLLESGGG LVQPGGSLRL SCAASGFTIS RYHMTWVRQA PGKGLEWIGH IYVNNDDTDY  60
ASSAKGRFTI SRDNSKNTLY LQMNSLRAED TATYFCARLD VGGGGAYIGD IWGQGTLVTV  120
SS                                                                 122

SEQ ID NO: 382          moltype = AA  length = 113
FEATURE                 Location/Qualifiers
source                  1..113
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 382
DIQMTQSPSS LSASVGDRVT ITCQSSQSVY NNNDLAWYQQ KPGKVPKLLI YYASTLASGV  60
PSRFSGSGSG TDFTLTISSL QPEDVATYYC AGGYDTDGLD TFAFGGGTKV EIK         113

SEQ ID NO: 383          moltype = AA  length = 116
FEATURE                 Location/Qualifiers
source                  1..116
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 383
QVQLQESGPG LVKPSQTLSL TCTVSGYAFT AYNIHWVRQA PGQGLEWMGS FDPYDGGSSY  60
NQKFKDRLTI SKDTSKNQVV LTMTNMDPVD TATYYCARGW YYFDYWGHGT LVTVSS      116

SEQ ID NO: 384          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 384
DIVMTQTPLS LPVTPGEPAS ISCRASKSIS KYLAWYQQKP GQAPRLLIYS GSTLQSGIPP  60
RFSGSGYGTD FTLTINNIES EDAAYYFCQQ HDESPYTFGE GTKVEIK                107

SEQ ID NO: 385          moltype = AA  length = 117
FEATURE                 Location/Qualifiers
source                  1..117
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 385
EVQLVESGGG LVQPGGSLRL SCAASGFTFS DYYMAWVRQA PGKGLEWVAS INYDGSSTYY  60
VDSVKGRFTI SRDNAKNSLY LQMNSLRAED TAVYYCARDR GYYFDYWGQG TTVTVSS     117

SEQ ID NO: 386          moltype = AA  length = 113
FEATURE                 Location/Qualifiers
source                  1..113
                        mol_type = protein
                        organism = sythetic construct
SEQUENCE: 386
DVVMTQTPLS LSVTPGQPAS ISCRSSQSLV HSNGNTYLHW YLQKPGQSPQ LLIYKVSNRF  60
SGVPDRFSGS GSGTDFTLKI SRVEAEDVGV YFCSQSTHVP PFTFGGGTKV EIK         113

SEQ ID NO: 387          moltype = AA  length = 119
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FEATURE                Location/Qualifiers
source                 1..119
                       mol_type = protein
                       organism = sythetic construct
SEQUENCE: 387
EVQLVESGGG LVQPGGSLRL SCAASGFDFS RYWMSWVRQA PGKGLEWIGE INPDSSTINY  60
APSLKDKFII SRDNAKNSLY LQMNSLRAED TAVYYCARPD GNYWYFDVWG QGTLVTVSS   119

SEQ ID NO: 388         moltype = AA  length = 107
FEATURE                Location/Qualifiers
source                 1..107
                       mol_type = protein
                       organism = sythetic construct
SEQUENCE: 388
DIQMTQSPSS LSASVGDRVT ITCKASQDVG IAVAWYQQKP GKVPKLLIYW ASTRHTGVPD  60
RFSGSGSGTD FTLTISSLQP EDVATYYCQQ YSSYPYTFGQ GTKVEIK                107
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What is claimed is:

1. A nucleic acid comprising a sequence encoding a decoy-resistant interleukin 18 (DR-18) polypeptide and a chimeric antigen receptor (CAR) sequence encoding a CAR, wherein the CAR is selected from:

i) an anti-5T4 CAR comprising an antigen binding domain (ABD) that specifically binds 5T4;
ii) an anti-AFP CAR comprising an ABD that specifically binds AFP;
iii) an anti-ALPP CAR comprising an ABD that specifically binds ALPP;
iv) an anti-ALPPL2 CAR comprising an ABD that specifically binds ALPPL2;
v) an anti-APN (CD13) CAR comprising an ABD that specifically binds APN (CD13);
vi) an anti-APRIL CAR comprising an ABD that specifically binds APRIL;
vii) an anti-AXL CAR comprising an ABD that specifically binds AXL;
viii) an anti-B7-H3 CAR comprising an ABD that specifically binds B7-H3;
ix) an anti-B7-H4 CAR comprising an ABD that specifically binds B7-H4;
x) an anti-BAFF-R CAR comprising an ABD that specifically binds BAFF-R;
xi) an anti-BCMA CAR comprising an ABD that specifically binds BCMA;
xii) an anti-BSG CAR comprising an ABD that specifically binds BSG;
xiii) an anti-CAIX CAR comprising an ABD that specifically binds CAIX;
xiv) an anti-CD123 CAR comprising an ABD that specifically binds CD123;
xv) an anti-CD133 CAR comprising an ABD that specifically binds CD133;
xvi) an anti-CD138 CAR comprising an ABD that specifically binds CD138;
xvii) an anti-CD19 CAR comprising an ABD that specifically binds CD19;
xviii) an anti-CD1A CAR comprising an ABD that specifically binds CD1A;
xix) an anti-CD2 CAR comprising an ABD that specifically binds CD2;
xx) an anti-CD20 CAR comprising an ABD that specifically binds CD20;
xxi) an anti-CD22 CAR comprising an ABD that specifically binds CD22;
xxii) an anti-CD24 CAR comprising an ABD that specifically binds CD24;
xxiii) an anti-CD276 CAR comprising an ABD that specifically binds CD276;
xxiv) an anti-CD3 CAR comprising an ABD that specifically binds CD3;
xxv) an anti-CD30 CAR comprising an ABD that specifically binds CD30;
xxvi) an anti-CD33 CAR comprising an ABD that specifically binds CD33;
xxvii) an anti-CD38 CAR comprising an ABD that specifically binds CD38;
xxviii) an anti-CD4 CAR comprising an ABD that specifically binds CD4;
xxix) an anti-CD43 CAR comprising an ABD that specifically binds CD43;
xxx) an anti-CD44 CAR comprising an ABD that specifically binds CD44;
xxxi) an anti-CD44v6 CAR comprising an ABD that specifically binds CD44v6;
xxxii) an anti-CD45 CAR comprising an ABD that specifically binds CD45;
xxxiii) an anti-CD5 CAR comprising an ABD that specifically binds CD5;
xxxiv) an anti-CD56 CAR comprising an ABD that specifically binds CD56;
xxxv) an anti-CD7 CAR comprising an ABD that specifically binds CD7;
xxxvi) an anti-CD70 CAR comprising an ABD that specifically binds CD70;
xxxvii) an anti-CD72 CAR comprising an ABD that specifically binds CD72;
xxxviii) an anti-CD83 CAR comprising an ABD that specifically binds CD83;
xxxix) an anti-CD84 CAR comprising an ABD that specifically binds CD84;
xl) an anti-CD99 CAR comprising an ABD that specifically binds CD99;
xli) an anti-CDCP1 CAR comprising an ABD that specifically binds CDCP1;
xlii) an anti-CDH17 CAR comprising an ABD that specifically binds CDH17;
xliii) an anti-CEA CAR comprising an ABD that specifically binds CEA;
xliv) an anti-CEACAM5 CAR comprising an ABD that specifically binds CEACAM5;

xlv) an anti-CLDN18.2 CAR comprising an ABD that specifically binds CLDN18.2;
xlvi) an anti-CLDN6 CAR comprising an ABD that specifically binds CLDN6;
xlvii) an anti-CLEC4K (CD207) CAR comprising an ABD that specifically binds CLEC4K (CD207);
xlviii) an anti-CLL-1 CAR comprising an ABD that specifically binds CLL-1;
xlix) an anti-CSPG4 CAR comprising an ABD that specifically binds CSPG4;
l) an anti-DLL3 CAR comprising an ABD that specifically binds DLL3;
li) an anti-DR5 CAR comprising an ABD that specifically binds DR5;
lii) an anti-EBV Protein CAR comprising an ABD that specifically binds EBV Protein;
liii) an anti-EGFR CAR comprising an ABD that specifically binds EGFR;
liv) an anti-EGFRvIII CAR comprising an ABD that specifically binds EGFRvII;
lv) an anti-EMR1 CAR comprising an ABD that specifically binds EMR1;
lvi) an anti-enkephalinase (CD10) CAR comprising an ABD that specifically binds enkephalinase (CD10);
lvii) an anti-EpCAM CAR comprising an ABD that specifically binds EpCAM;
lviii) an anti-EphA2 CAR comprising an ABD that specifically binds EphA2;
lix) an anti-EphA3 CAR comprising an ABD that specifically binds EphA3;
lx) an anti-FAP CAR comprising an ABD that specifically binds FAP;
lxi) an anti-FGFR4 CAR comprising an ABD that specifically binds FGFR4;
lxii) an anti-FLT3 CAR comprising an ABD that specifically binds FLT3;
lxiii) an anti-FOLR1 CAR comprising an ABD that specifically binds FOLR1;
lxiv) an anti-FSHR CAR comprising an ABD that specifically binds FSHR;
lxv) an anti-GC-C CAR comprising an ABD that specifically binds GC-C;
lxvi) an anti-GD2 CAR comprising an ABD that specifically binds GD2;
lxvii) an anti-GFRA4 CAR comprising an ABD that specifically binds GFRA4;
lxviii) an anti-Globo H CAR comprising an ABD that specifically binds Globo H;
lxix) an anti-GM2 (ganglioside M2) CAR comprising an ABD that specifically binds GM2 (ganglioside M2);
lxx) an anti-gp100 CAR comprising an ABD that specifically binds gp100;
lxxi) an anti-GPC1 CAR comprising an ABD that specifically binds GPC1;
lxxii) an anti-GPC2 CAR comprising an ABD that specifically binds GPC2;
lxxiii) an anti-GPC3 CAR comprising an ABD that specifically binds GPC3;
lxxiv) an anti-HAAH (ASPH) CAR comprising an ABD that specifically binds HAAH (ASPH);
lxxv) an anti-HER2 CAR comprising an ABD that specifically binds HER2;
lxxvi) an anti-HGF CAR comprising an ABD that specifically binds HGF;
lxxvii) an anti-HIV envelope protein gp120 CAR comprising an ABD that specifically binds HIV envelope protein gp120;
lxxviii) an anti-HIV-1 pol CAR comprising an ABD that specifically binds HIV-1 pol;
lxxix) an anti-HLA-A2 CAR comprising an ABD that specifically binds HLA-A2;
lxxx) an anti-HLA-G CAR comprising an ABD that specifically binds HLA-G;
lxxxi) an anti-HSP70 heat-shock protein CAR comprising an ABD that specifically binds HSP70 heat-shock protein;
lxxxii) an anti-ICAM-1 CAR comprising an ABD that specifically binds ICAM-1;
lxxxiii) an anti-IL-10R CAR comprising an ABD that specifically binds IL-10R;
lxxxiv) an anti-IL-13Rα2 CAR comprising an ABD that specifically binds IL-13Rα2;
lxxxv) an anti-integrin αvβ6 CAR comprising an ABD that specifically binds integrin αvβ6;
lxxxvi) an anti-ITGB7 CAR comprising an ABD that specifically binds ITGB7;
lxxxvii) an anti-KKLC1 CAR comprising an ABD that specifically binds KKLC1;
lxxxviii) an anti-KMA CAR comprising an ABD that specifically binds KMA;
lxxxix) an anti-L1CAM CAR comprising an ABD that specifically binds L1CAM;
xc) an anti-Lewis-Y antigen CAR comprising an ABD that specifically binds Lewis-Y antigen;
xci) an anti-LGR5 CAR comprising an ABD that specifically binds LGR5;
xcii) an anti-LILRB4 CAR comprising an ABD that specifically binds LILRB4;
xciii) an anti-LMP1 CAR comprising an ABD that specifically binds LMP1;
xciv) an anti-MAGEA3 CAR comprising an ABD that specifically binds MAGEA3;
xcv) an anti-M-CSF CAR comprising an ABD that specifically binds M-CSF;
xcvi) an anti-MG7 CAR comprising an ABD that specifically binds MG7;
xcvii) an anti-MICA CAR comprising an ABD that specifically binds MICA;
xcviii) an anti-MMP2 CAR comprising an ABD that specifically binds MMP2;
xcix) an anti-MSLN CAR comprising an ABD that specifically binds MSLN;
c) an anti-MUC1/MUC16 CAR comprising an ABD that specifically binds MUC1/MUC16;
ci) an anti-NCR3LG1 CAR comprising an ABD that specifically binds NCR3LG1;
cii) an anti-NECTIN2 CAR comprising an ABD that specifically binds NECTIN2;
ciii) an anti-nectin-4 CAR comprising an ABD that specifically binds nectin-4;
civ) an anti-NKG2D CAR comprising an ABD that specifically binds NKG2D;
cv) an anti-NKG2DL CAR comprising an ABD that specifically binds NKG2DL;
cvi) an anti-NR2F6 CAR comprising an ABD that specifically binds NR2F6;
cvii) an anti-NY-ESO-1 CAR comprising an ABD that specifically binds NY-ESO-1;

cviii) an anti-OPCML CAR comprising an ABD that specifically binds OPCML;
cix) an anti-PD-1 CAR comprising an ABD that specifically binds PD-1;
cx) an anti-PDGFRα CAR comprising an ABD that specifically binds PDGFRα;
cxi) an anti-PDL1 CAR comprising an ABD that specifically binds PDL1;
cxii) an anti-PLA2R CAR comprising an ABD that specifically binds PLA2R;
cxiii) an anti-PR1 CAR comprising an ABD that specifically binds PR1;
cxiv) an anti-PSCA CAR comprising an ABD that specifically binds PSCA;
cxv) an anti-PTK7 CAR comprising an ABD that specifically binds PTK7;
cxvi) an anti-PVR CAR comprising an ABD that specifically binds PVR;
cxvii) an anti-ROBO1 CAR comprising an ABD that specifically binds ROBO1;
cxviii) an anti-ROR1 CAR comprising an ABD that specifically binds ROR1;
cxix) an anti-ROR2 CAR comprising an ABD that specifically binds ROR2;
cxx) an anti-SEMA4A CAR comprising an ABD that specifically binds SEMA4A;
cxxi) an anti-SLAMF7 CAR comprising an ABD that specifically binds SLAMF7;
cxxii) an anti-TAG-72 CAR comprising an ABD that specifically binds TAG-72;
cxxiii) an anti-TGF-β CAR comprising an ABD that specifically binds TGF-β;
cxxiv) an anti-TM4SF1 CAR comprising an ABD that specifically binds TM4SF1;
cxxv) an anti-TRAIL CAR comprising an ABD that specifically binds TRAIL;
cxxvi) an anti-TRBC1 CAR comprising an ABD that specifically binds TRBC1;
cxxvii) an anti-TRBC2 CAR comprising an ABD that specifically binds TRBC2;
cxxviii) an anti-Trop-2 CAR comprising an ABD that specifically binds Trop-2;
cxxix) an anti-TSHR CAR comprising an ABD that specifically binds TSHR;
cxxx) an anti-TSLPR CAR comprising an ABD that specifically binds TSLPR;
cxxxi) an anti-ULBP1 CAR comprising an ABD that specifically binds ULBP1;
cxxxii) an anti-αvβ3 integrin CAR comprising an ABD that specifically binds αvβ3 integrin; and
cxxxiii) an anti-K light chain of human immunoglobulin CAR comprising an ABD that specifically binds K light chain of human immunoglobulin.

2. The nucleic acid of claim **1**, wherein the CAR is selected from:
a) inaticabtagene autoleucel;
b) actalycabtagene autoleucel;
c) relmacabtagene autoleucel;
d) lisocabtagene maraleucel;
e) brexucabtagene autoleucel;
f) axicabtagene ciloleucel;
g) tisagenlecleucel;
h) obecabatagene autoleucel;
i) azercabtagene zapreleucel;
j) rapcabtagene autoleucel; and
k) idecabtagene vicleucel.

3. The nucleic acid of claim **1**, wherein the CAR is an anti-EGFRvIII CAR comprising a means for specifically binding EGFRvIII.

4. The nucleic acid of claim **3**, wherein the anti-EGFRvIII CAR comprises an antigen binding domain comprising one or more of light chain complementary determining region 1 (LC CDR1), light chain complementary determining region 2 (LC CDR2), and light chain complementary determining region 3 (LC CDR3) of SEQ ID NO:110, and one or more of heavy chain complementary determining region 1 (HC CDR1), heavy chain complementary determining region 2 (HC CDR2), and heavy chain complementary determining region 3 (HC CDR3) of SEQ ID NO:111.

5. The nucleic acid of claim **4**, wherein the antigen binding domain comprises LC CDR1 (SEQ ID NO:112), LC CDR2 (SEQ ID NO:113), and LC CDR3 (SEQ ID NO:114); and HC CDR1 (SEQ ID NO:115), HC CDR2 (SEQ ID NO:116), and HC CDR3 (SEQ ID NO:117).

6. The nucleic acid of claim **5**, wherein the antigen binding domain comprises a light chain variable region comprising sequence SEQ ID NO:110 and a heavy chain variable region comprising sequence SEQ ID NO:111.

7. The nucleic acid of claim **1**, wherein the DR-18 polypeptide comprises the following amino acid sequence: XFGKXESXLSVIRNLNDQVLFIDQGNR-PLFEDMTDSDXRDNAPRTI-FIISXYXDXXXRXXAVTISV KXEKISTLSXXNKIIS-FKEMNPPDNIKDTKSDIIFFXRXVPGHXXKXQFESSS YEGYFLAXEKERD LFKLILKKEDELGDR-SIMFTXQXED (SEQ ID NO: 66), wherein the X at pos. 1 is Y, R or H; the X at pos. 5 is L, H, I or Y; the X at pos. 8 is K, Q or R; the X at pos. 38 is C or S; the X at pos. 51 is M, T, K, D, N, E or R; the X at pos. 53 is K, R, G, S or T; the X at pos. 55 is S, K or R; the X at pos. 56 is Q, E, A, R, V, G, K, L or R; the X at pos. 57 is P, L, G, A or K; the X at pos. 59 is G, A or T; the X at pos. 60 is M, K, Q, R or L; the X at pos. 68 is C, S, G, A, V, D, E or N; the X at pos. 76 is C or S; the X at pos. 77 is E or D; the X at pos. 103 is Q, E, K, P, A or R; the X at pos. 105 is S, D, N, R, K or A; the X at pos. 110 is D, K, H, N, Q, E, S or G; the X at pos. 111 is N, H, Y, D, R, S or G; the X at pos. 113 is M, V, R, T or K; the X at pos. 127 is C or S; the X at pos. 153 is V, I, T or A; and the X at pos. 155 is N, K or H.

8. The nucleic acid of claim **7**, wherein the sequence encoding the DR-18 polypeptide is codon optimized for expression in humans.

9. The nucleic acid of claim **8**, wherein the sequence encoding the DR-18 polypeptide comprises a DR-18 encoding sequence selected from SEQ ID NOs: 67-84.

10. A cell comprising the nucleic acid, or an expression vector comprising the nucleic acid, of claim **1**.

11. A method of treating a subject for cancer, the method comprising administering to the subject an effective amount of the nucleic acid of claim **1** or an expression vector or a cell comprising the nucleic acid.

12. A method, the method comprising contacting a human cell with the nucleic acid, or an expression vector comprising the nucleic acid, of claim **1** under conditions sufficient for delivery of the nucleic acid or expression vector into the human cell.

13. A nucleic acid, or expression vector comprising or consisting of the nucleic acid, comprising a sequence encoding a decoy-resistant interleukin 18 (DR-18) polypeptide, wherein the sequence encoding the DR-18 polypeptide is codon optimized for expression in human cells, and the DR-18 polypeptide comprises the following amino acid sequence: XFGKXESXLSVIRNLNDQVLFIDQGNRPLFEDMTDSDXRDNAPRTIFIISXYXDXXXRXXAVTISV KXEKISTLSXXNKIISFKEMNPPDNIKDTKSDIIFFXRXVPGHXXKXQFESSS YEGYFLAXEKERD LFKLILKKEDELGDRSIMFTXQXED (SEQ ID NO: 66), wherein the X at pos. 1 is Y, R or H; the X at pos. 5 is L, H, I or Y; the X at pos. 8 is K, Q or R; the X at pos. 38 is C or S; the X at pos. 51 is M, T, K, D, N, E or R; the X at pos. 53 is K, R, G, S or T; the X at pos. 55 is S, K or R; the X at pos. 56 is Q, E, A, R, V, G, K, L or R; the X at pos. 57 is P, L, G, A or K; the X at pos. 59 is G, A or T; the X at pos. 60 is M, K, Q, R or L; the X at pos. 68 is C, S, G, A, V, D, E or N; the X at pos. 76 is C or S; the X at pos. 77 is E or D; the X at pos. 103 is Q, E, K, P, A or R; the X at pos. 105 is S, D, N, R, K or A; the X at pos. 110 is D, K, H, N, Q, E, S or G; the X at pos. 111 is N, H, Y, D, R, S or G; the X at pos. 113 is M, V, R, T or K; the X at pos. 127 is C or S; the X at pos. 153 is V, I, T or A; and the X at pos. 155 is N, K or H.

14. The nucleic acid or expression vector of claim **13**, wherein the sequence encoding the DR-18 polypeptide comprises a DR-18 encoding sequence selected from SEQ ID NOs: 67-84.

15. The nucleic acid or expression vector of claim **13** further comprising a chimeric antigen receptor (CAR) sequence encoding a CAR, wherein the CAR is an anti-EGFRvIII CAR comprising a means for specifically binding EGFRvIII.

16. The nucleic acid or expression vector of claim **15**, wherein the anti-EGFRvIII CAR comprises an antigen binding domain comprising one or more of light chain complementary determining region 1 (LC CDR1), light chain complementary determining region 2 (LC CDR2), and light chain complementary determining region 3 (LC CDR3) of SEQ ID NO: 110, and one or more of heavy chain complementary determining region 1 (HC CDR1), heavy chain complementary determining region 2 (HC CDR2), and heavy chain complementary determining region 3 (HC CDR3) of SEQ ID NO:111.

17. The nucleic acid or expression vector of claim **16**, wherein the antigen binding domain comprises LC CDR1 (SEQ ID NO:112), LC CDR2 (SEQ ID NO:113), and LC CDR3 (SEQ ID NO:114); and HC CDR1 (SEQ ID NO:115), HC CDR2 (SEQ ID NO:116), and HC CDR3 (SEQ ID NO: 117), optionally wherein the antigen binding domain comprises a light chain variable region comprising sequence SEQ ID NO:110 and a heavy chain variable region comprising sequence SEQ ID NO:111.

18. A cell comprising the nucleic acid or expression vector of claim **17**.

19. A method of treating a subject for cancer, the method comprising administering to the subject an effective amount of the nucleic acid or expression vector of claim **17**, or a cell comprising the nucleic acid or expression vector.

20. A method of making a population of therapeutic cells, the method comprising:

a) contacting a human cell with a nucleic acid comprising a chimeric antigen receptor (CAR) sequence encoding a CAR and a sequence encoding a decoy-resistant interleukin 18 (DR-18) polypeptide; wherein the nucleic acid is configured to result in expression and secretion of the DR-18 by the human cell ex vivo;

b) culturing the contacted human cell ex vivo in the presence of the secreted DR-18 polypeptide, thereby producing a population of therapeutic cells, wherein the population of therapeutic cells is enhanced as compared to a corresponding population of cells generated without the sequence encoding the DR-18 polypeptide.

* * * * *